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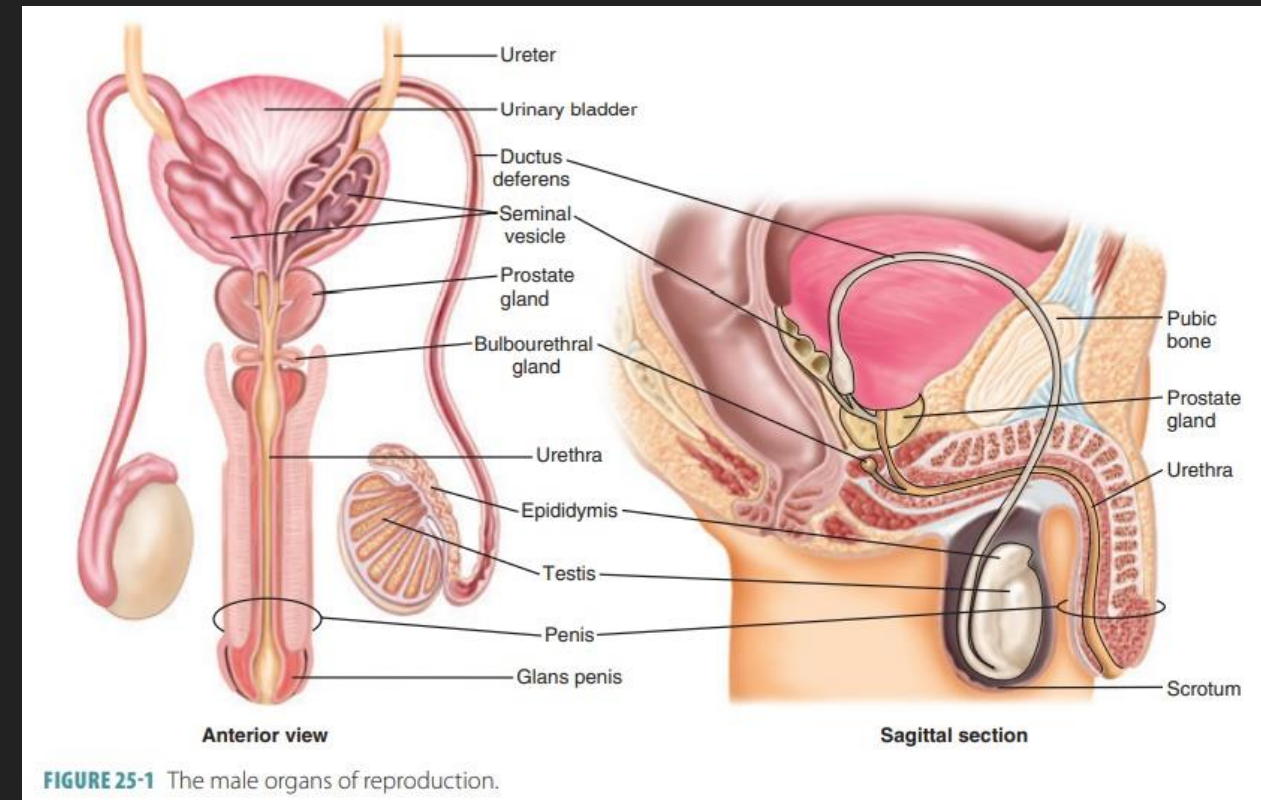
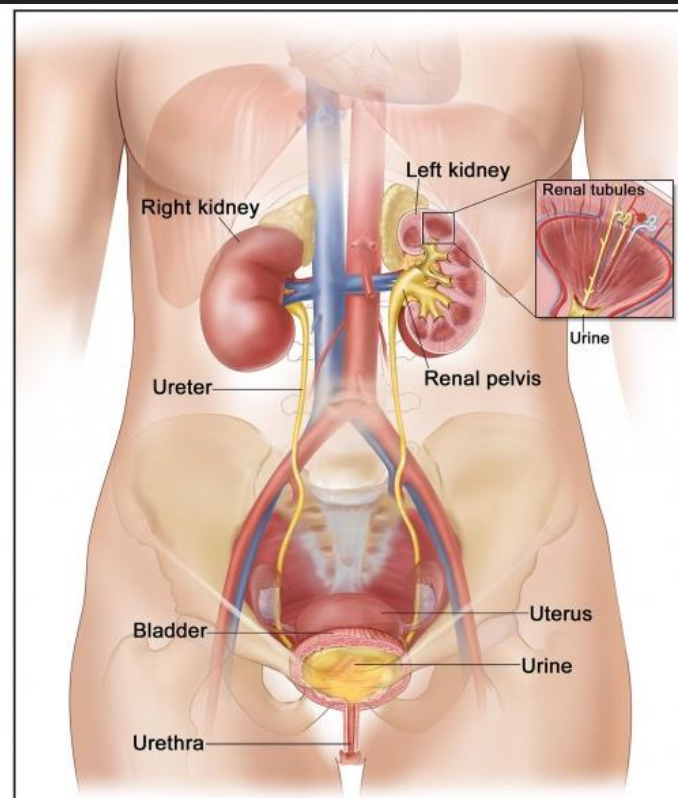
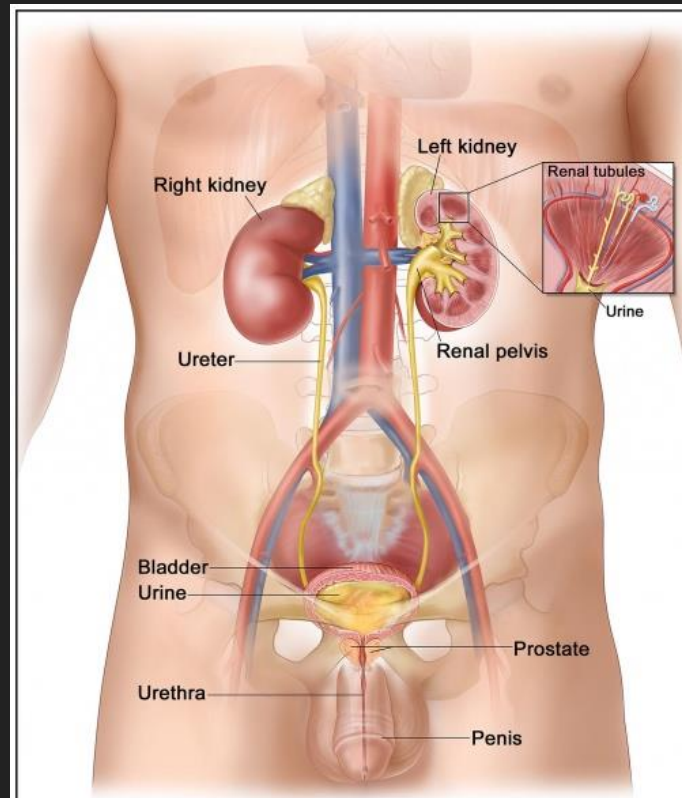
ANATOMY AND PHYSIOLOGY

URINARY TRACT

UROLOGY

- Urinary system, KUB system
- Male reproductive system

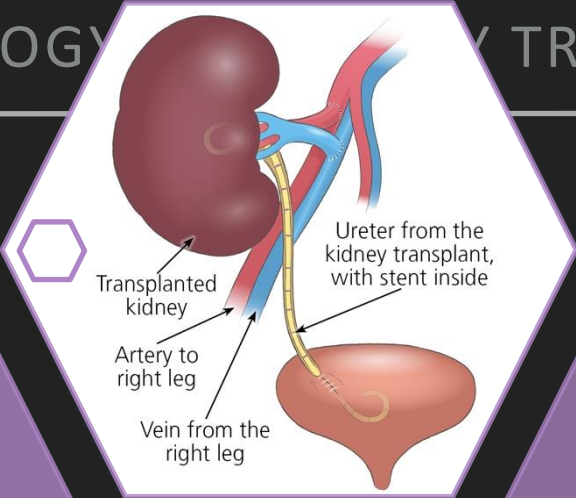
สาขาวิชาศัลยศาสตร์ยูโรวิทยา
สาขาวิชาศัลยศาสตร์ทางเดินปัสสาวะ และระบบสืบพันธุ์เพศชาย



UROLOGY: SUBSPECIALTIES

- ▶ Pediatric urology
- ▶ Onco-urology
- ▶ Endo-urology
- ▶ Female urology
- ▶ Men's health
- ▶ Transplantation
- ▶ Neuro-urology
- ▶ Minimally invasive surgery
- ▶ Reconstructive urology
- ▶ Transgender urology
- ▶ Geriatric urology

BASIC ANATOMY AND PHYSIOLOGY OF UROLOGICAL TRACT



Urologic
Oncology

Renal
transplantation

Endourology &
Stone disease

MIS/Robotic Sx

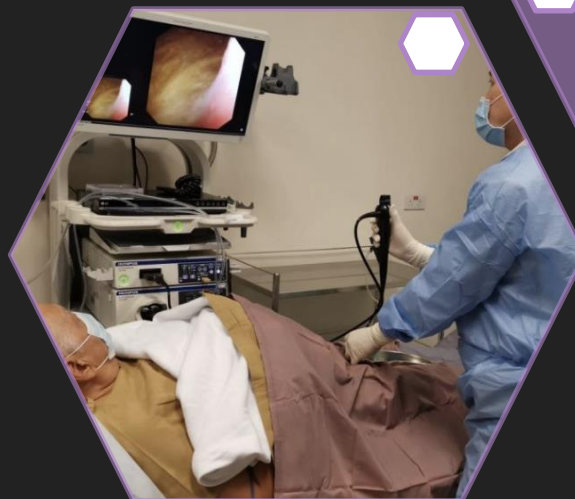
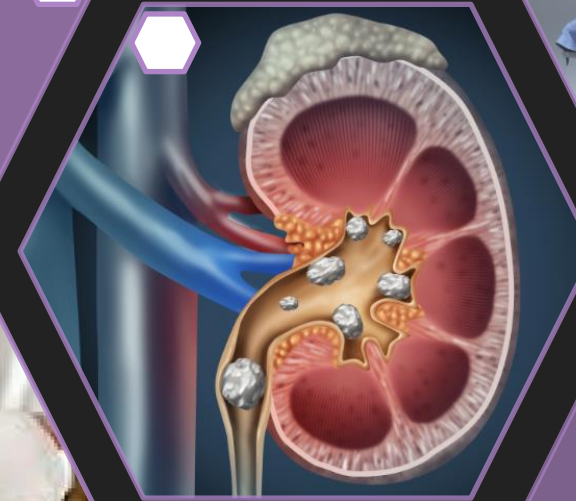
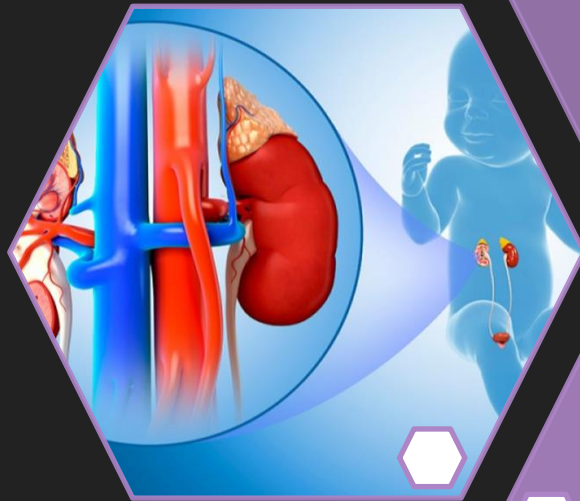
Female Pelvic
Medicine &
Reconstructive
Surgery
(FPMRS)

Pediatric
Urology

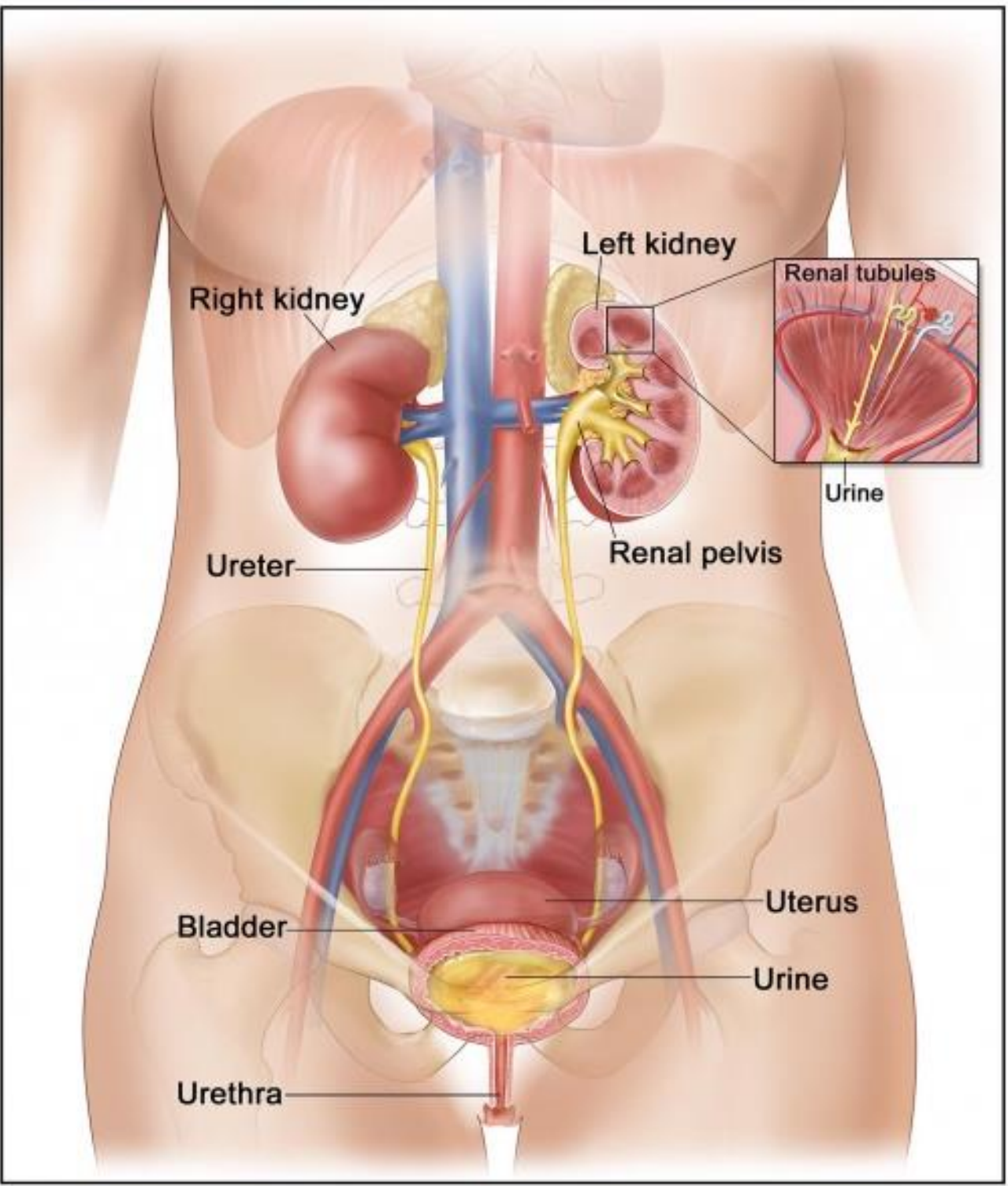
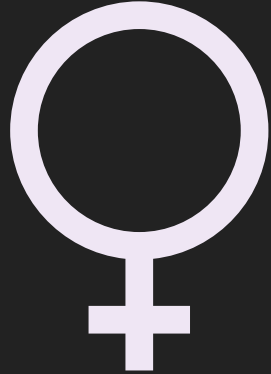
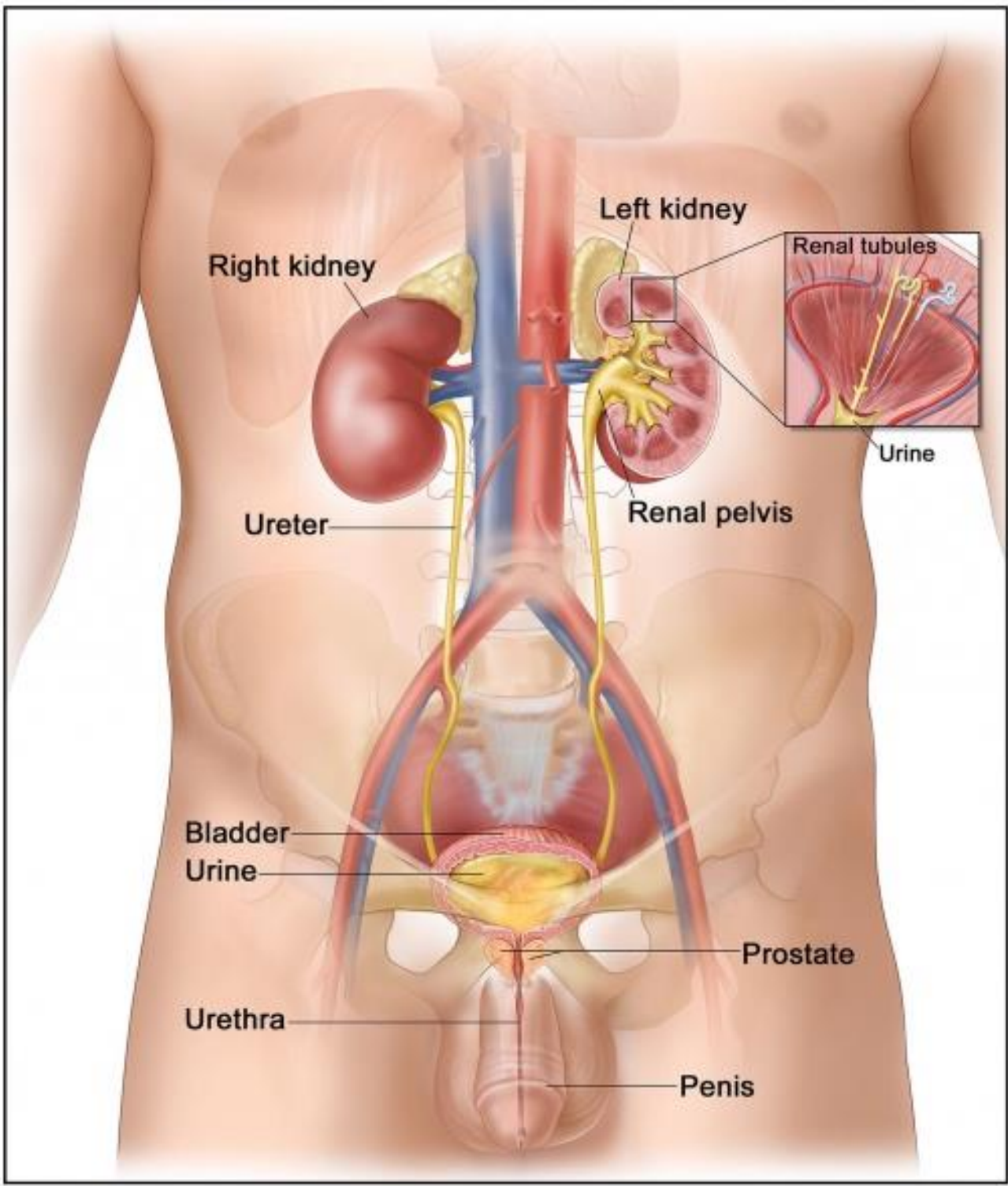
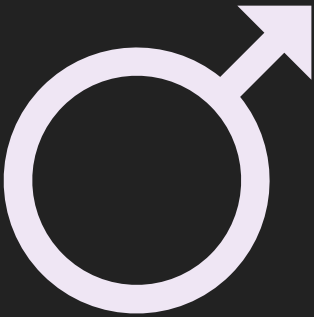
Neurourology

Male infertility
& Andrology

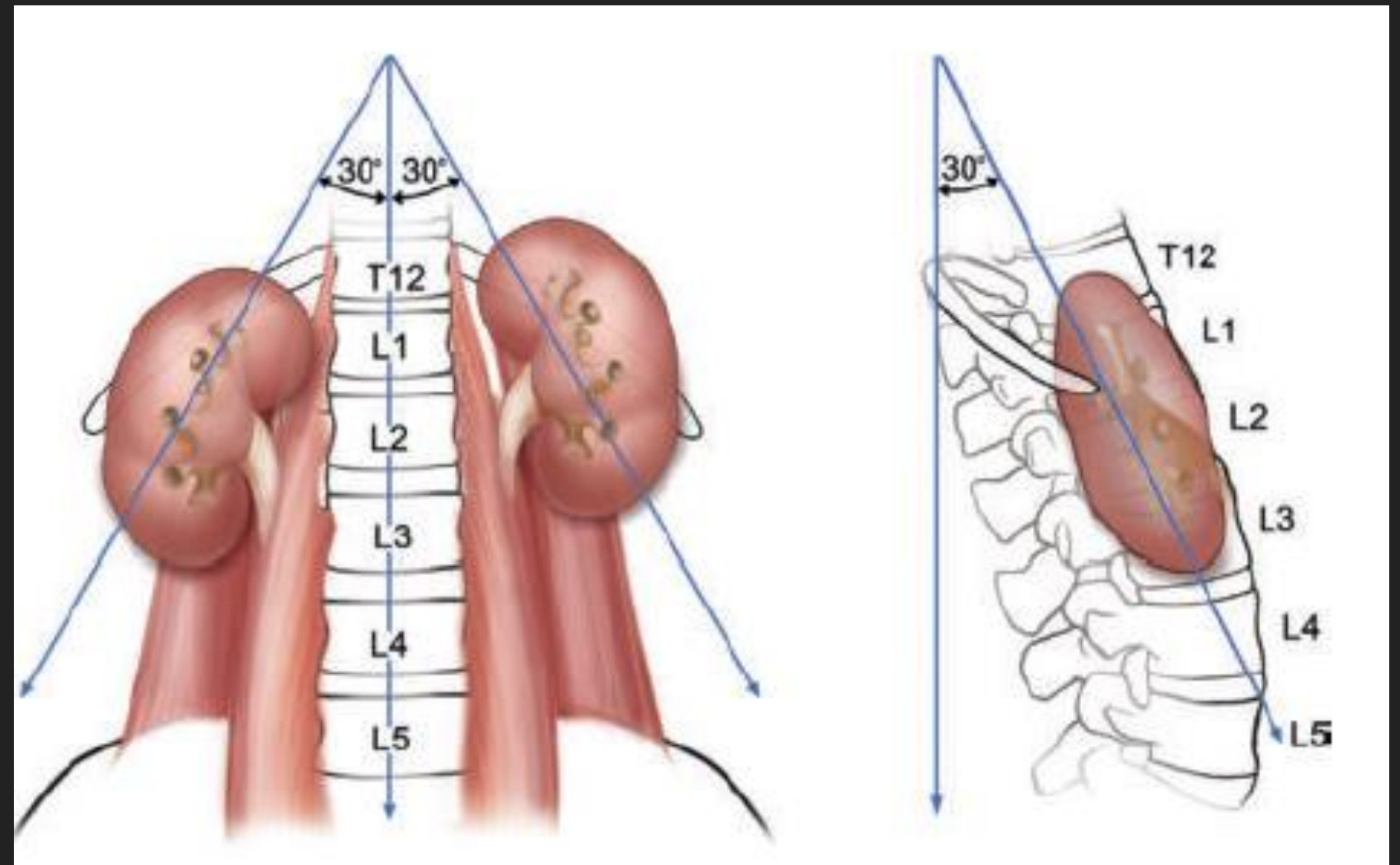
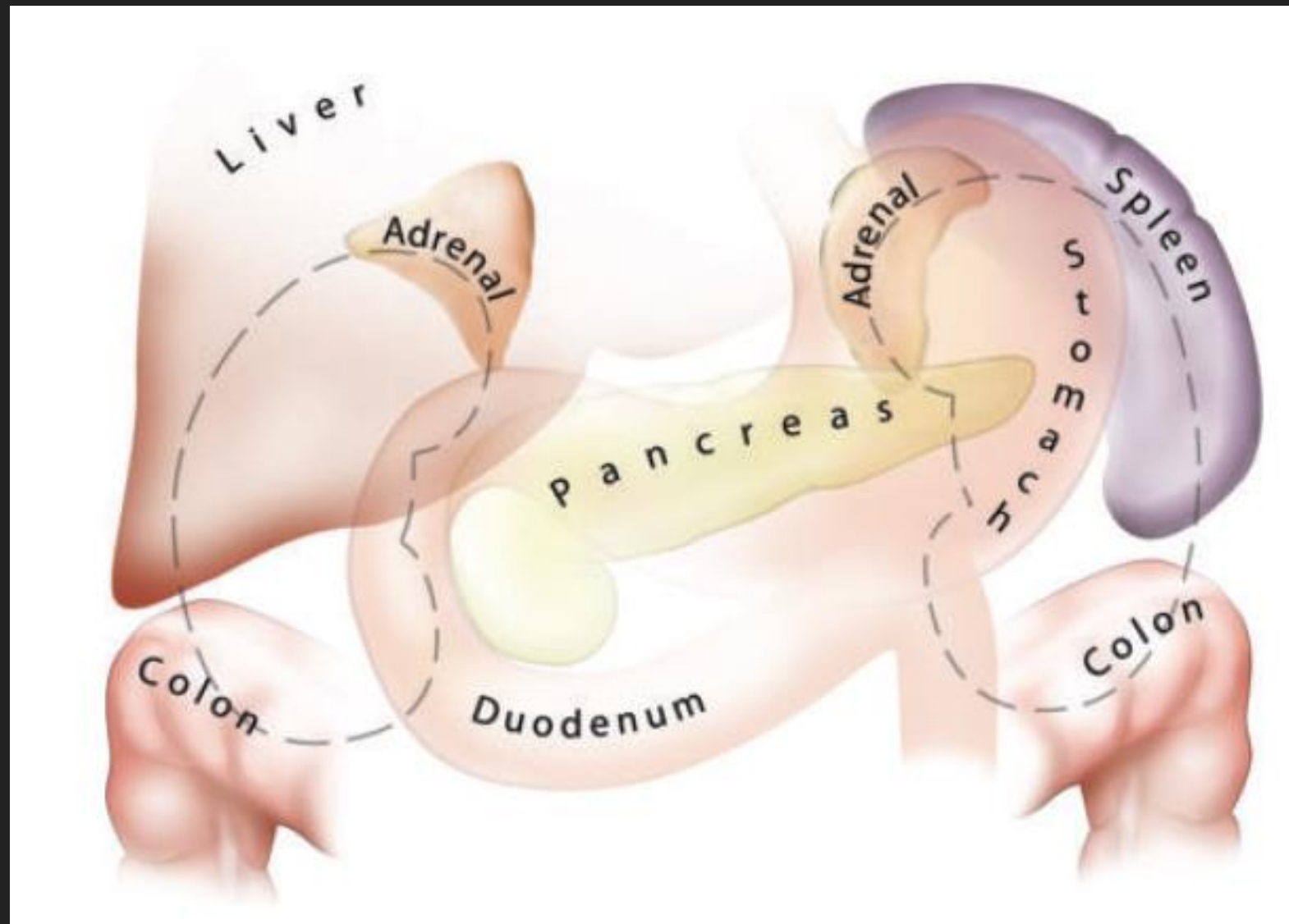
Reconstructive
Urology



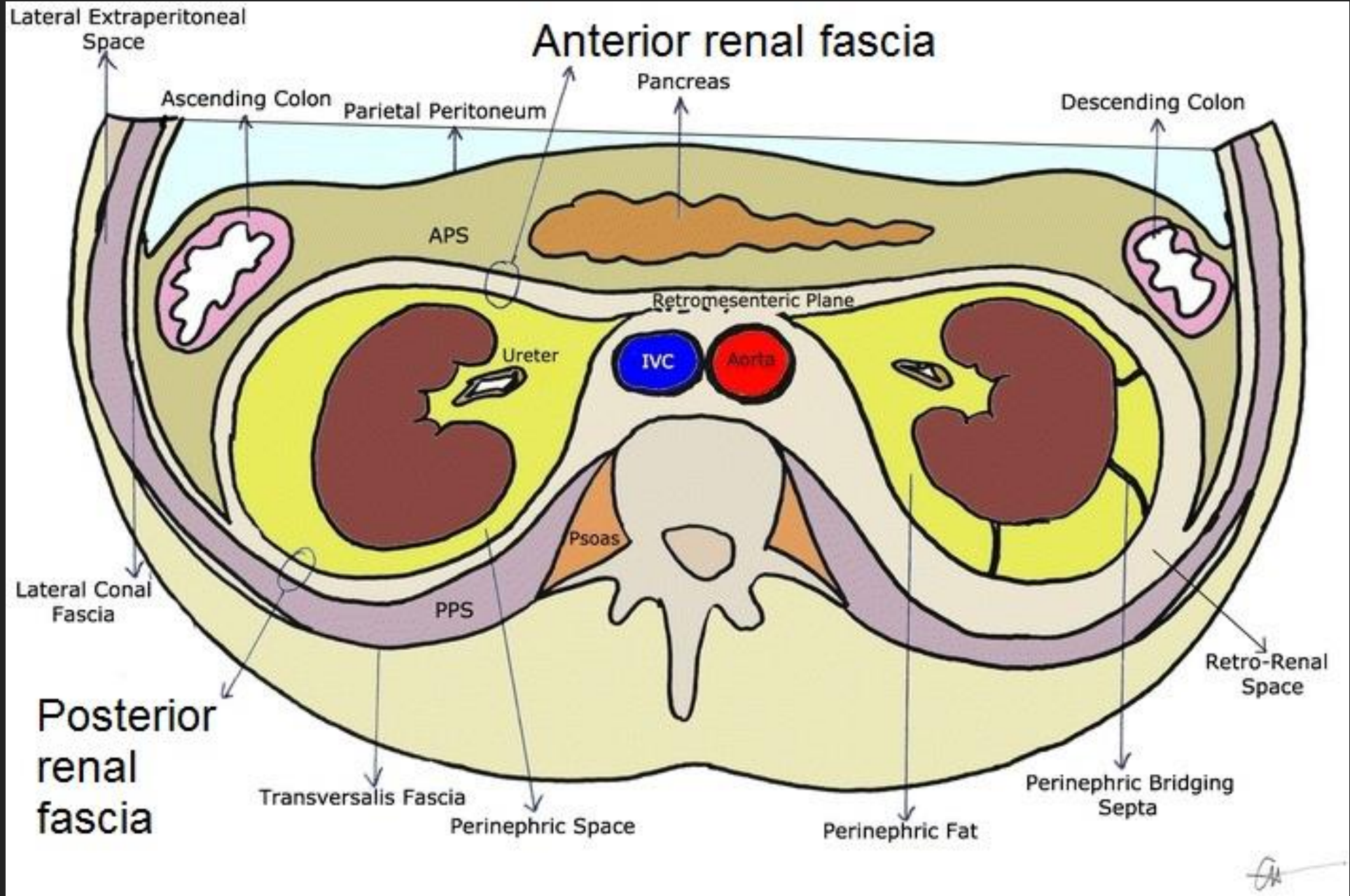
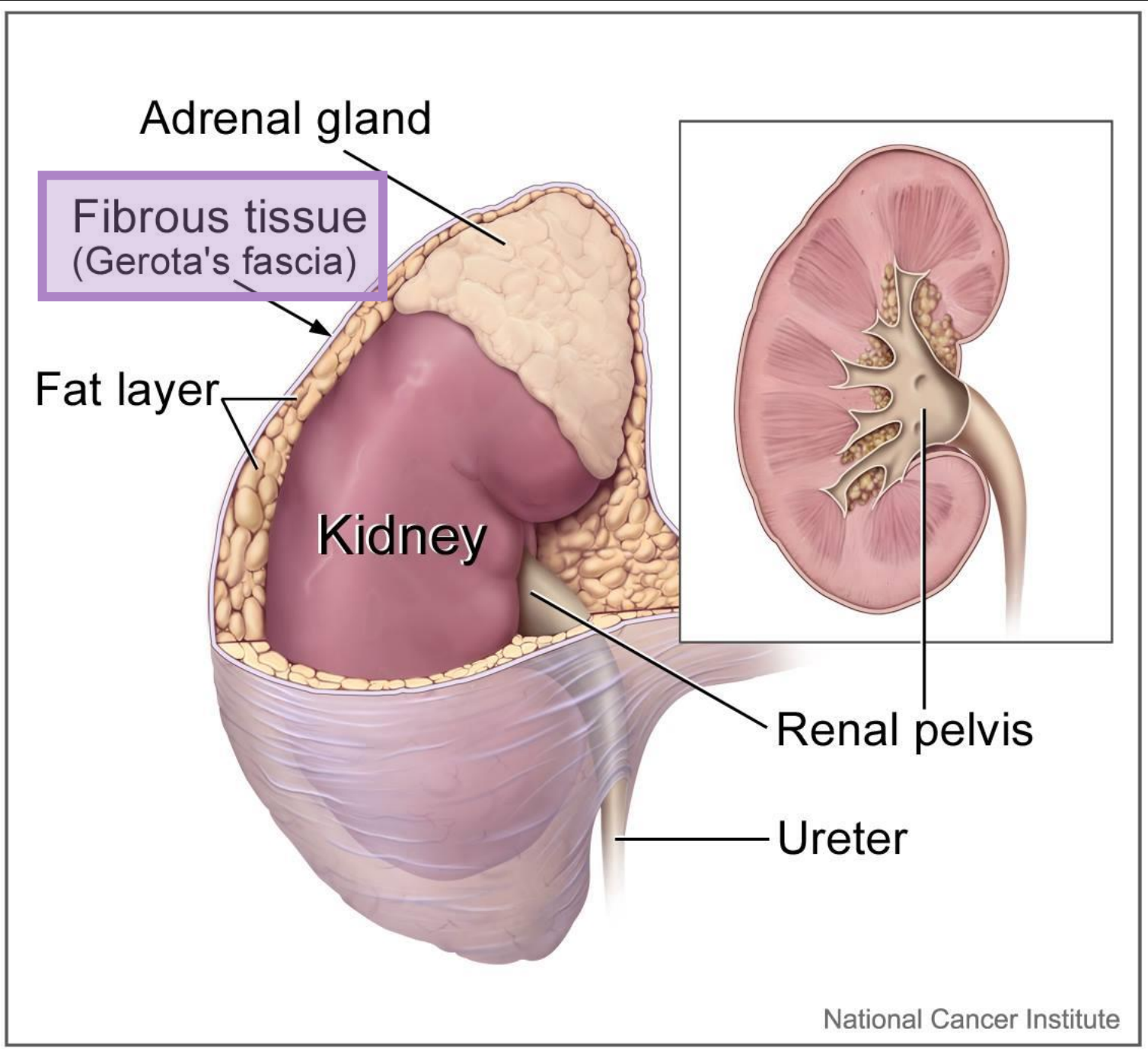
URINARY TRACT ANATOMY



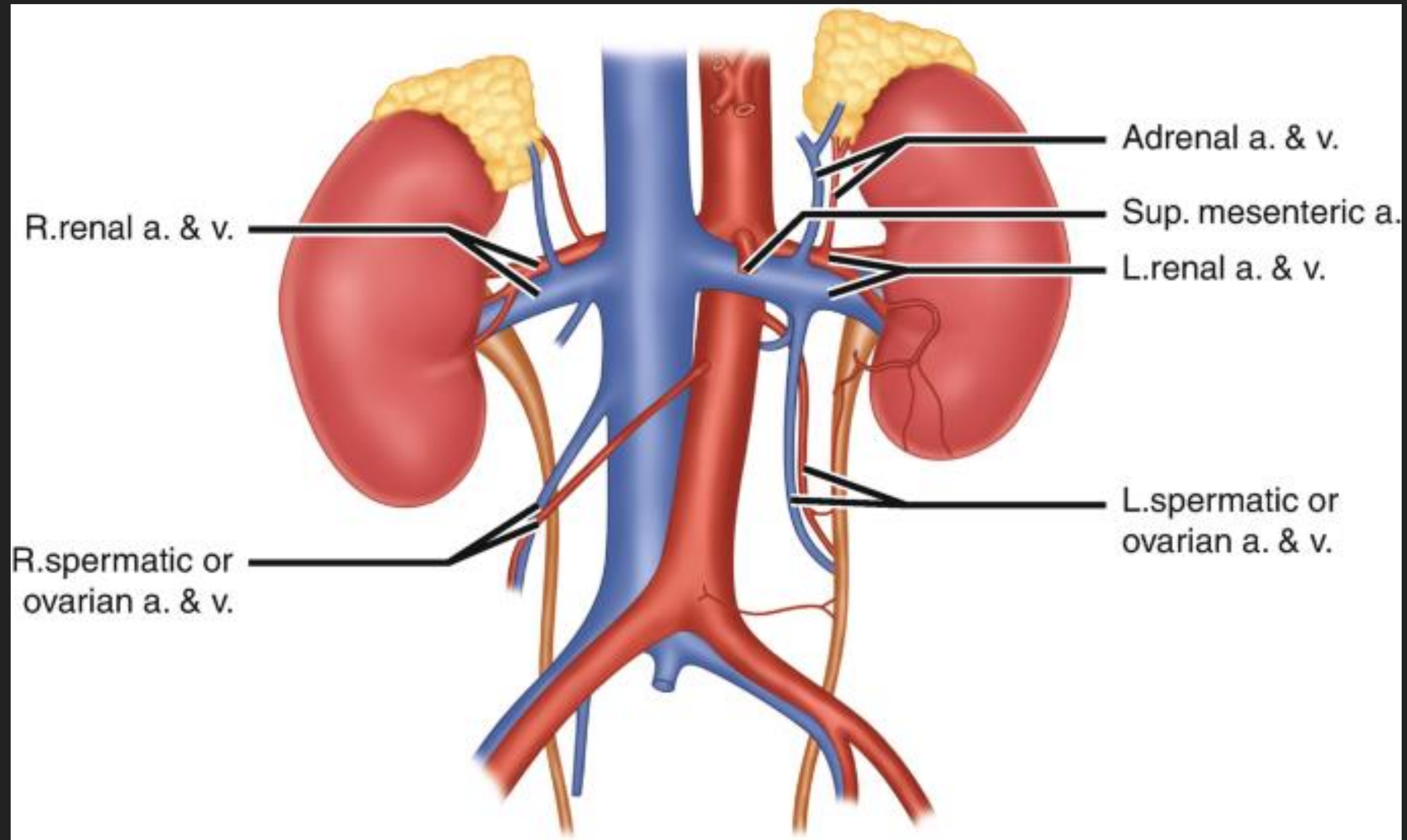
KIDNEY: ANATOMY



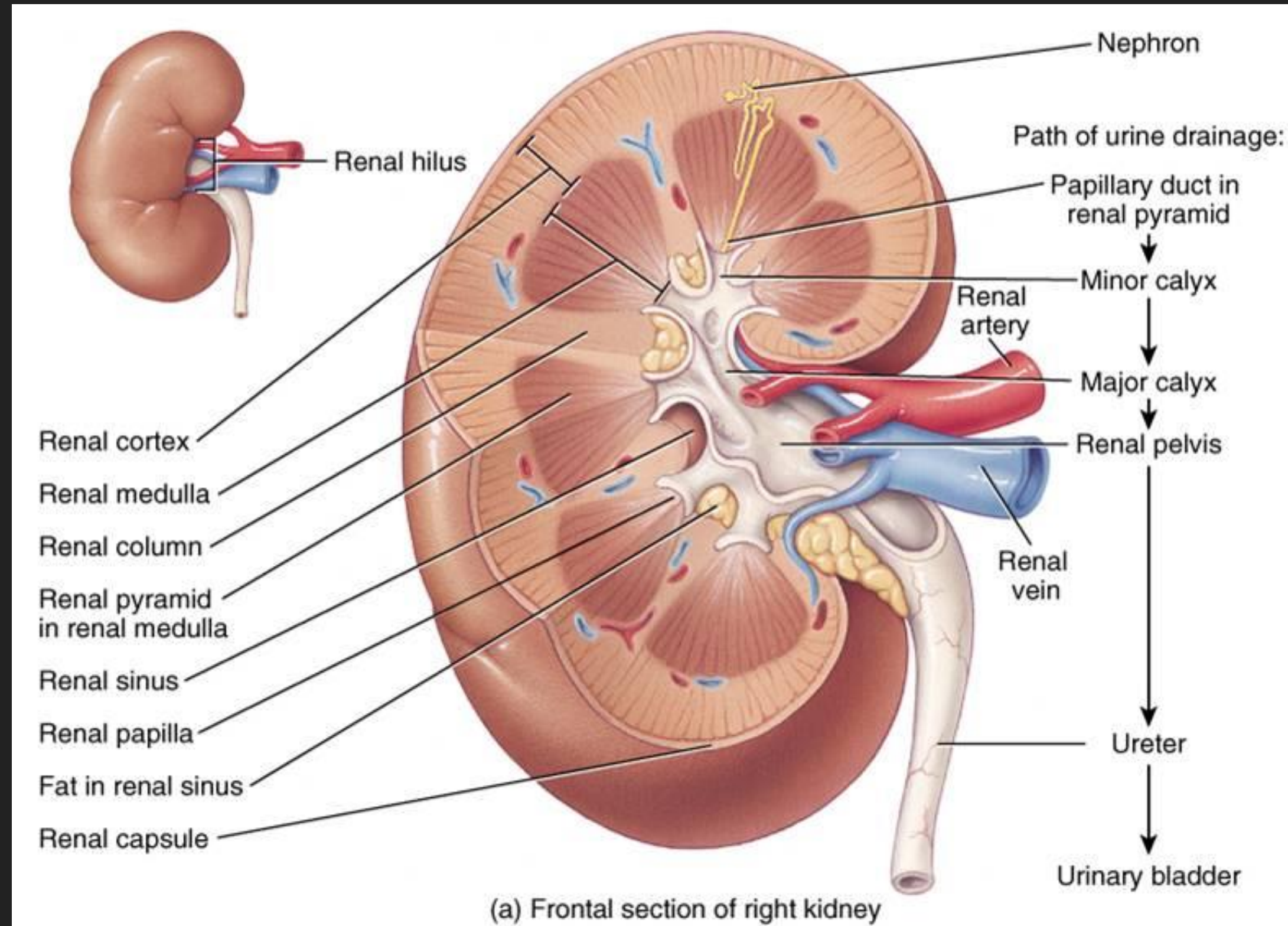
KIDNEY: ANATOMY



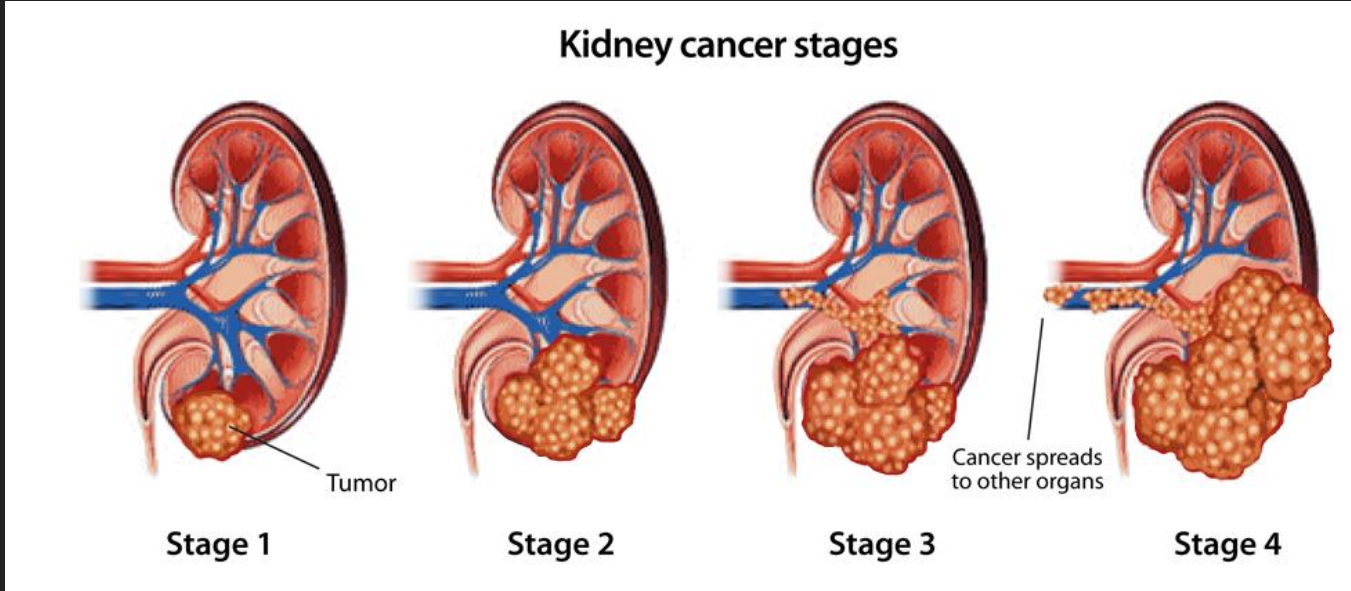
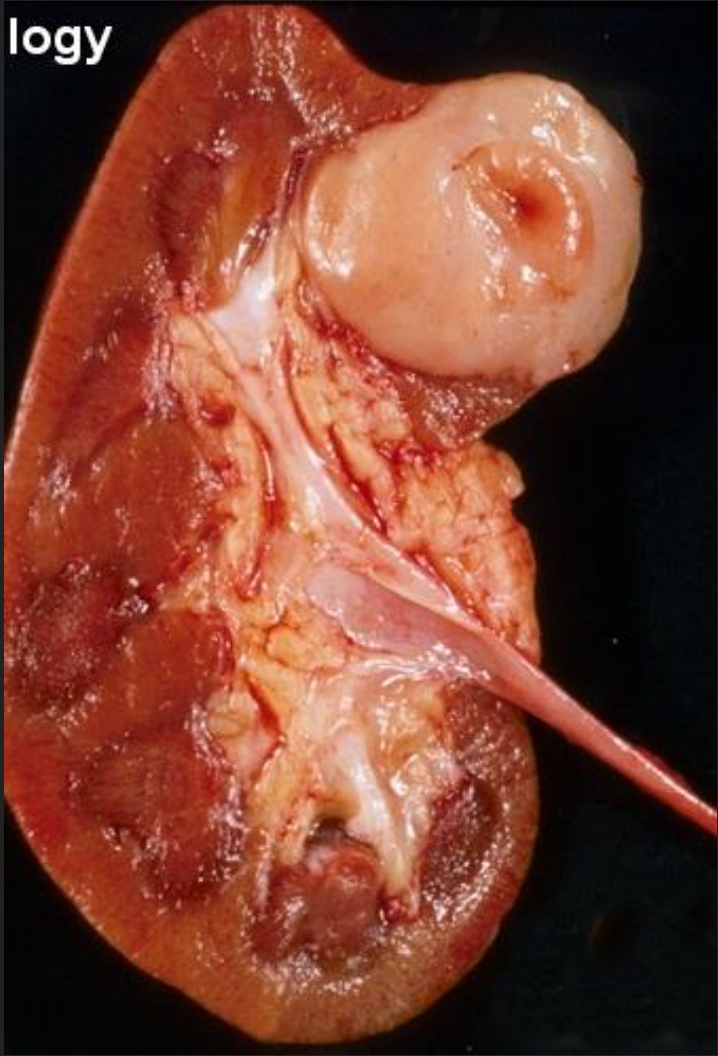
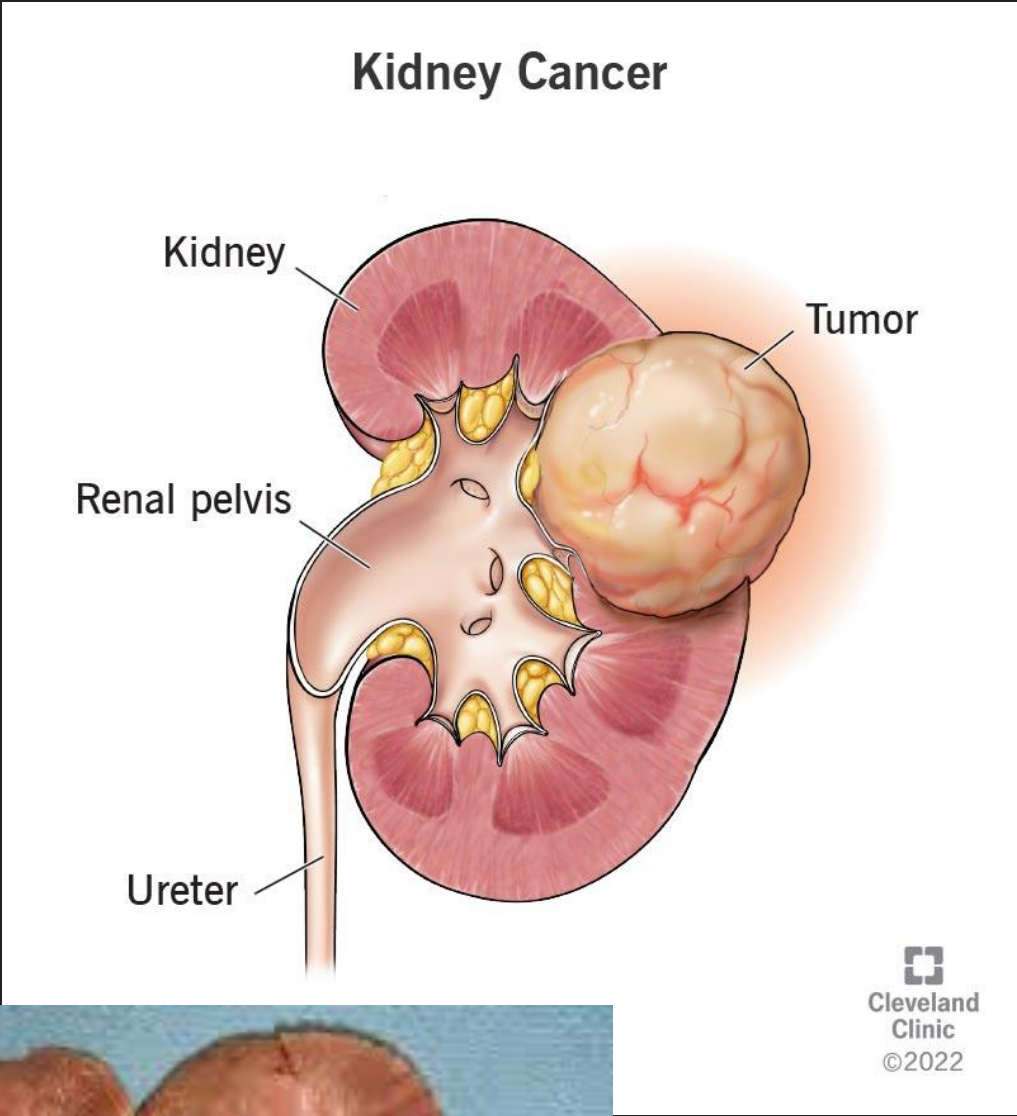
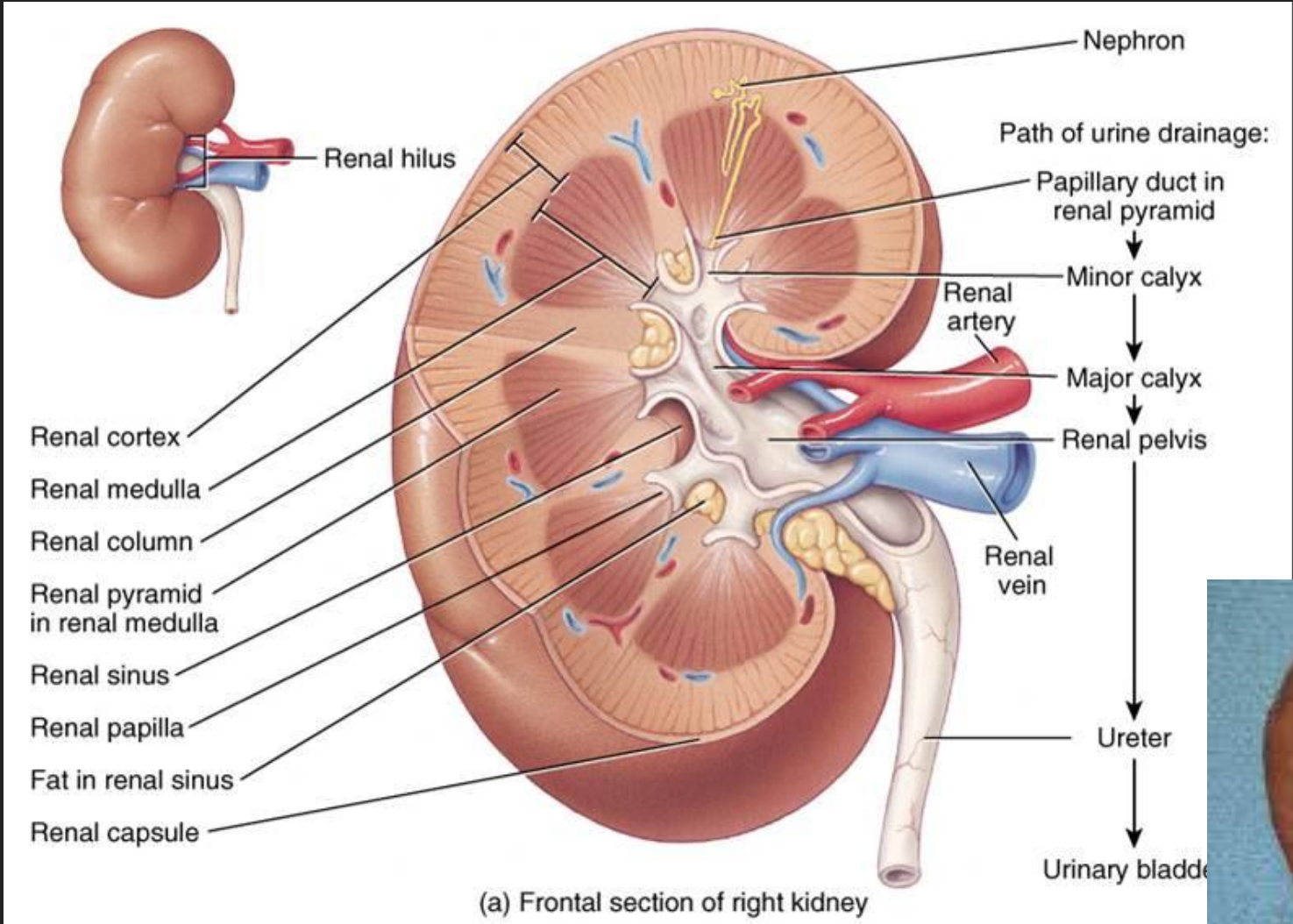
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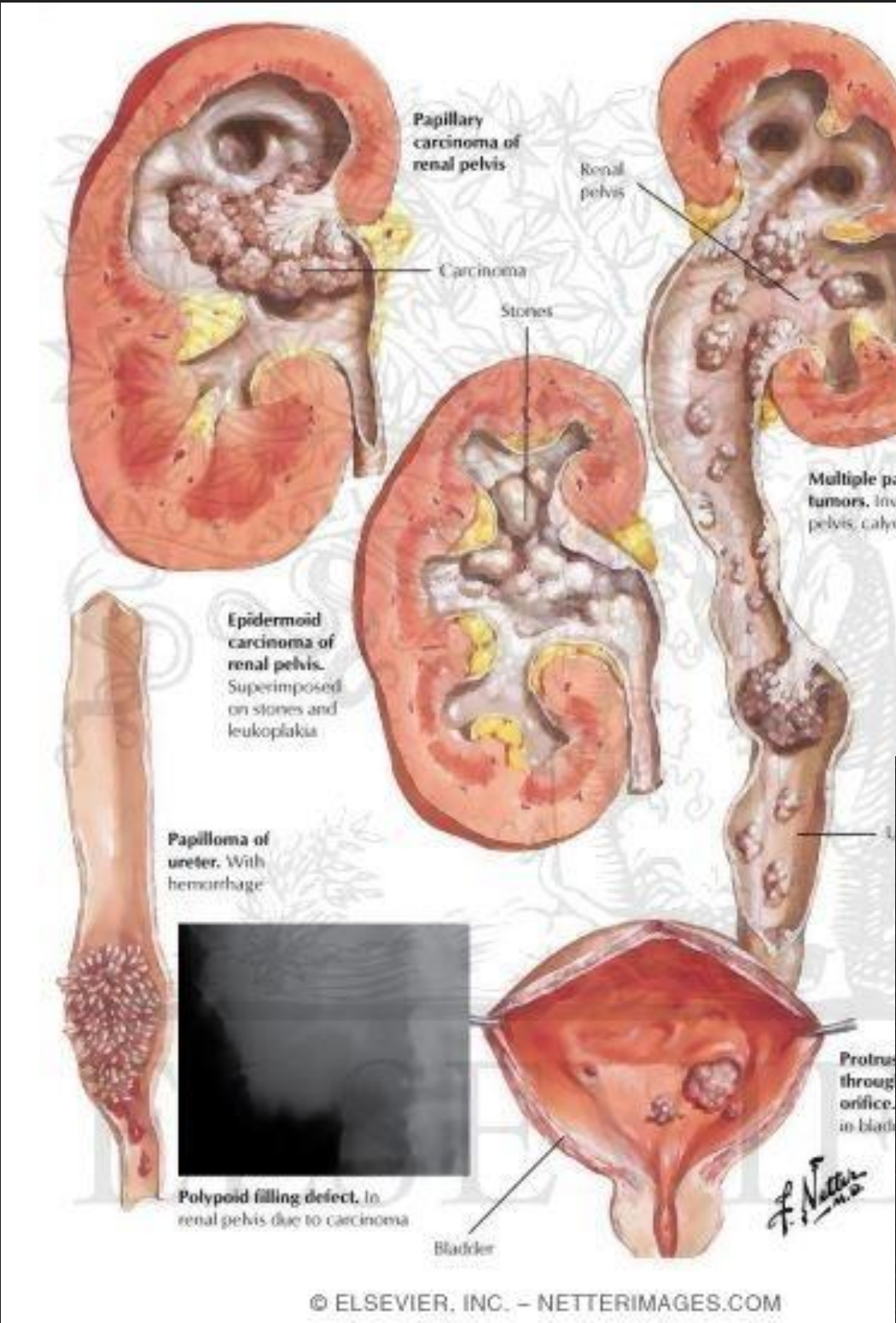
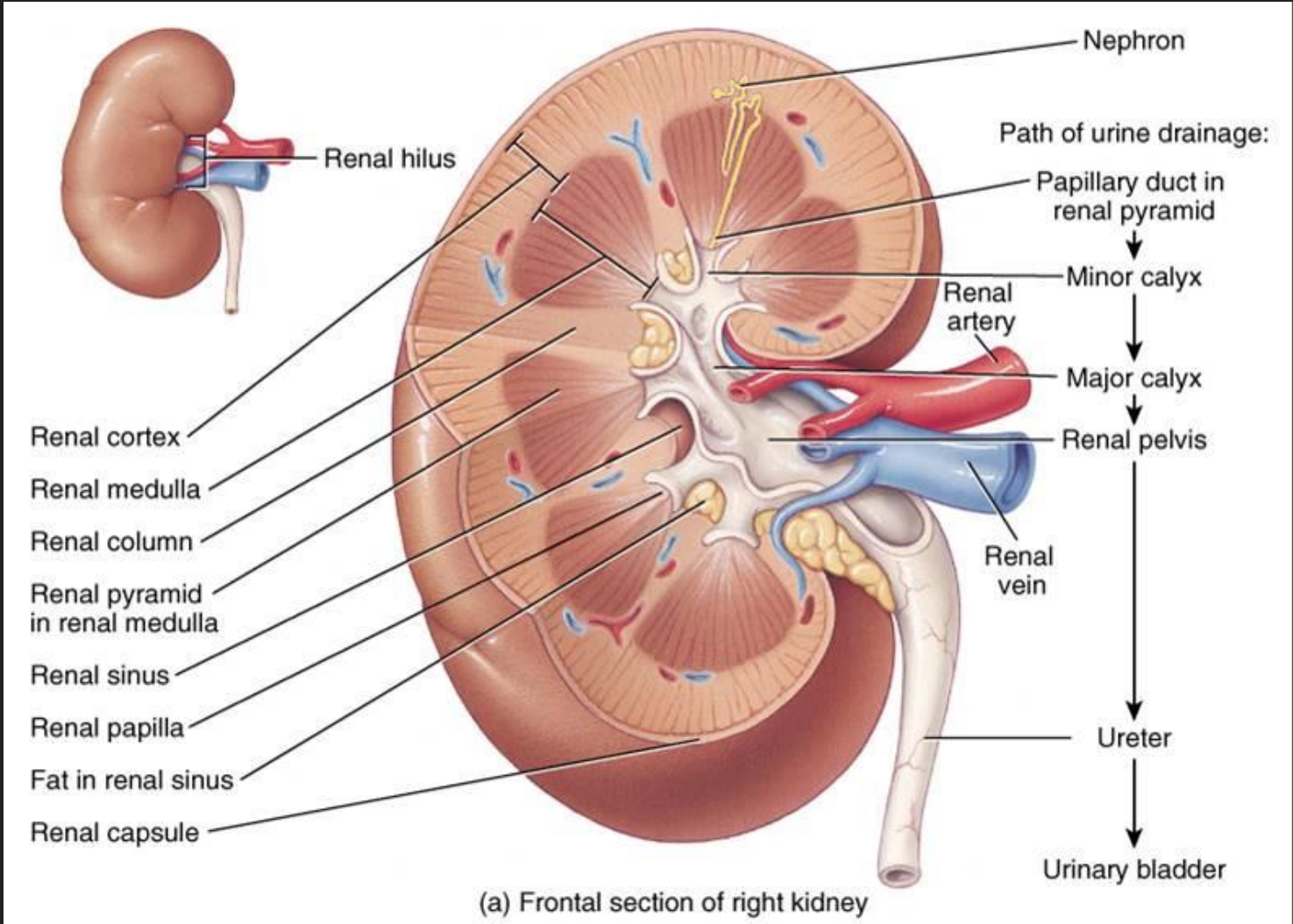
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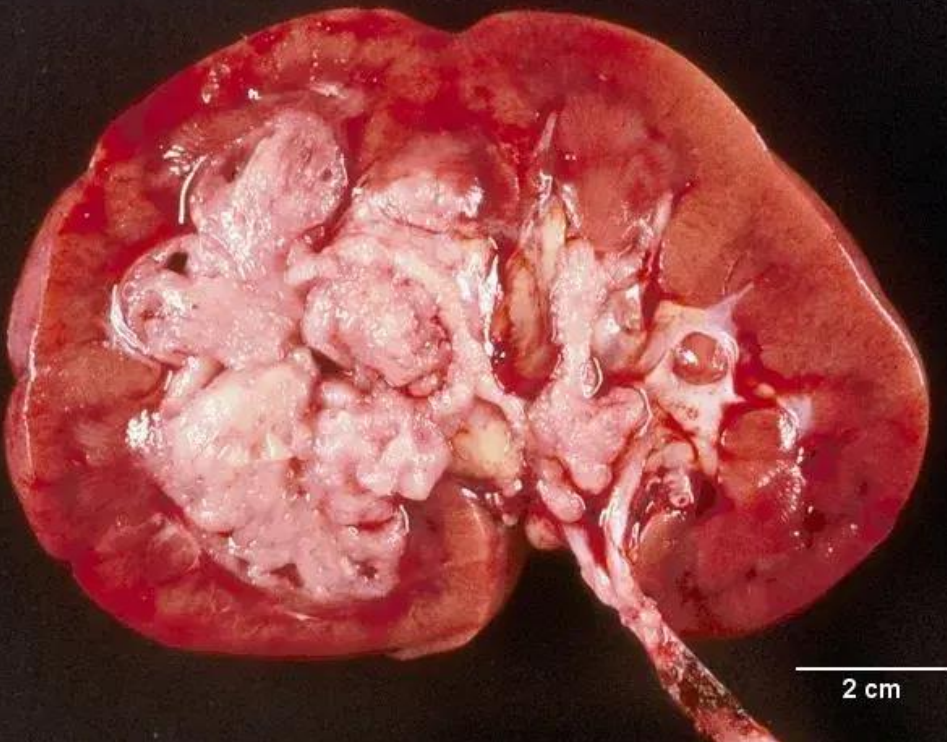
KIDNEY: ANATOMY



KIDNEY: ANATOMY

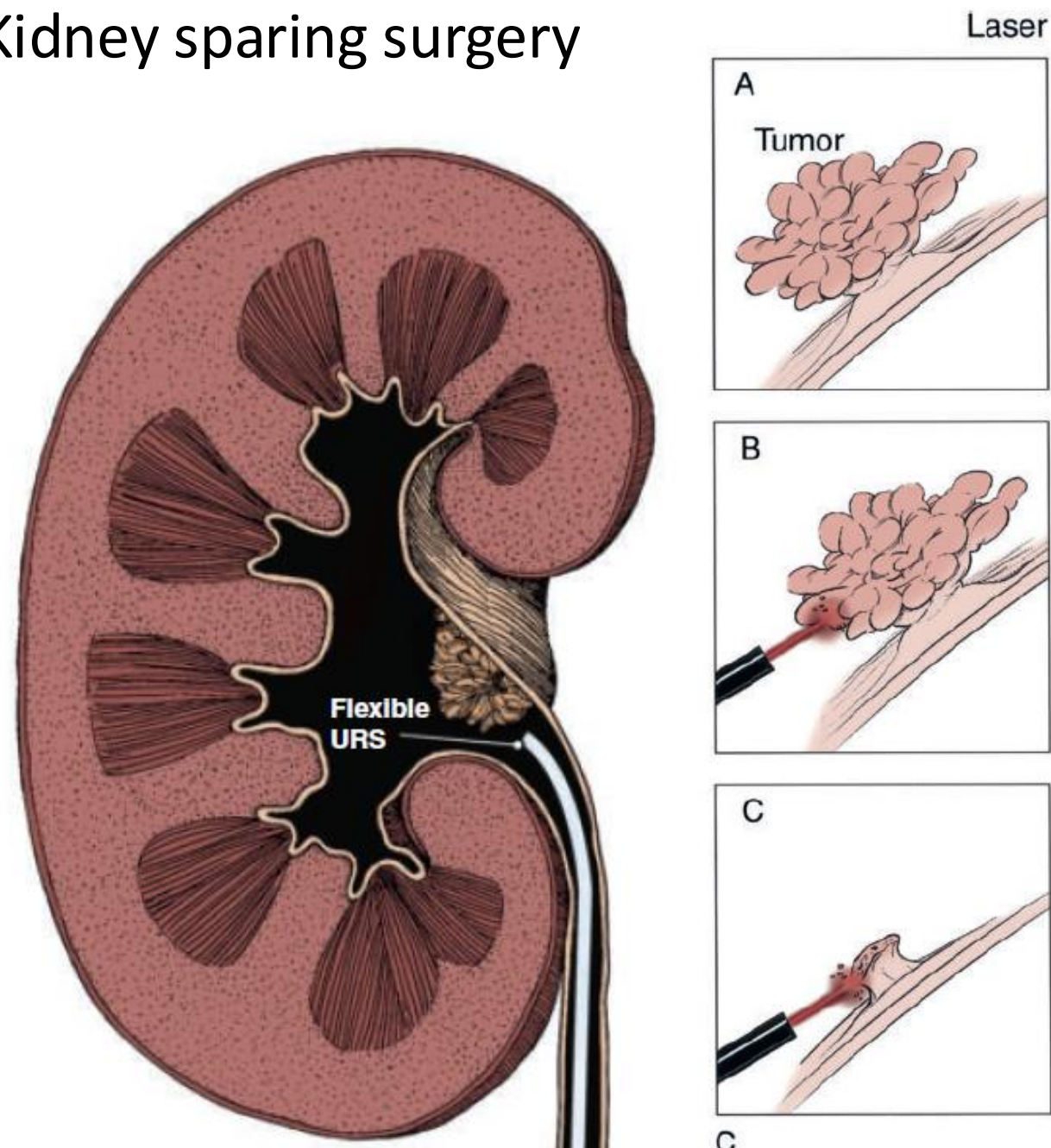


©WebPathology

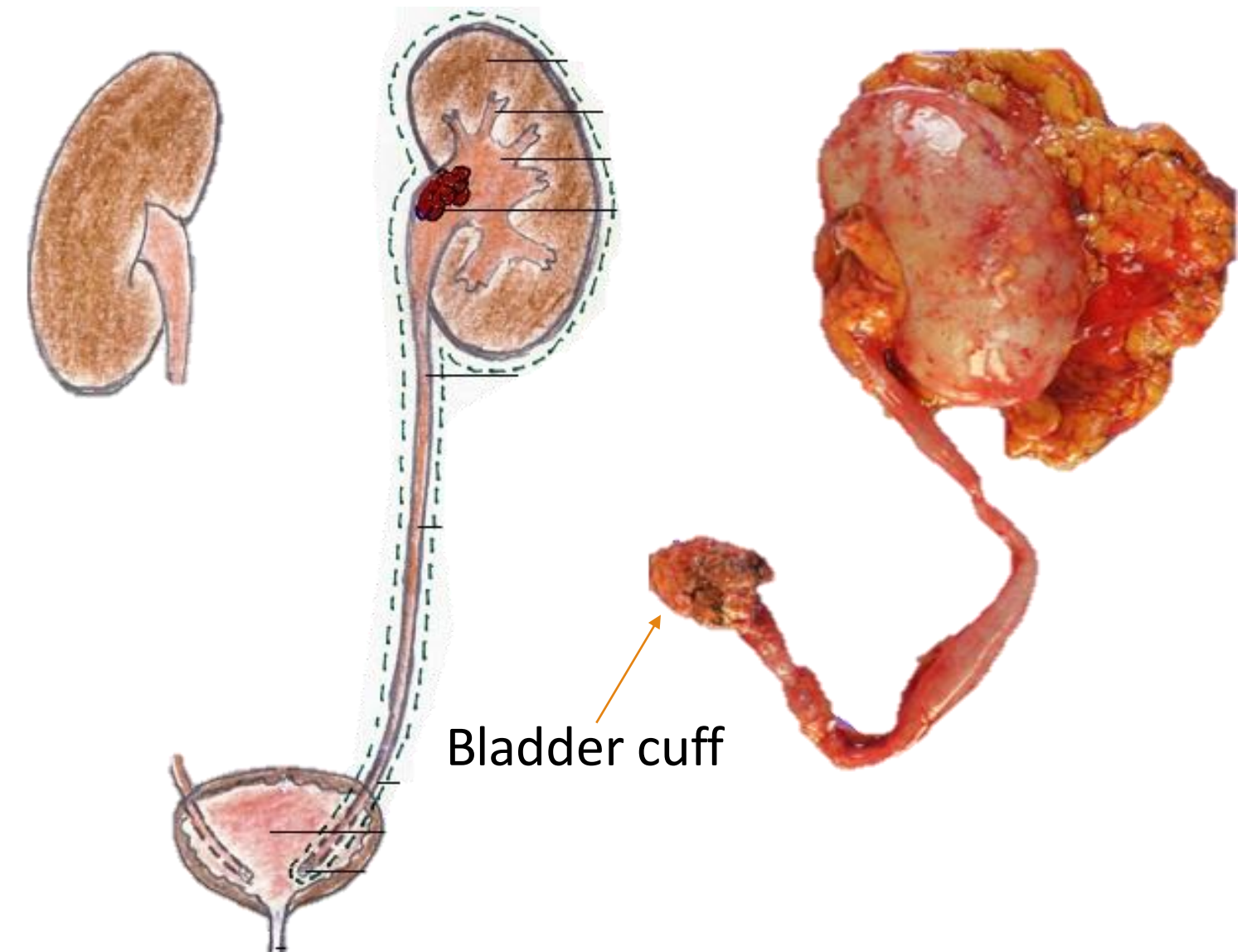


Upper urinary tract Cancer (UTUC) Management

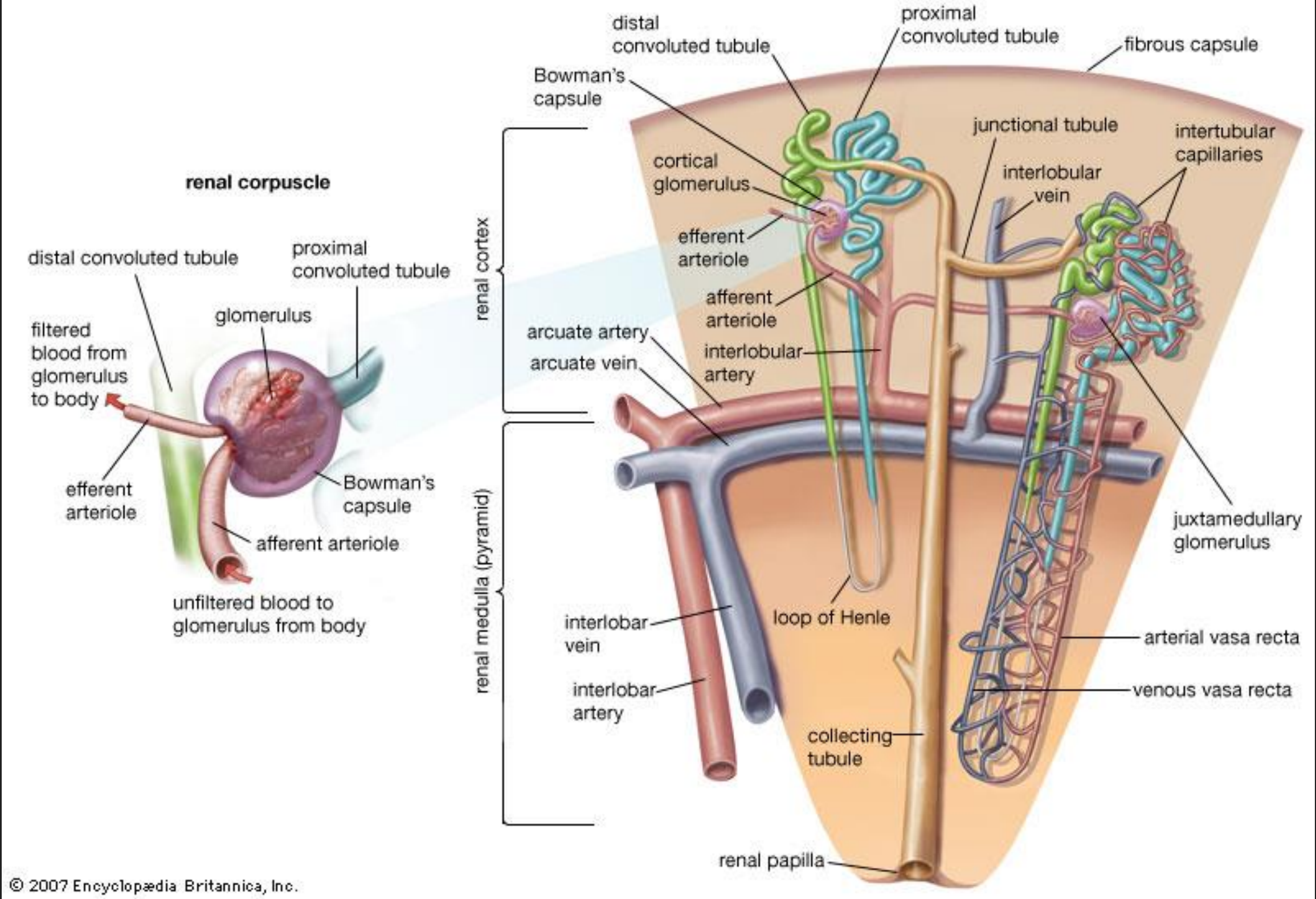
Kidney sparing surgery



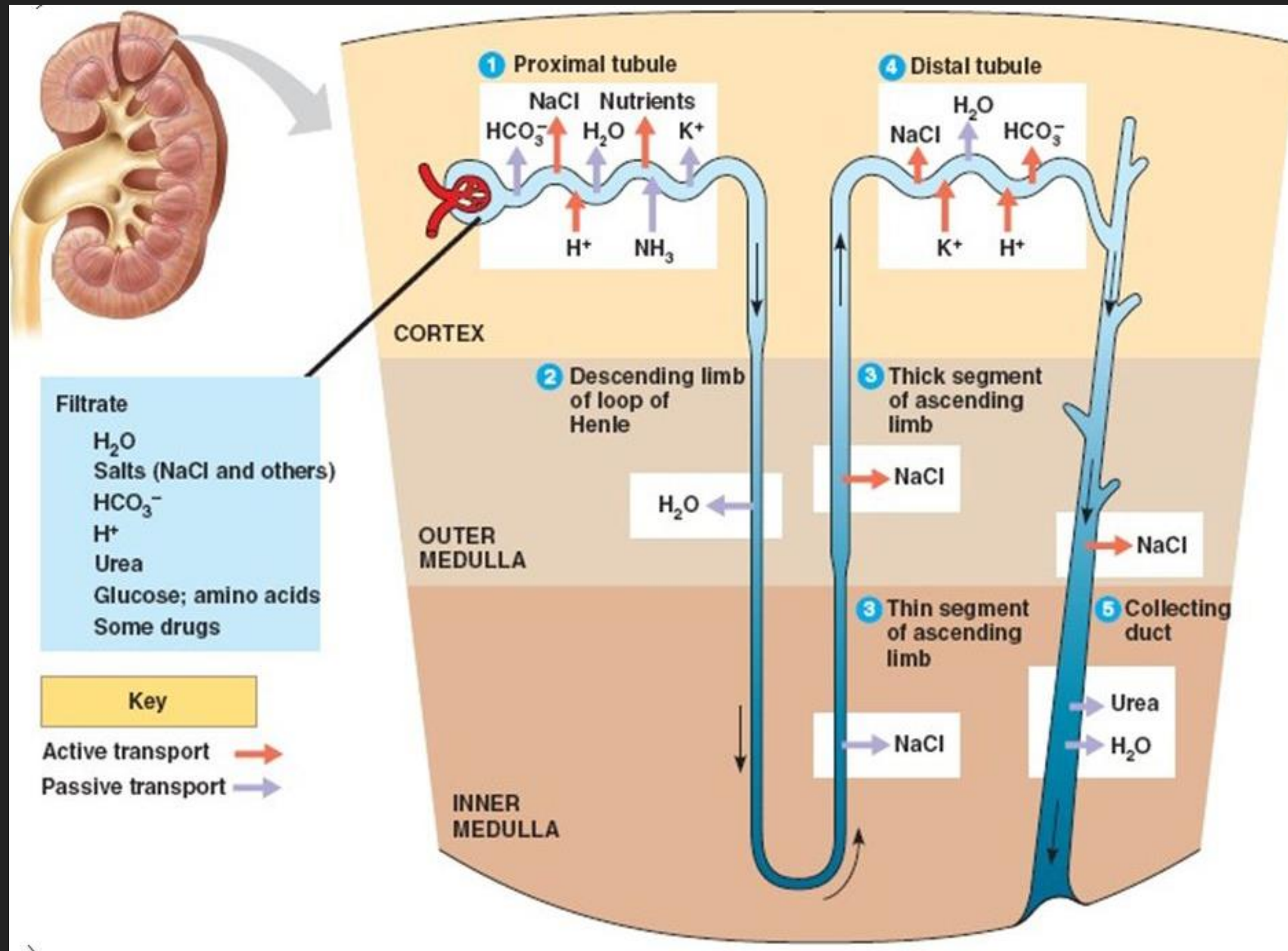
❖ *Nephroureterectomy with bladder cuff excision*



NEPHRON



NEPHRON



KIDNEY FUNCTION

Urine production

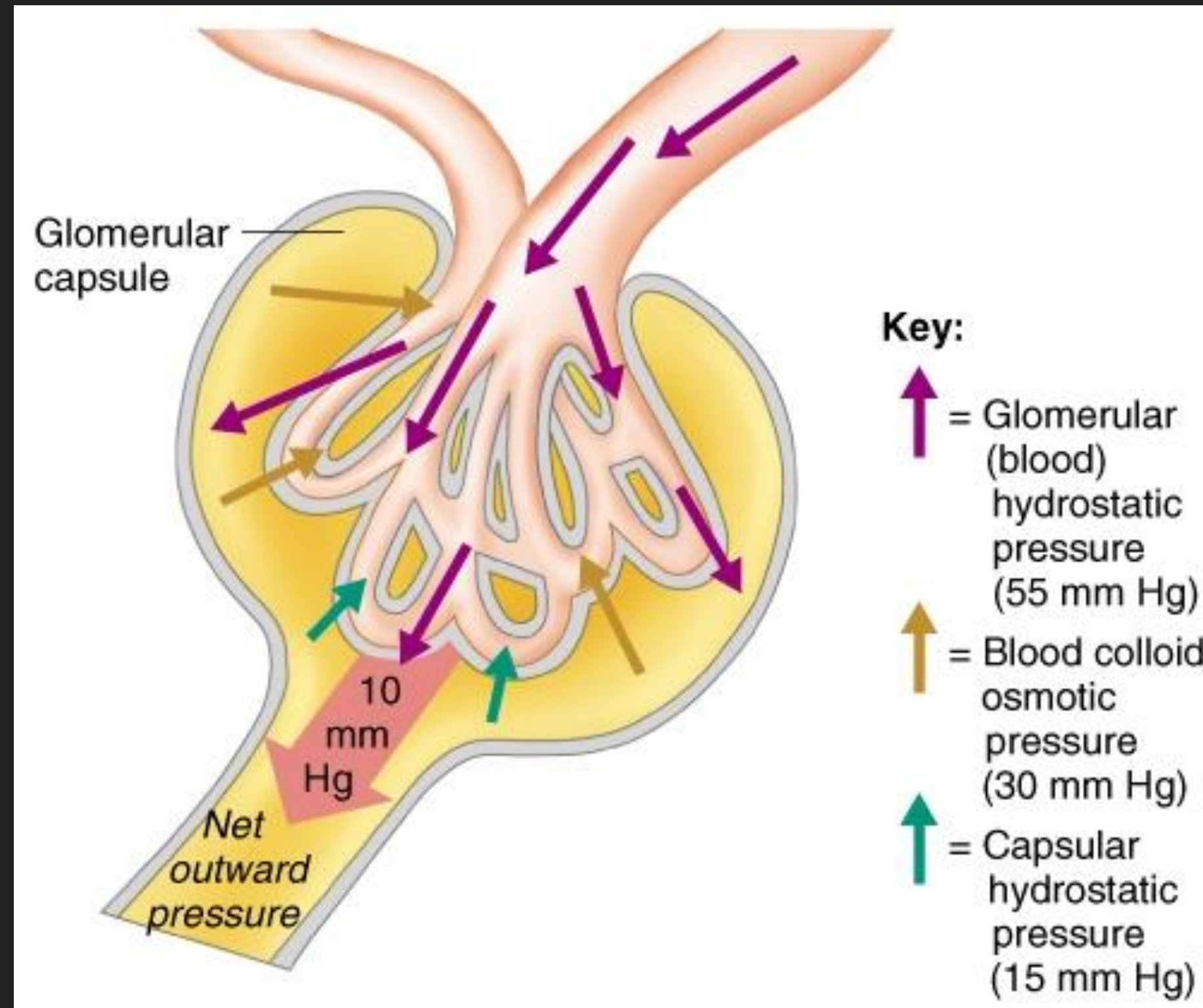
Acid-Base and Electrolyte balance

- Glomerular filtration
- Tubular reabsorption
- Tubular secretion

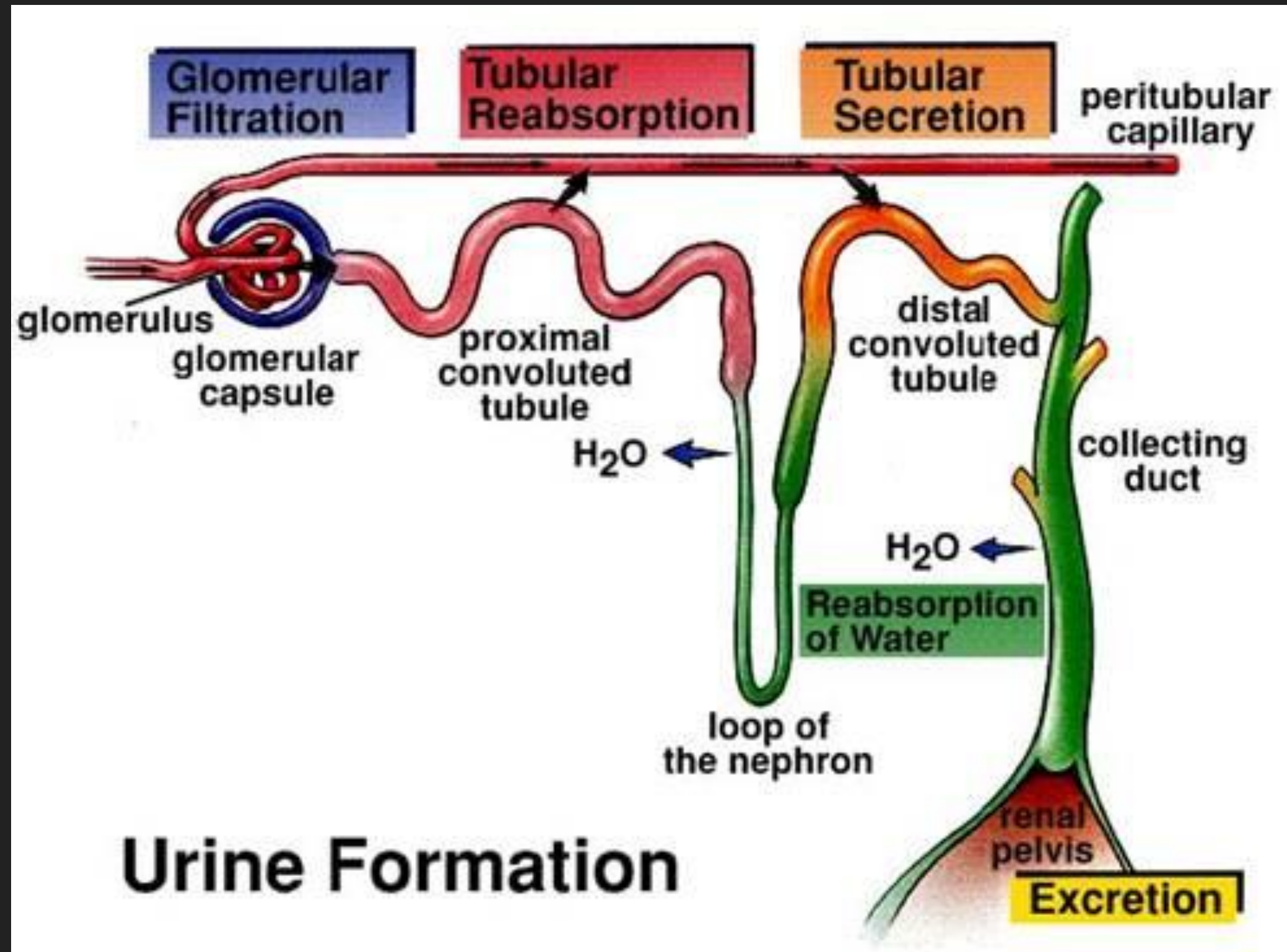
Hormonal function

- Control of renal vascular tone
 - *Vasoconstrictors*: Angiotensin II, Endothelin, Norepinephrine, Atrial natriuretic peptide(ANP), Vasopressin
 - *Vasodilators*: Nitric oxide(NO), Carbon monoxide
- Erythropoiesis: Erythropoietin
- Bone mineral regulation: PTH, Vit.D regulation
- Anti-diuretic hormone: ADH regulation

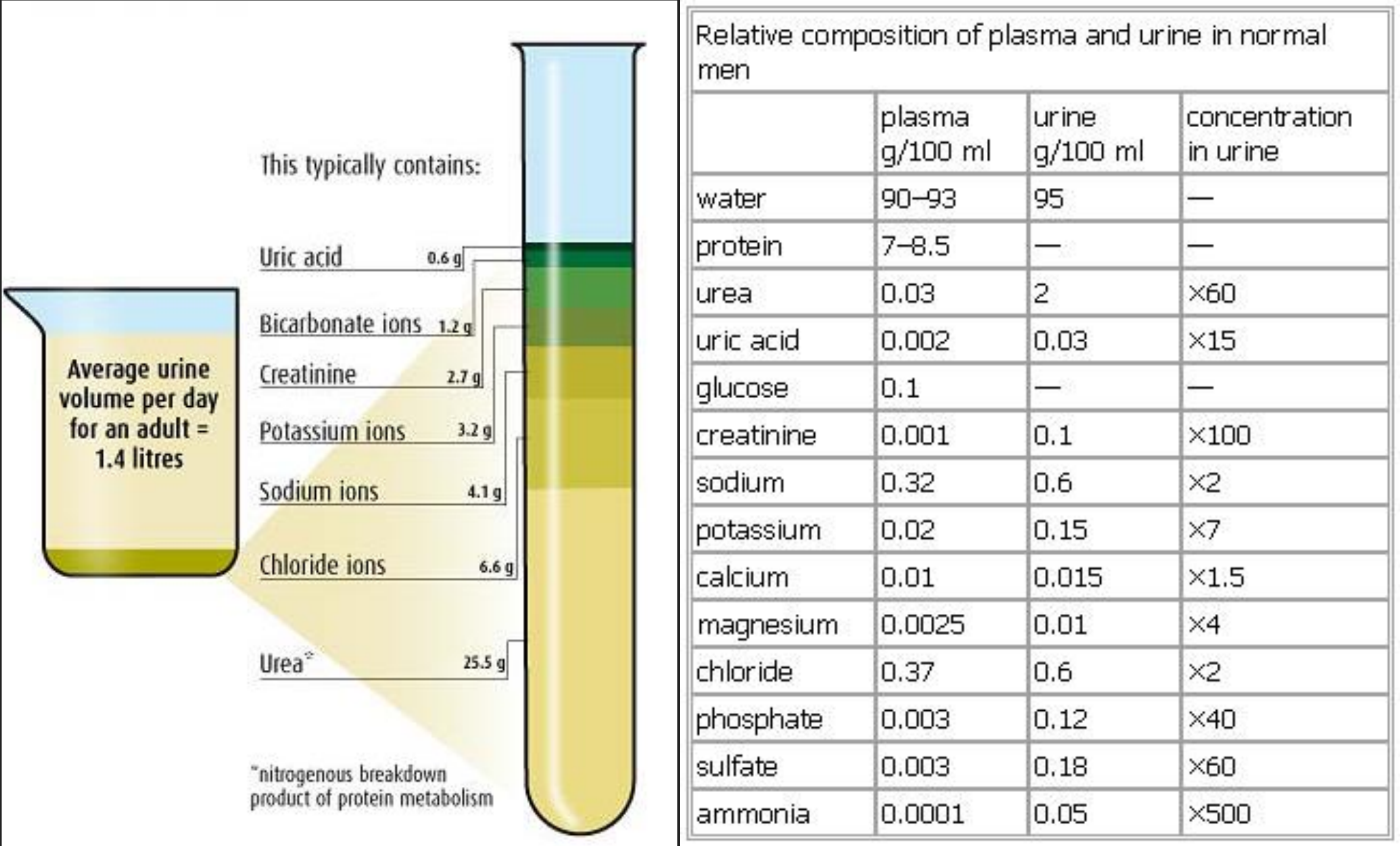
KIDNEY FUNCTION: URINE PRODUCTION



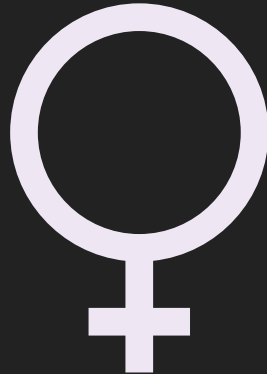
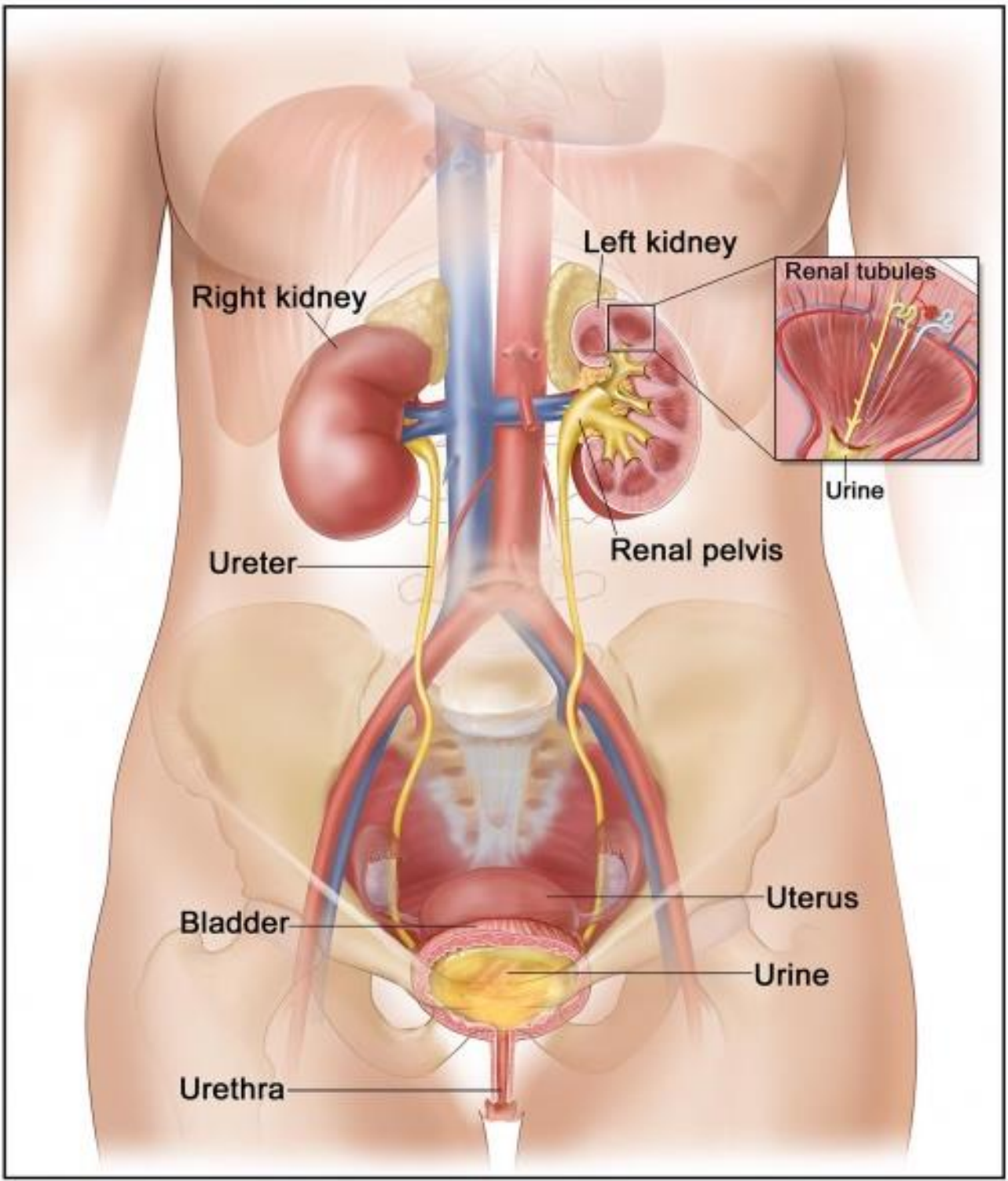
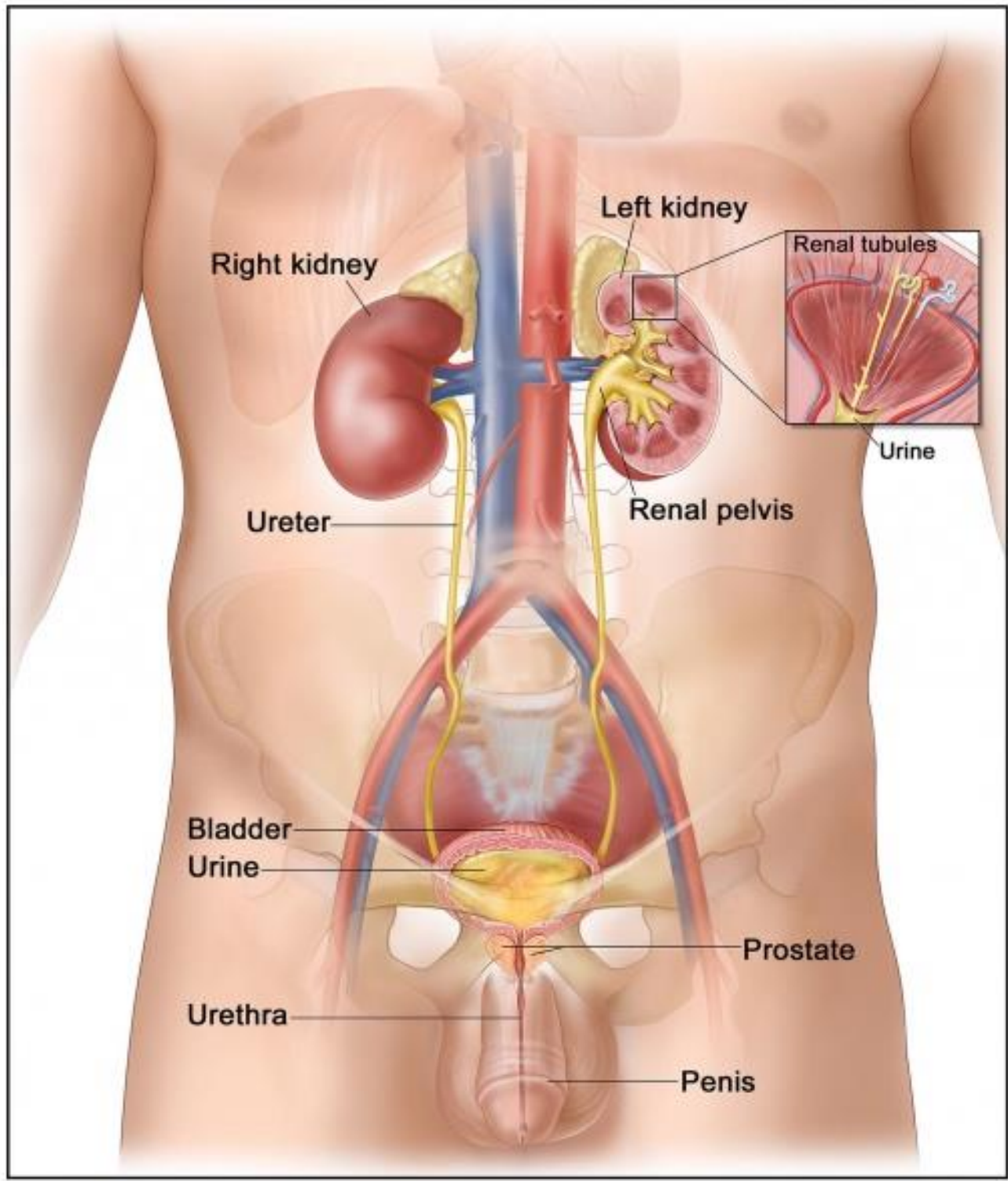
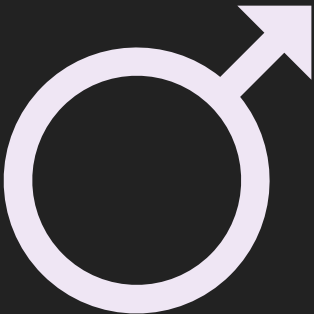
KIDNEY FUNCTION: URINE PRODUCTION



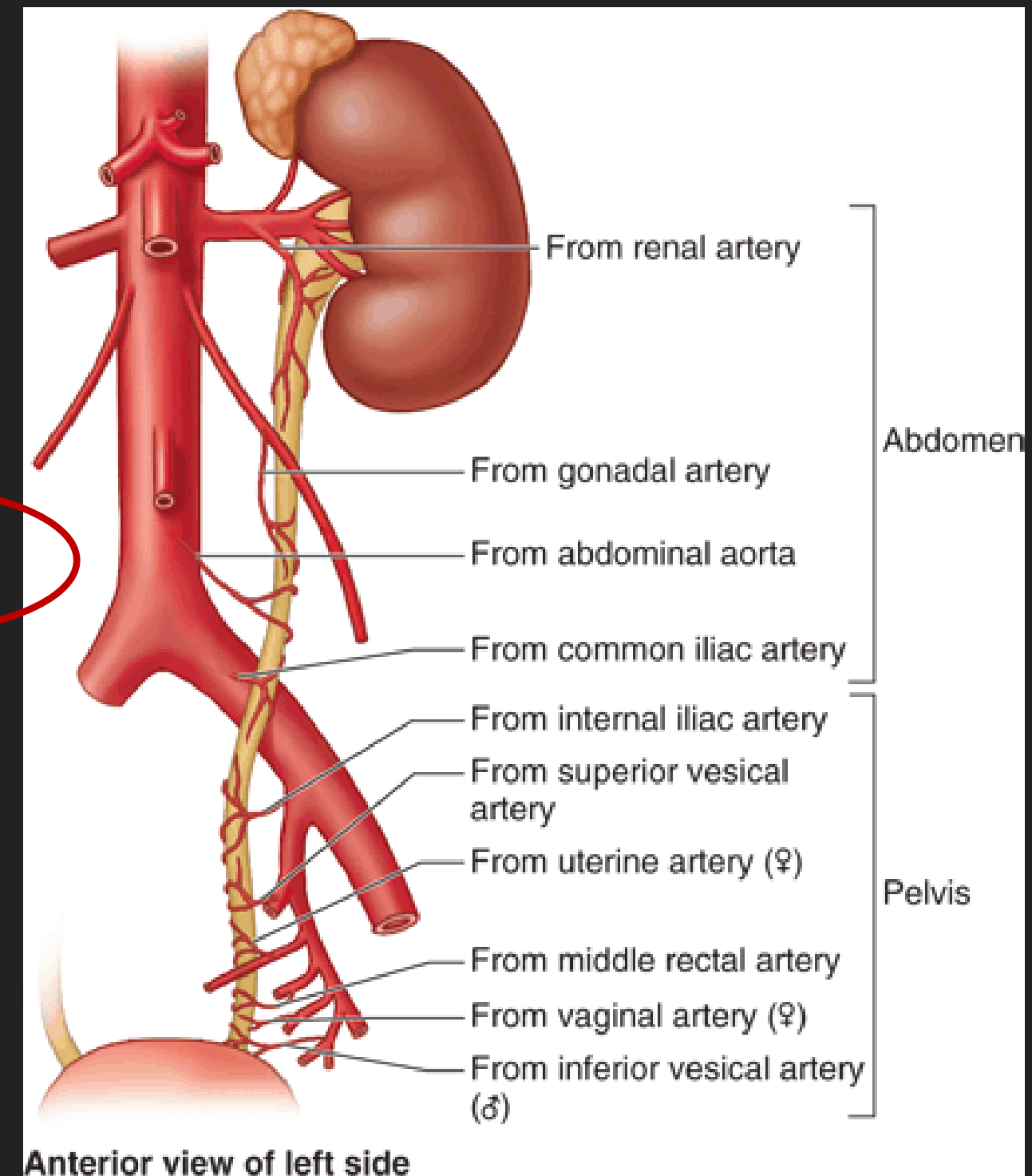
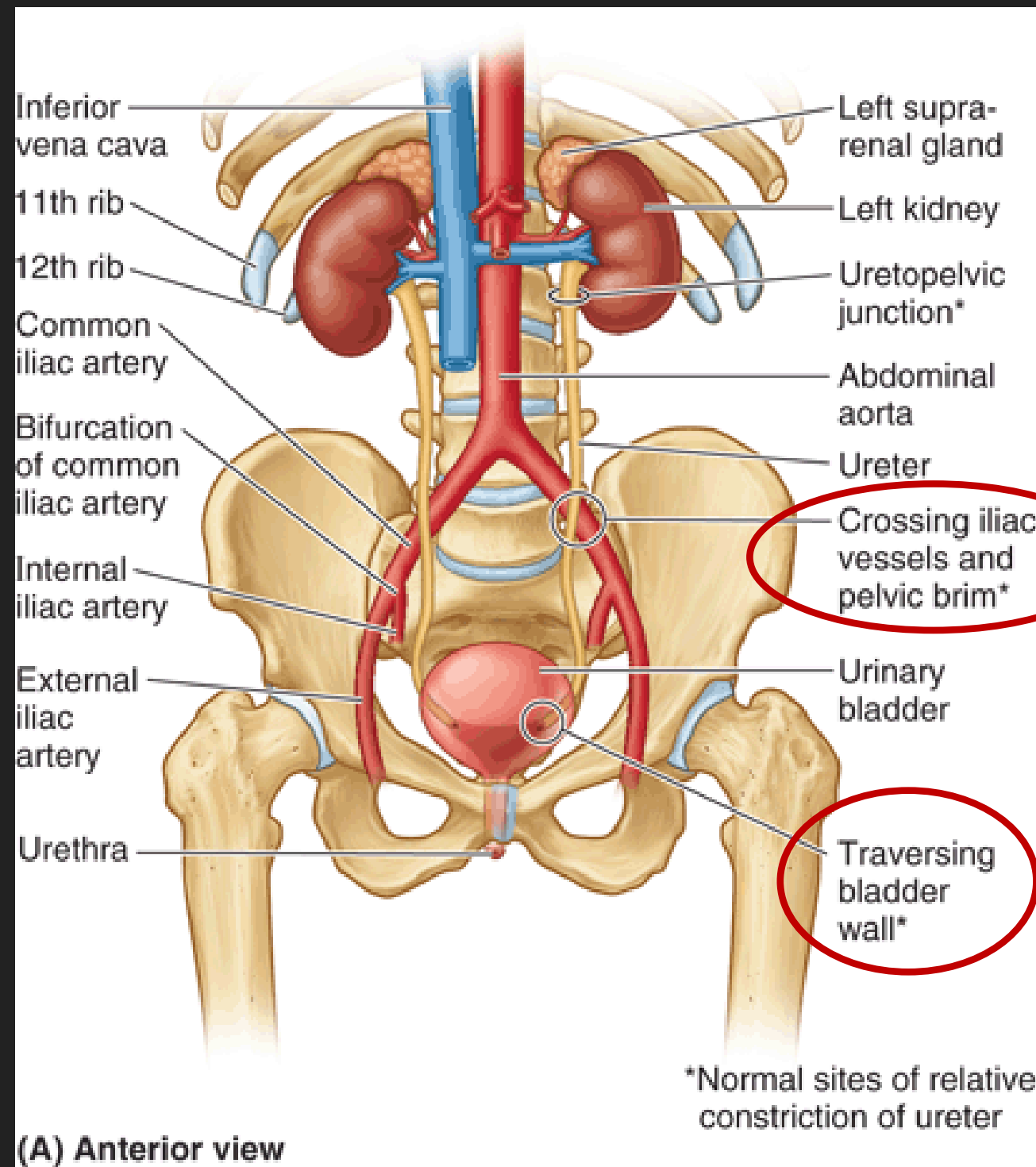
KIDNEY FUNCTION: URINE PRODUCTION



URINARY TRACT ANATOMY: URETER

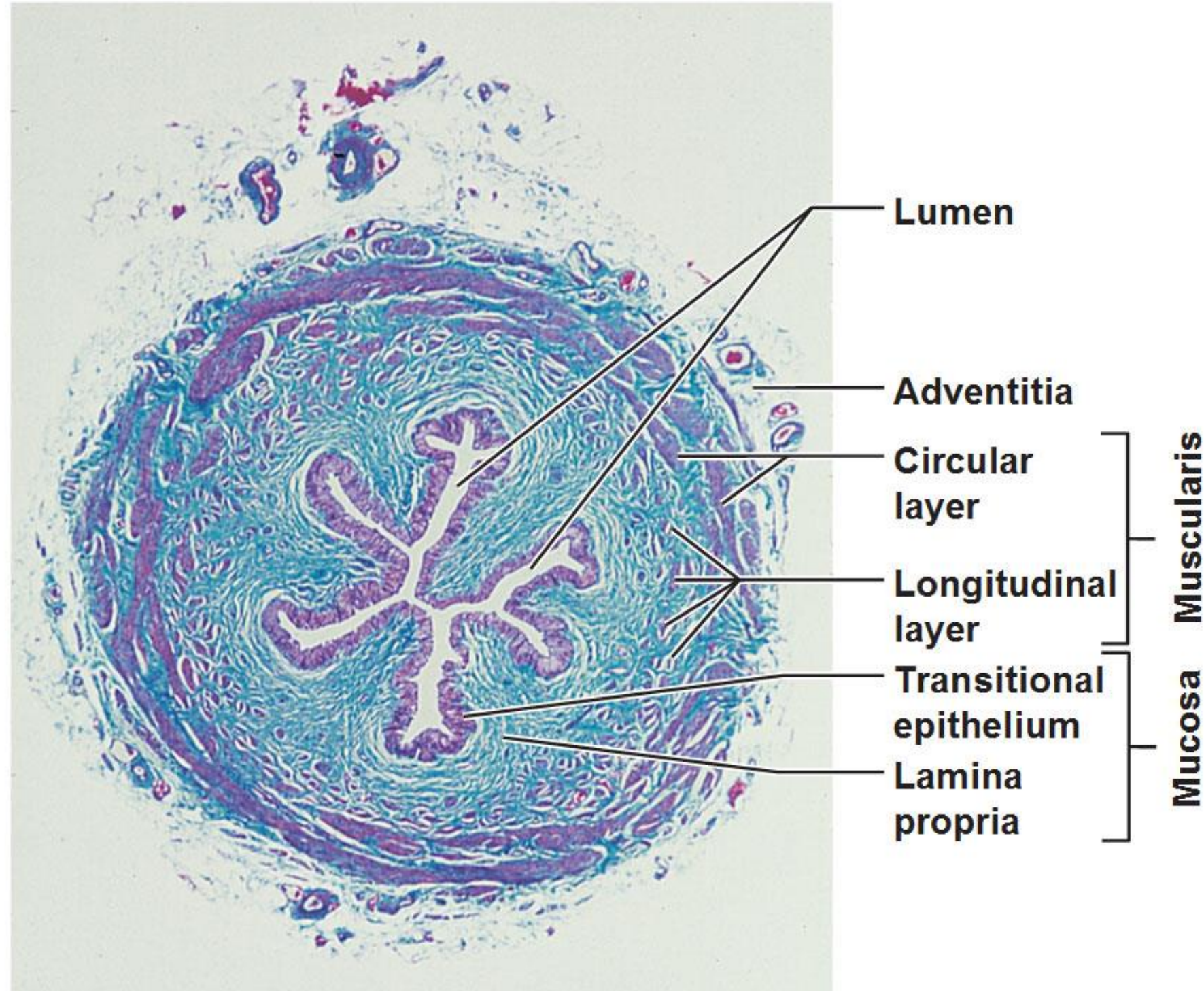


URETER

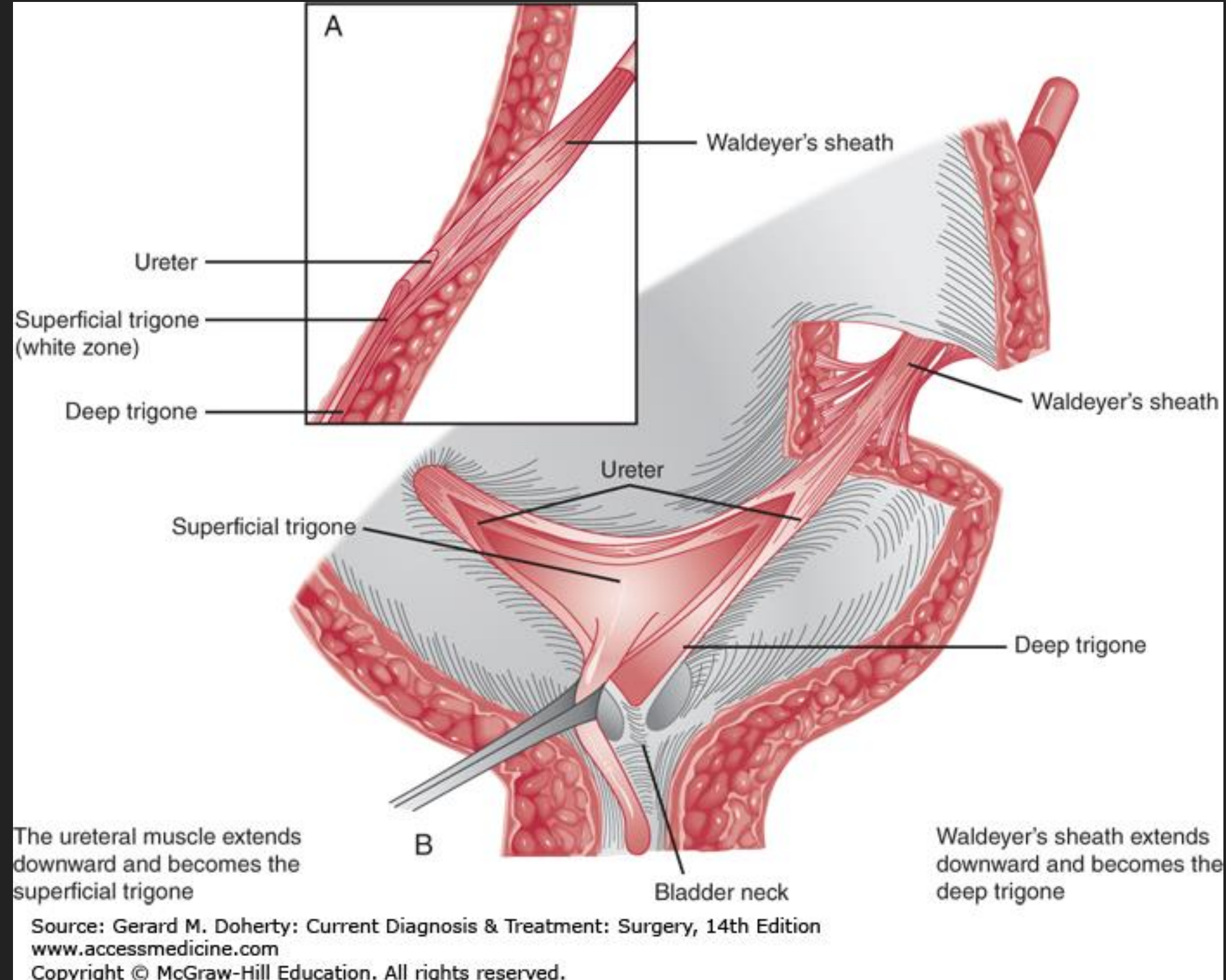


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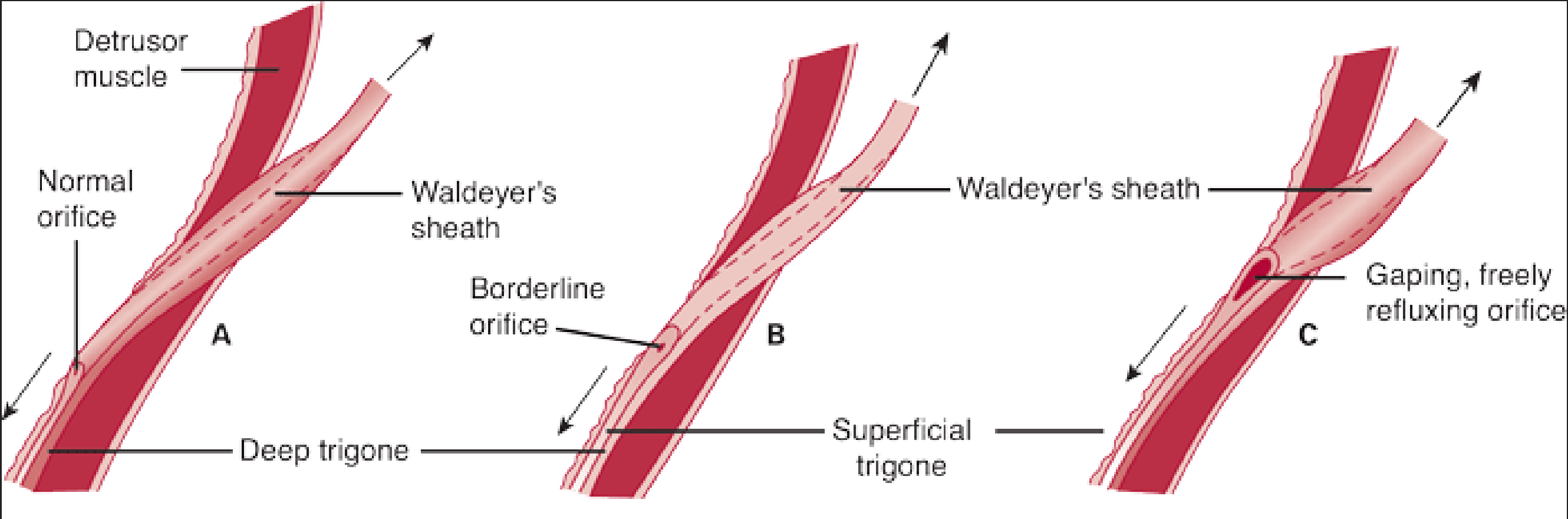
Microscopic Structure of the Ureter



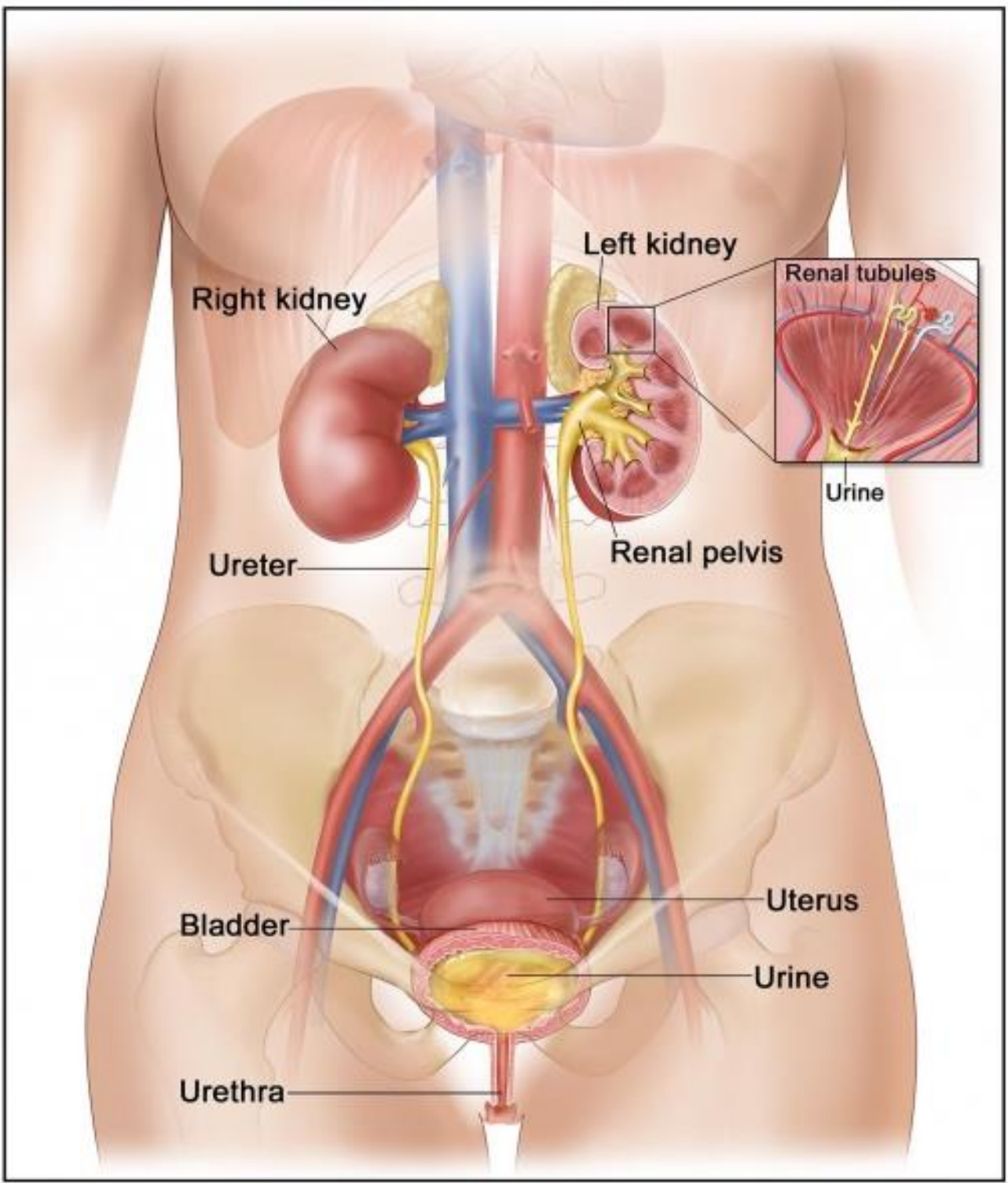
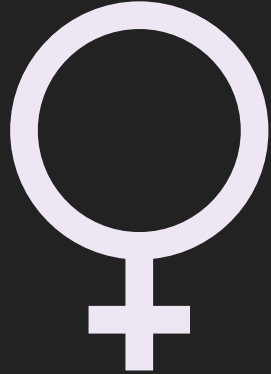
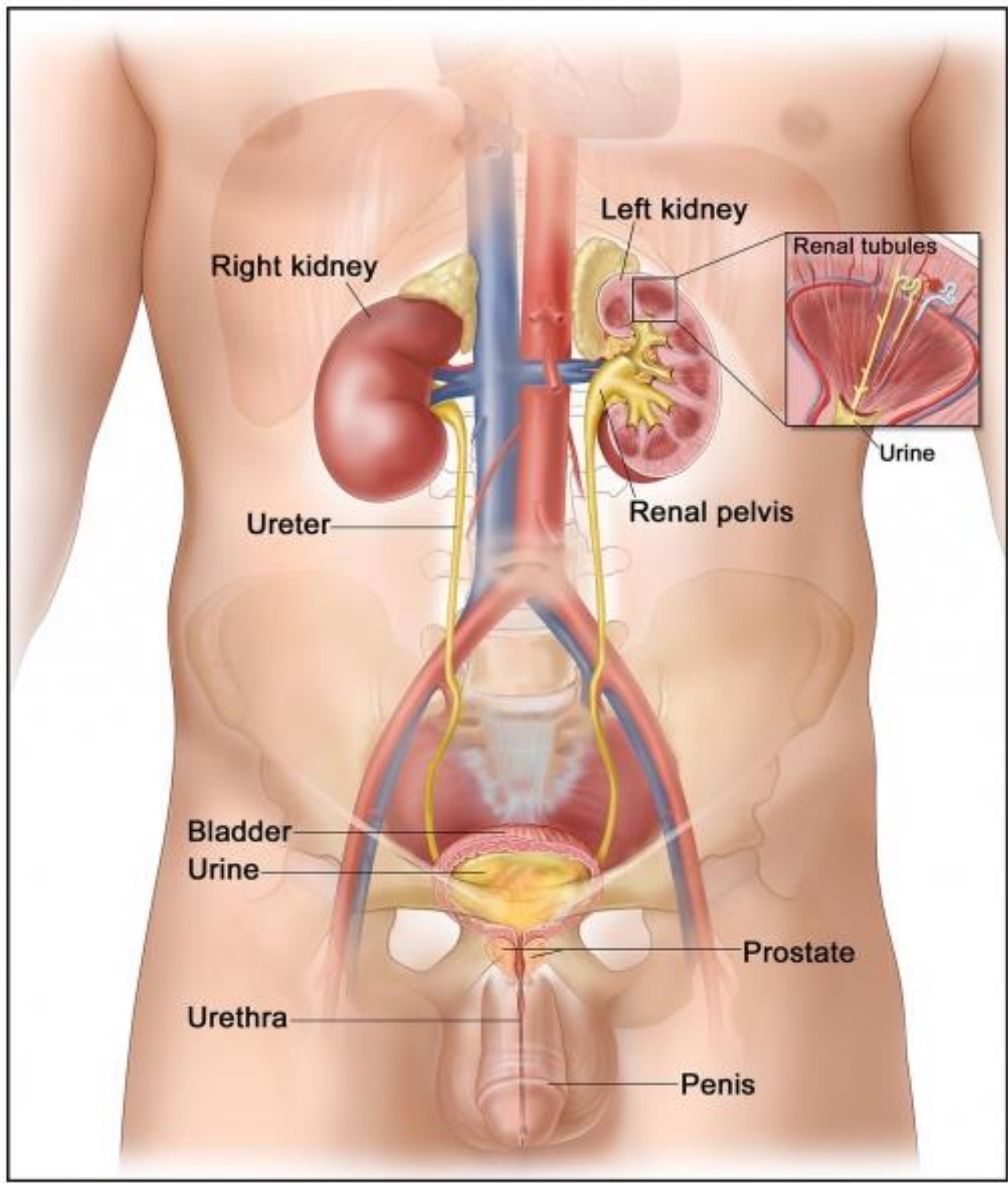
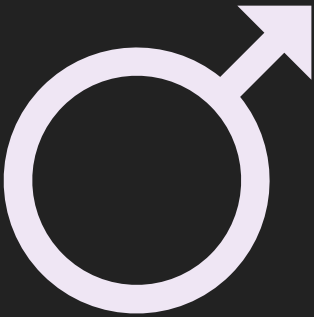
URETER



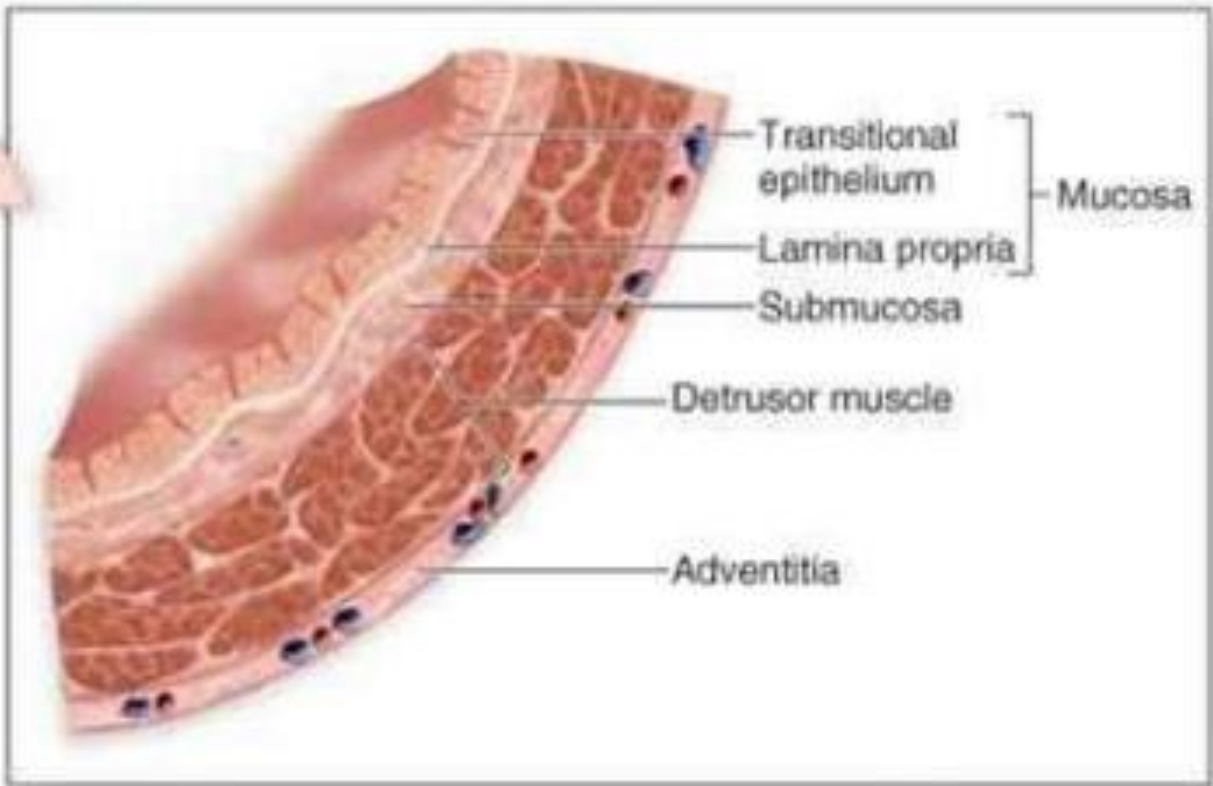
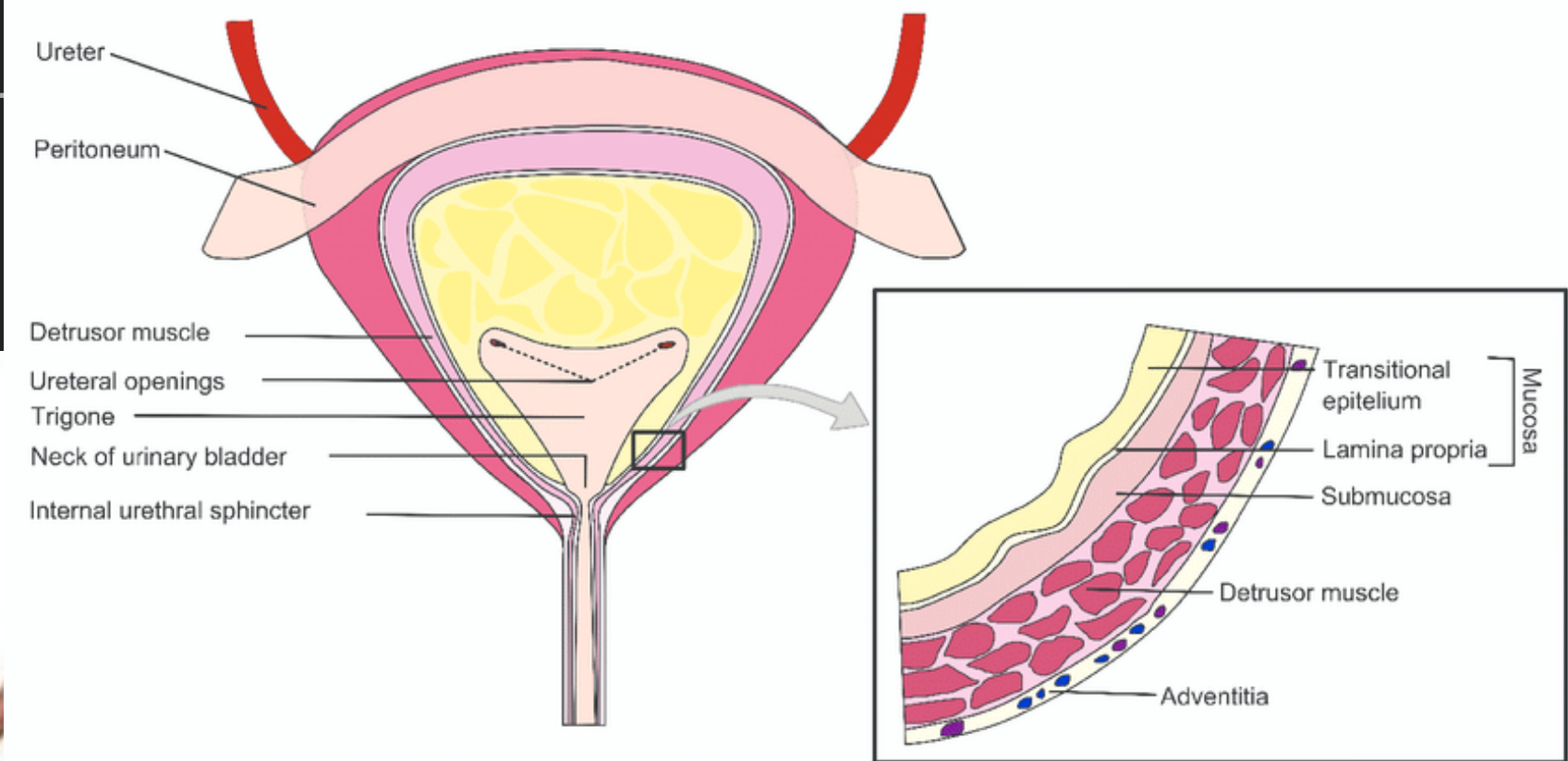
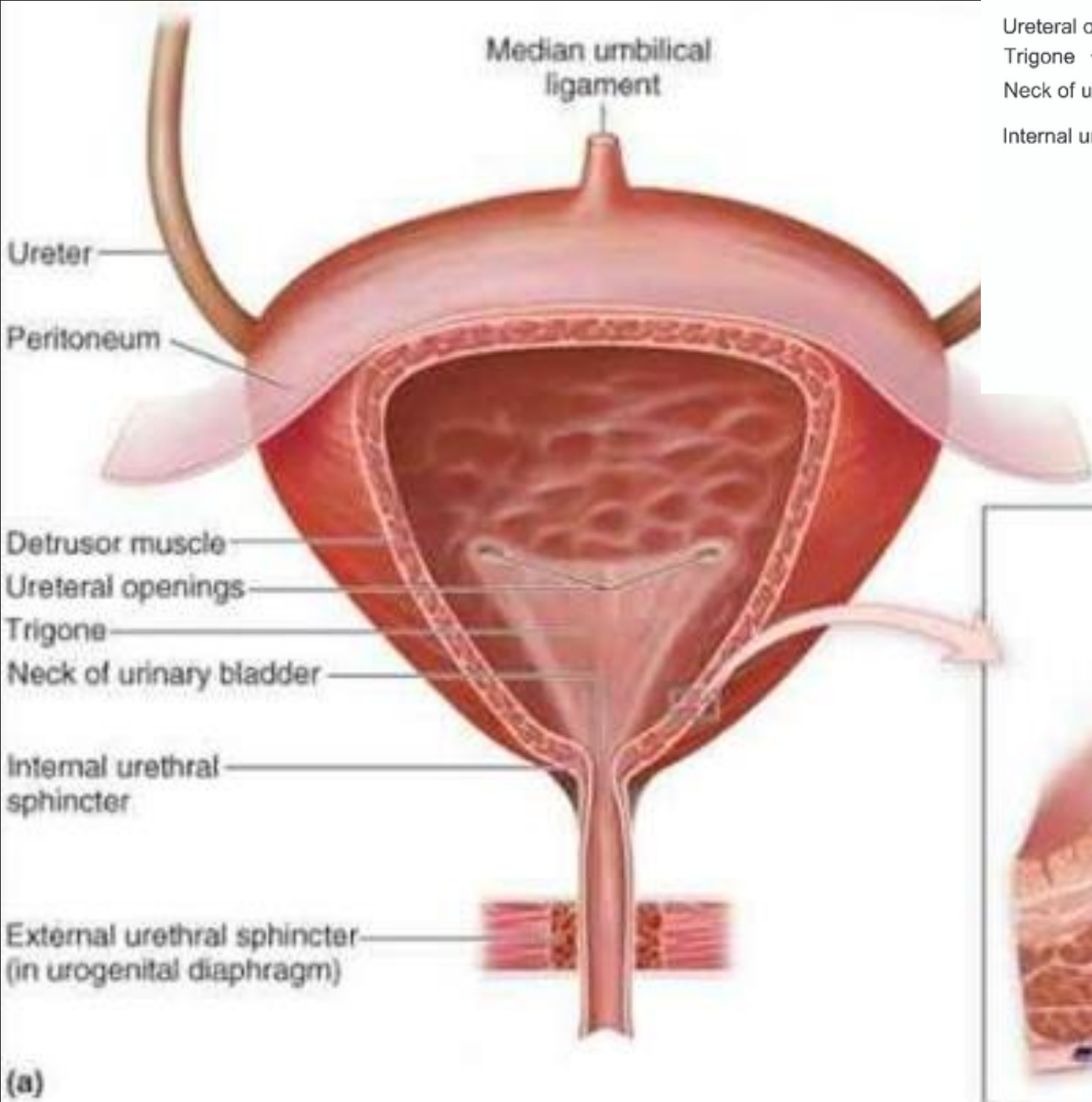
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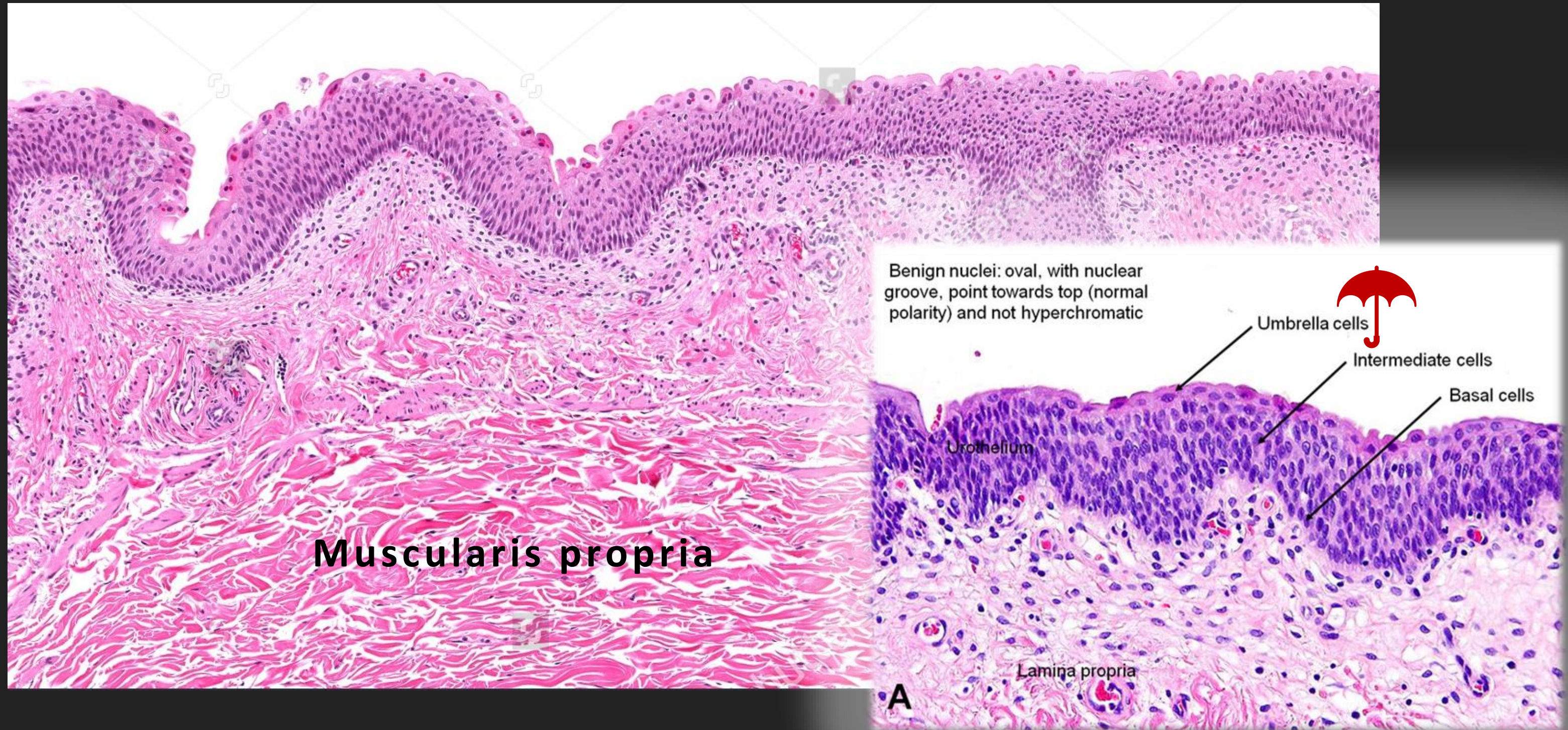
URINARY TRACT ANATOMY: BLADDER



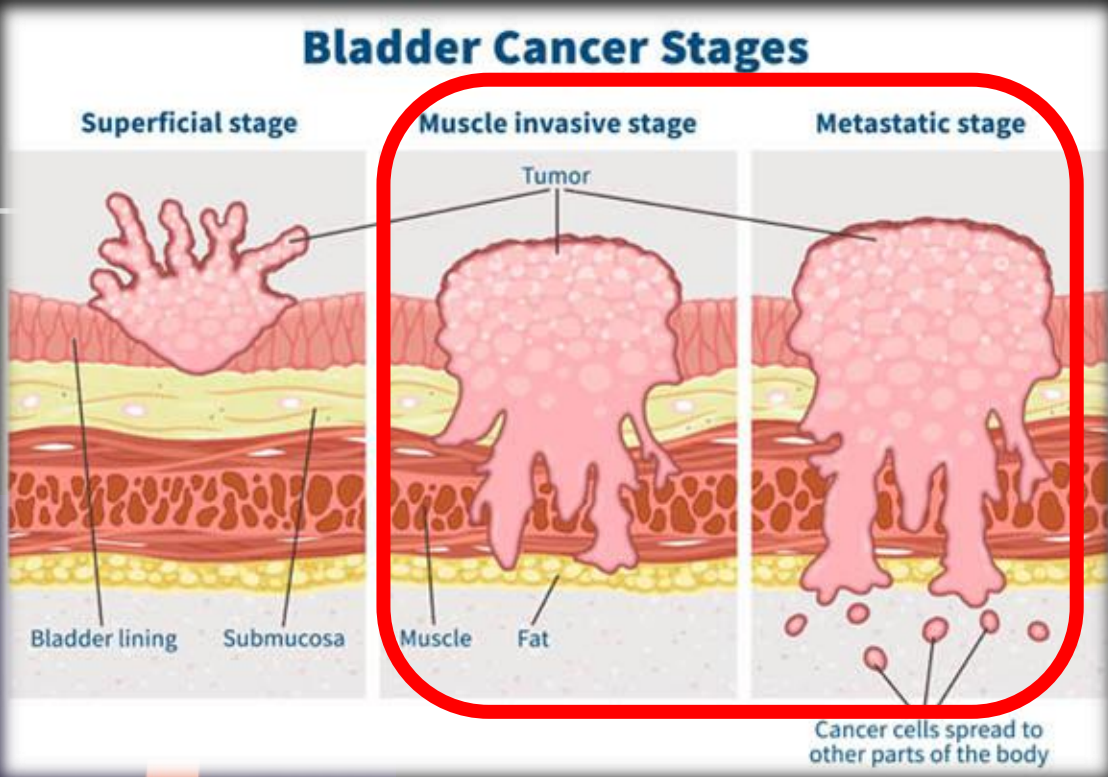
BLADDER: ANATOMY



BLADDER: MUCOSA



BLADDER: ANATOMY



Stages of Bladder Cancer

Stage 0
Precancerous lesions on inner lining

Stage I
Cancerous tumors have spread into connective tissue layer

Stage II
Tumors have spread into muscular wall

Stage III
Cancer has started to spread into local or regional organs, or into pelvic lymph nodes

Stage IV
Cancer has spread to major organs



BLADDER: ANATOMY

BLADDER CANCER

- Cystoscopy

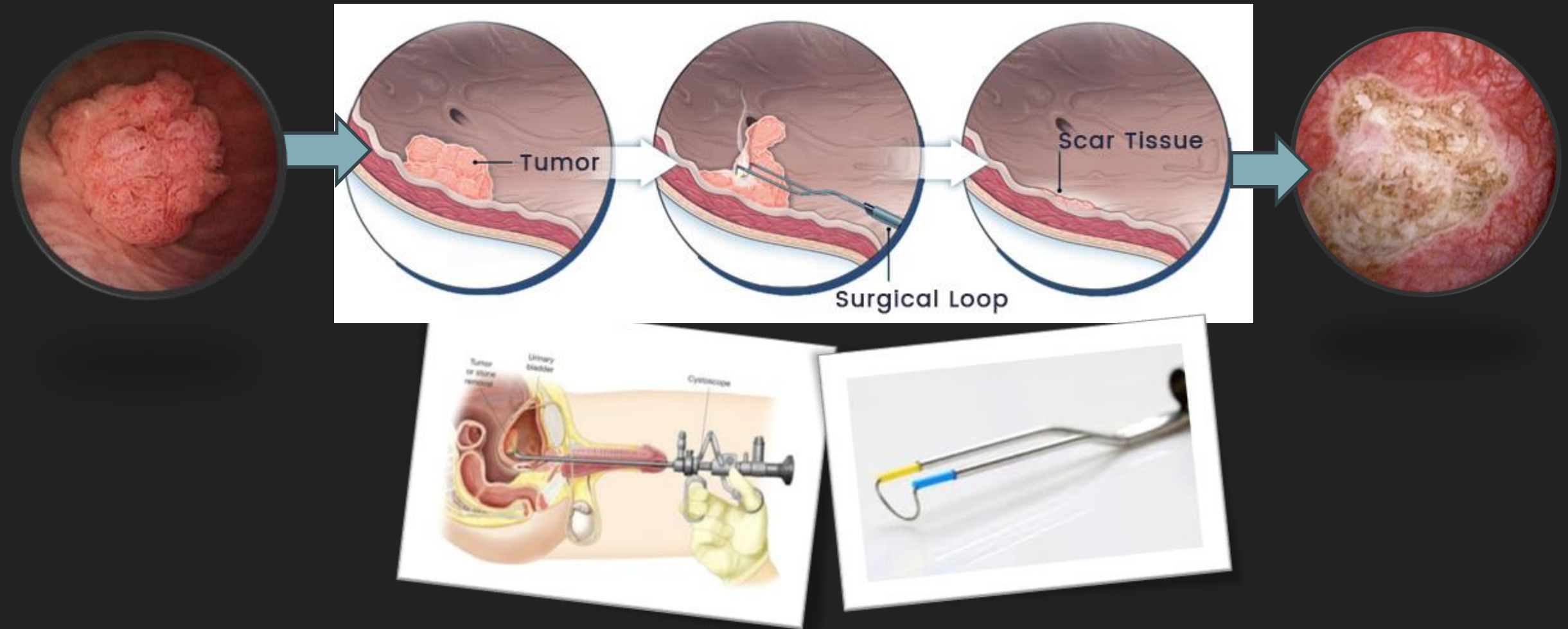


BLADDER: ANATOMY

BLADDER CANCER

- Management

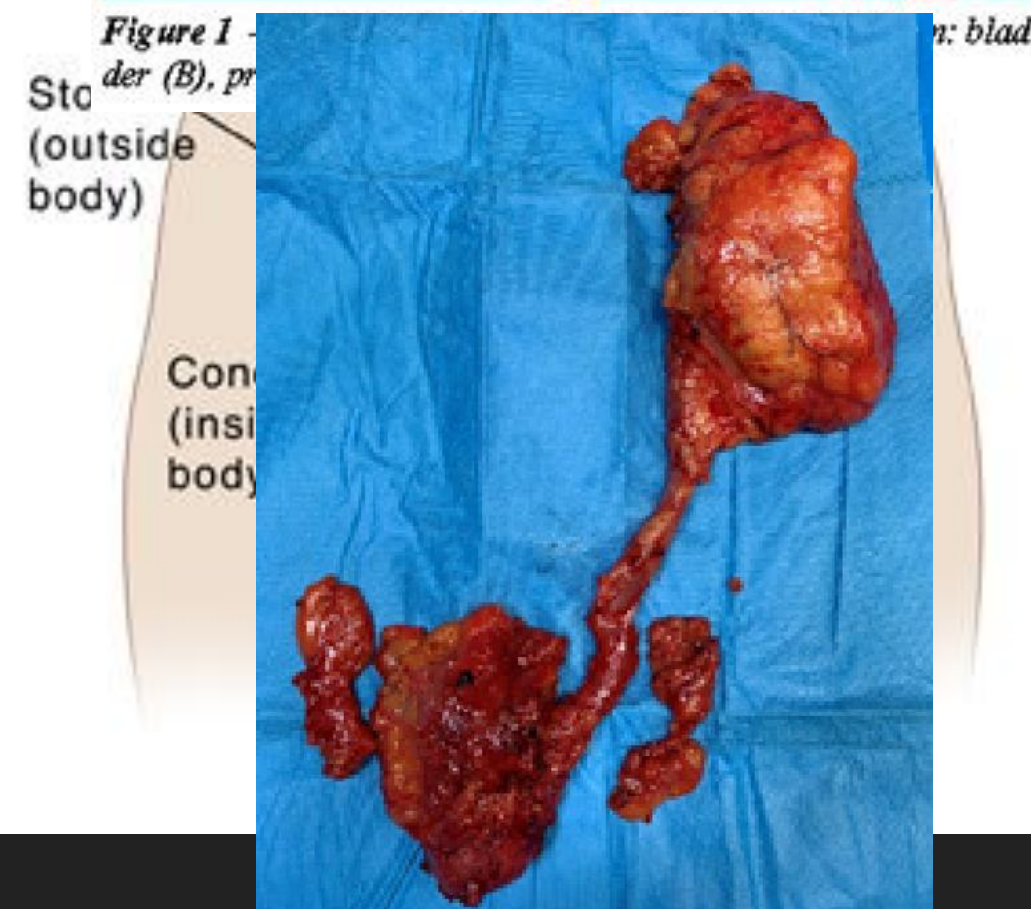
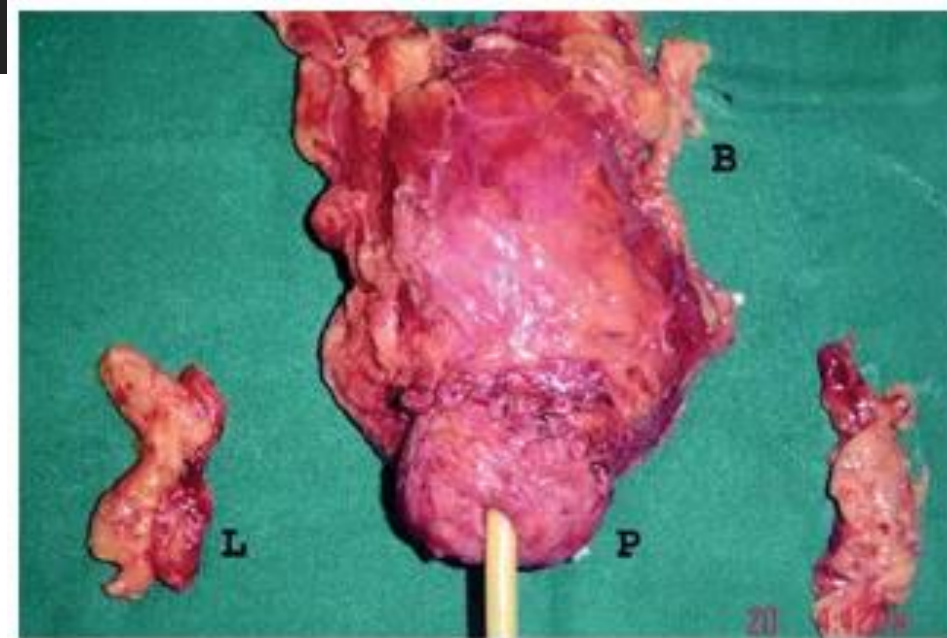
Transurethral resection of bladder tumor (TUR-BT)



BLADDER: ANATOMY

BLADDER CANCER

- Management



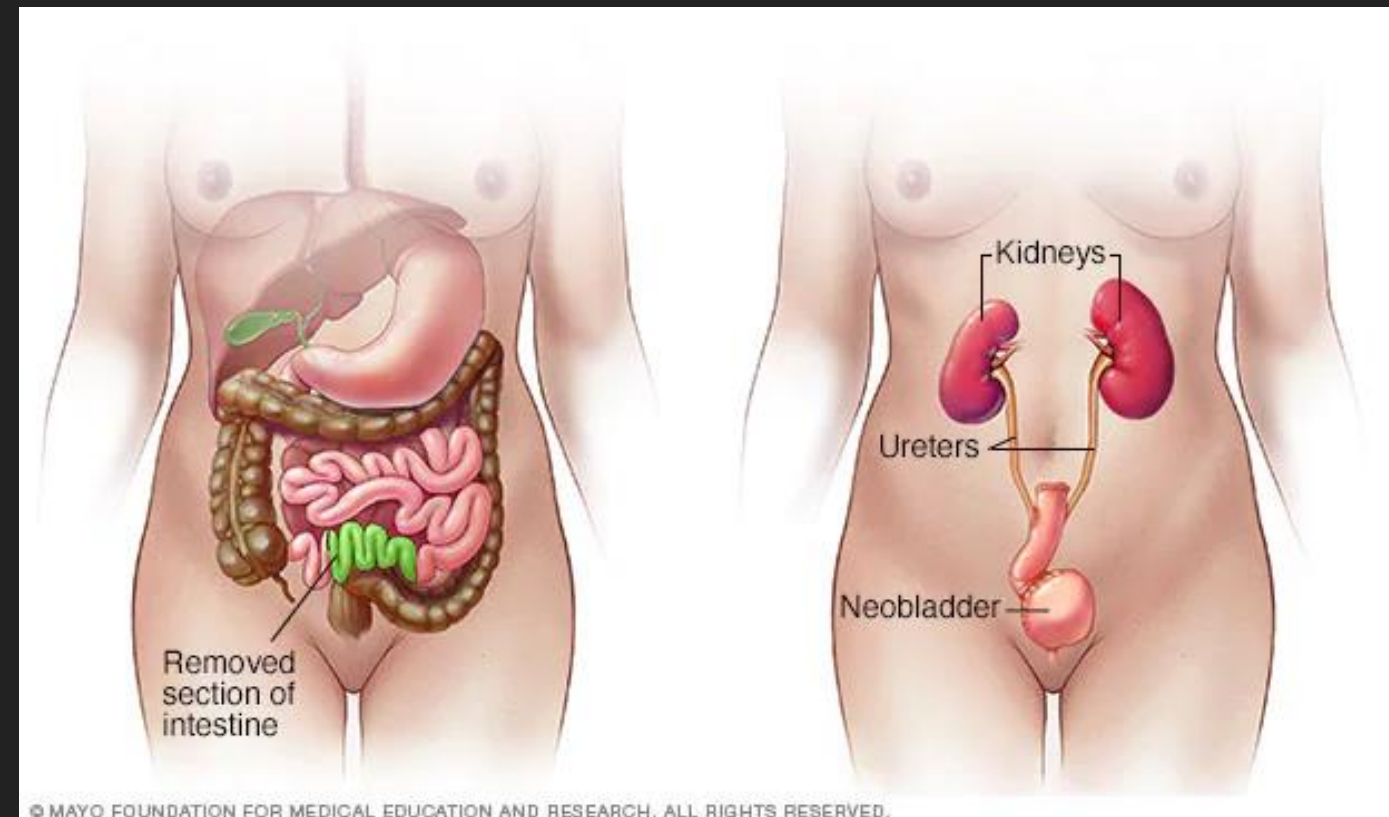
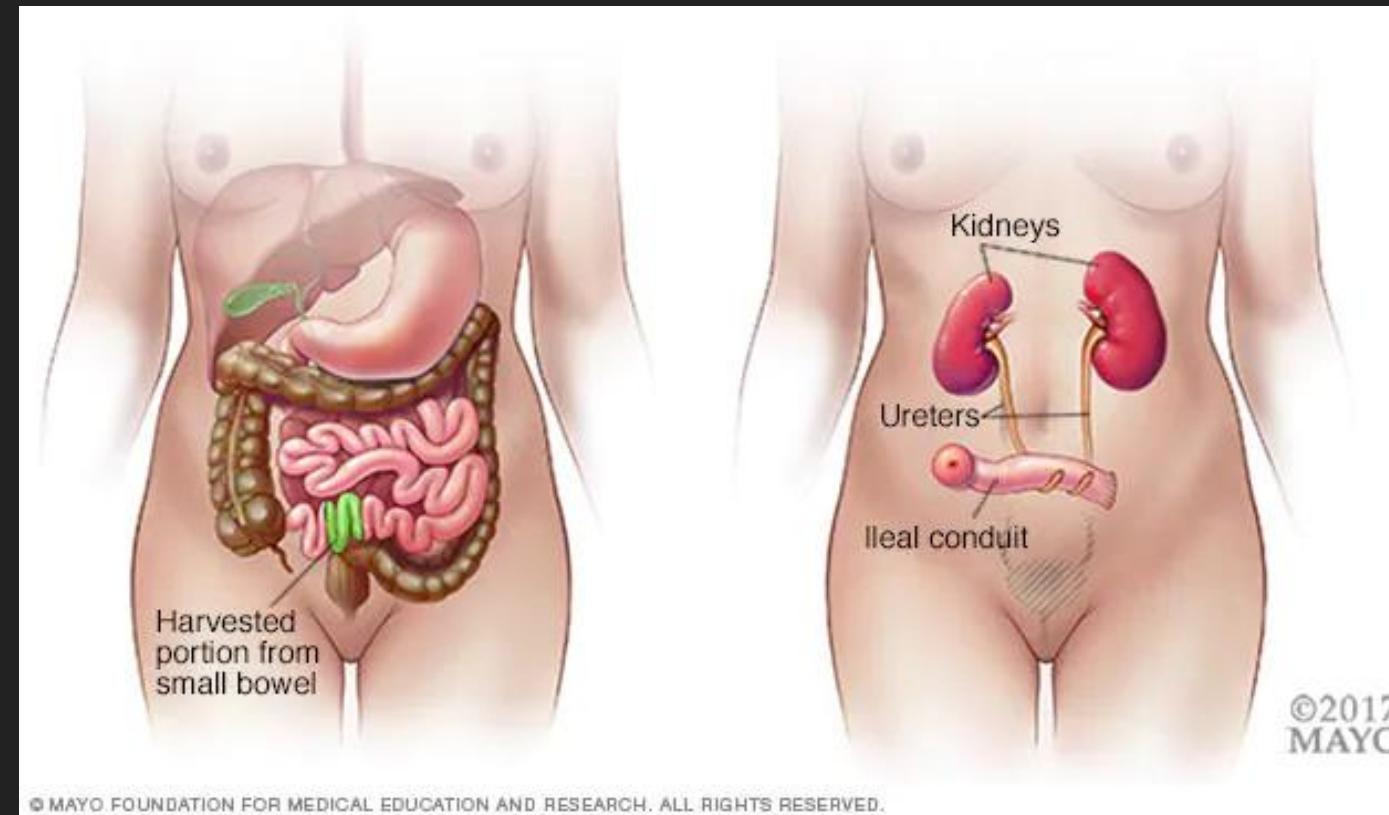
Radical cystectomy

- ❖ Remove bladder and
 - Prostate in male
 - Anterior vagina/Uterus in female
- ❖ Concern !!
 - Major operation with long op-time
 - Bowel preparation for conduit
 - Healthy patient
 - Nutrition

BLADDER: ANATOMY

BLADDER CANCER

- Management



BLADDER: FUNCTION

Normal lower urinary tract function



Storage phase

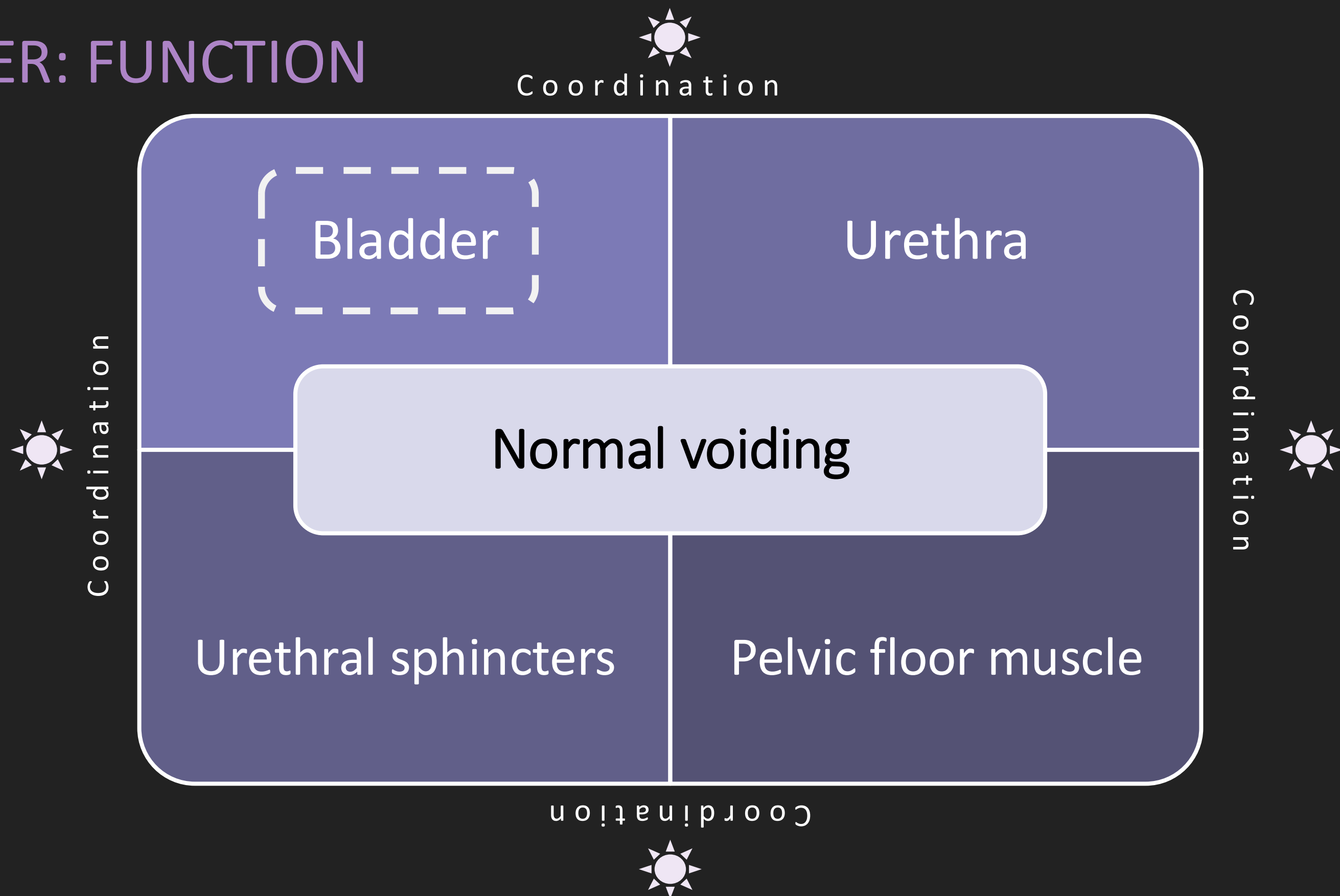
- Low pressure reservoir
- **Continence**
- Appropriate bladder capacity to hold urine (long period of time)
- No pain

Voiding phase

- Complete emptying
- No obstruction
- Short period of time
- No pain

At socially appropriate and convenient times

BLADDER: FUNCTION

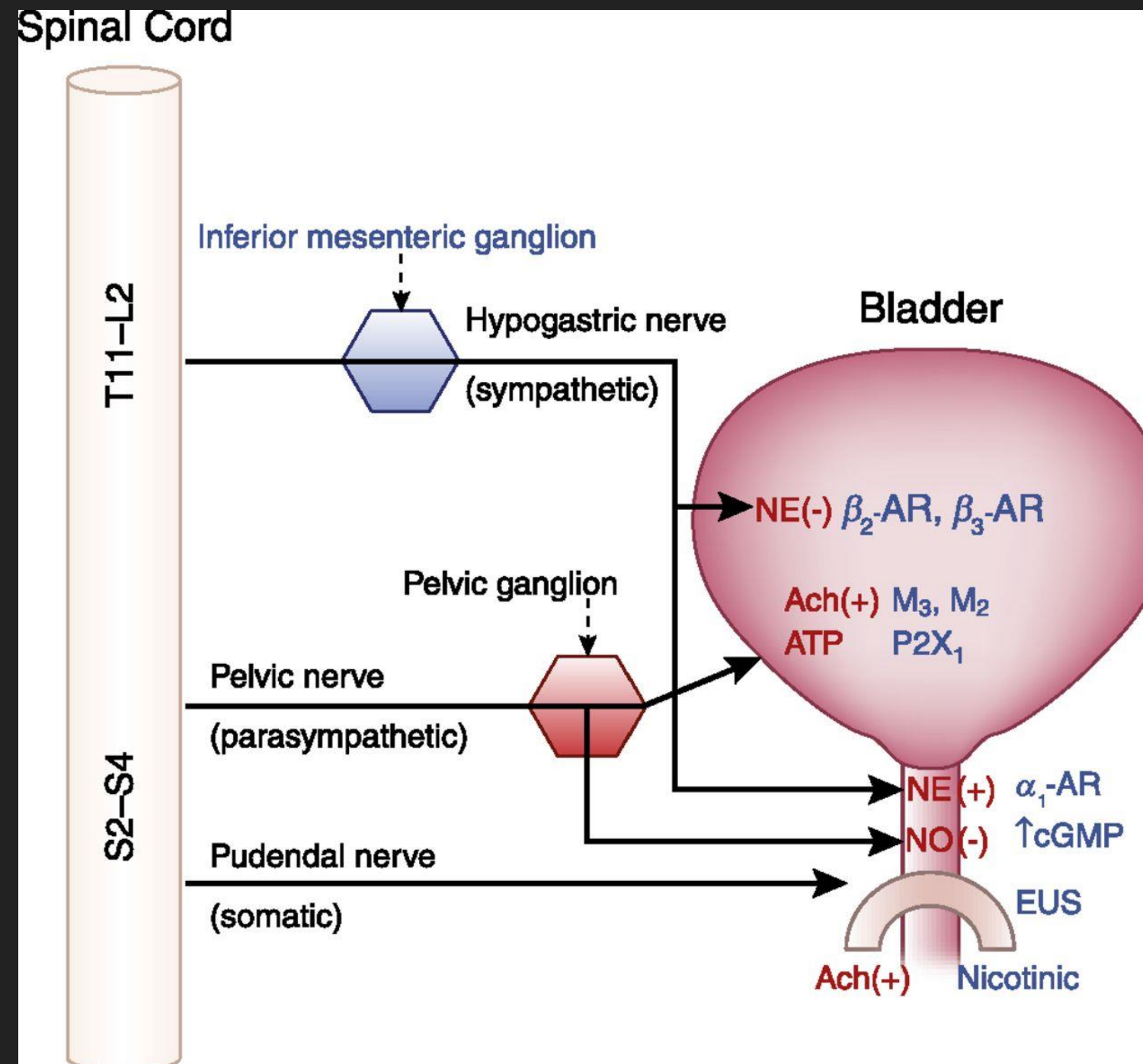


BLADDER: FUNCTION

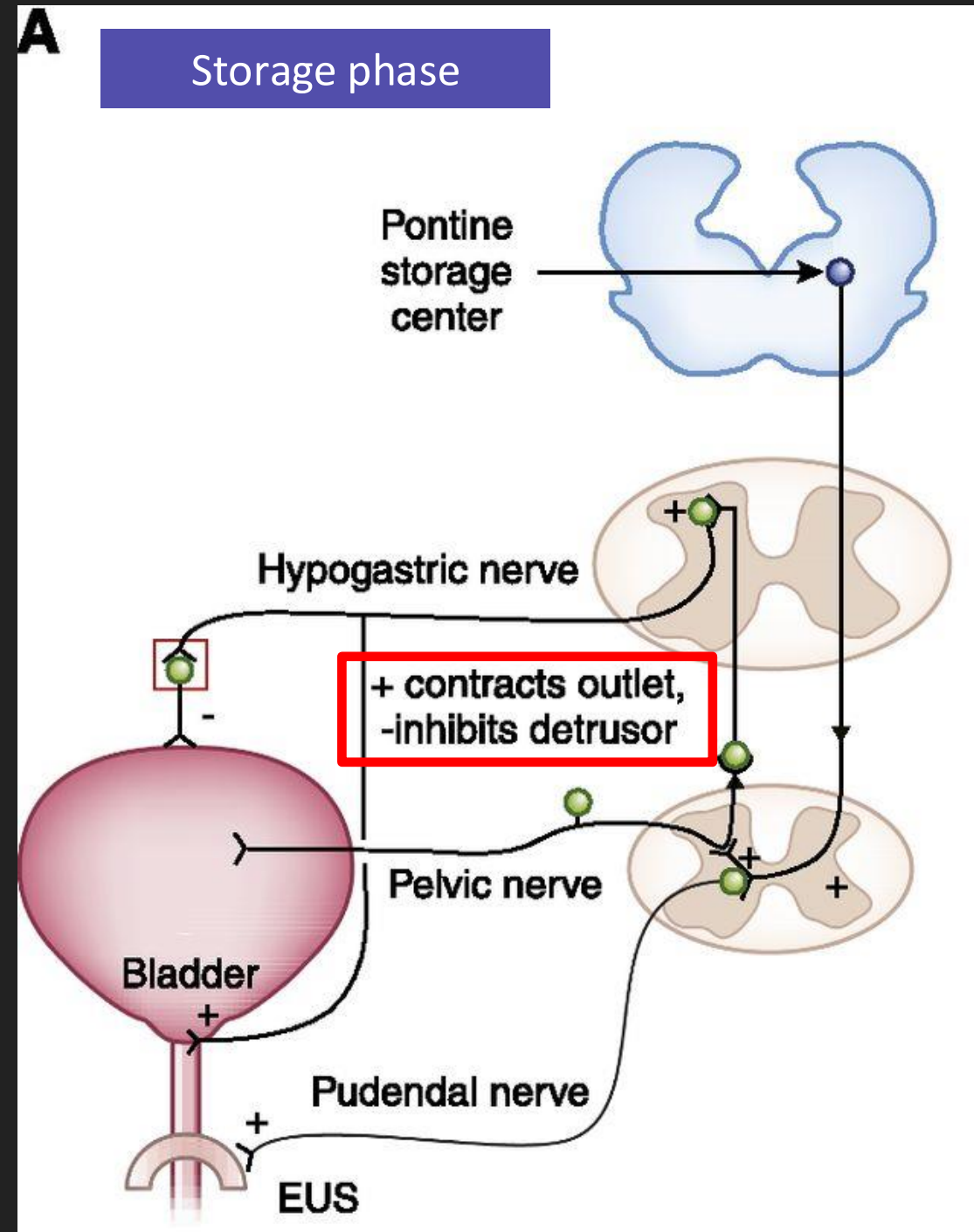
Sympathetic:
Hypogastric n.

Parasympathetic:
Pelvic n.

Somatic:
Pudendal n.



BLADDER: FUNCTION

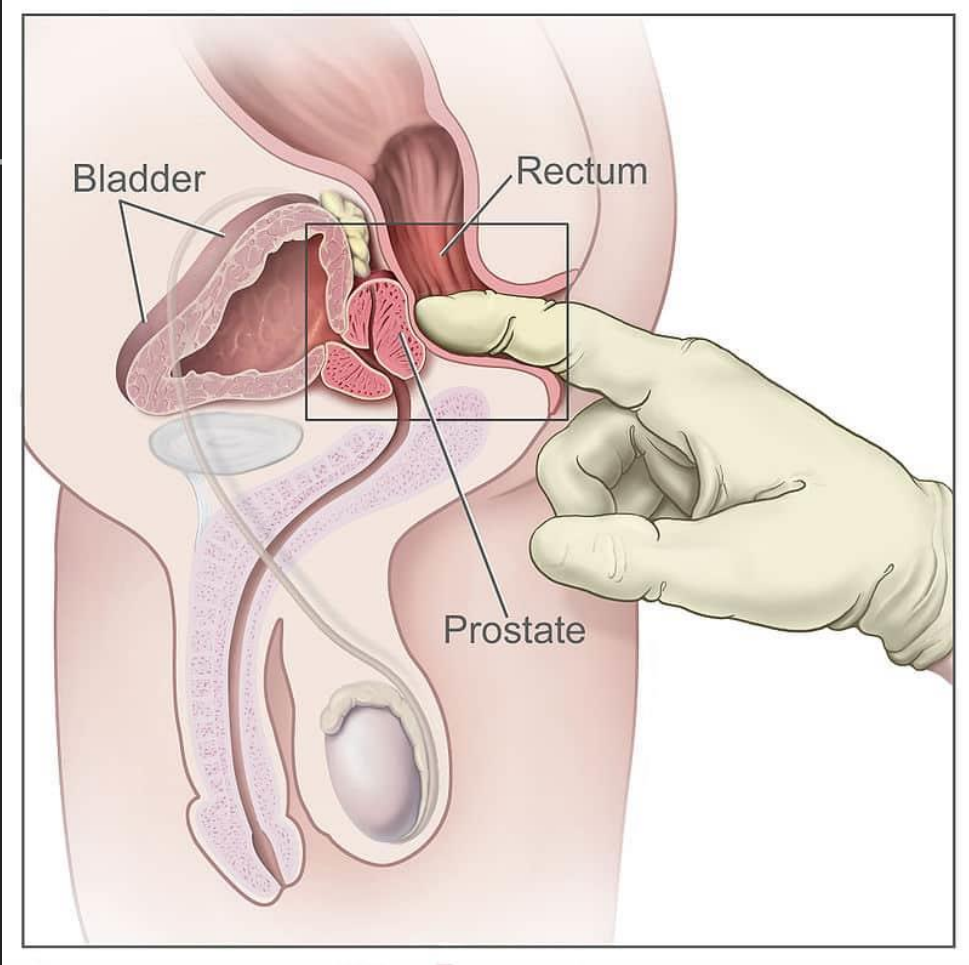


BLADDER: FUNCTION

- ▶ 2 Phases
 - ▶ Storage (filling) phase: low intravesical pressure without leakage
 - ▶ Voiding phase: complete emptying
- ▶ Co-ordination between bladder and sphincter (outlet)

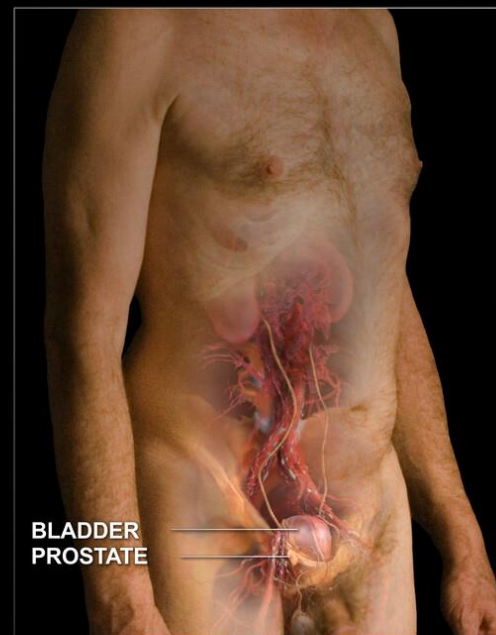
	Bladder	Sphincter
Filling	Relax	Contract
Voiding	Contract	Relax

PROSTATE GLAND AND SEMINAL VESICLES

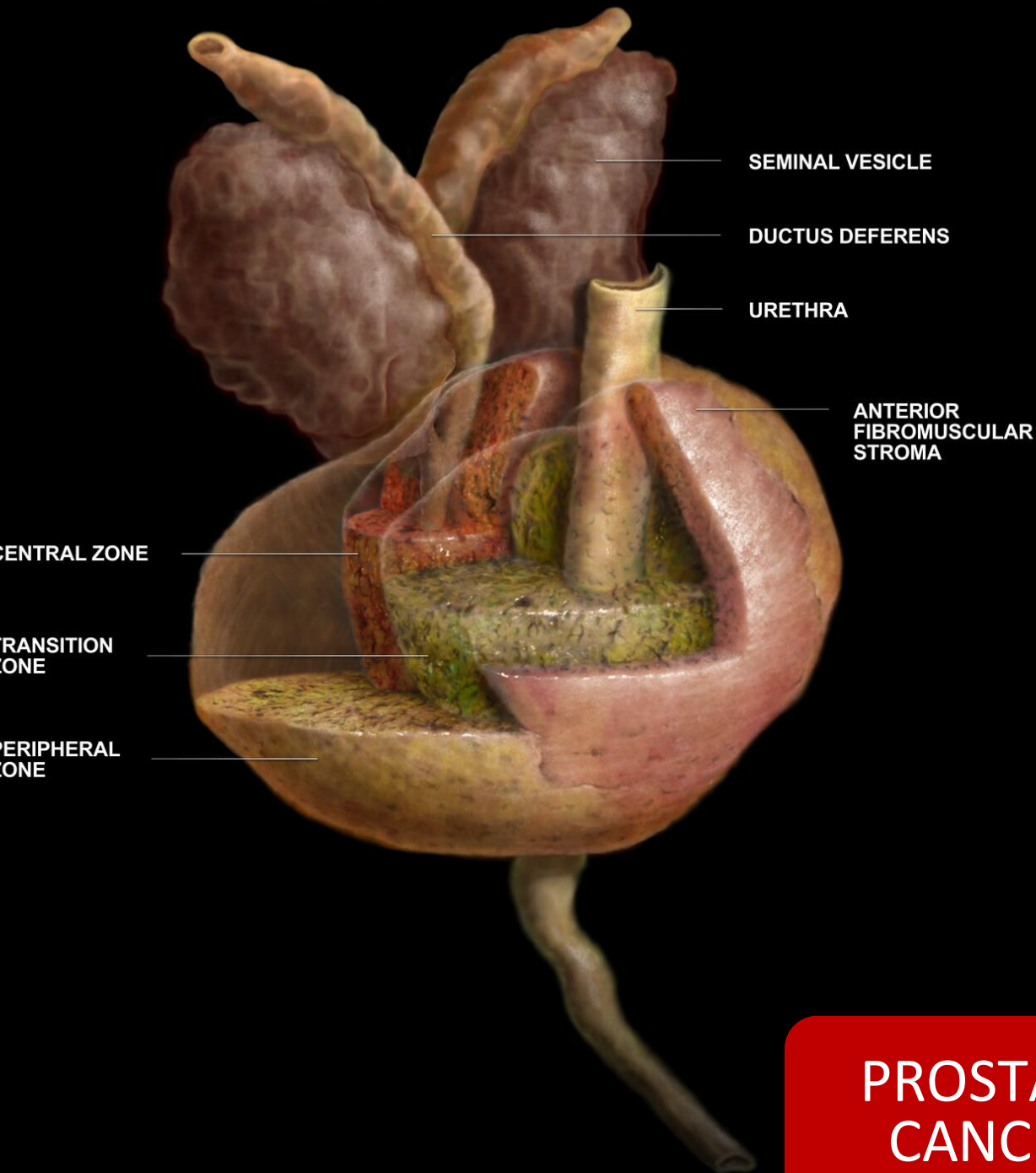
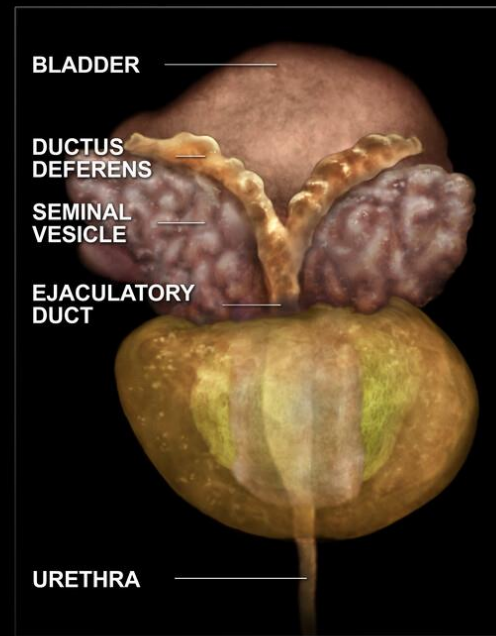


Prostate Anatomy By Zones

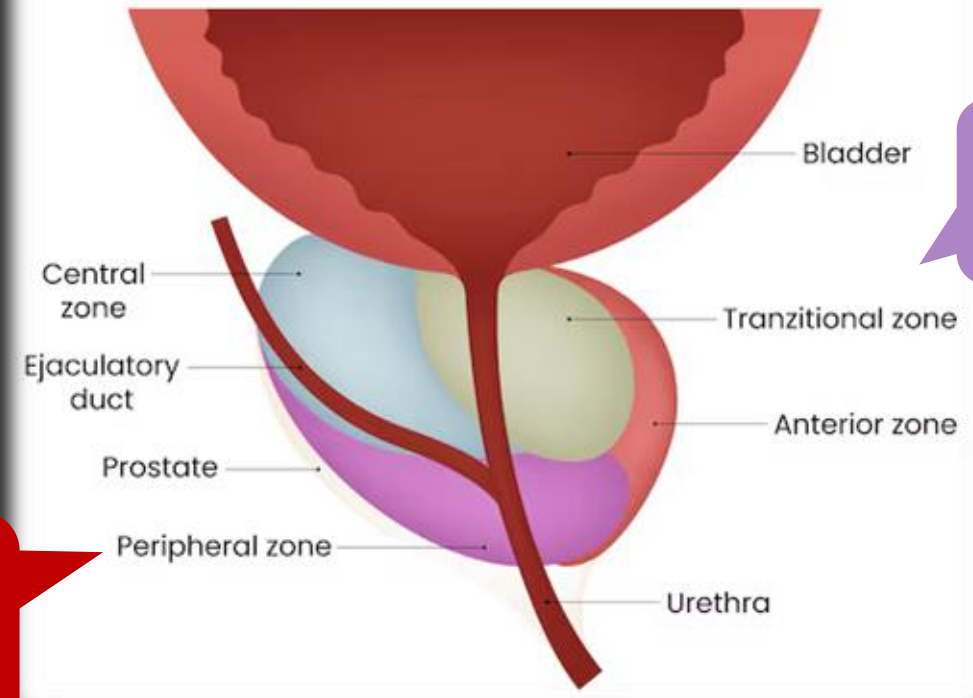
3/4 Anterior View



Posterior View



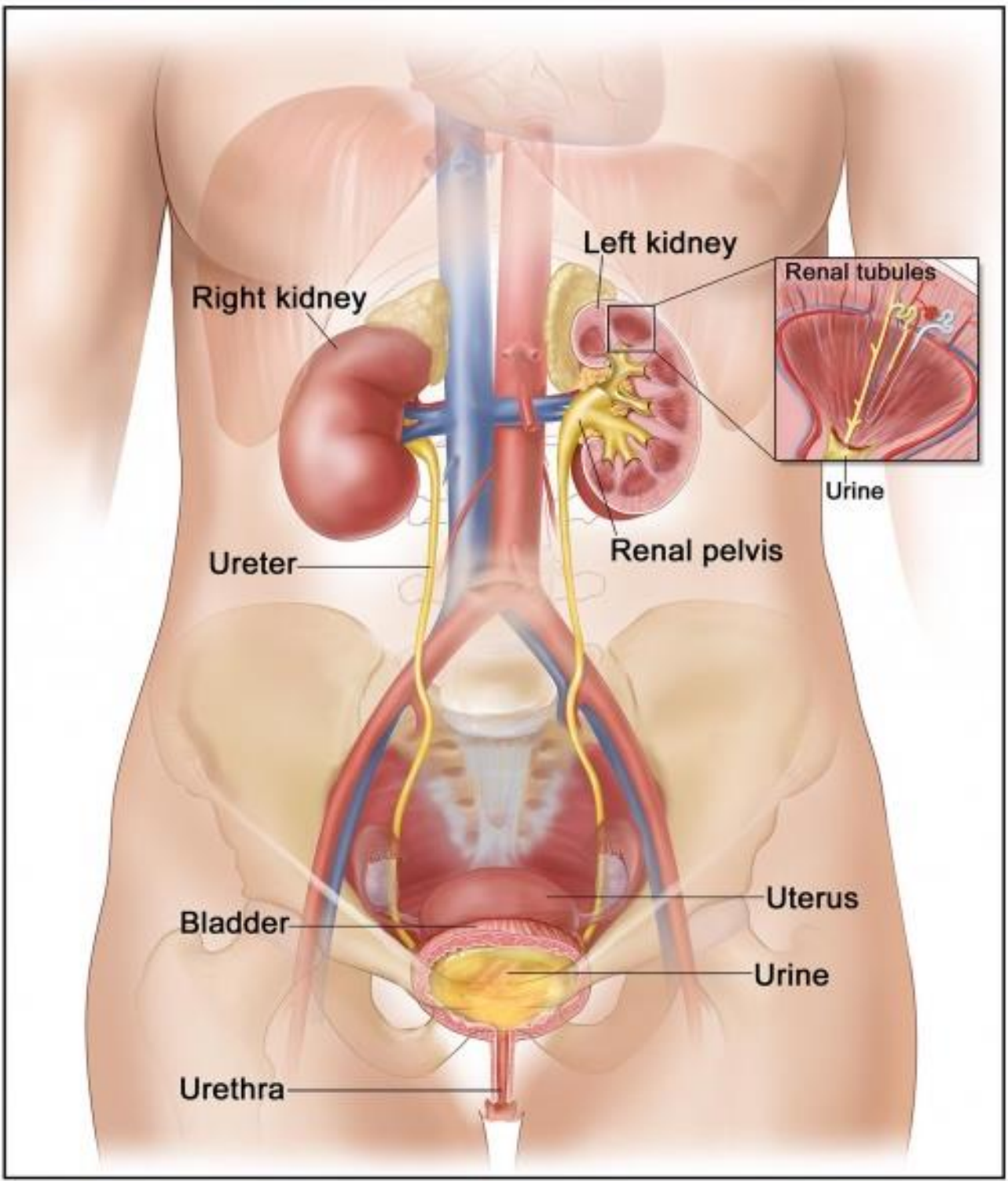
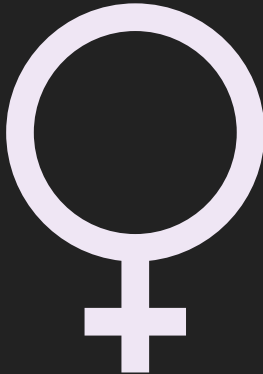
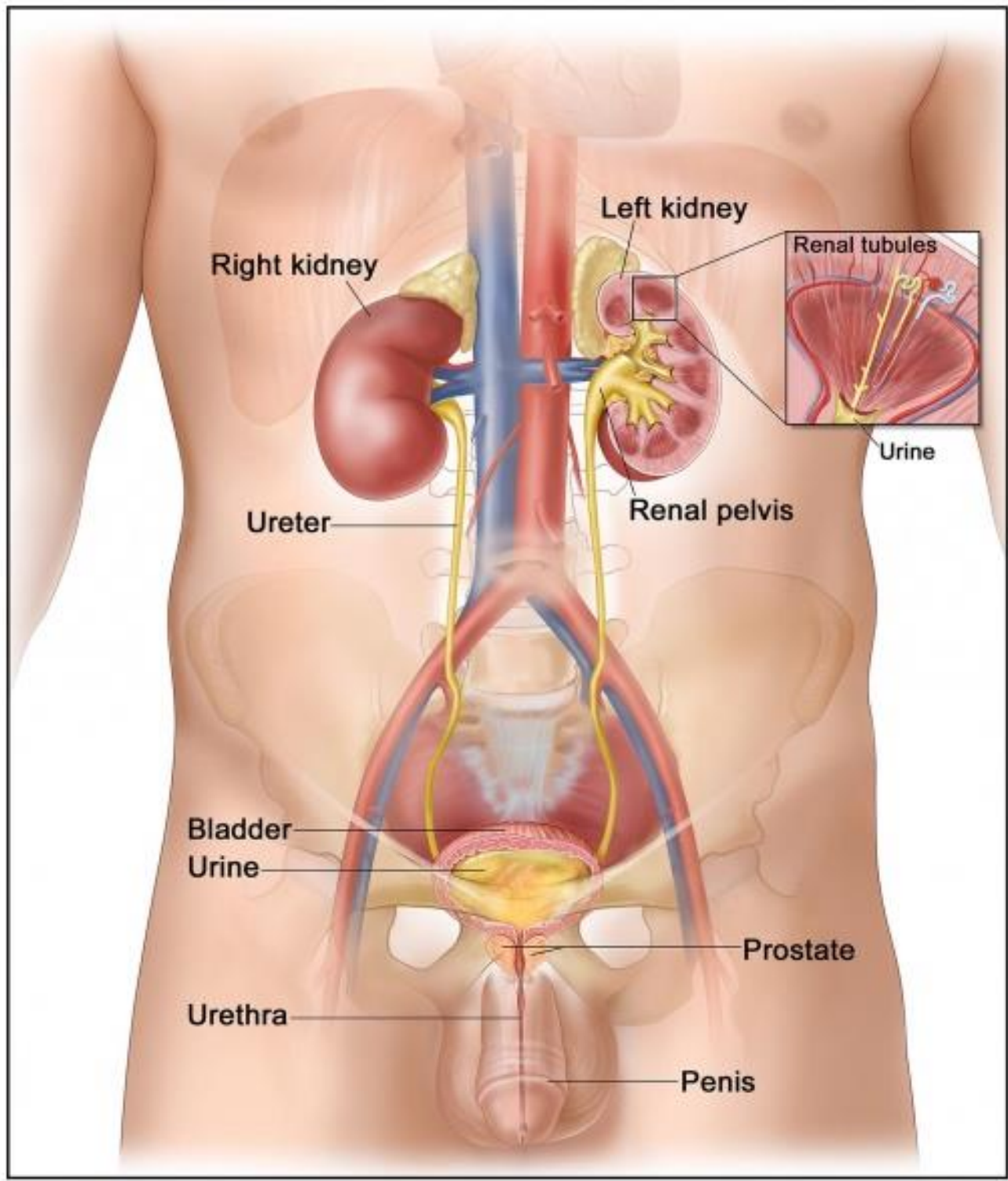
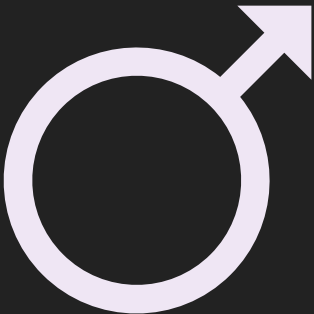
Prostate Zones



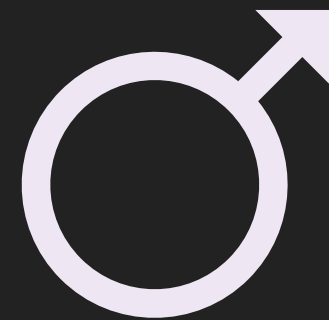
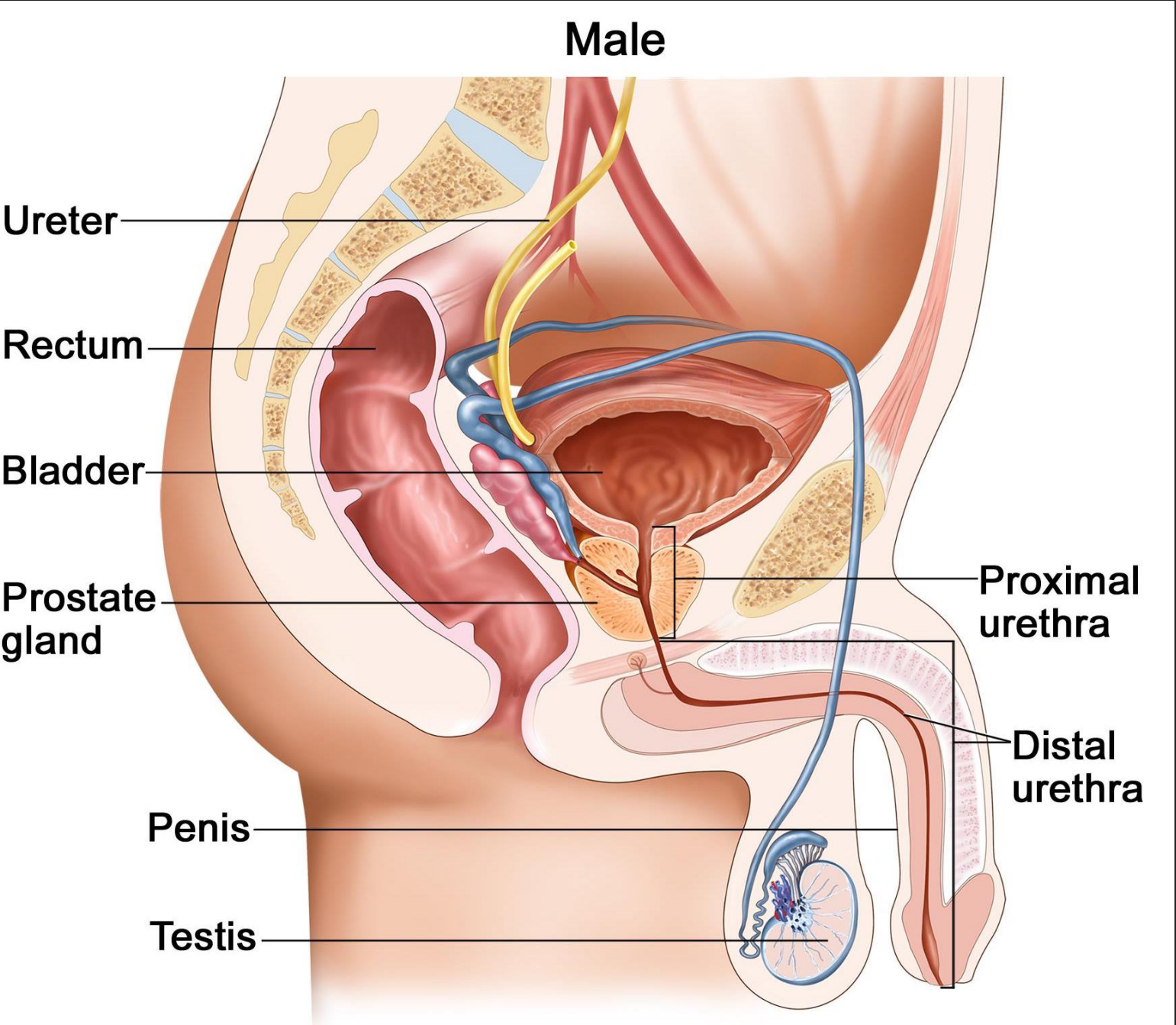
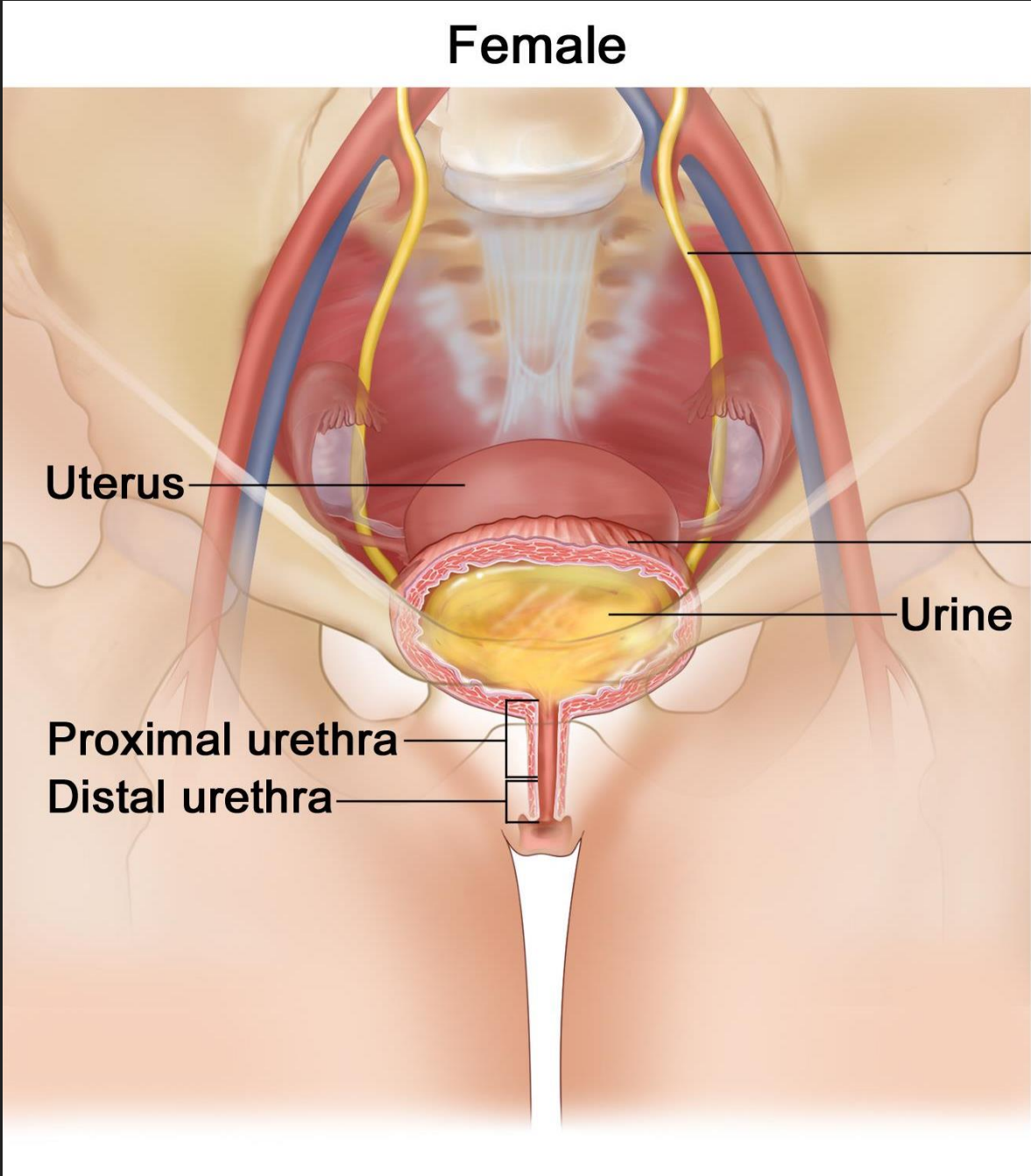
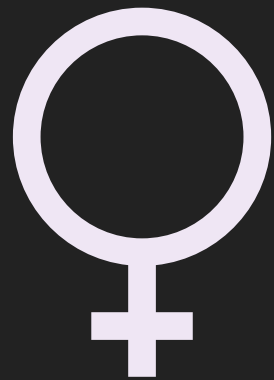
BPH

PROSTATE
CANCER

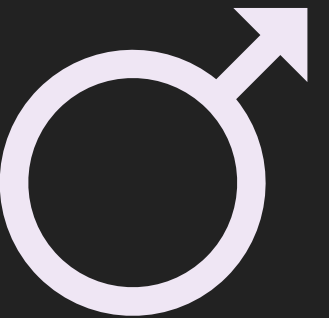
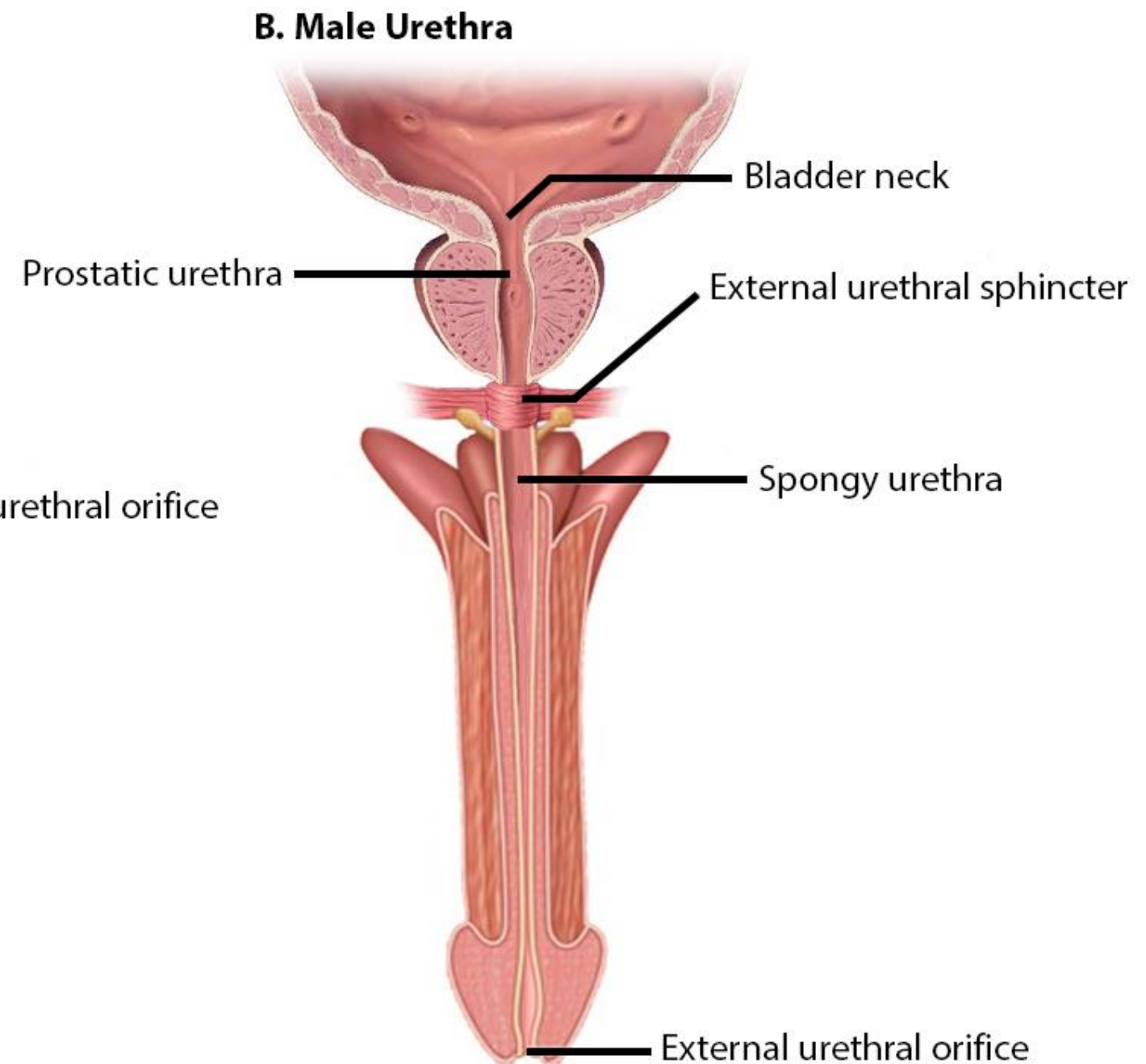
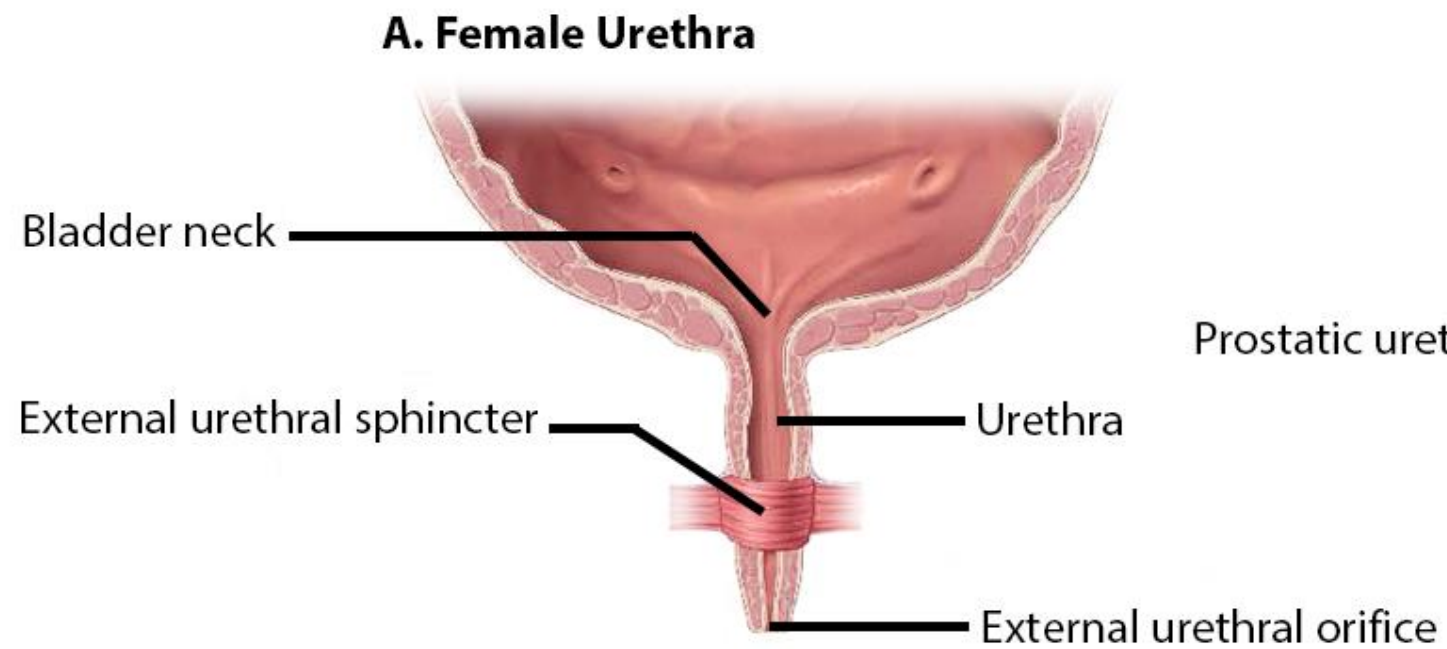
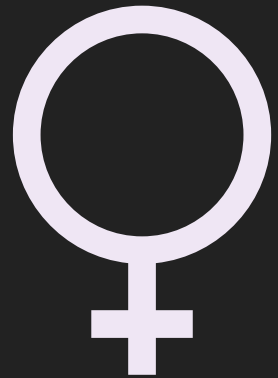
URINARY TRACT ANATOMY: URETHRA



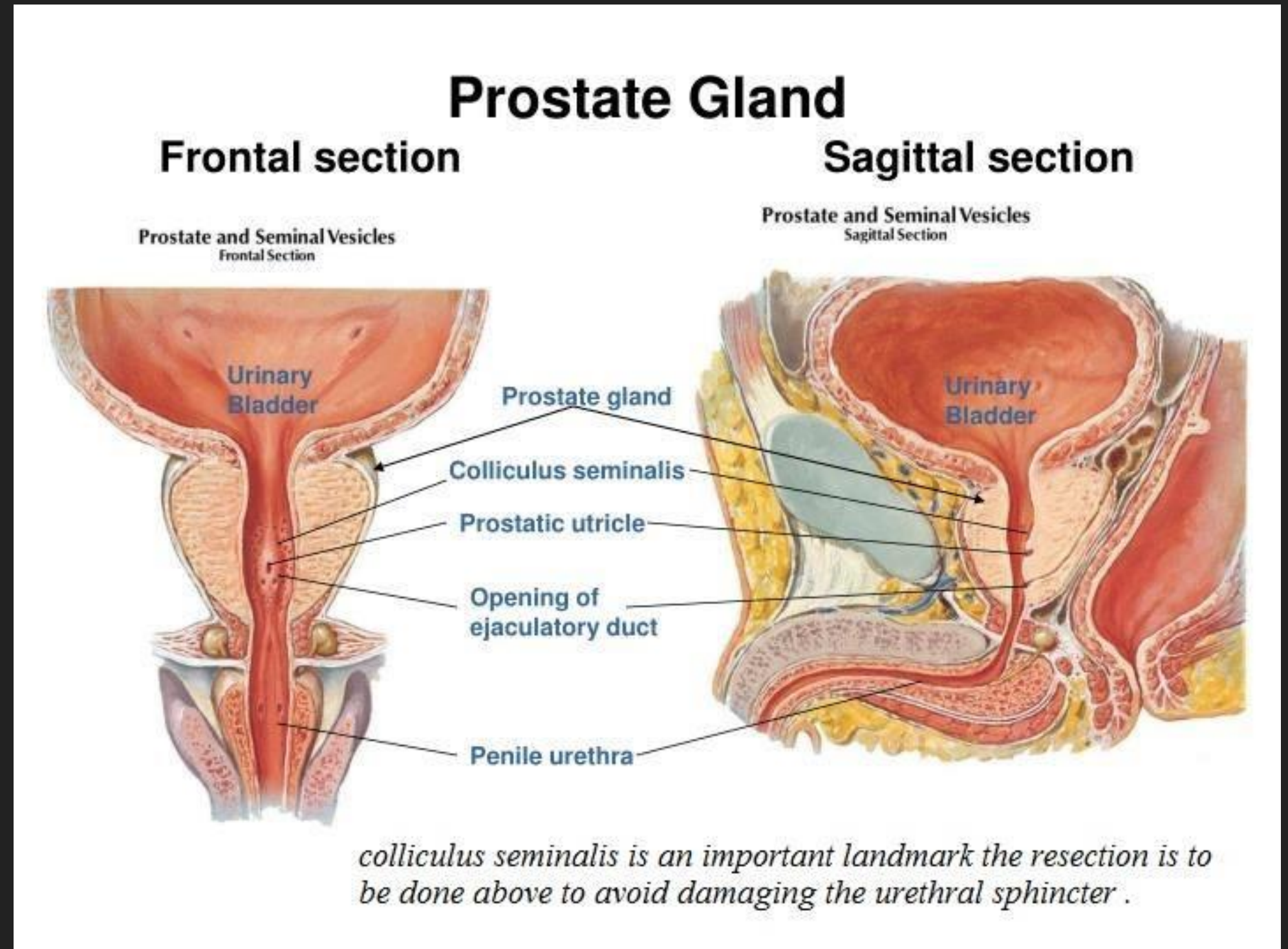
URETHRA



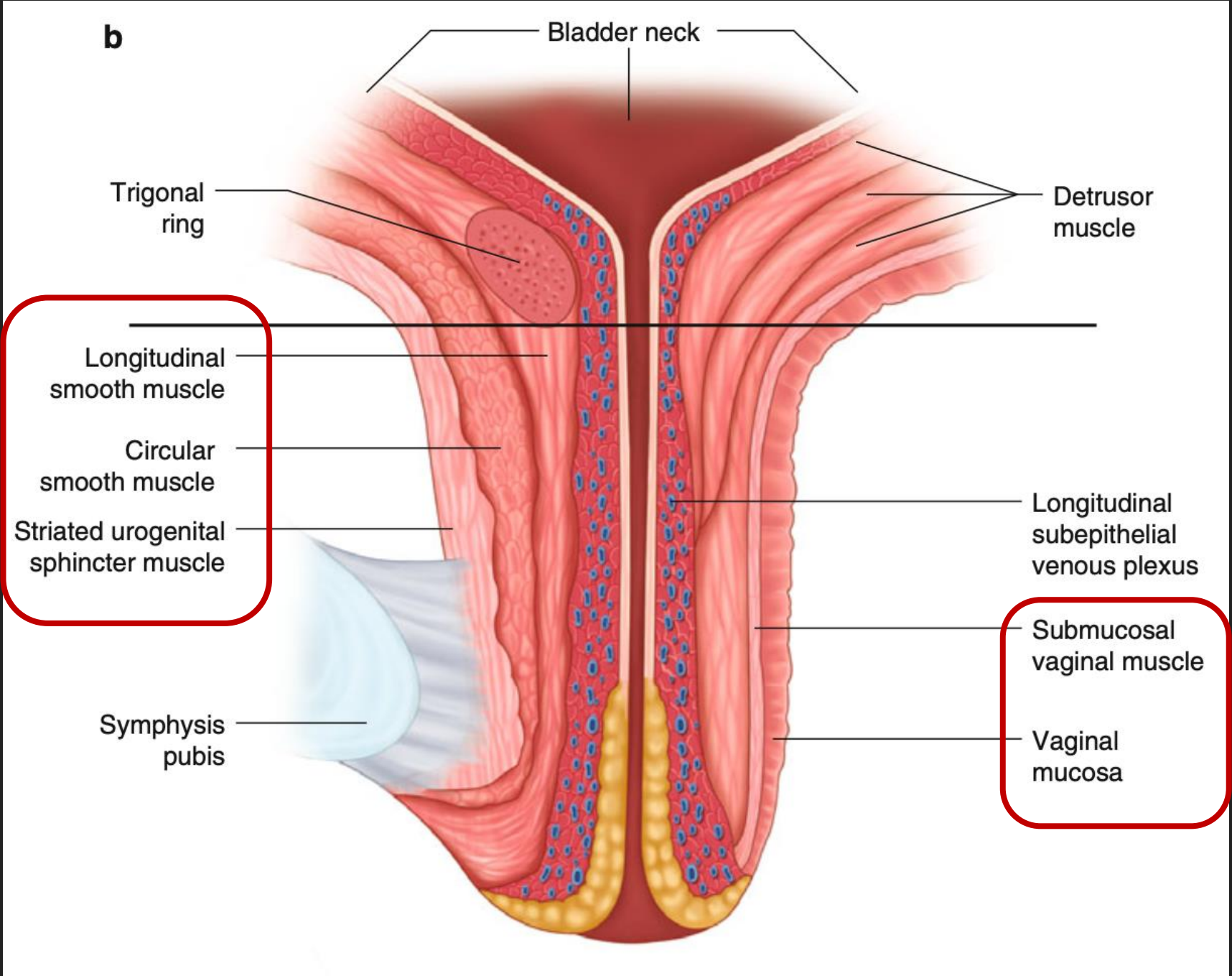
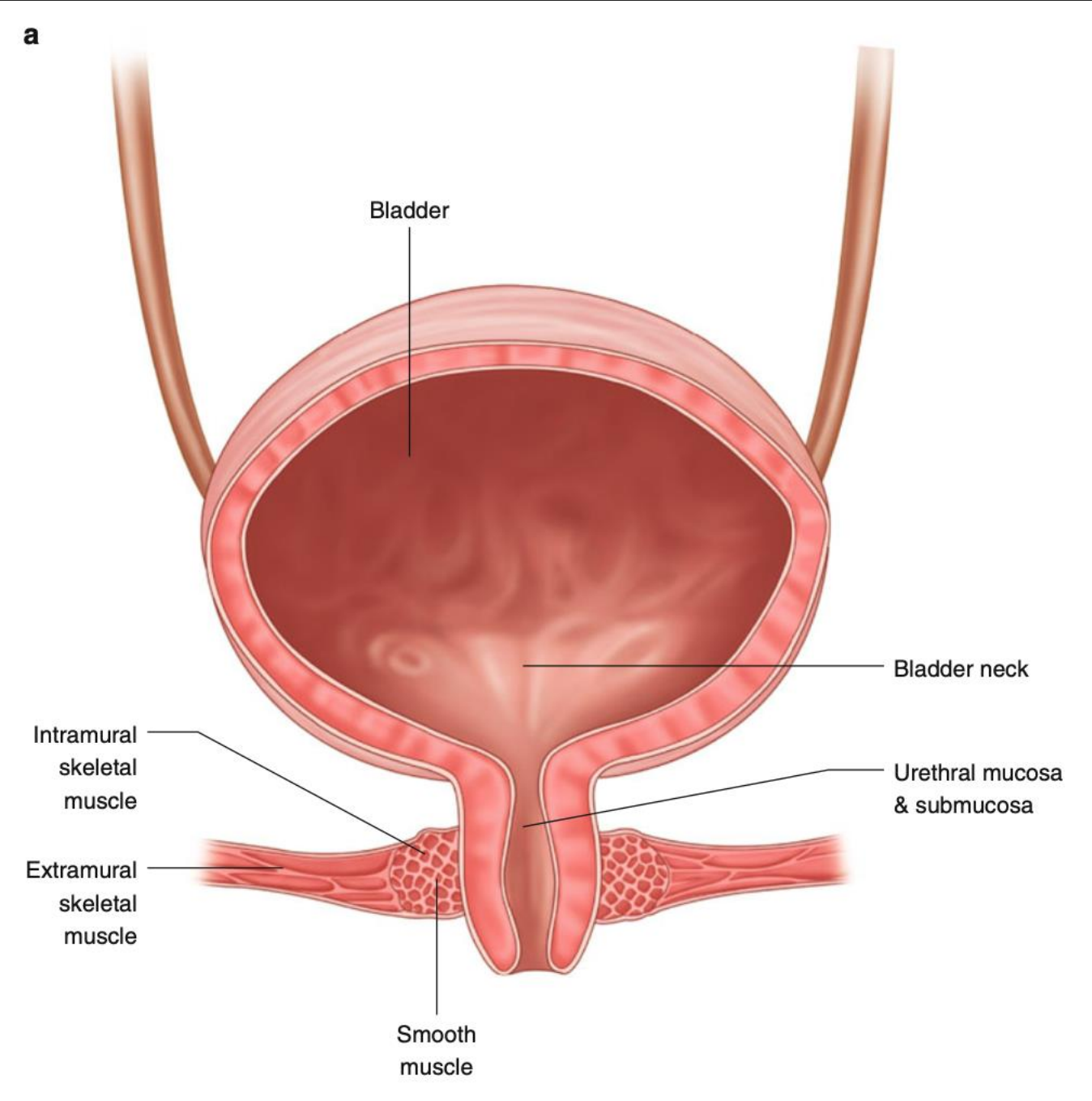
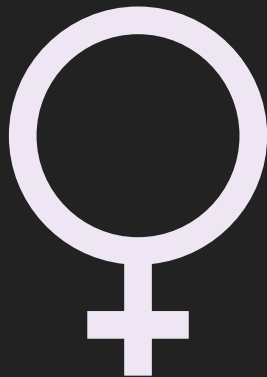
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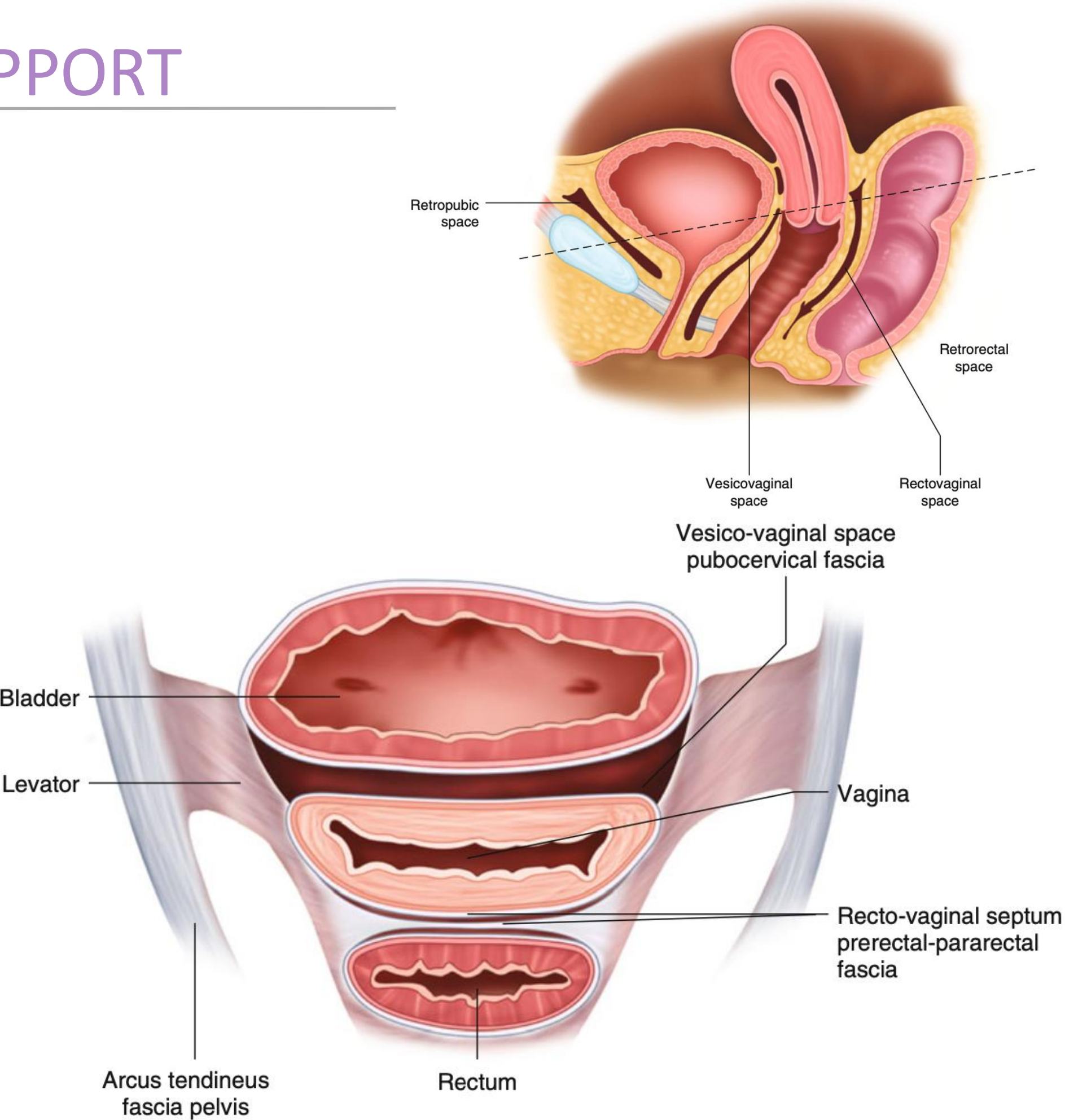
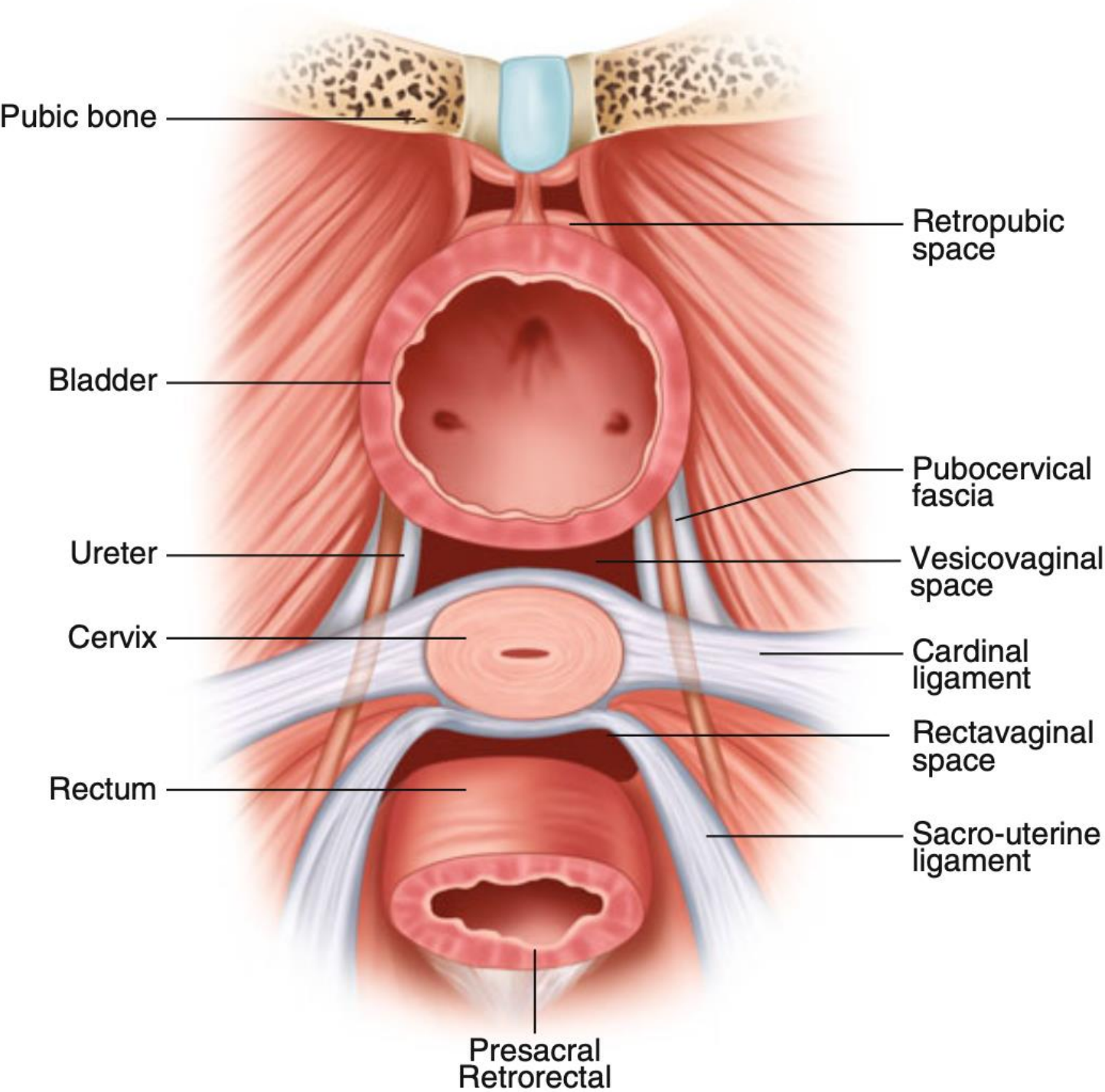
URETHRA ♂



URETHRA



LEVELS OF VAGINAL WALL SUPPORT BY DELANCEY (1992)



TESTIS, EPIDIDYMITIS AND VAS DEFERENCE

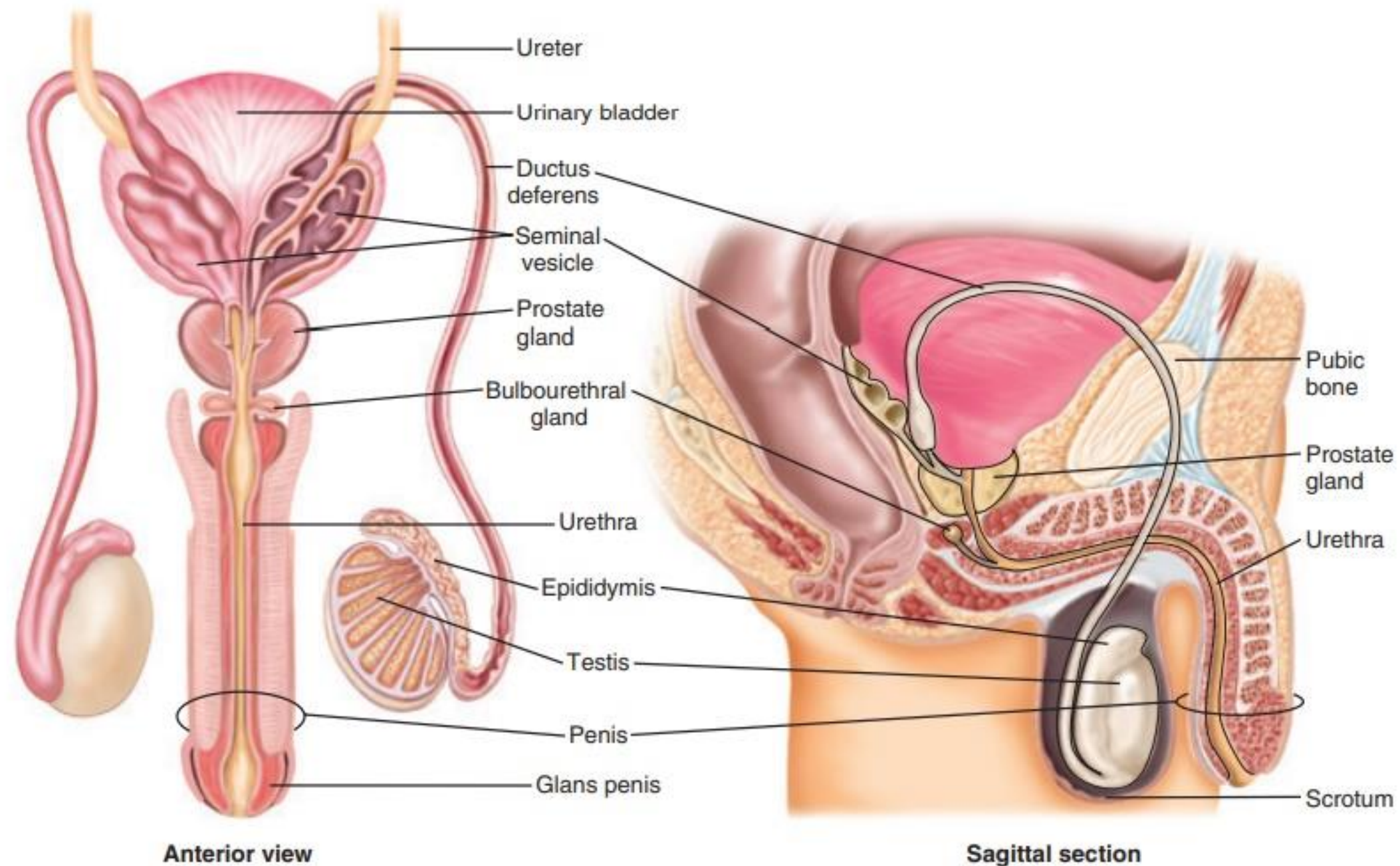
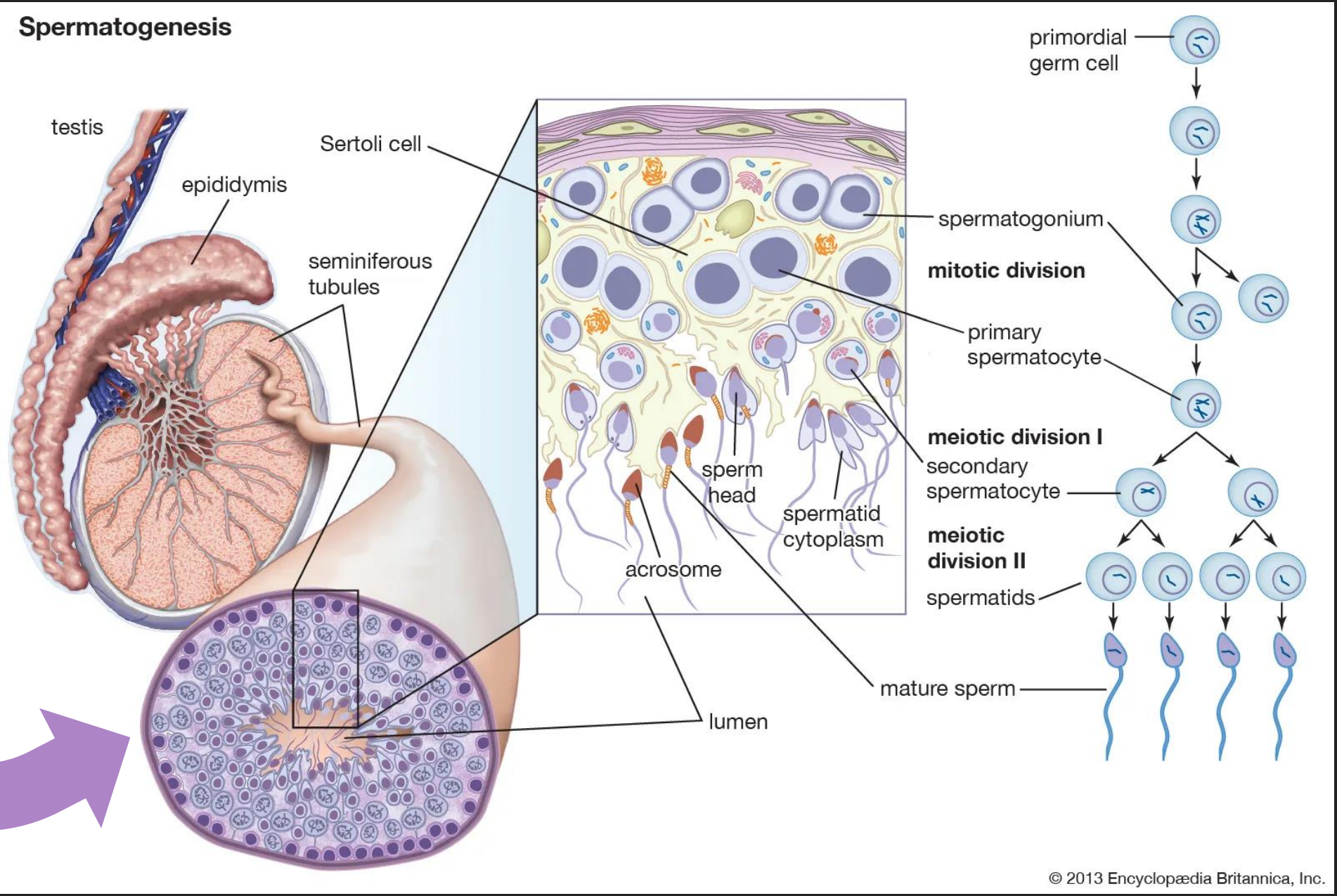
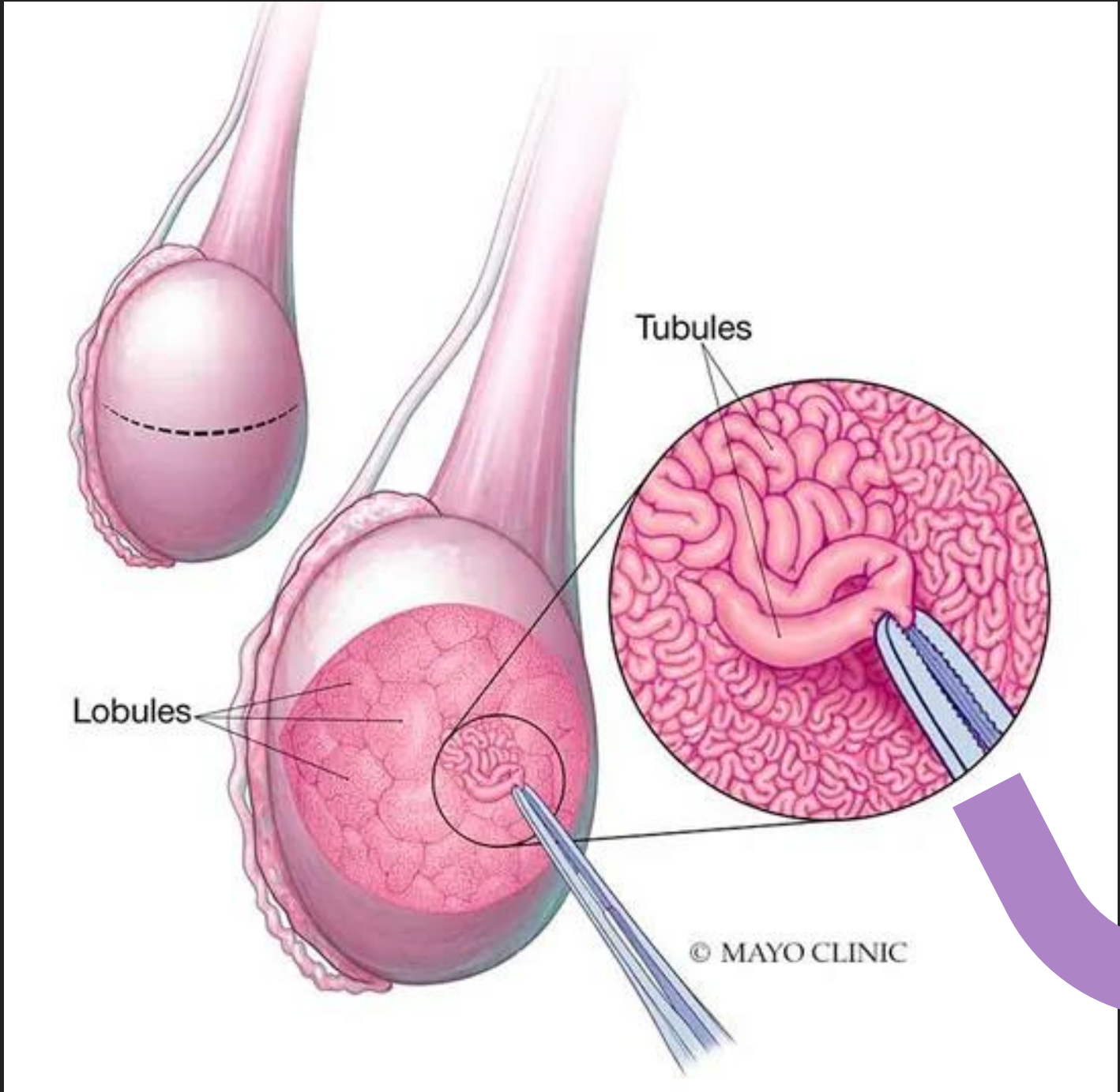
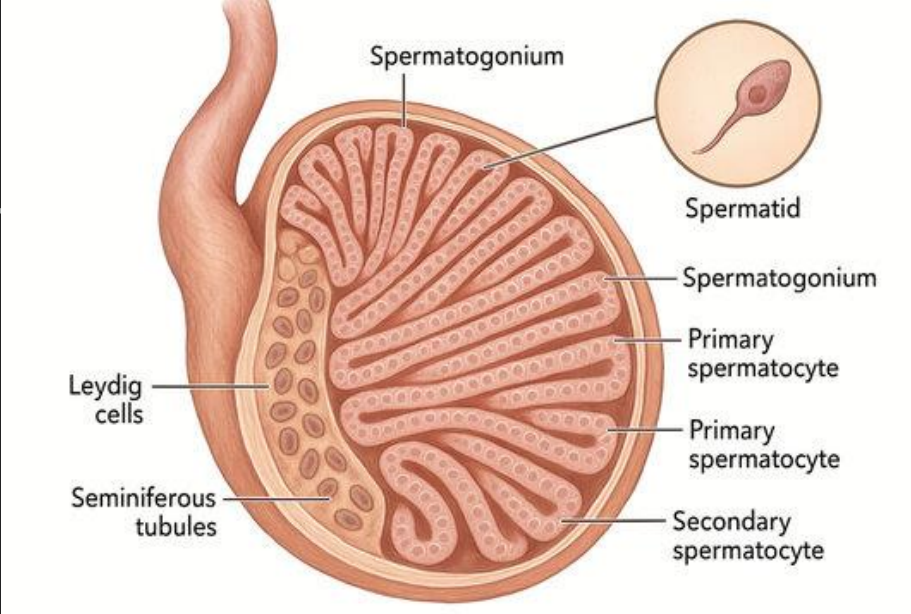
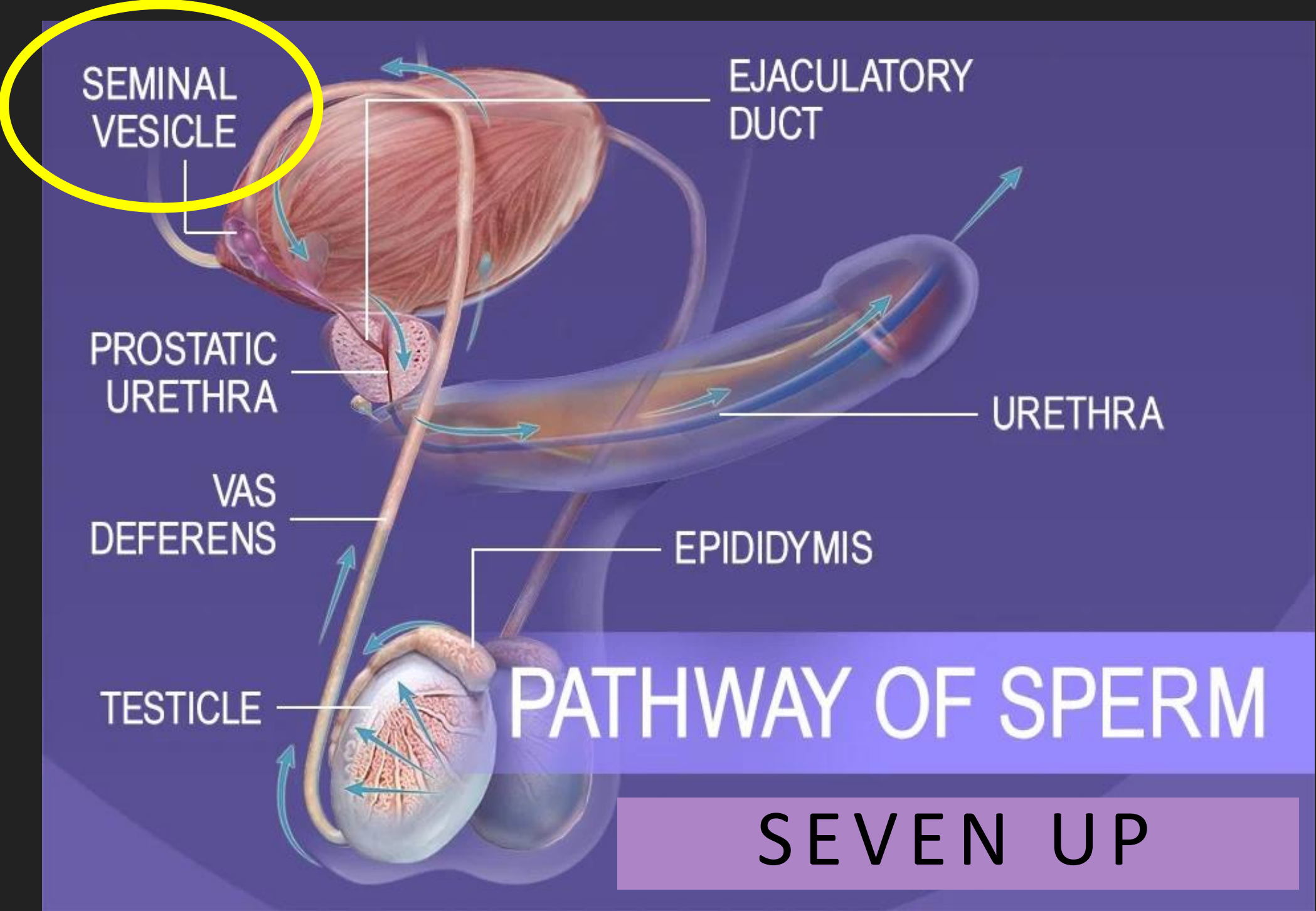


FIGURE 25-1 The male organs of reproduction.

TESTIS, EPIDIDYMITIS AND VAS DEFERENCE

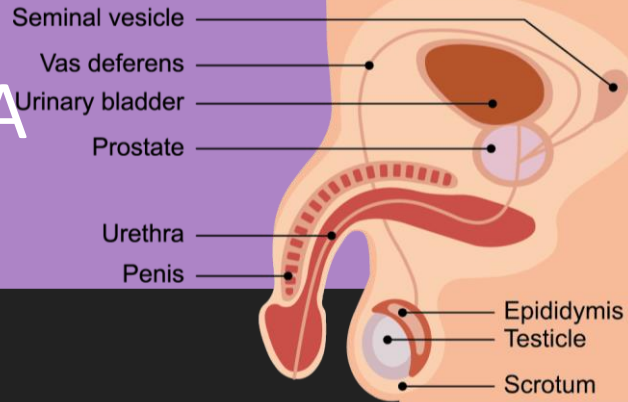


TESTIS, EPIDIDYMITIS AND VAS DEFERENCE



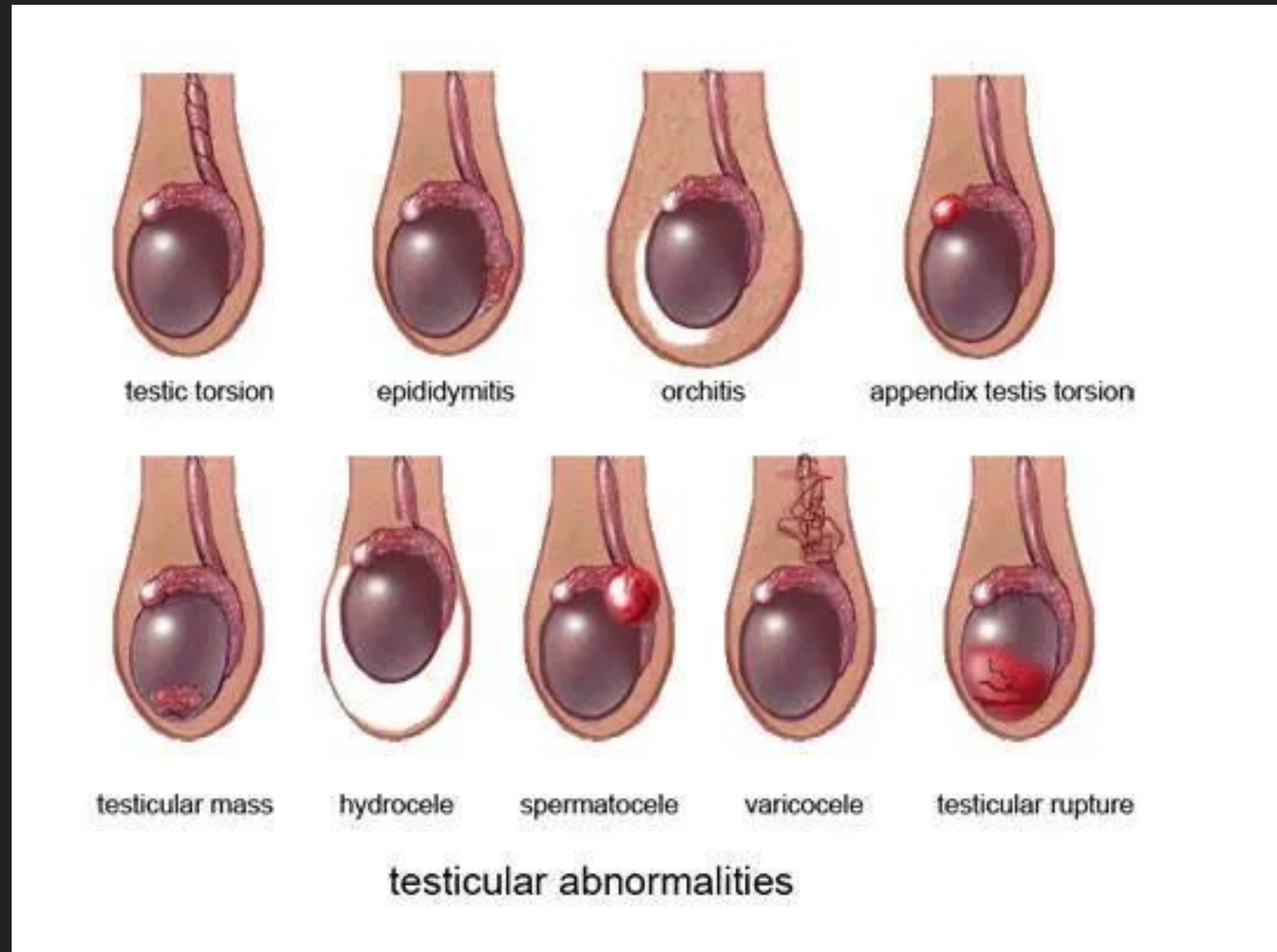
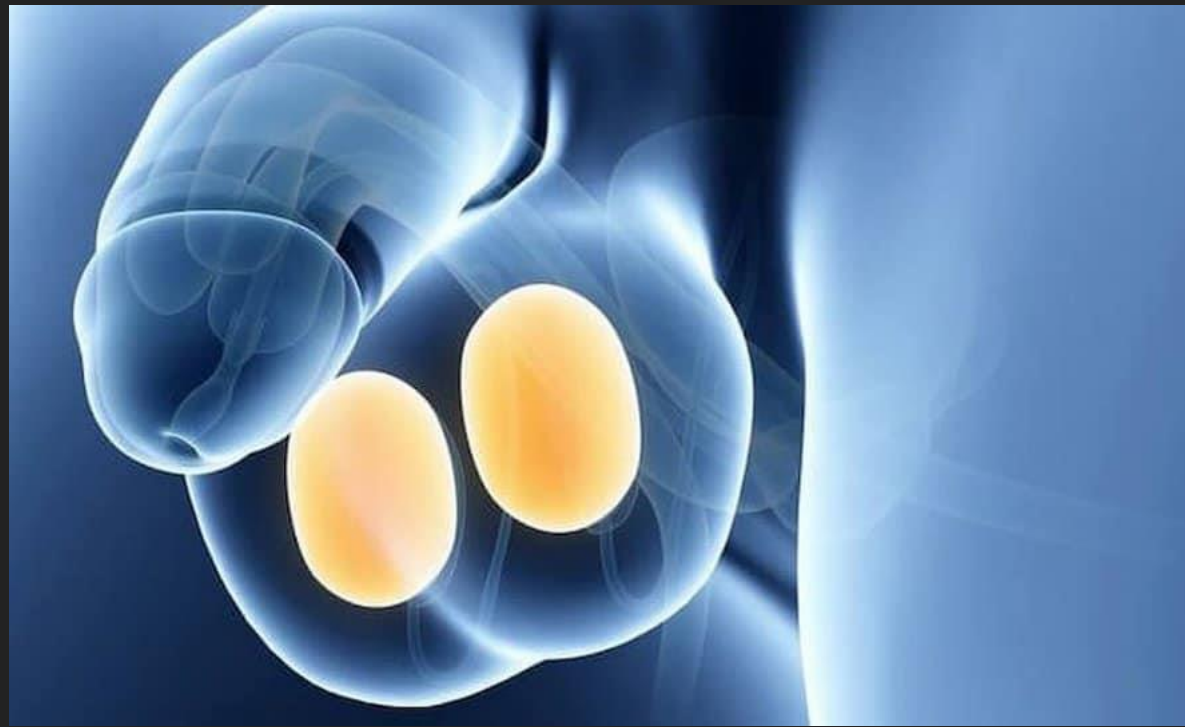
S SEMINIFEROUS TUBULE
E EPIDIDYMIS
V VAS DEFERENS
E EJACULATORY DUCT
N -NONE-

U URETHRA
P PENIS

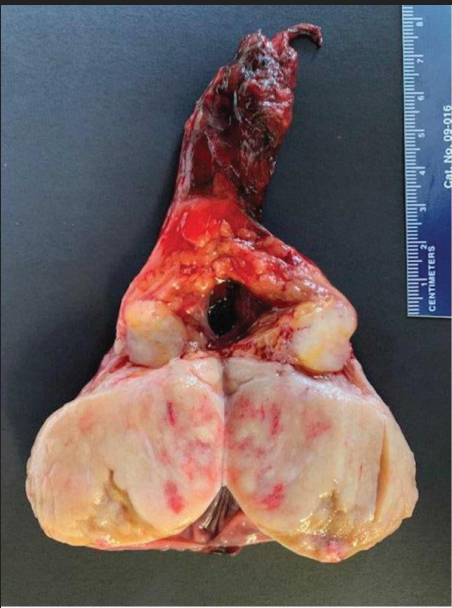
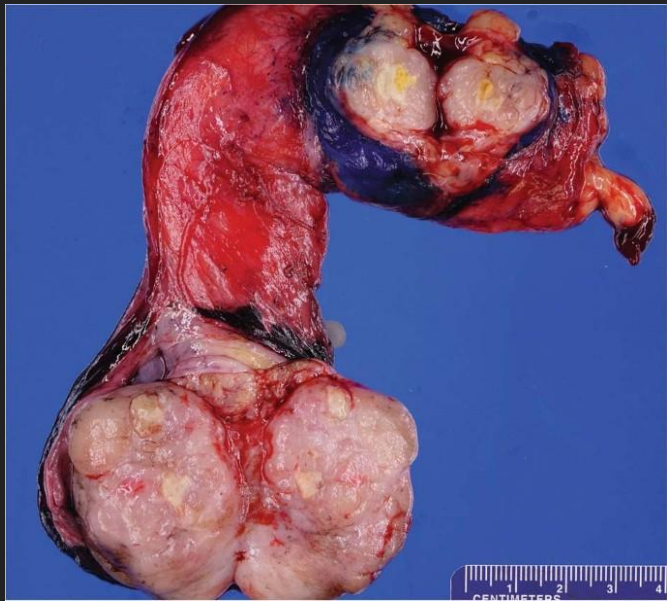
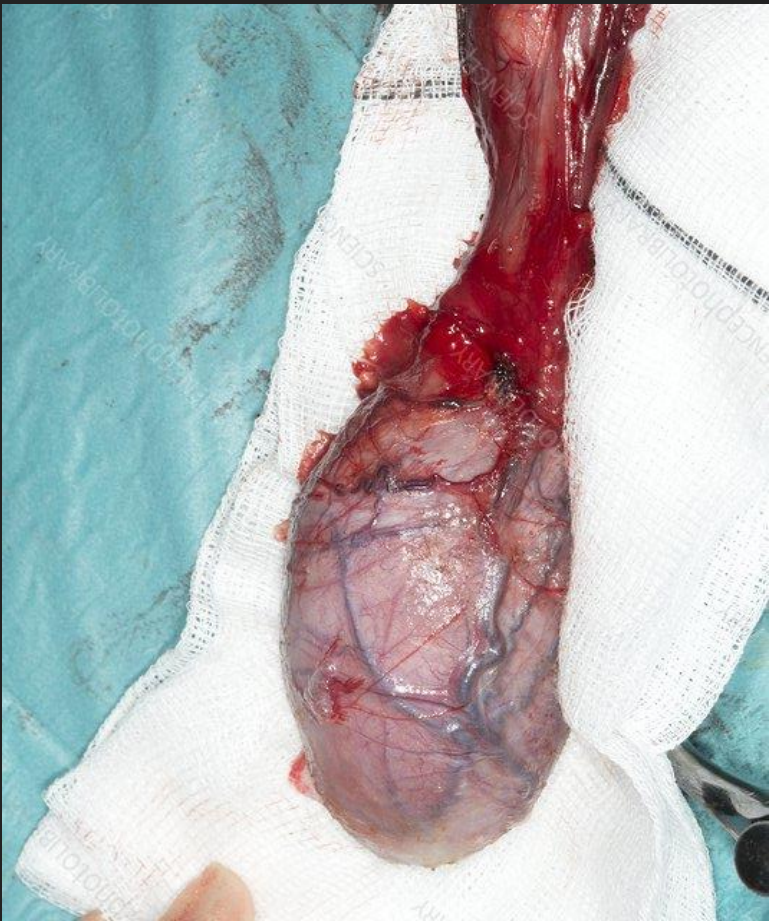
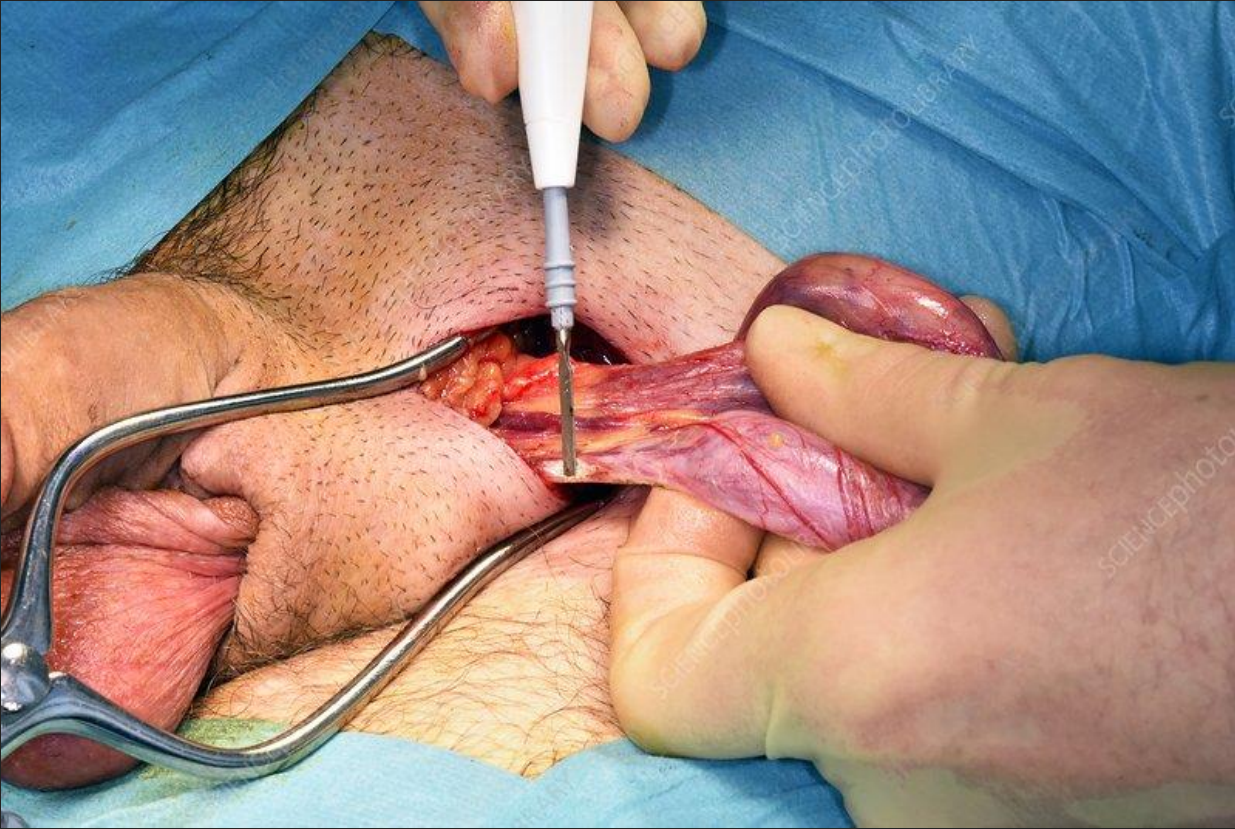
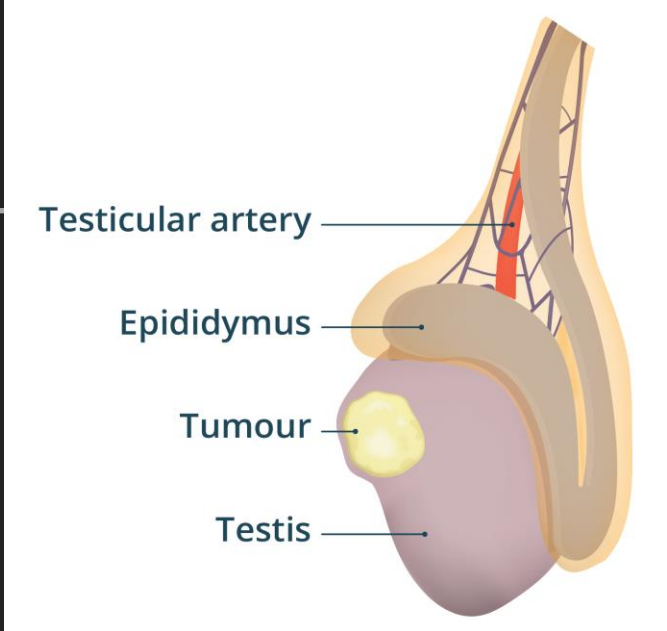


- **Produce seminal fluid** -Produce around 65–75% of the fluid that makes up semen.
- **Nourish sperm:** The fluid contains fructose.
- **Aid in fertilization:** Prostaglandins

TESTIS, EPIDIDYMITIS AND VAS DEFERENCE



TESTIS, EPIDIDYMITIS AND VAS DEFERENCE



[illegible]

UROLOGICAL

OPERATIONS

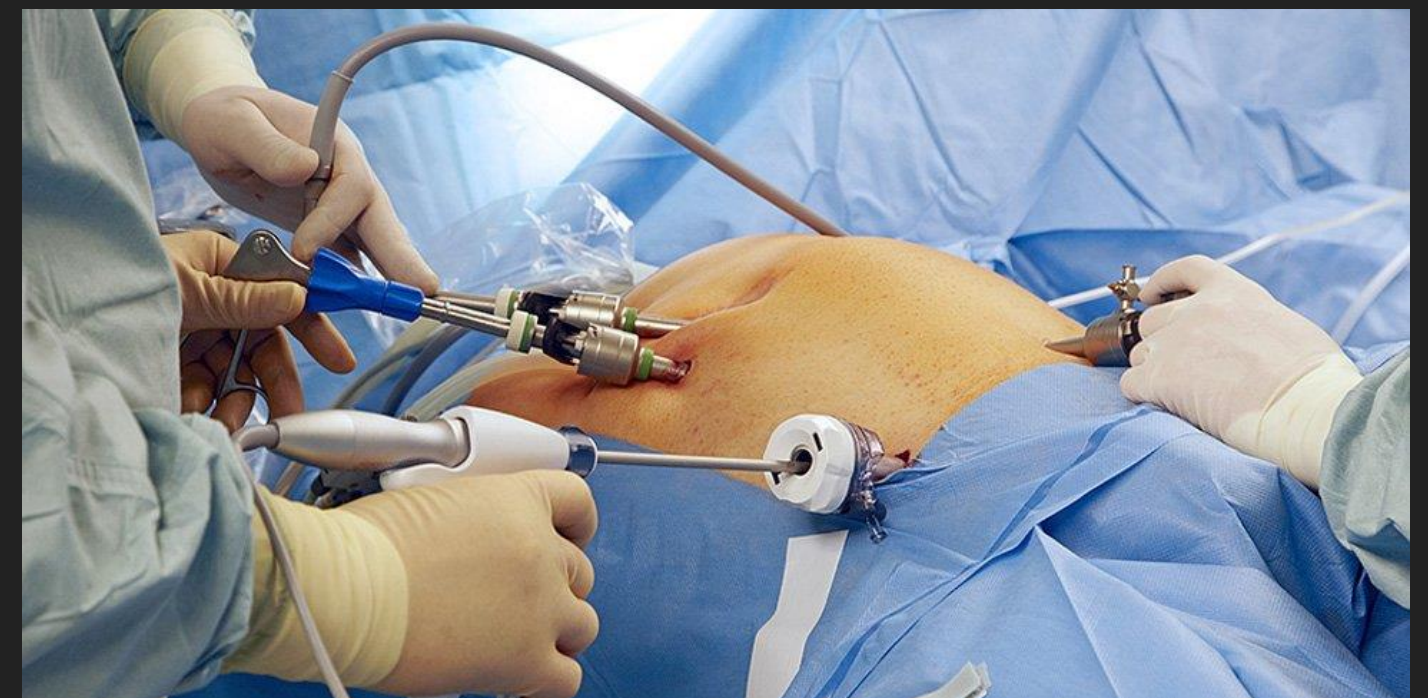
UROLOGICAL OPERATIONS

► **Classification:** *Approach*

- Open procedures
- Laparoscopic & Robot-Assisted Laparoscopic procedures
- Endoscopic procedures
- Microscopic procedures
- Misc.: ESWL

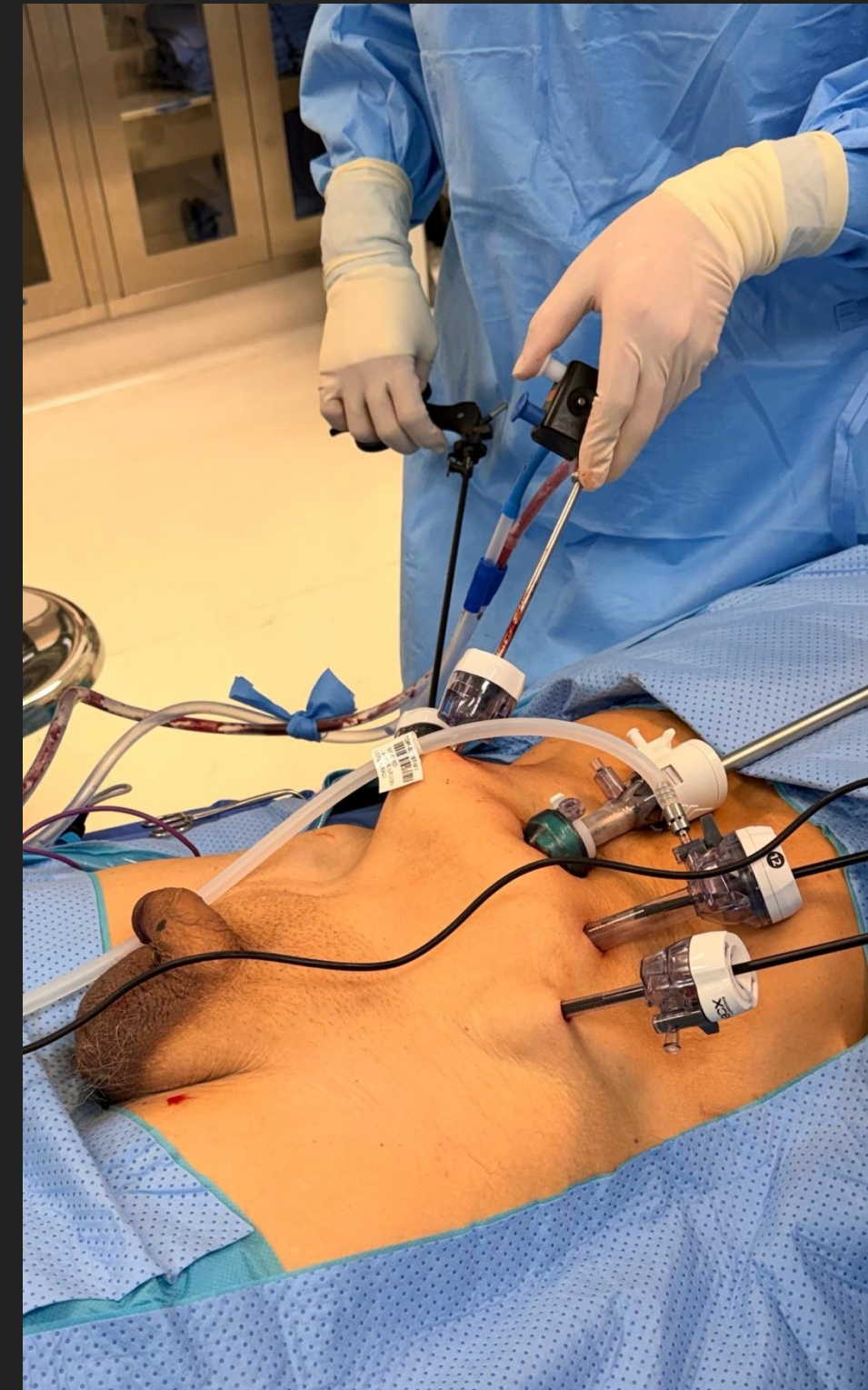
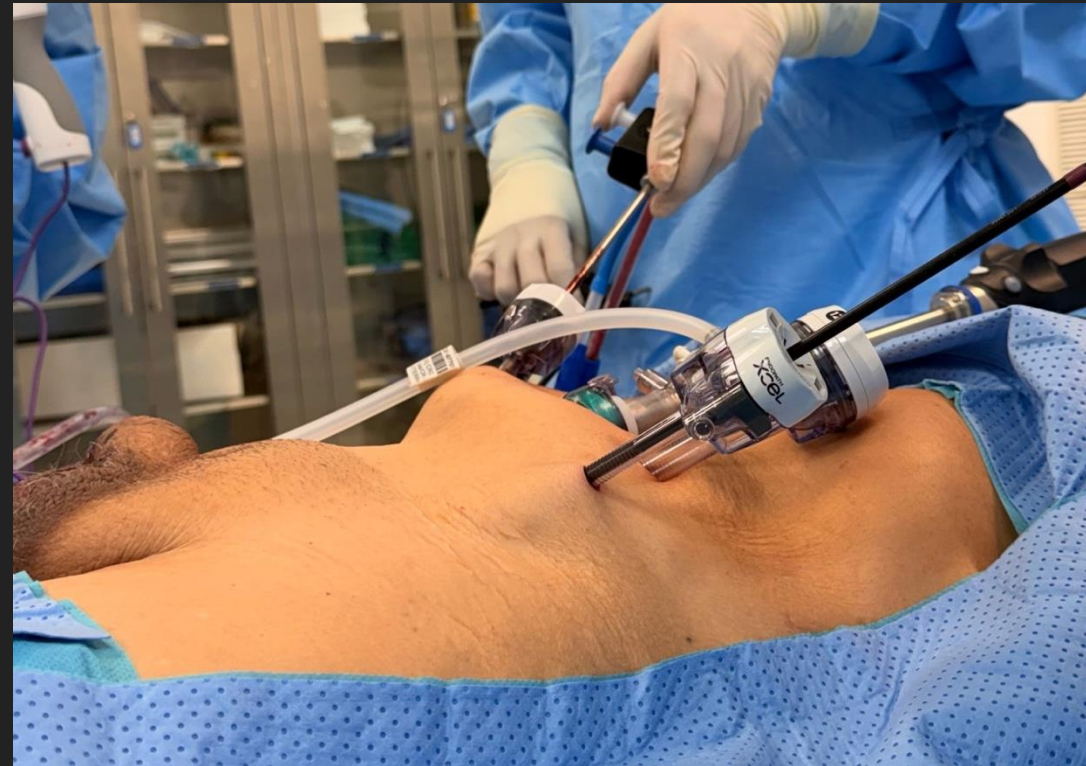
UROLOGICAL OPERATIONS

- ▶ **Benefit of *Laparoscopic surgery***
 - ▶ Less pain
 - ▶ Less time to recover
 - ▶ Less length of hospital stay
 - ▶ Better cosmetic outcome
 - ▶ Less intra-operative bleeding
(depend on procedure/surgeon's skill)



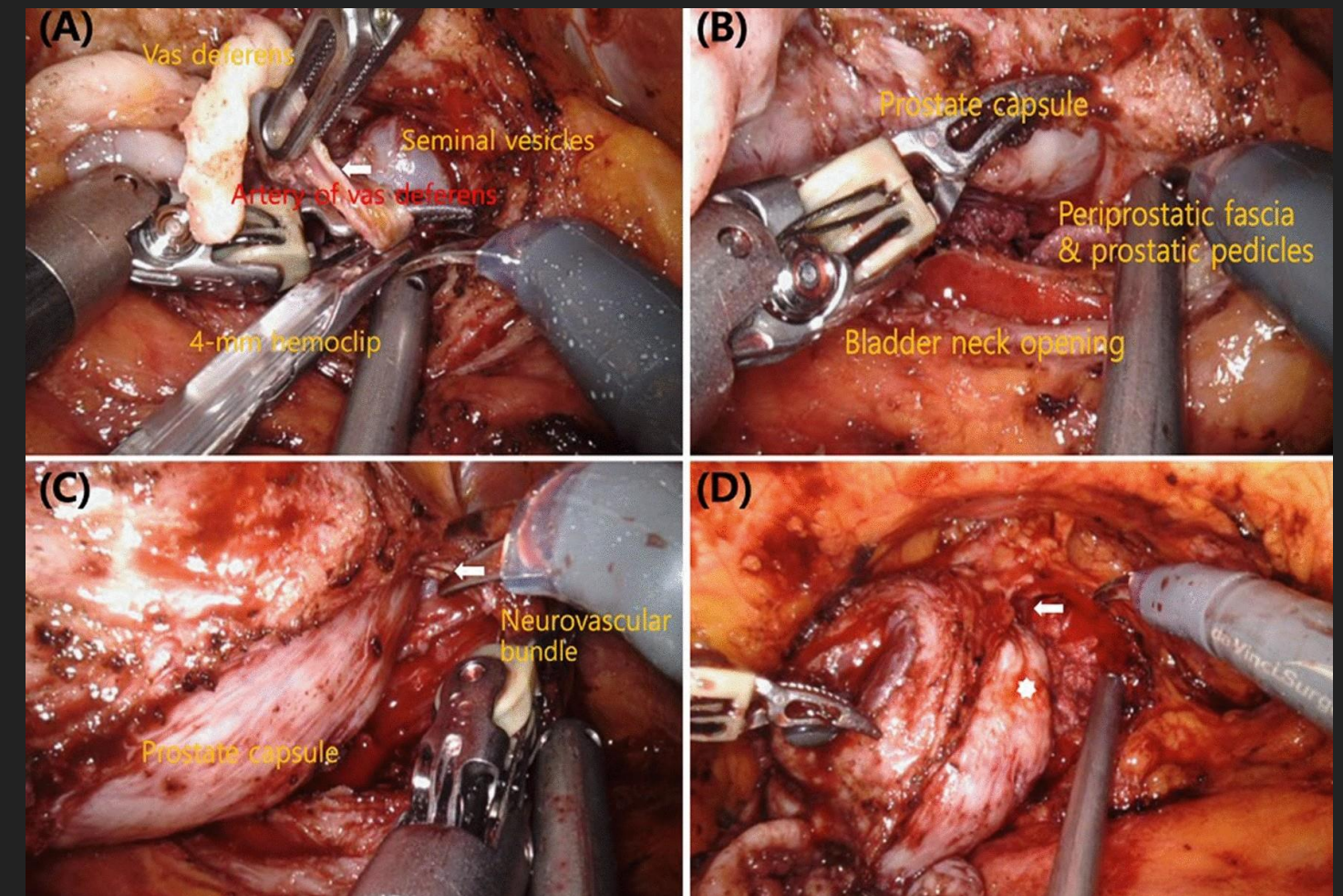
UROLOGICAL OPERATIONS

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 - ▶ Less length of hospital stay
 - ▶ Better cosmetic outcome
 - ▶ Less intra-operative bleeding
(depend on procedure/surgeon's skill)



UROLOGICAL OPERATIONS

- ▶ Benefit of *Robot-Assisted* Laparoscopic surgery
 - ▶ Ergonomic work station
 - ▶ More degree of freedom
 - ▶ No surgical tremor
 - ▶ Scale motion to allow precise microsurgery
 - ▶ Better vision; 3D



UROLOGICAL OPERATIONS

► Benefit of *Robot-Assisted* Laparoscopic surgery

- Ergonomic work station
- More degree of freedom
- No surgical tremor
- Scale motion to allow precise microsurgery
- Better vision; 3D



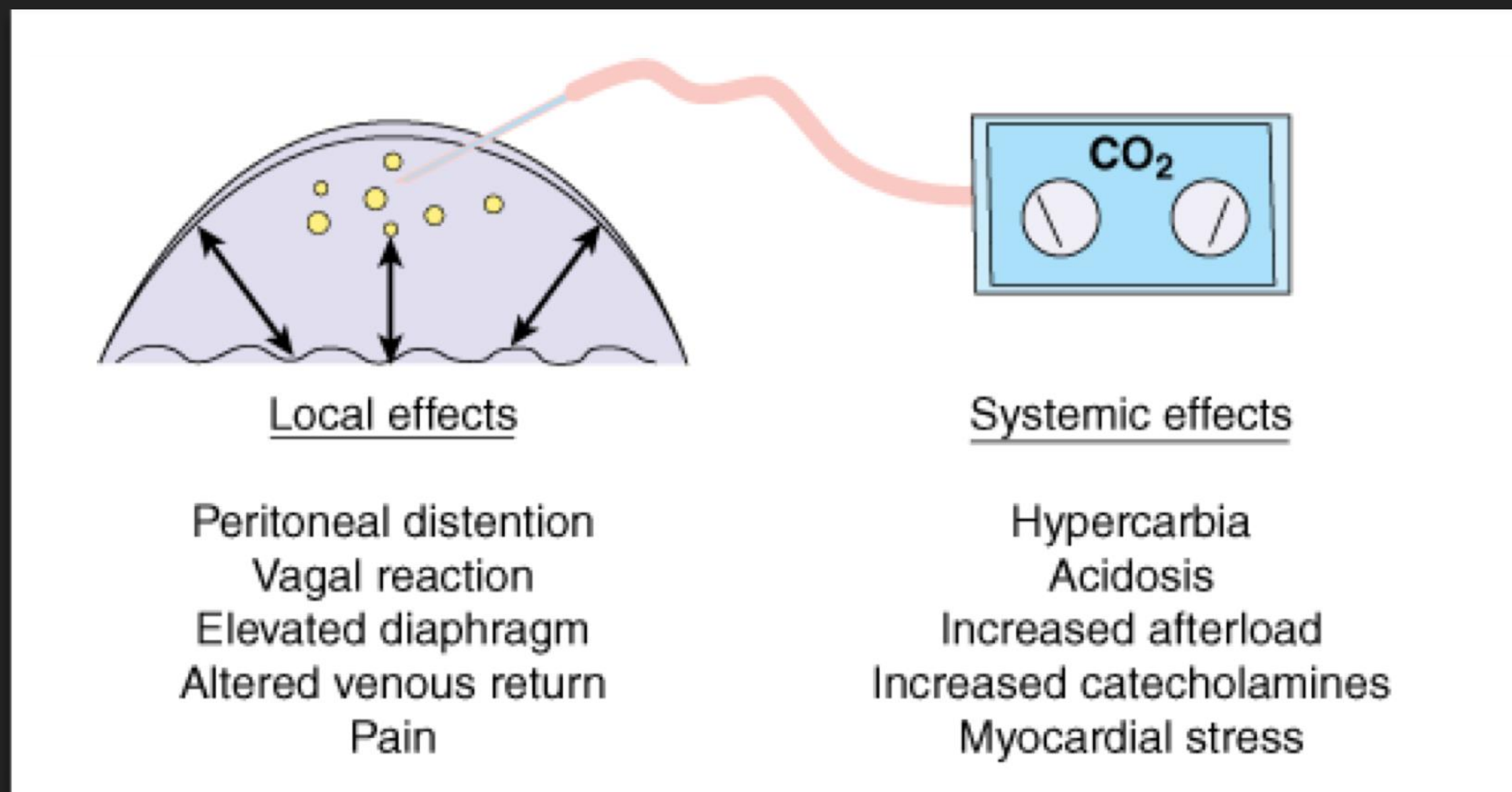
UROLOGICAL OPERATIONS

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 - ▶ Scale motion to allow precise microsurgery
 - ▶ Better vision; 3D



UROLOGICAL OPERATIONS

► Effects of CO₂



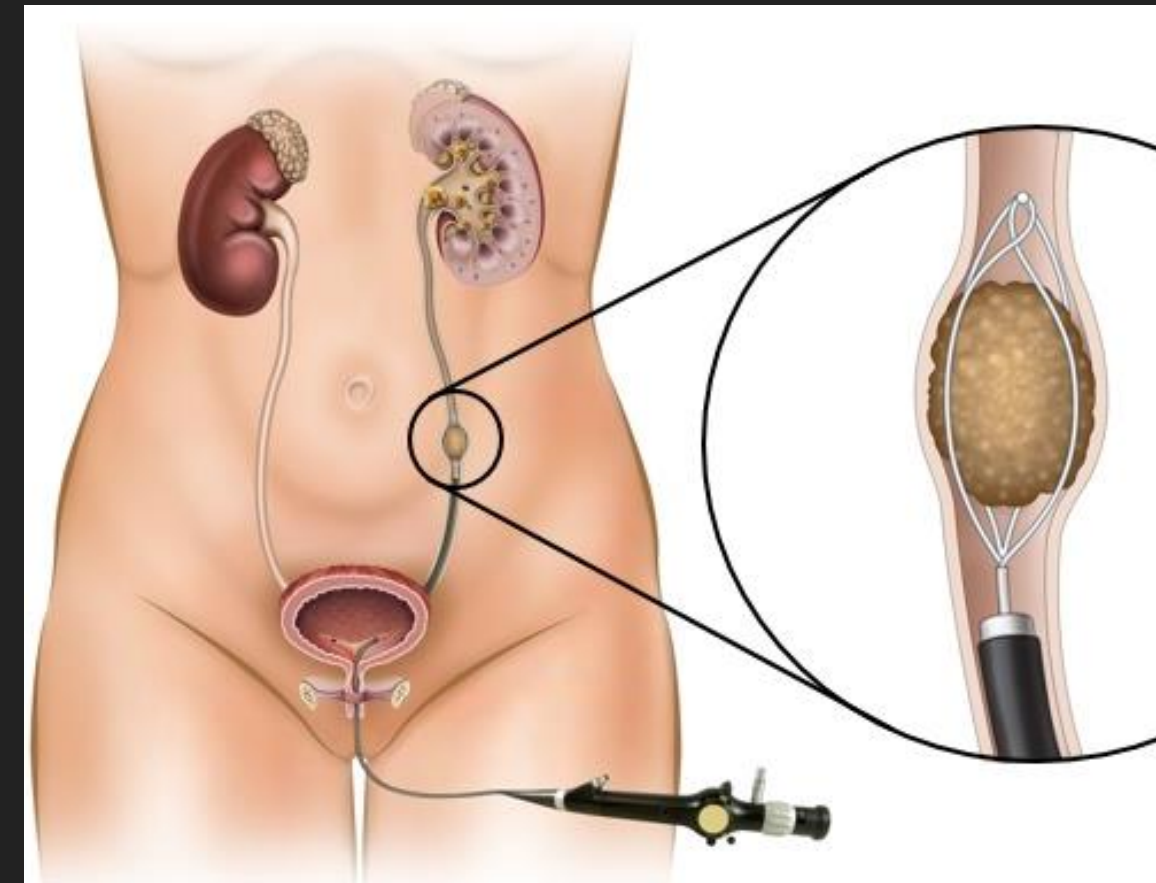
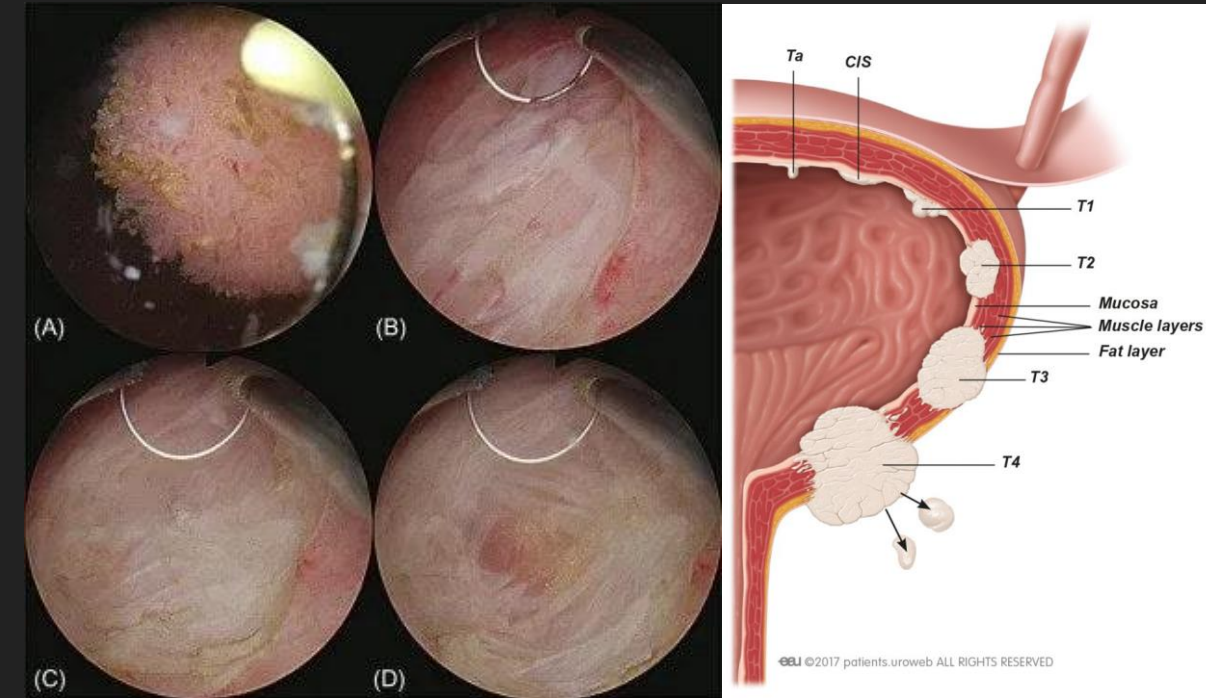
UROLOGICAL OPERATIONS

- **Effects of CO₂**
 - **Gas emboli:**
 - Hypotension during insufflation
 - Mill wheel murmur
 - *Rx: left lateral decubitus position with the head down to trap the gas in the apex of the right ventricle*
 - *!! Stop inflation of CO₂ !!*

UROLOGICAL OPERATIONS

► *Endoscopic procedures*

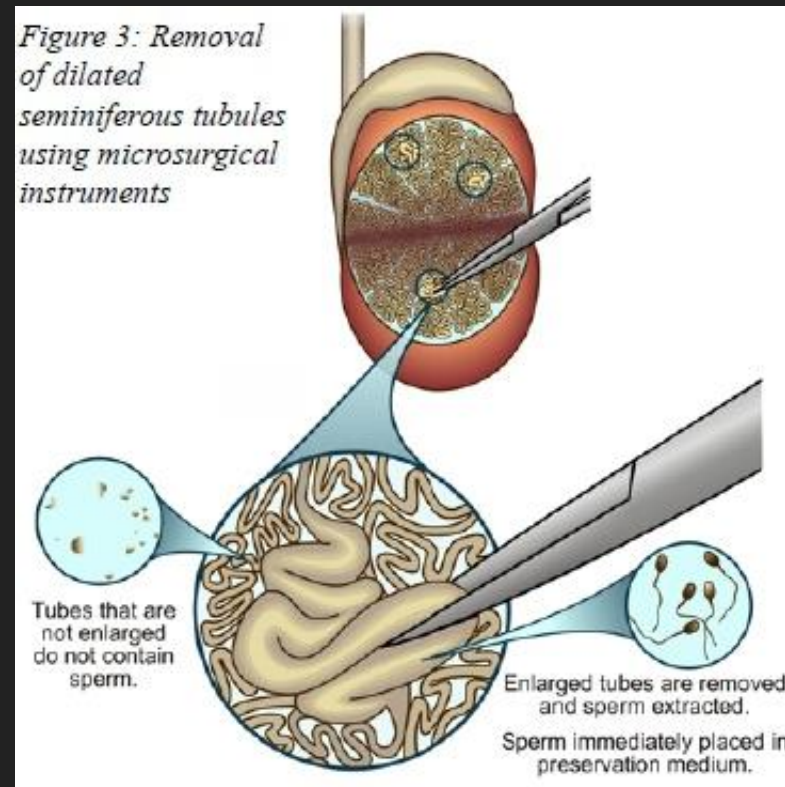
- Transurethral/natural orifice surgeries (and through puncture site)
- Mostly **retrograde** approach
- Irrigating fluid: **Water, NSS**



UROLOGICAL OPERATIONS

► *Microscopic procedures*

- Varicocelelectomy
- Vaso-vasostomy
- Micro-TESE

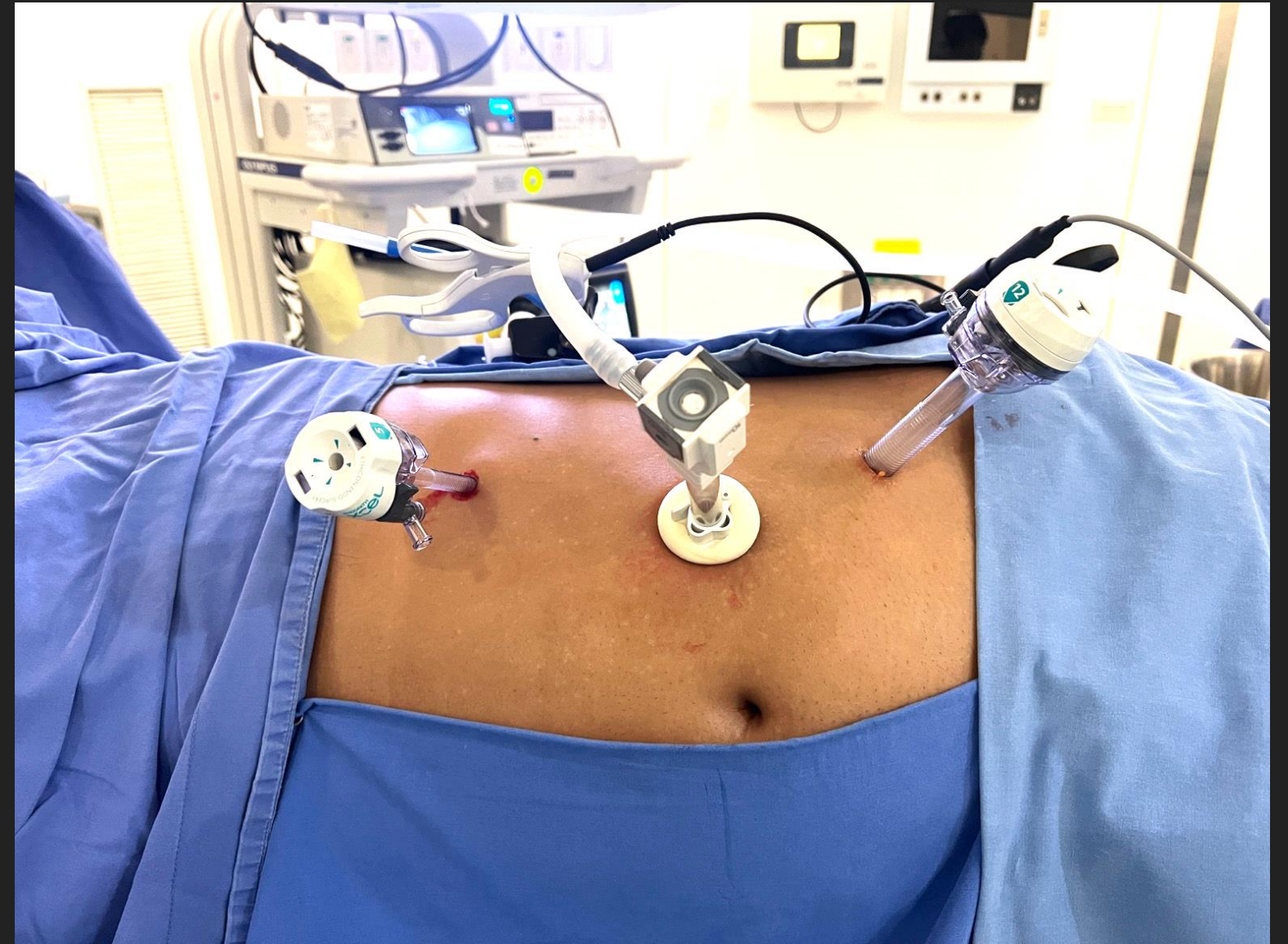
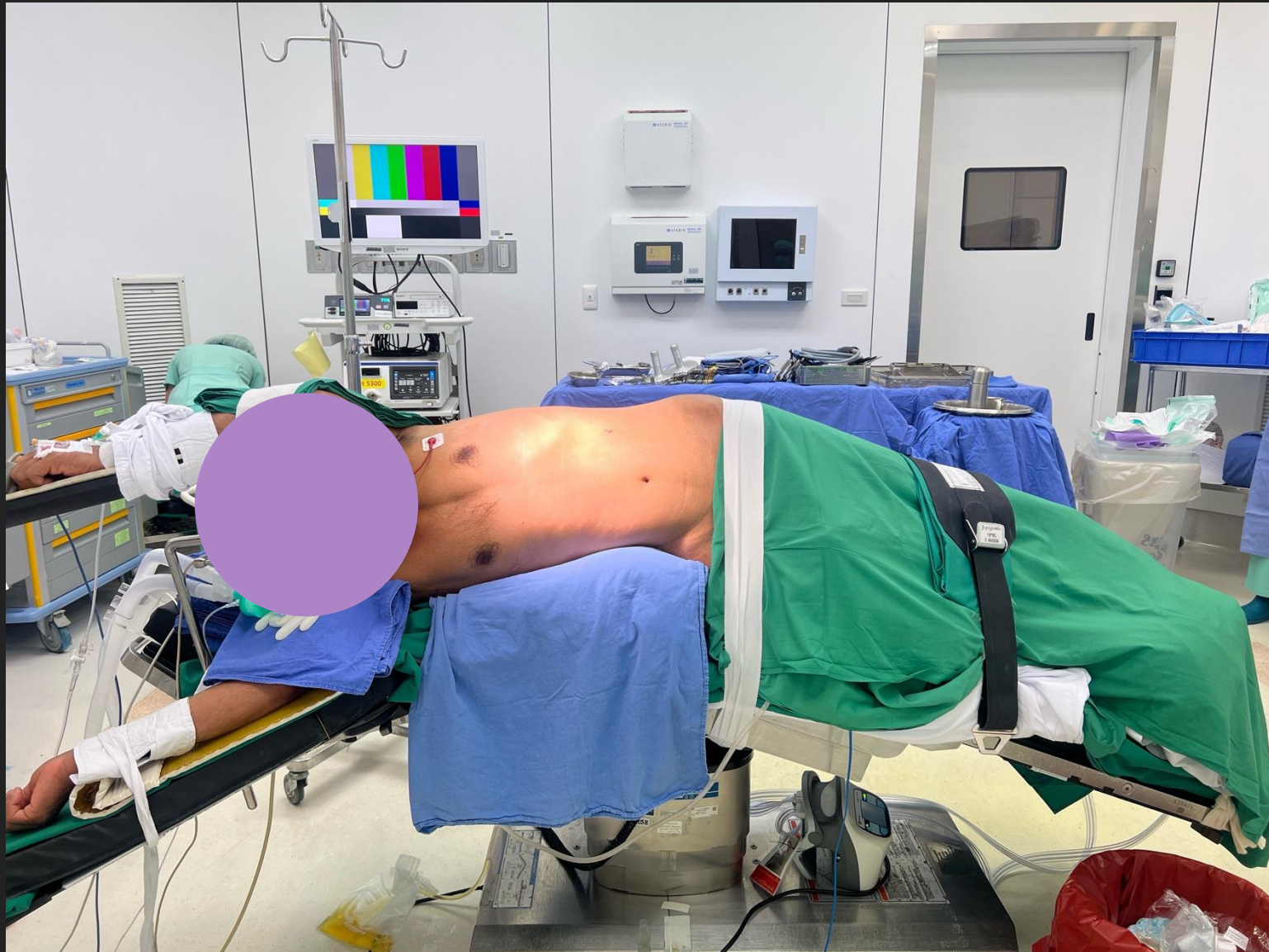


KIDNEY AND URETER

- ▶ Nephrectomy: radical, simple, subcapsular, partial
- ▶ Anatomic nephrolithotomy
- ▶ Deroof renal cyst
- ▶ Kidney transplantation
- ▶ Renal pelvis: pyeloplasty, pyelolithotomy, pyelostomy
- ▶ Ureter: ureterolithotomy, uretero-ureterostomy, ureteroneocystostomy
- ▶ Endoscopic: URS, URSL, RIRS, endopyelotomy, endoureterotomy

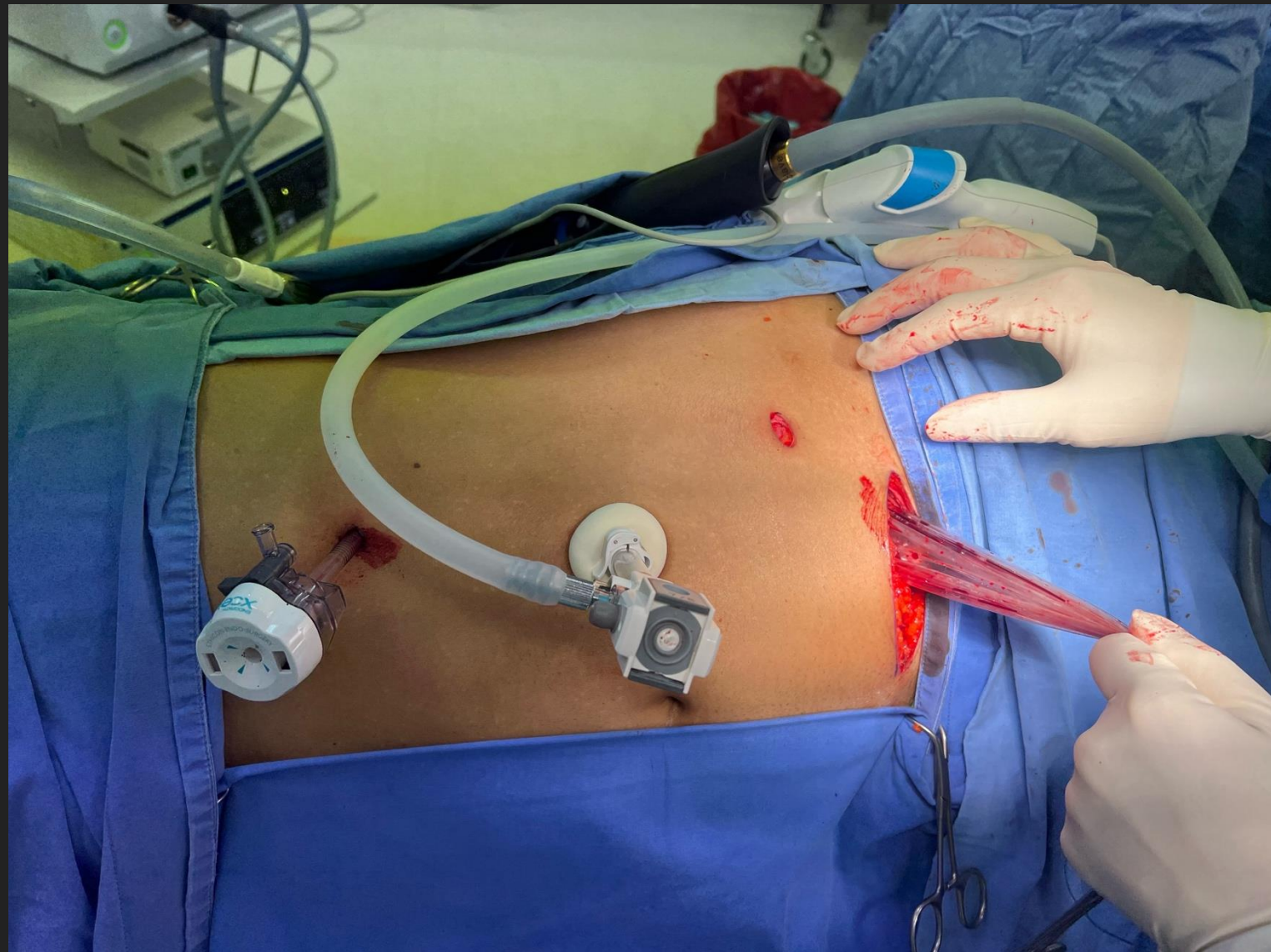
KIDNEY AND URETER

- Nephrectomy: radical, simple, subcapsular, partial



KIDNEY AND URETER

- Nephrectomy: radical, simple, subcapsular, partial



KIDNEY AND URETER

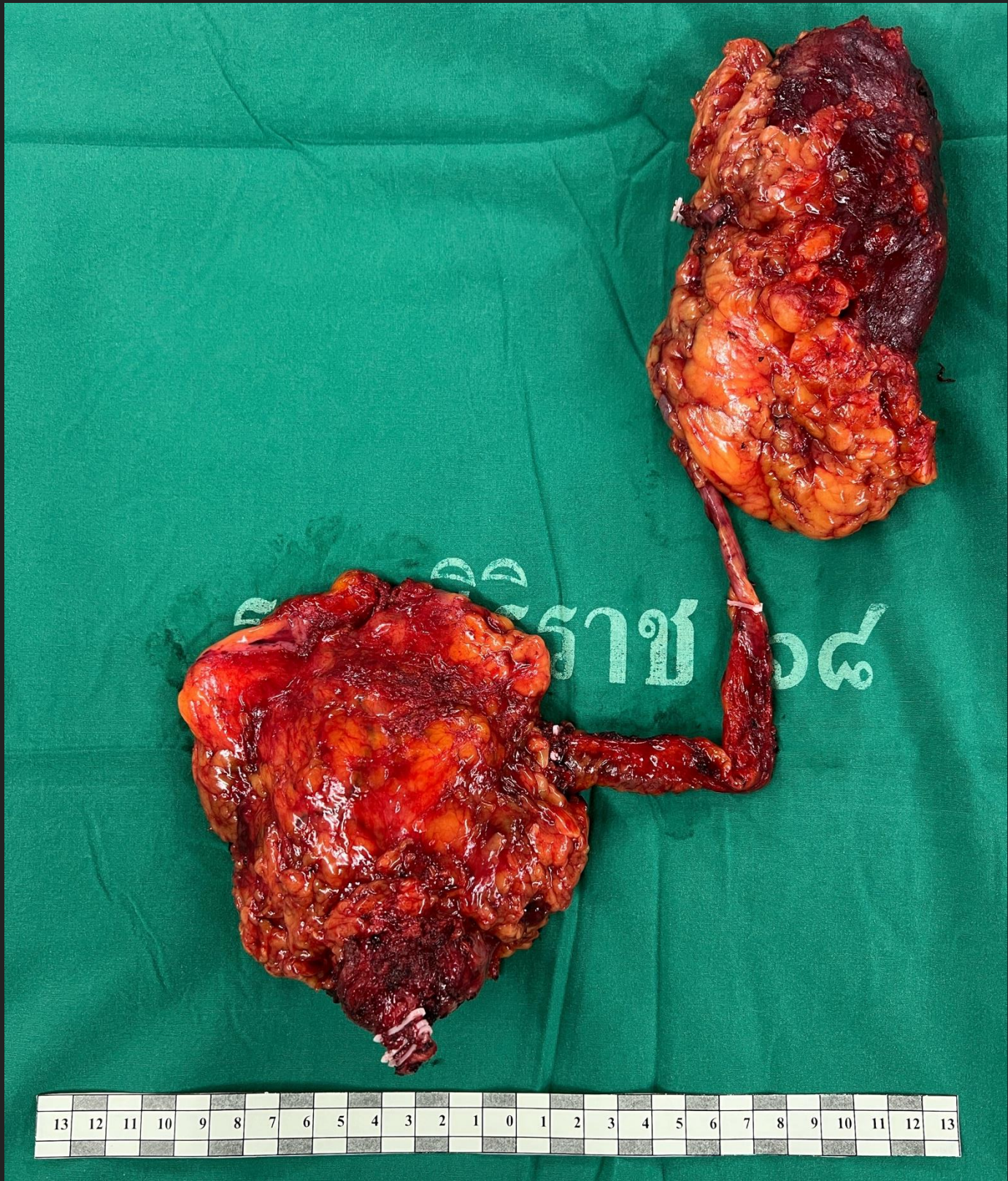
- Nephrectomy: radical, simple, subcapsular, partial



BLADDER

- ▶ Cystectomy: partial, radical
- ▶ Cystoplasty/ Augmentation cystoplasty
- ▶ Cystolithotomy
- ▶ Bladder diverticulectomy
- ▶ Ureterocele excision
- ▶ Vesicostomy
- ▶ Boari flap and Psoas hitch
- ▶ Endoscopic: TUR-BT, TUI-BN, Cystolitholapaxy/Cystolithotripsy

BLADDER



PROSTATE GLAND AND URETHRA

- ▶ Prostatectomy: simple, radical
- ▶ Endoscopic: TUR-P, TUI-P, Enucleation, TRUS-Bx
- ▶ Urethrectomy
- ▶ Urethroplasty
- ▶ Urethral diverticulectomy
- ▶ Urethrolysis, Bladder neck suspension, Midurethral sling
- ▶ Endoscopic: Internal urethrotomy

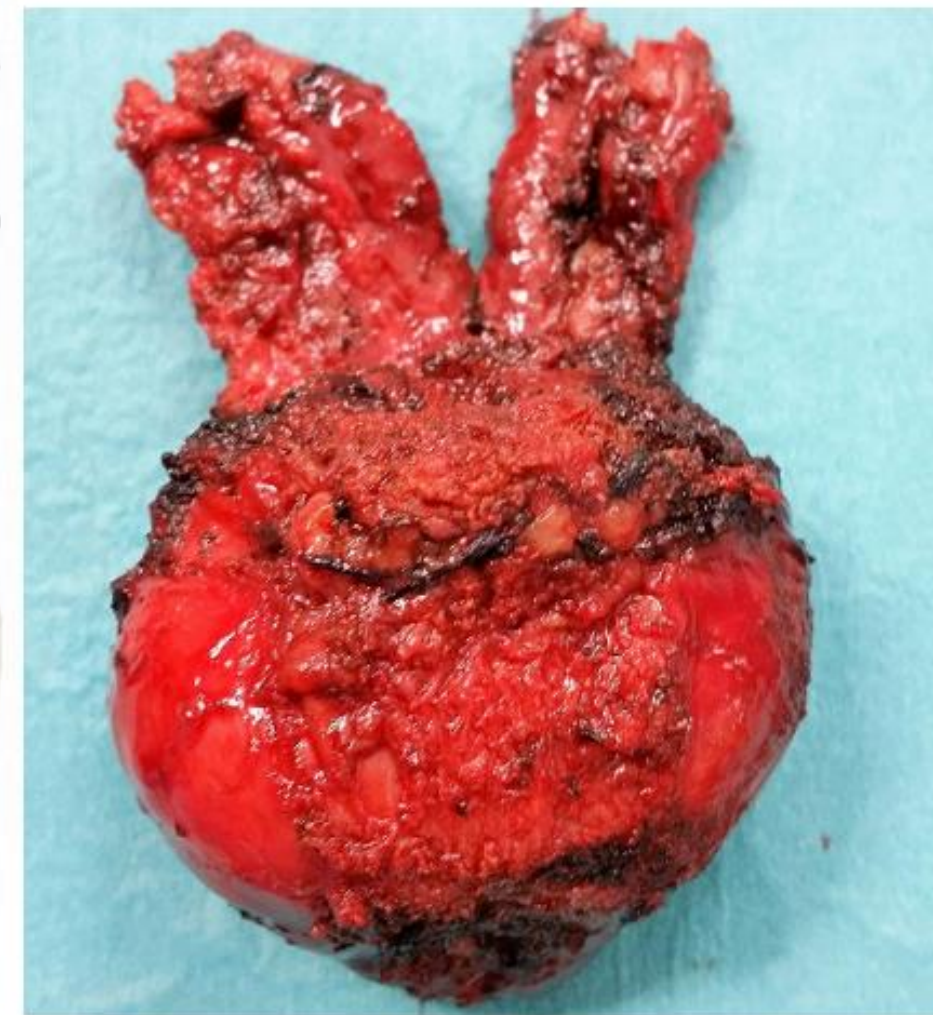
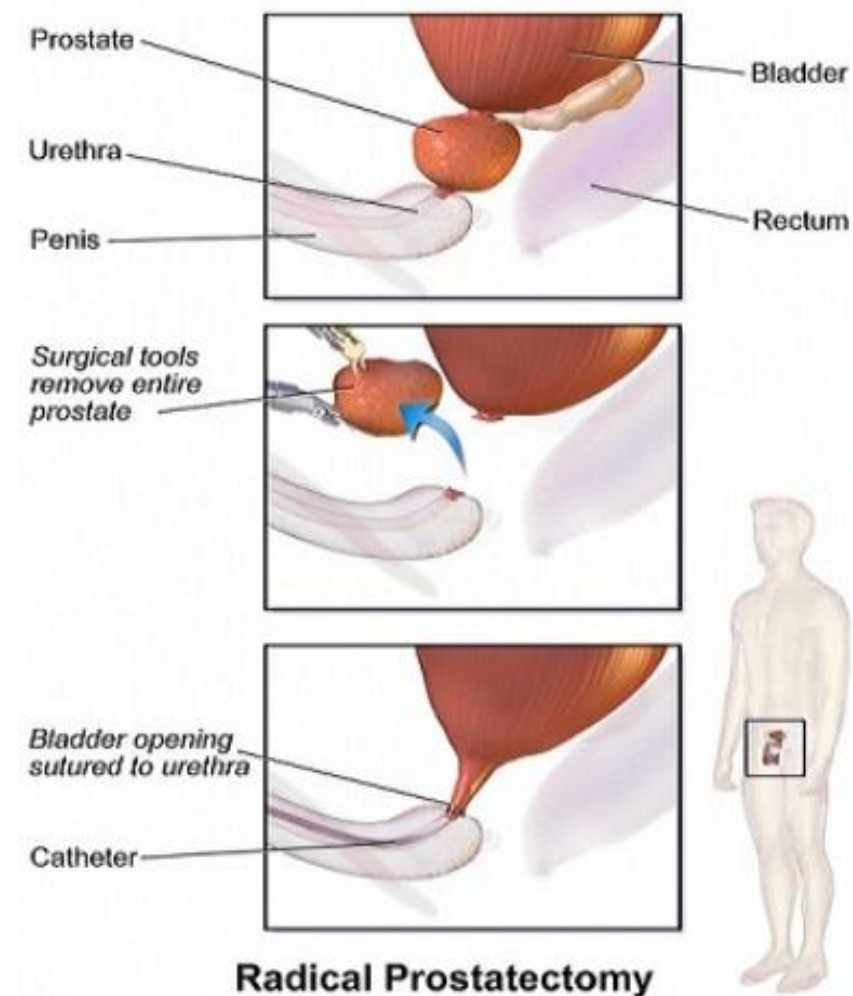
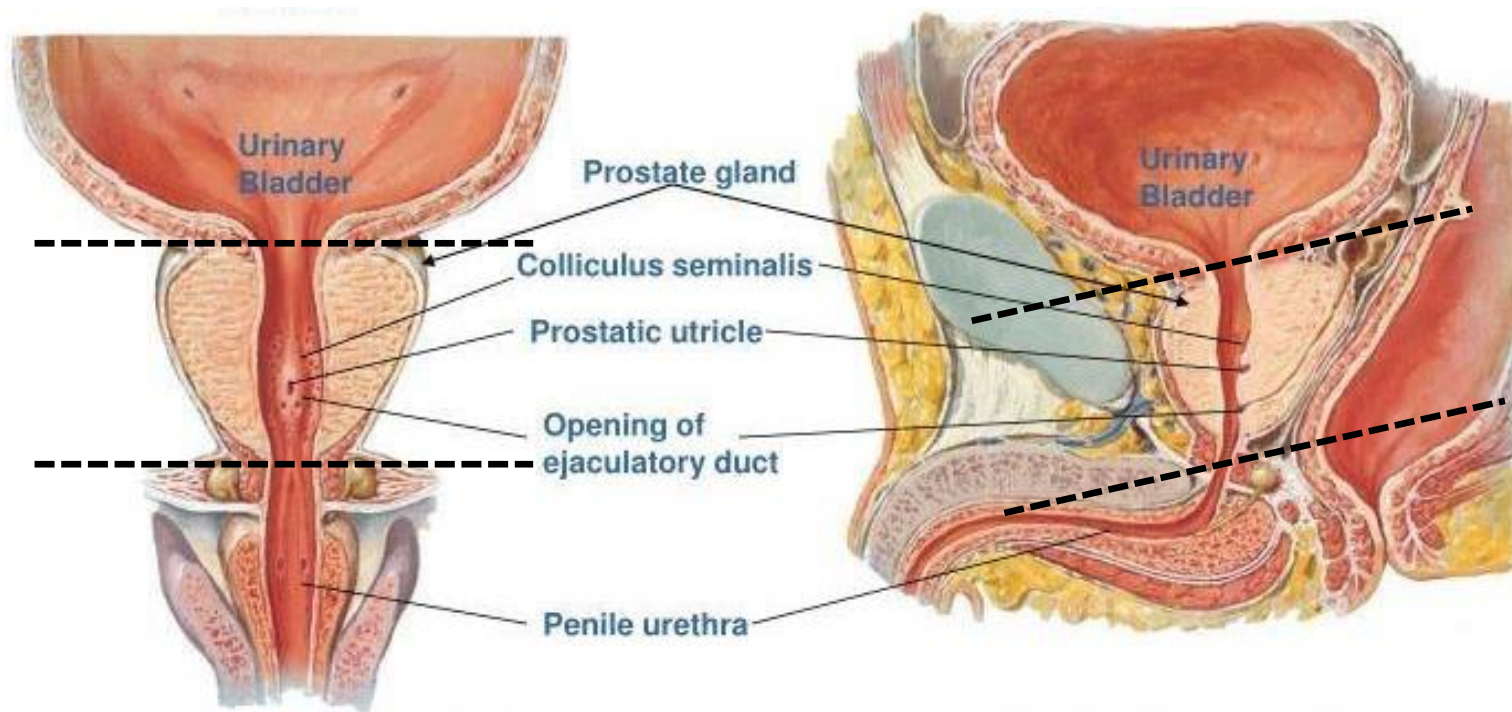
What is prostatectomy?

Radical prostatectomy

Remove *entire prostate with seminal vesical*

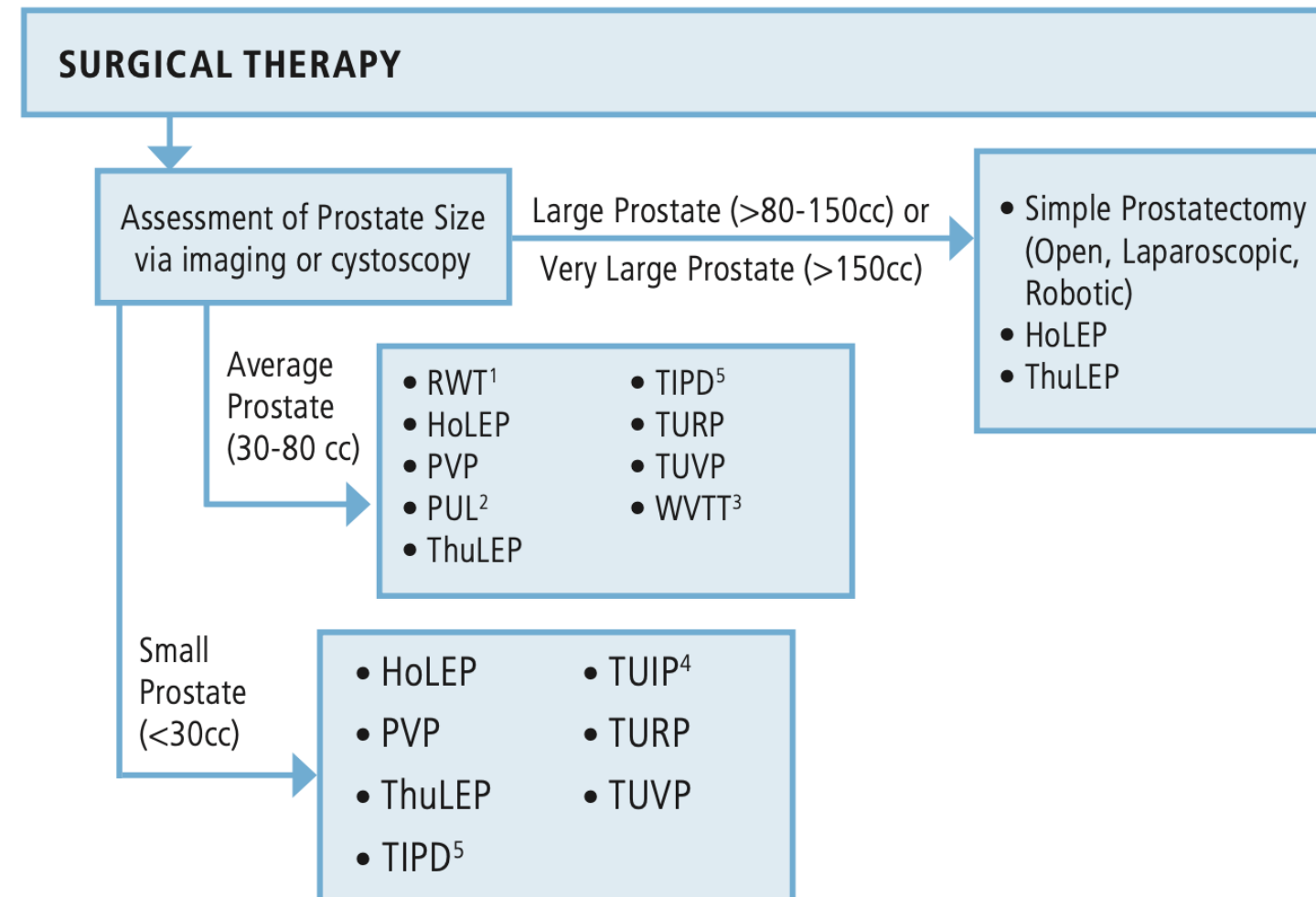
Treatment option in CA prostate

Aim: En bloc resection → remove all tumor



What is prostatectomy?

Surgical Management of Lower Urinary Tract Symptoms Attributed to Benign Prostatic Hyperplasia



Patients concerned with preservation of erectile and ejaculatory function may be offered PUL or WVT as data indicate that both therapies provide a greater likelihood of preservation of sexual function.

Partial prostatectomy

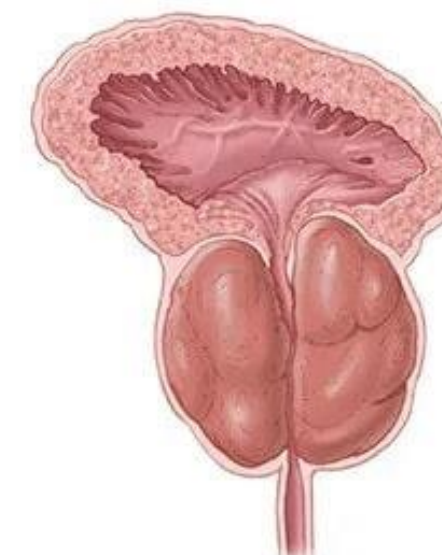
Remove **transitional zone** of prostate

Treatment option in **BPH**

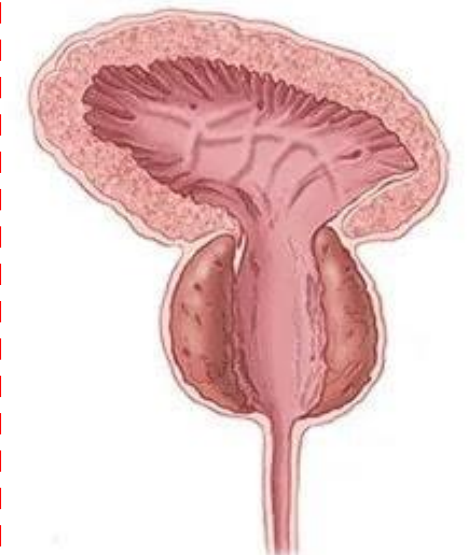
Aim: Improve voiding



Normal Prostate



Enlarged Prostate



After Procedure

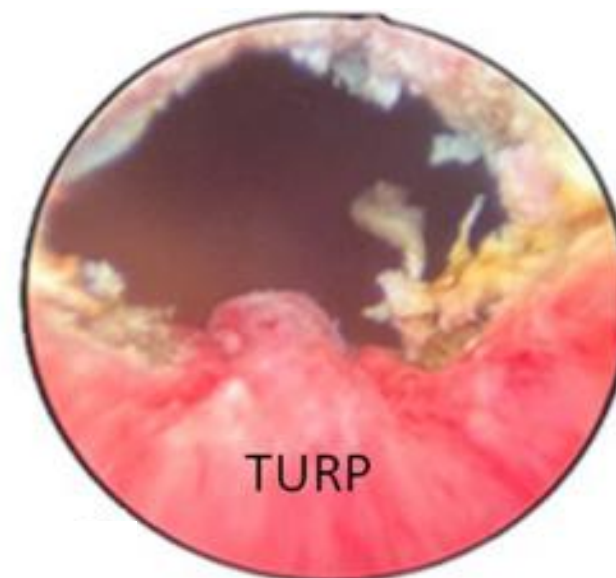
Male continence mechanism

Internal sphincter: located at bladder neck

External sphincter: located at membranous urethra
(between bulbous urethra and prostatic urethra)

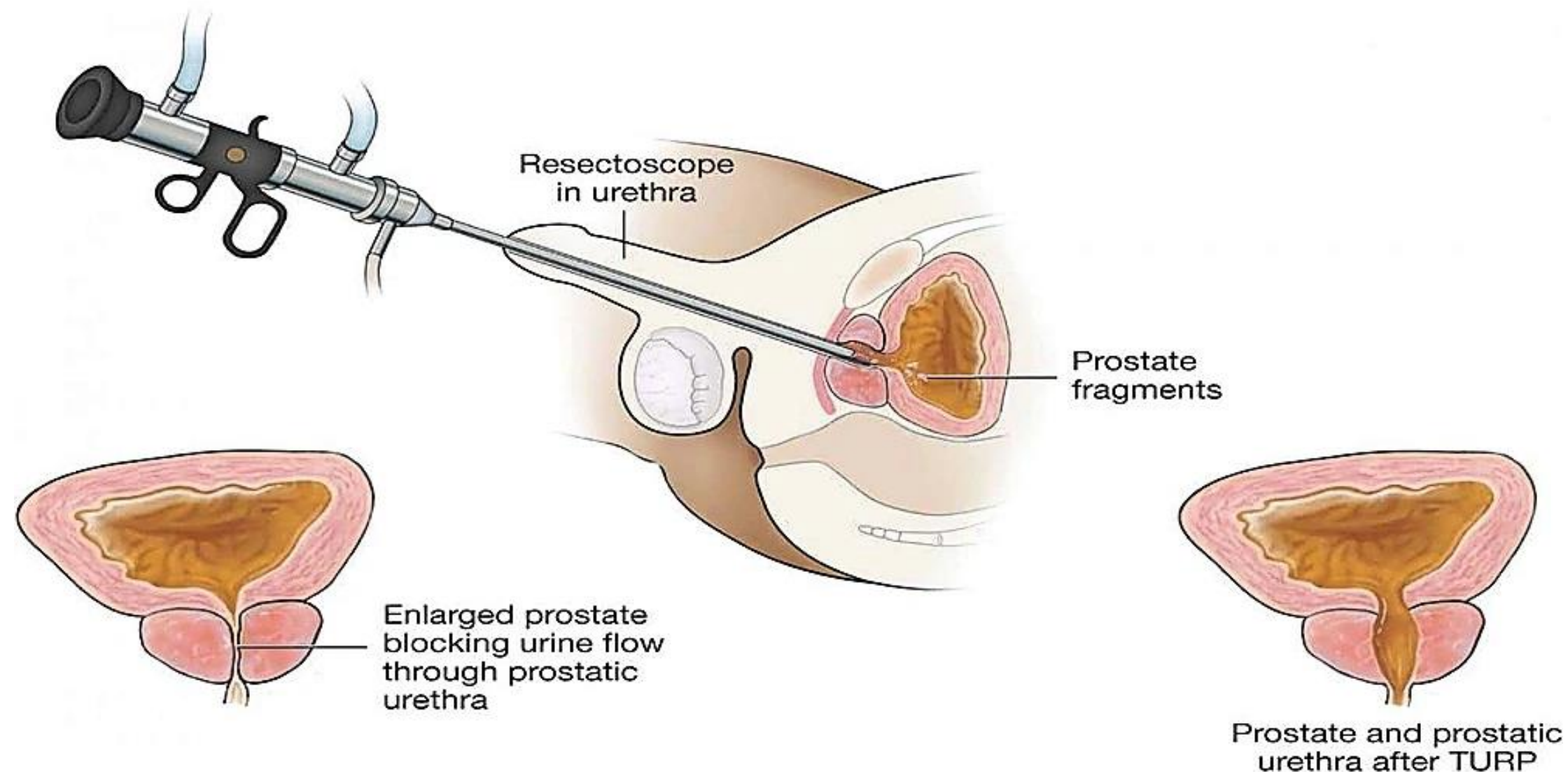
TUR-P → usually *damaged internal sphincter* (bladder neck)

*** Surgeon should be aware of damaging external sphincter



Male continence mechanism

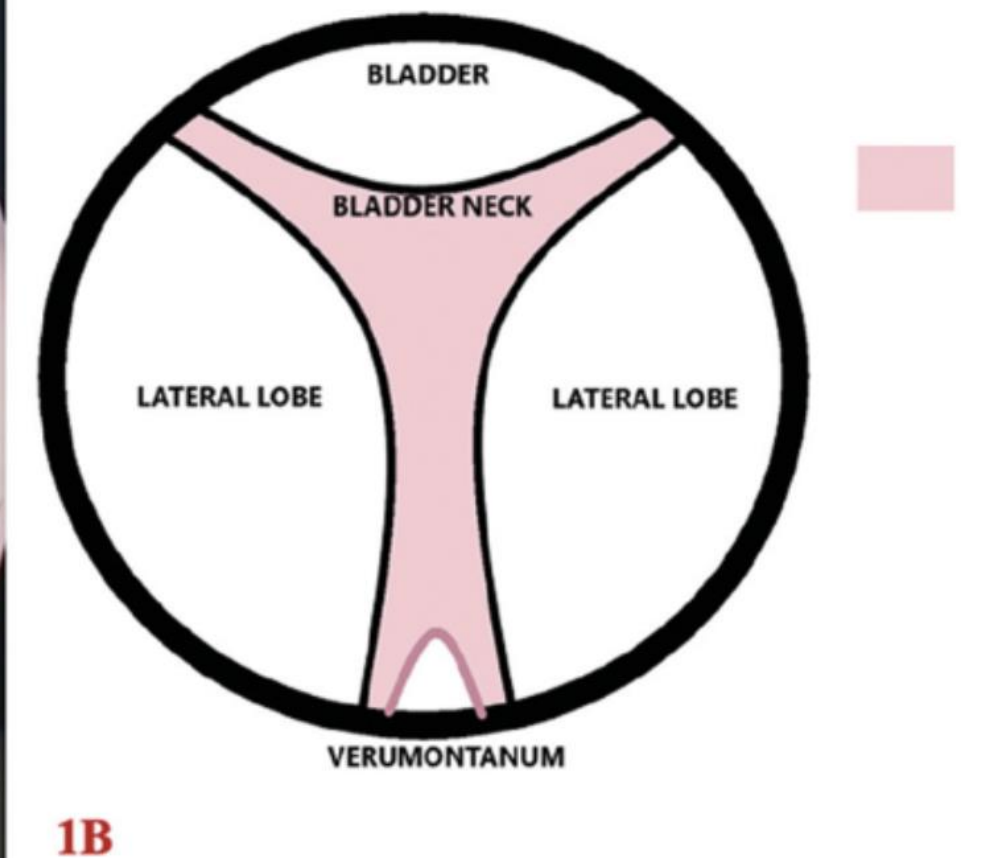
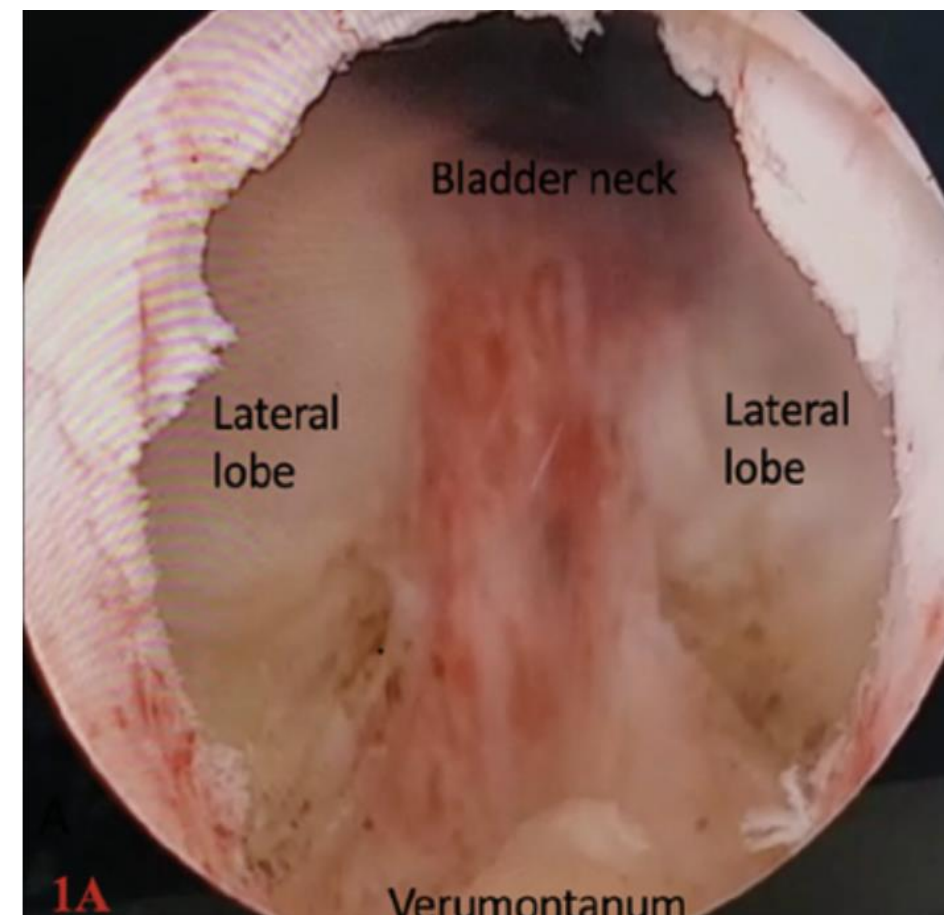
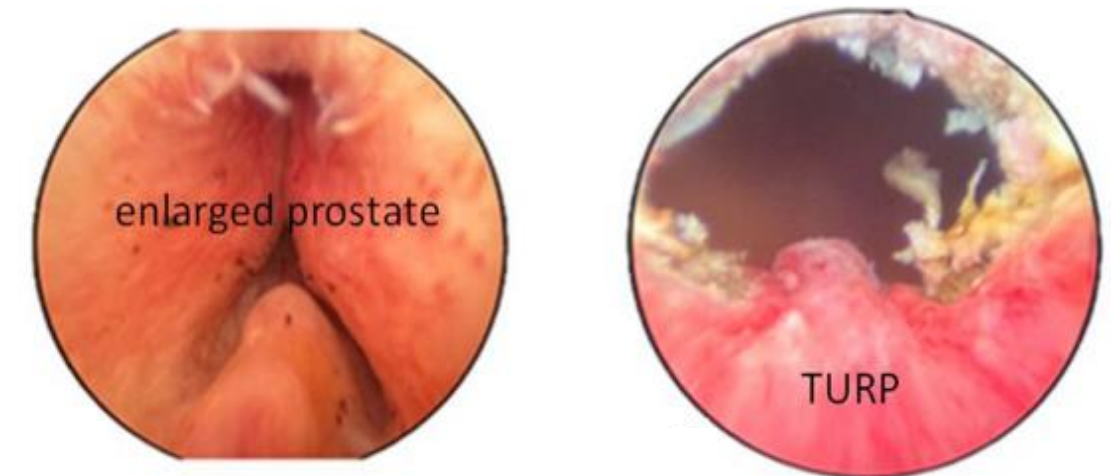
TUR-P → usually *damaged internal sphincter* (bladder neck)
*** Surgeon should beware damage external sphincter



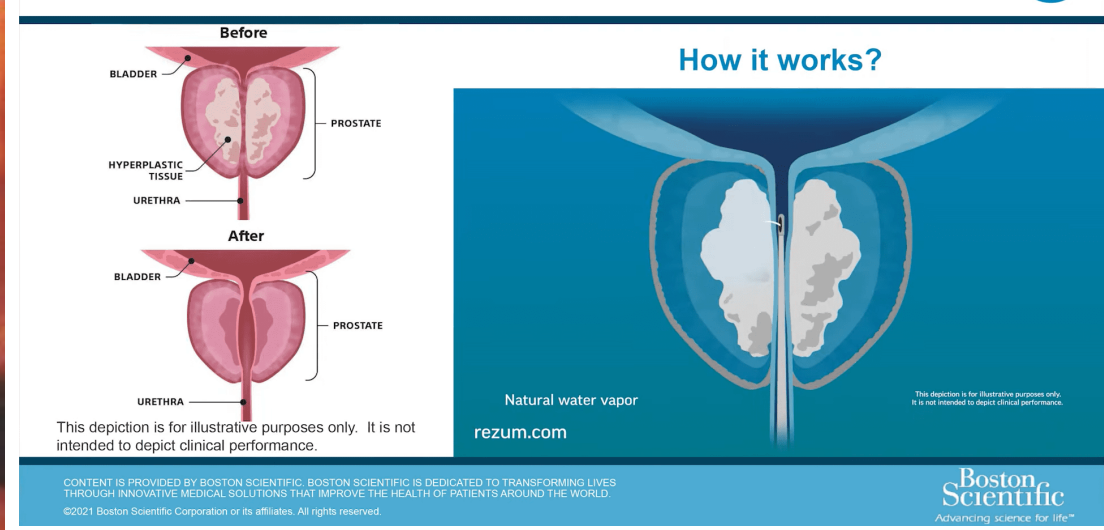
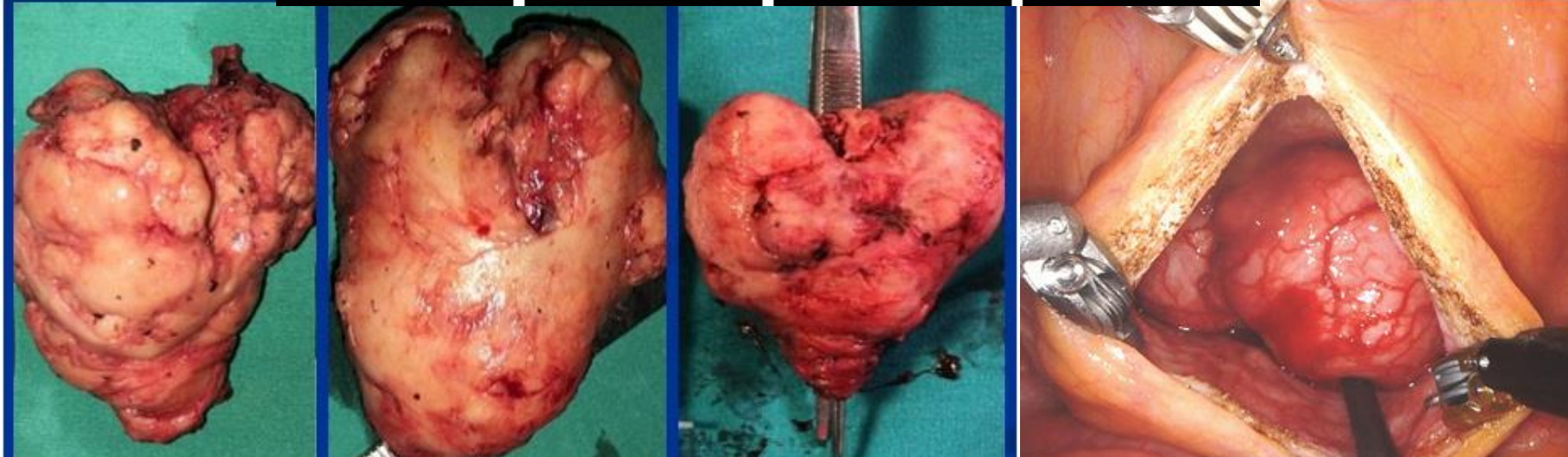
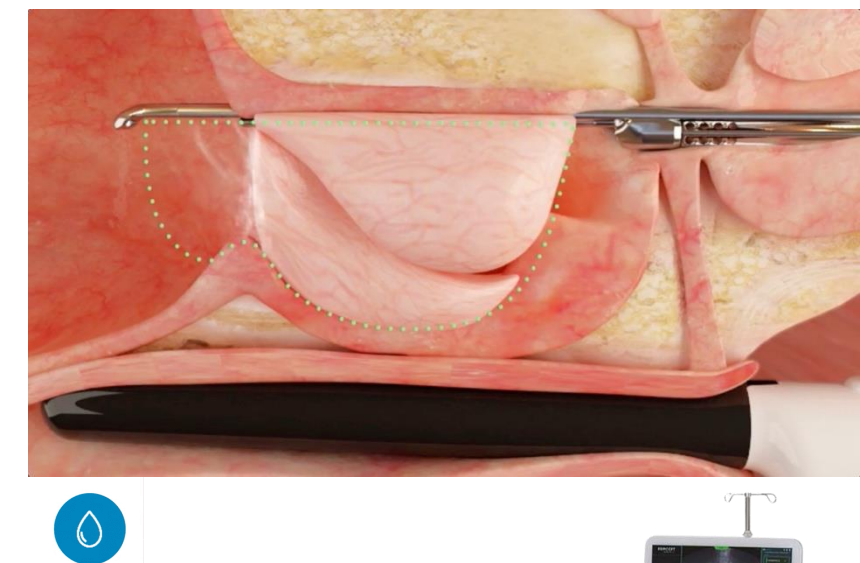
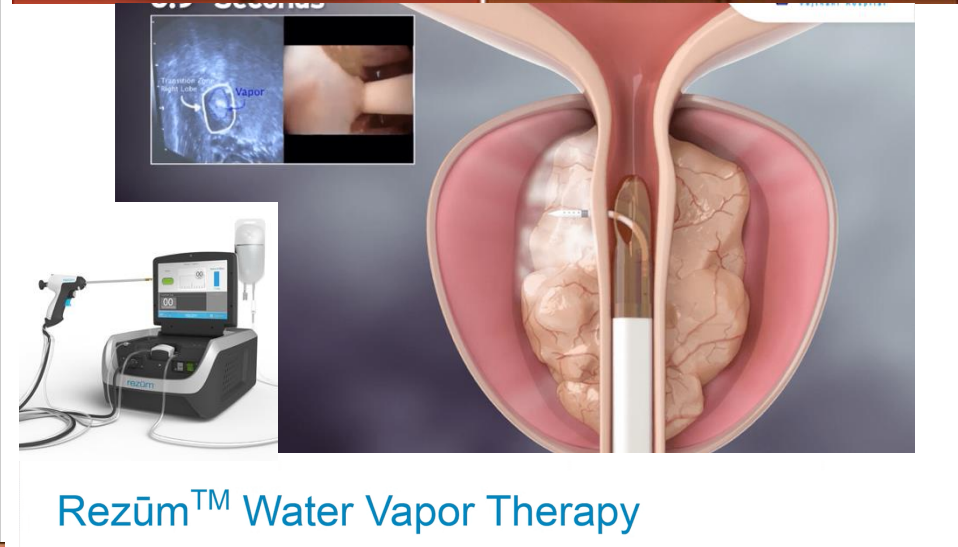
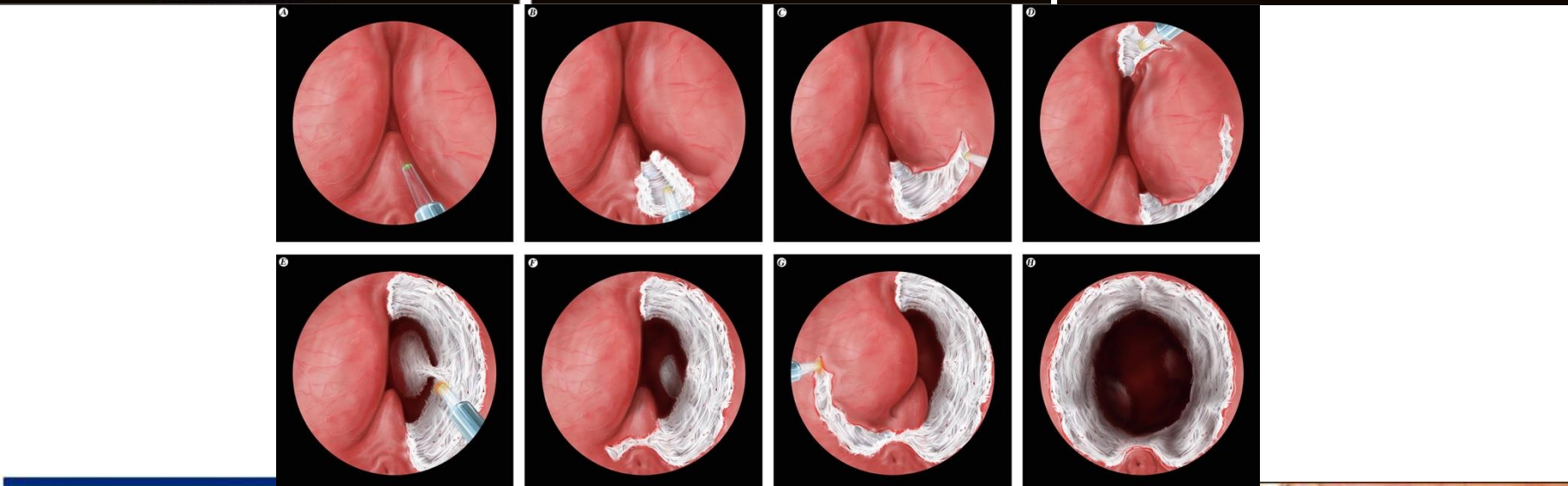
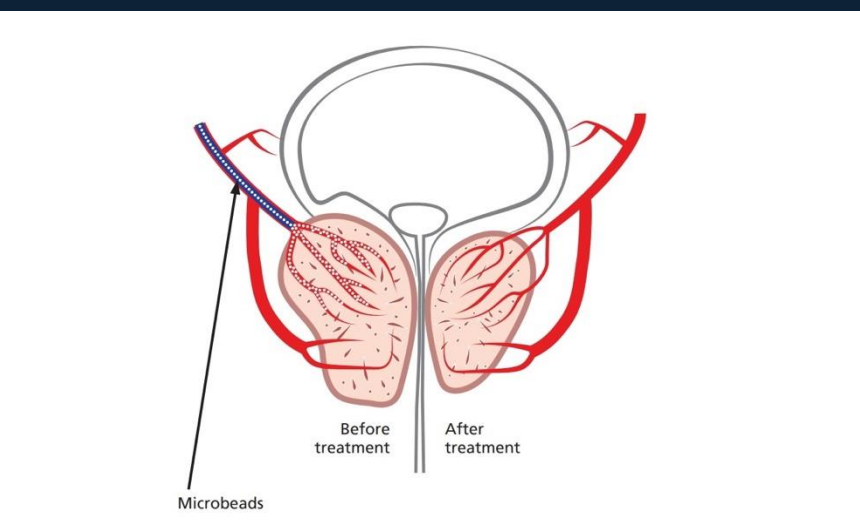
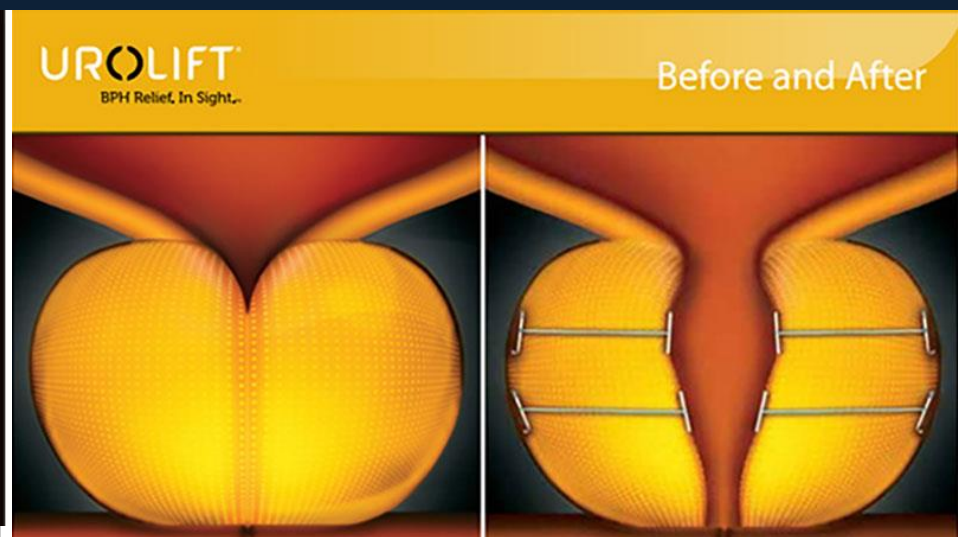
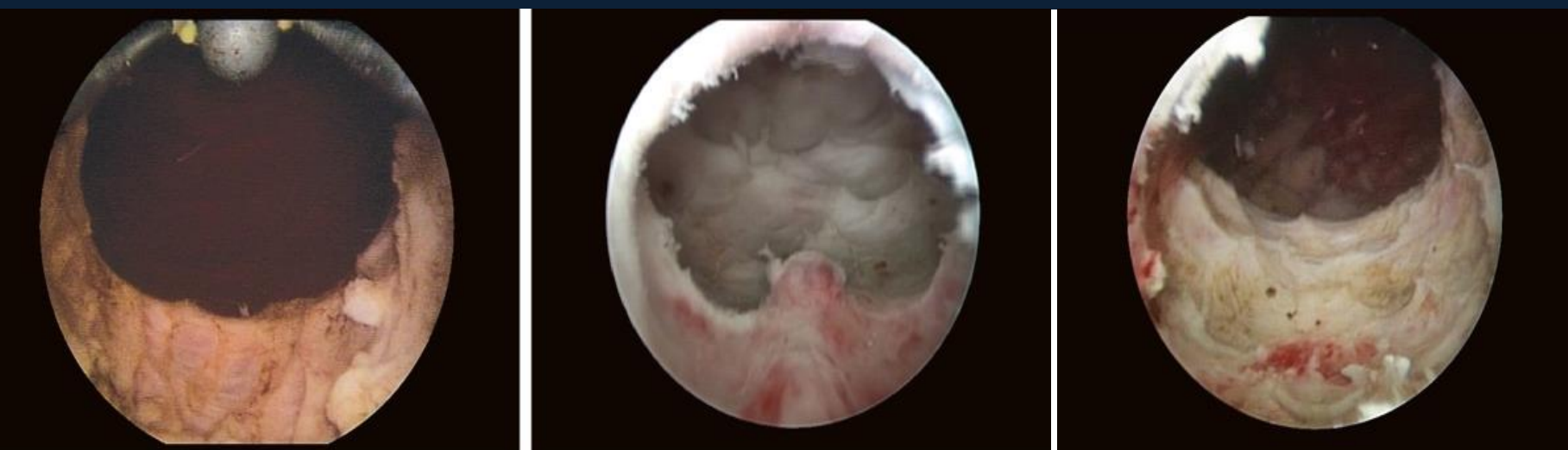
What is the surgical landmark of TURP?

Identify

- **Verumontanum** (Distal landmark)
- **Bladder neck** (Proximal landmark)
- **Surgical capsule** (Depth of resection)



Surgical options in BPH

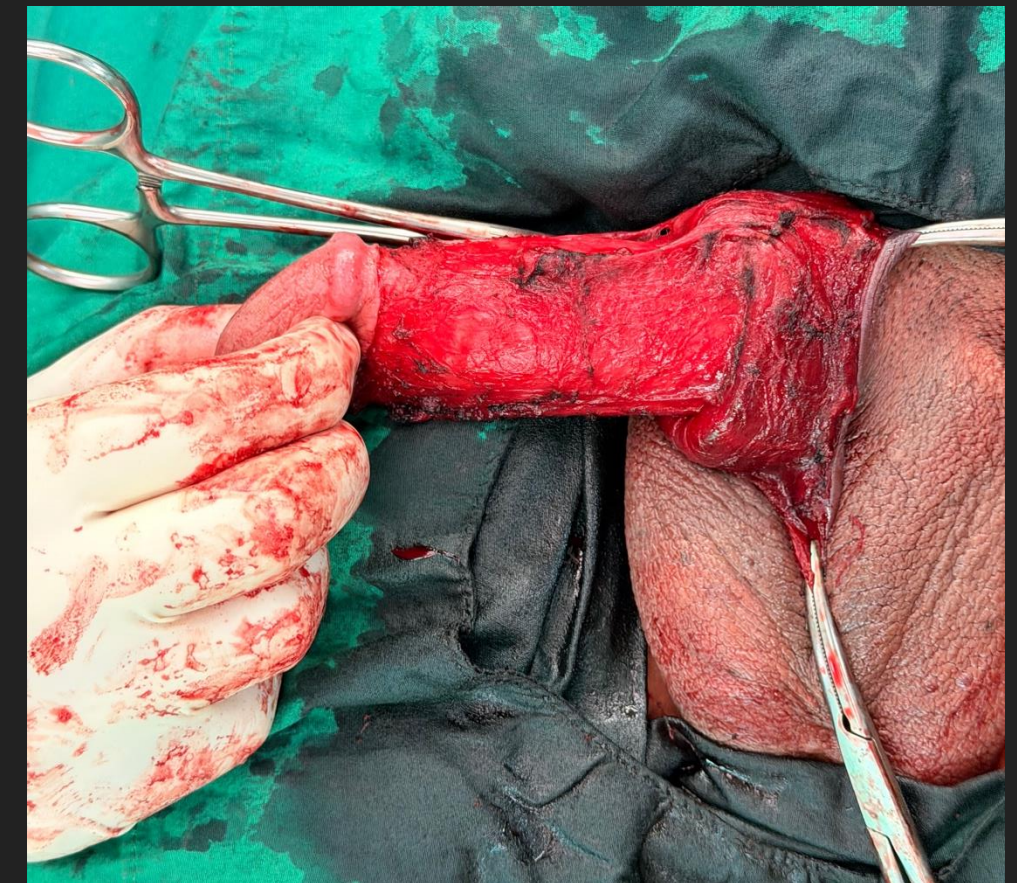


GENITALIA

- ▶ Orchiectomy: simple, radical
- ▶ Orchidoplexy
- ▶ Epididymectomy
- ▶ Circumcision
- ▶ Hypospadias/Epispadias repair
- ▶ Penile rupture repair
- ▶ Penile augmentation
- ▶ Vasectomy, Vaso-Vasostomy

GENITALIA

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ADRENAL GLAND AND RETROPERITONEUM

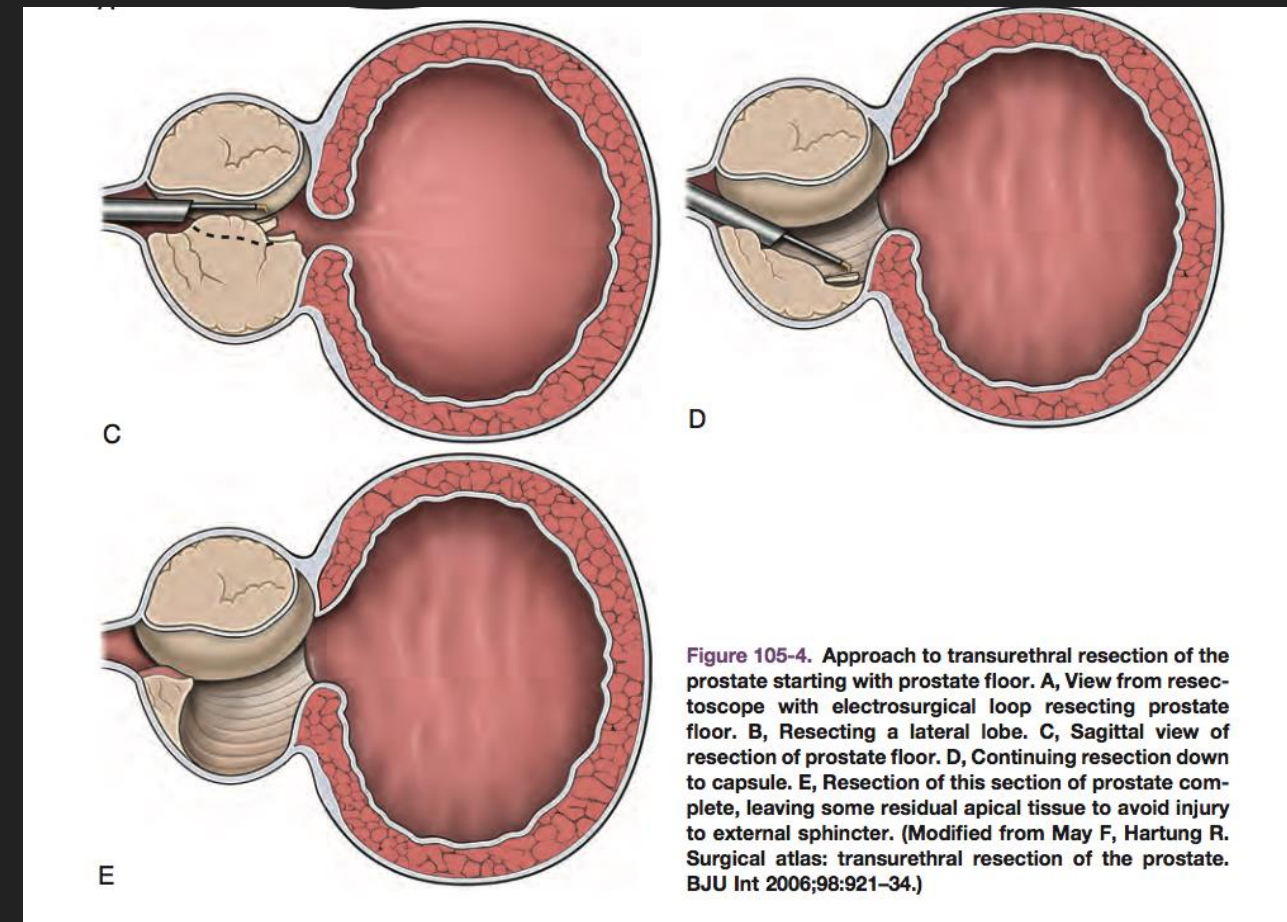
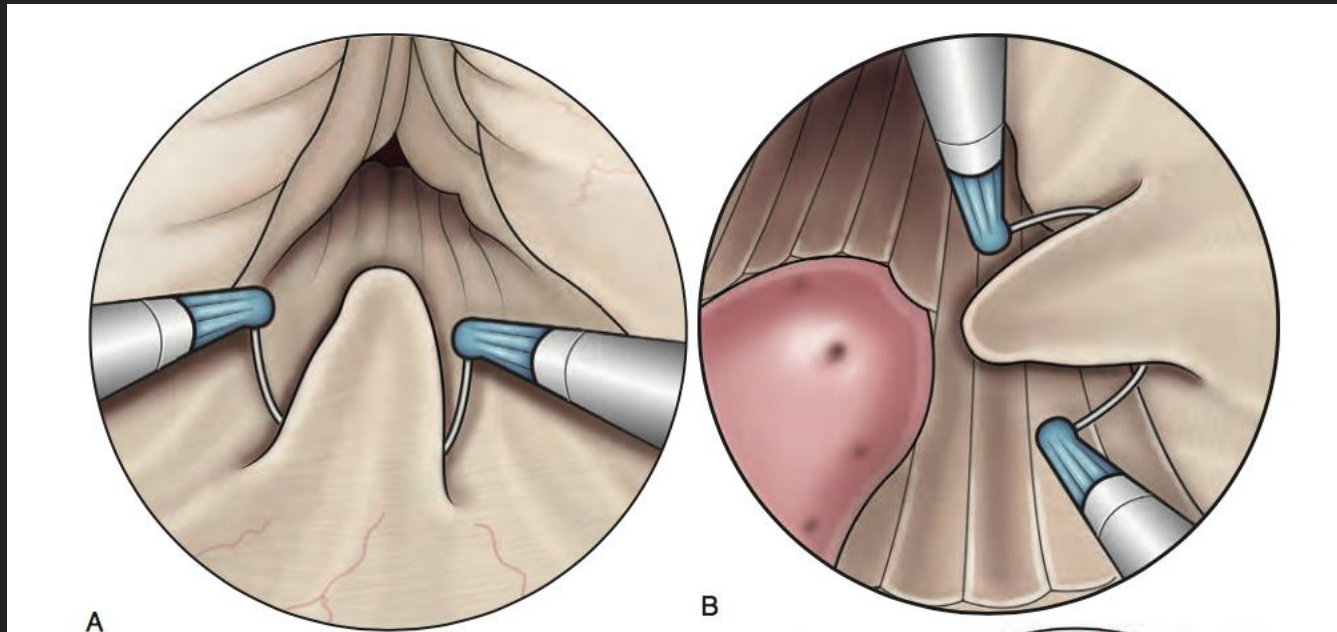
- ▶ Adrenalectomy: total, partial
- ▶ Retroperitoneal lymph node dissection
- ▶ Retroperitoneal tumor removal

BOWEL SEGMENT FOR URINARY DIVERSION

- ▶ Conduit: ileum, colon
- ▶ Neobladder: ileum, colon
- ▶ Continent pouch: stomach, ileum, colon
- ▶ Os for continent pouch: appendix, ileum, fallopian tube
- ▶ Ureteral substitution: ileum

COMMON UROLOGICAL PROCEDURES: TUR-P

- ▶ ***Transurethral resection of prostate gland (TUR-P)***
 - ▶ Anesthesia: Spinal block, General anesthesia
 - ▶ Position: Lithotomy



COMMON UROLOGICAL PROCEDURES: TUR-P



COMMON UROLOGICAL PROCEDURES: TUR-P



COMMON UROLOGICAL PROCEDURES: TUR-P

► Peri-operative complications

► Bleeding

► **TUR syndrome**: Hypervolemia and dilutional hyponatremia

Risk factors

- Gland > 45 g
- Water height > 60 cm from pubic symphysis
- Resection time > 90 minutes

COMMON UROLOGICAL PROCEDURES: TUR-P

- ▶ **Post-operative care**
 - ▶ Foley catheter indwelling
 - ▶ +/- Traction
 - ▶ +/- Continuous bladder irrigation



QUESTION?



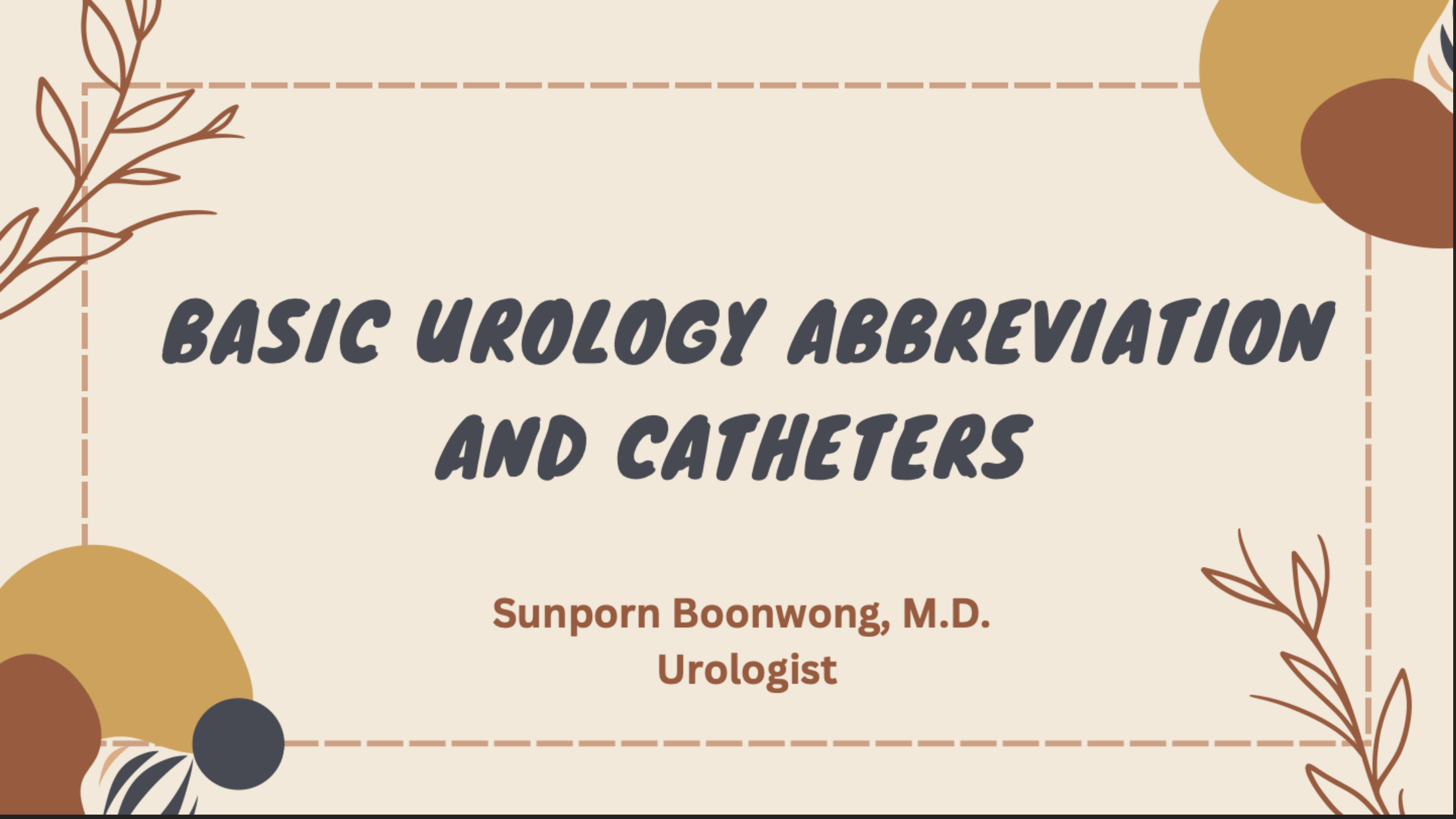
urosiriraj



Division of Urology, Siriraj

sunporn.boo@mahidol.ac.th

Thank you



BASIC UROLOGY ABBREVIATION AND CATHETERS

Sunporn Boonwong, M.D.
Urologist

AUR



- ปัสสาวะไม่ออก = ออกเองไม่ได้
และคนไข้มี **full bladder**
- หลังจากใส่สายสวนปัสสาวะ **ควรวัด Retention volume ของ urine** ที่ออกมาหลังจากใส่สายสวนไปสักพัก

Acute Urinary Retention

BOO

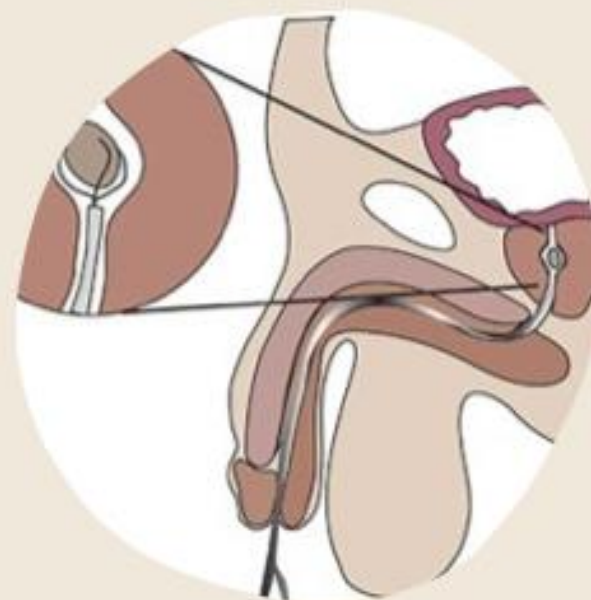
Bladder Outlet Obstruction



BPH



Urethral
stricture



Urethral
stone

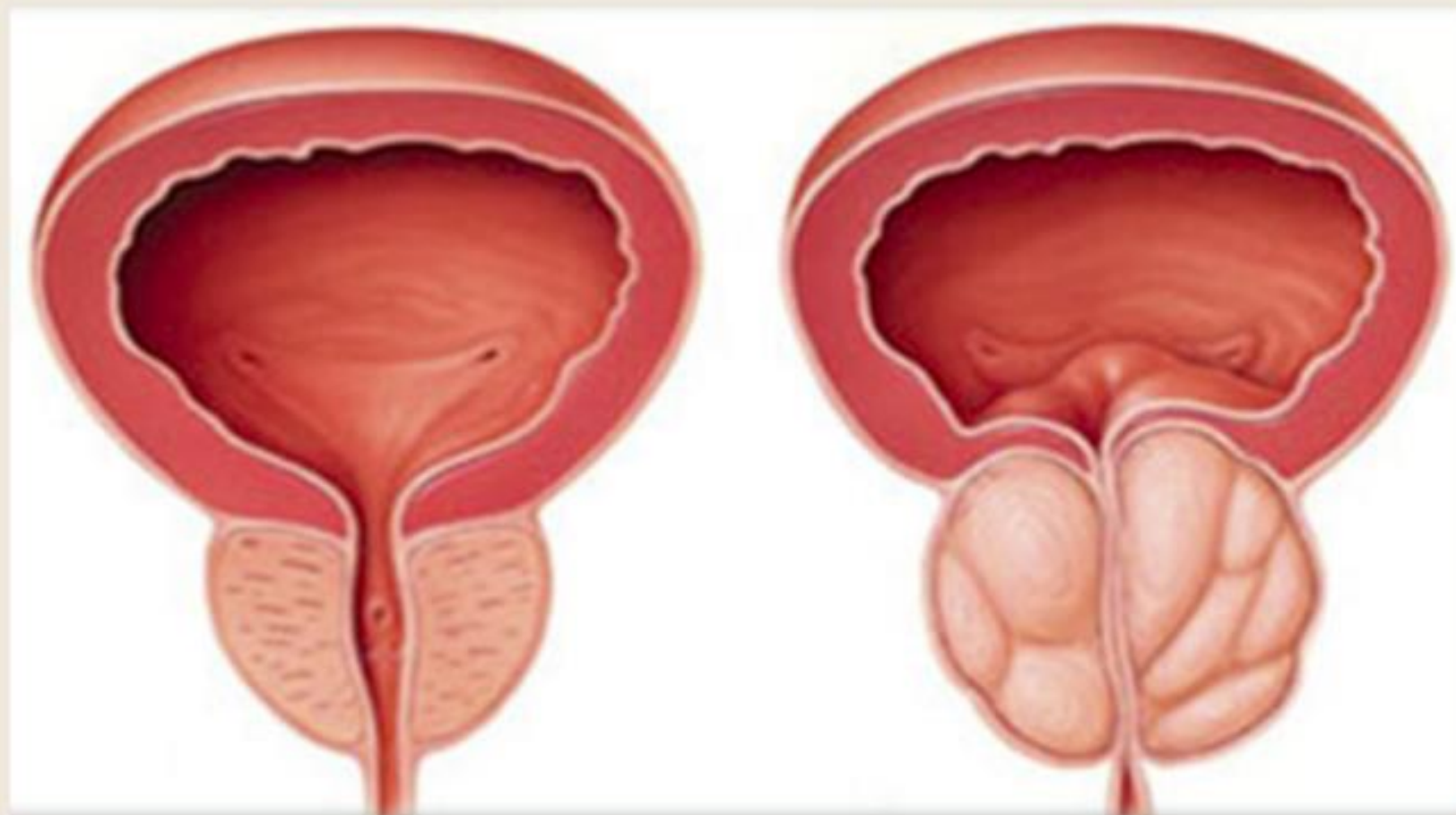


Phimosis

ภาวะทางออกของกระเพาะปัสสาวะอุดตัน

เป็นการเรียกรวมๆ เกิดได้จากหลายสาเหตุ

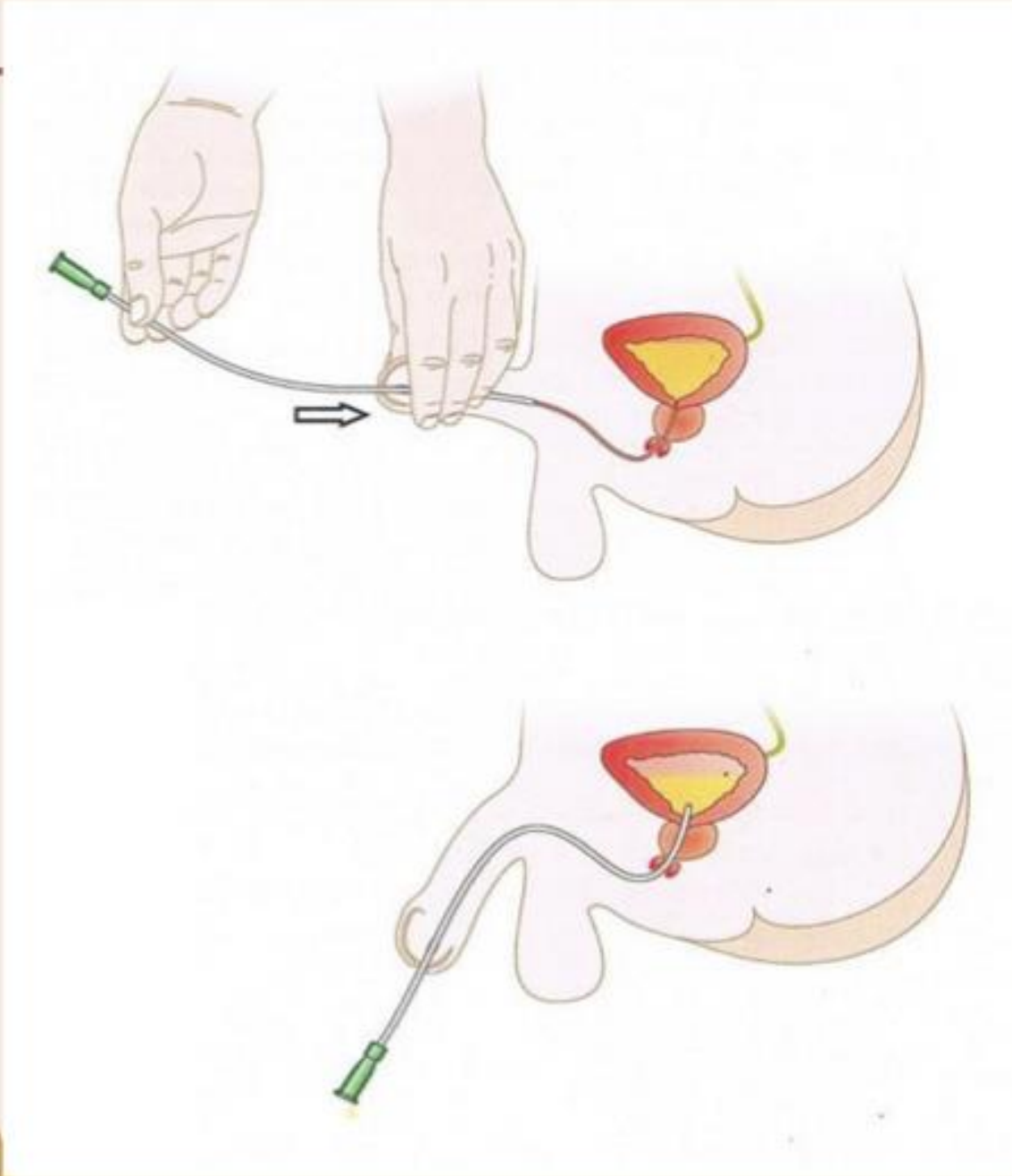
BPH



Benign Prostatic Hyperplasia
ภาวะต่อมลูกหมากโต

CIC

Clean Intermittent Catheterization



- การให้ผู้ป่วย หรือญาติสวนปัสสาวะ
ออกเองที่บ้านเป็นครั้งๆ โดยไม่ต้องคา
สายสวนปัสสาวะ
- เช่น ในผู้ป่วยกระเพาะปัสสาวะพิการ
(Neurogenic bladder)

LUTS

Lower Urinary Tract Symptoms

อาการแสดงของทางเดินปัสสาวะส่วนล่าง

เช่น ฉี่บ่อย ฉี่ลำบาก ฉี่แสบขัด ฉี่ไม่พุ่ง ฉี่นาน ฉี่ต้องเบ่ง ตื่นมาฉี่กลางคืน กลั้นปัสสาวะไม่ได้ เป็นต้น

ปัสสาวะลำบาก



ปัสสาวะไม่พุ่ง



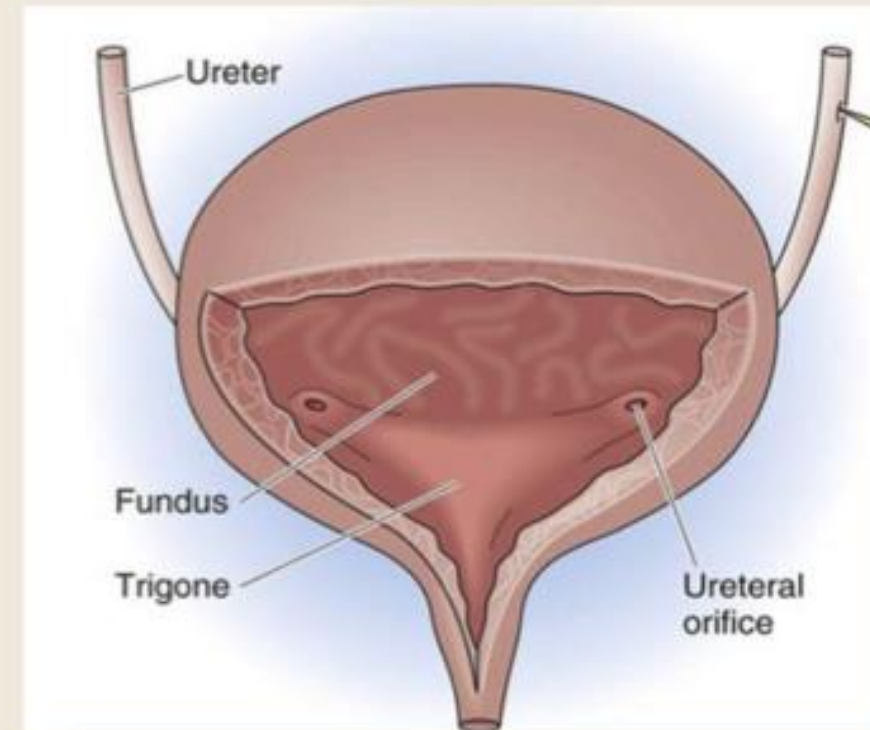
ปัสสาวะบ่อย
ในเวลากลางคืน



RP : Retrograde Pyelography



- การส่องกล้องเข้าไปในกระเพาะปัสสาวะ แล้วใส่สาย ureteric catheter เข้าไปในท่อไต
- ฉีดสารทึบรังสี (contrast) ผ่านทาง ureteric catheter พร้อมกับ X-ray real time โดยใช้ fluoroscopy
- ดูพยาธิสภาพของกรวยไต และท่อไต (collecting system)



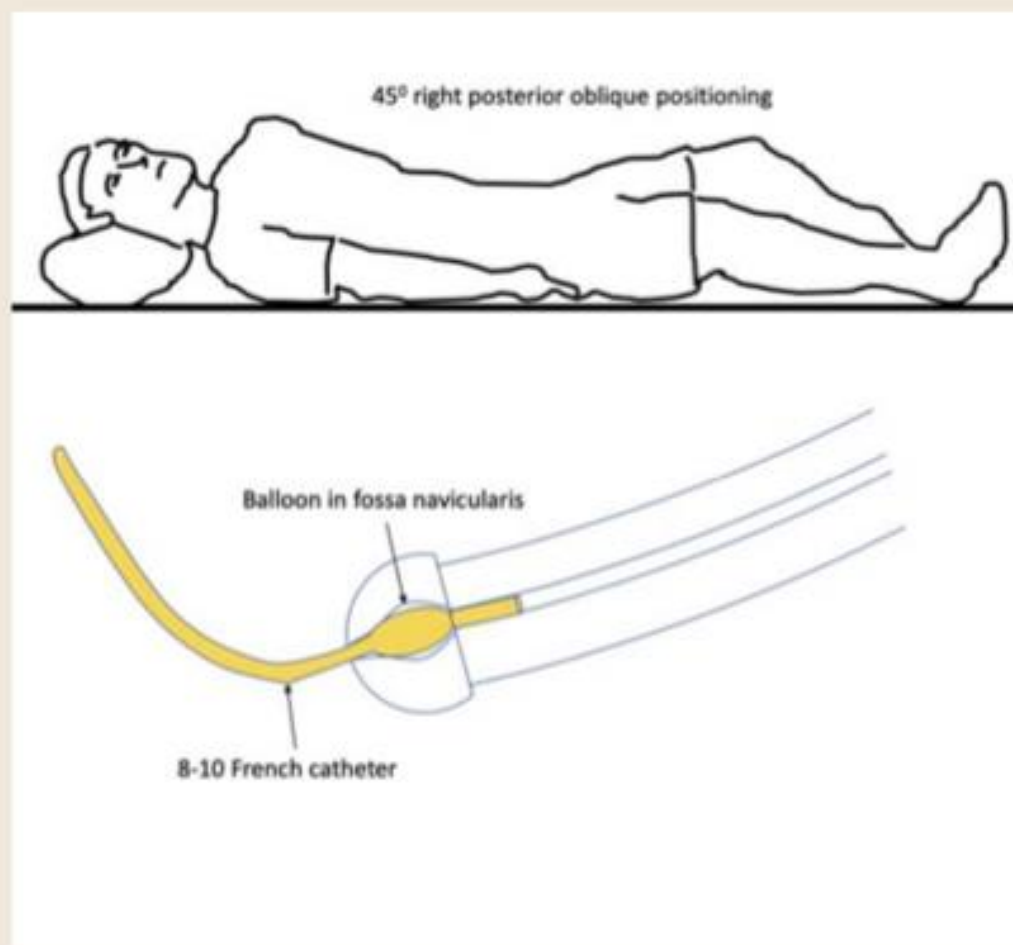
AP : Antegrade Pyelography



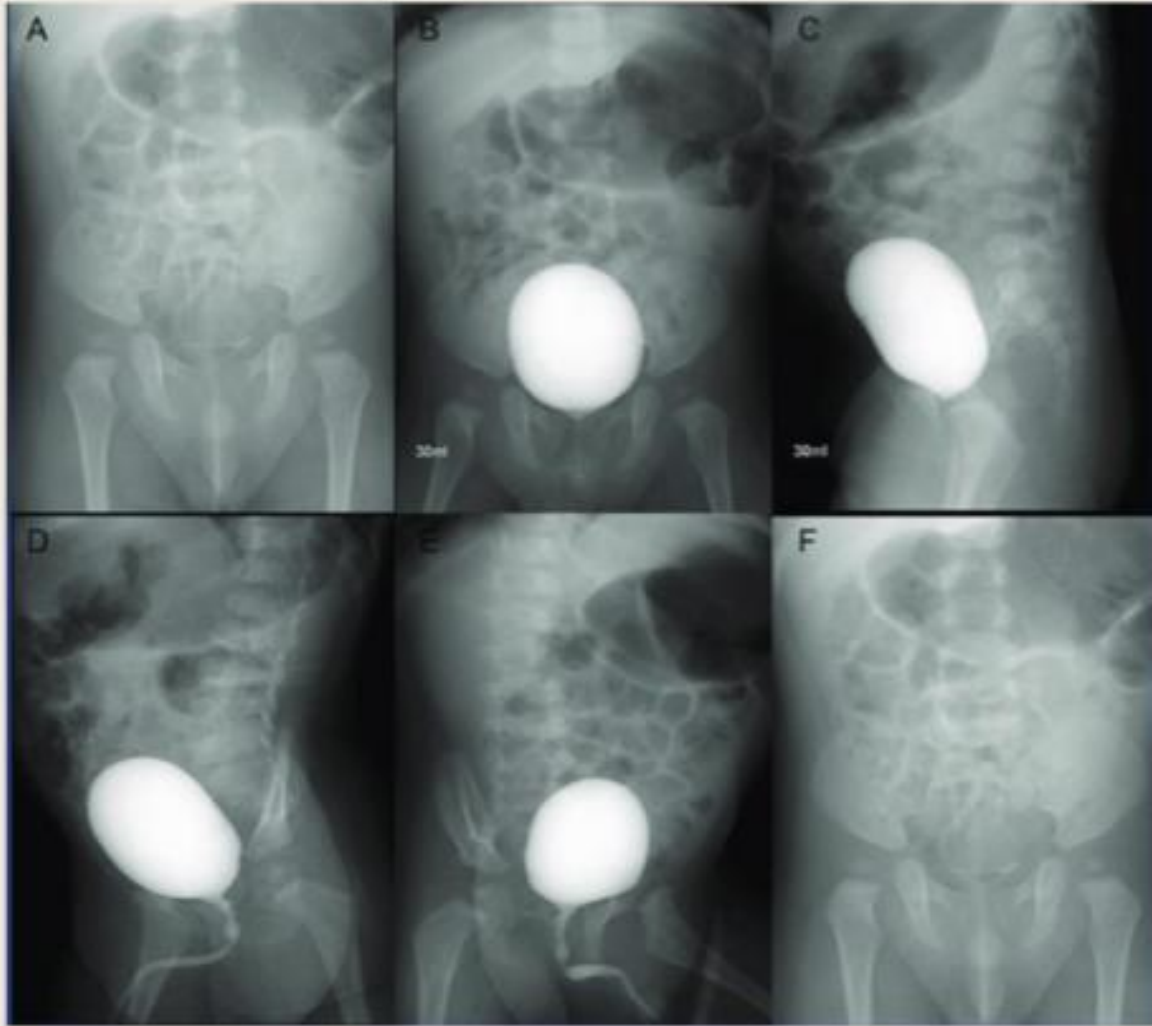
- การฉีดสารทึบรังสี (contrast) ผ่านทาง Nephrostomy tube (PCN) พร้อมกับ X-ray real time โดยใช้ fluoroscopy
- ดูพยาธิสภาพของกรวยไต และท่อไต (collecting system)
- (ไม่ต้องส่องกล้องทางกระเพาะปัสสาวะ)

RUG : Retrograde urethrography

การฉีดสารทึบรังสีเข้าทางท่อปัสสาวะ และ X-ray real time (fluoroscopy)
เพื่อประเมินความผิดปกติของท่อปัสสาวะ



VCUG

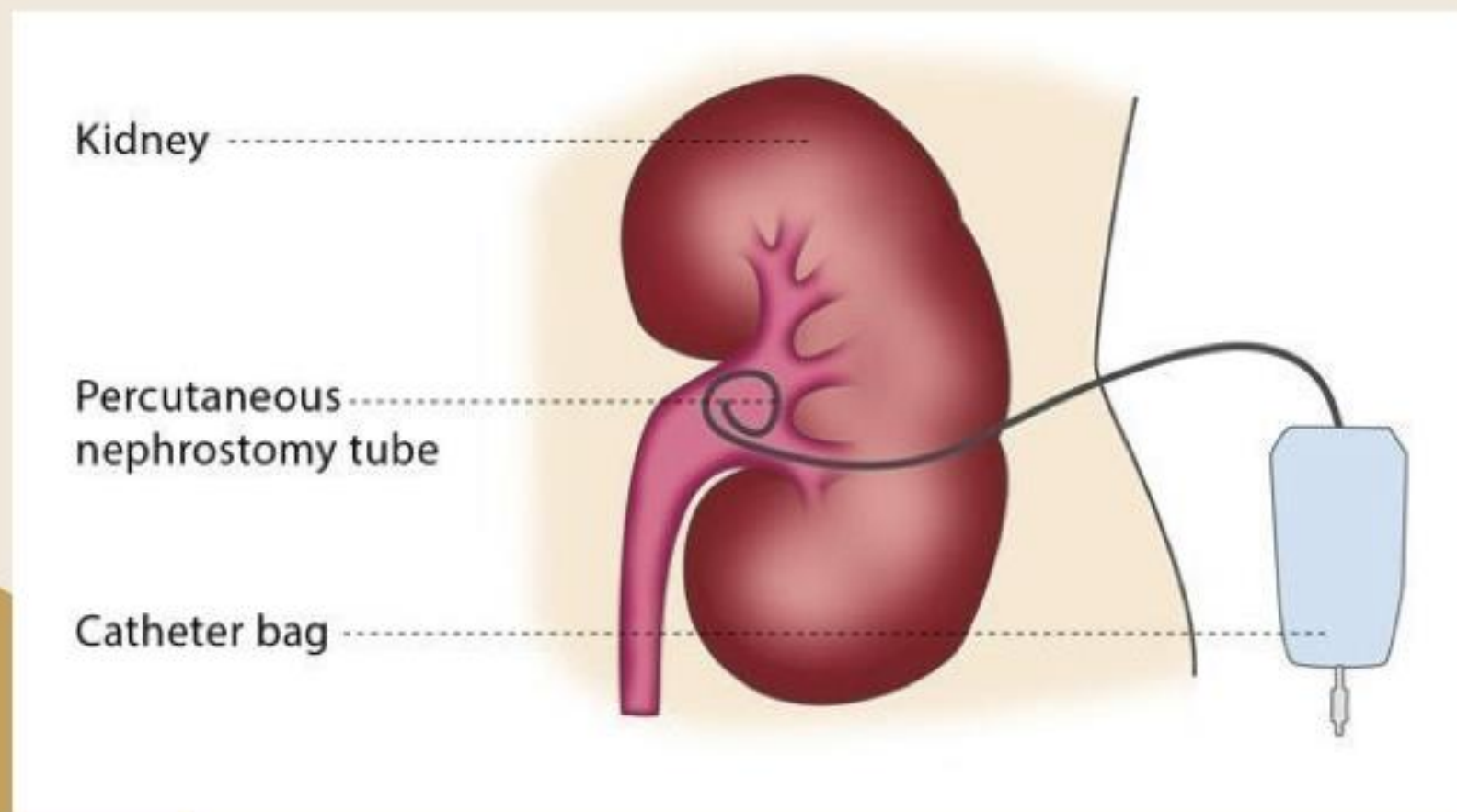


- ทำโดยรังสีแพทย์ (ต้องเขียนใบ request)
- ใส่สายสวนปัสสาวะเข้าไปในกระเพาะปัสสาวะ แล้วใส่สารทึบรังสี (contrast) จน full bladder
- หลังจากนั้นนำสายสวนปัสสาวะออก แล้วให้คนไข้ปัสสาวะเอง
- โดยมีการถ่ายภาพ Xray เป็นระยะ
- ประเมินพยาธิสภาพของกระเพาะปัสสาวะ ท่อปัสสาวะ

Voiding CystoUrethroGram

PCN

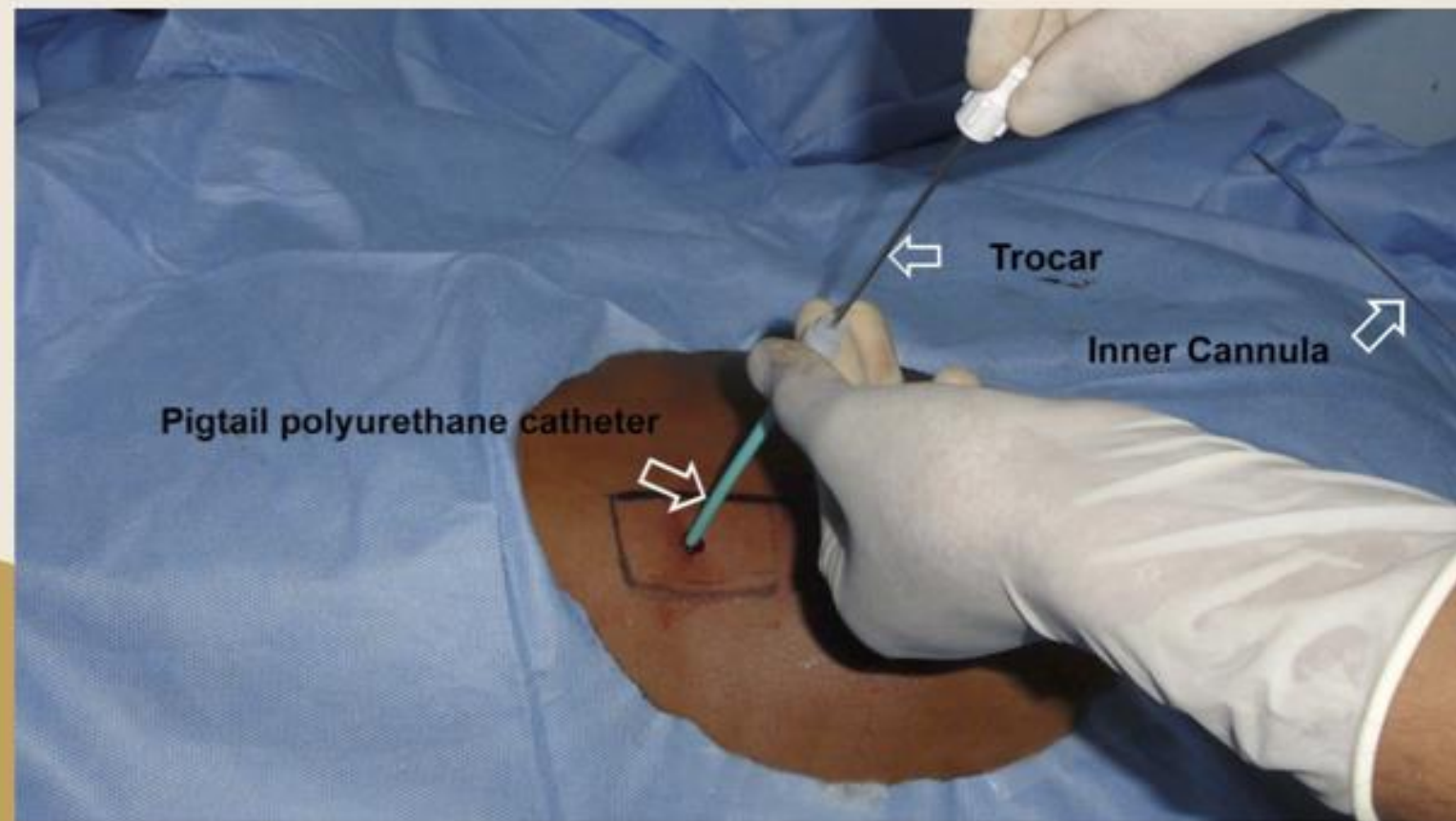
Percutaneous nephrostomy



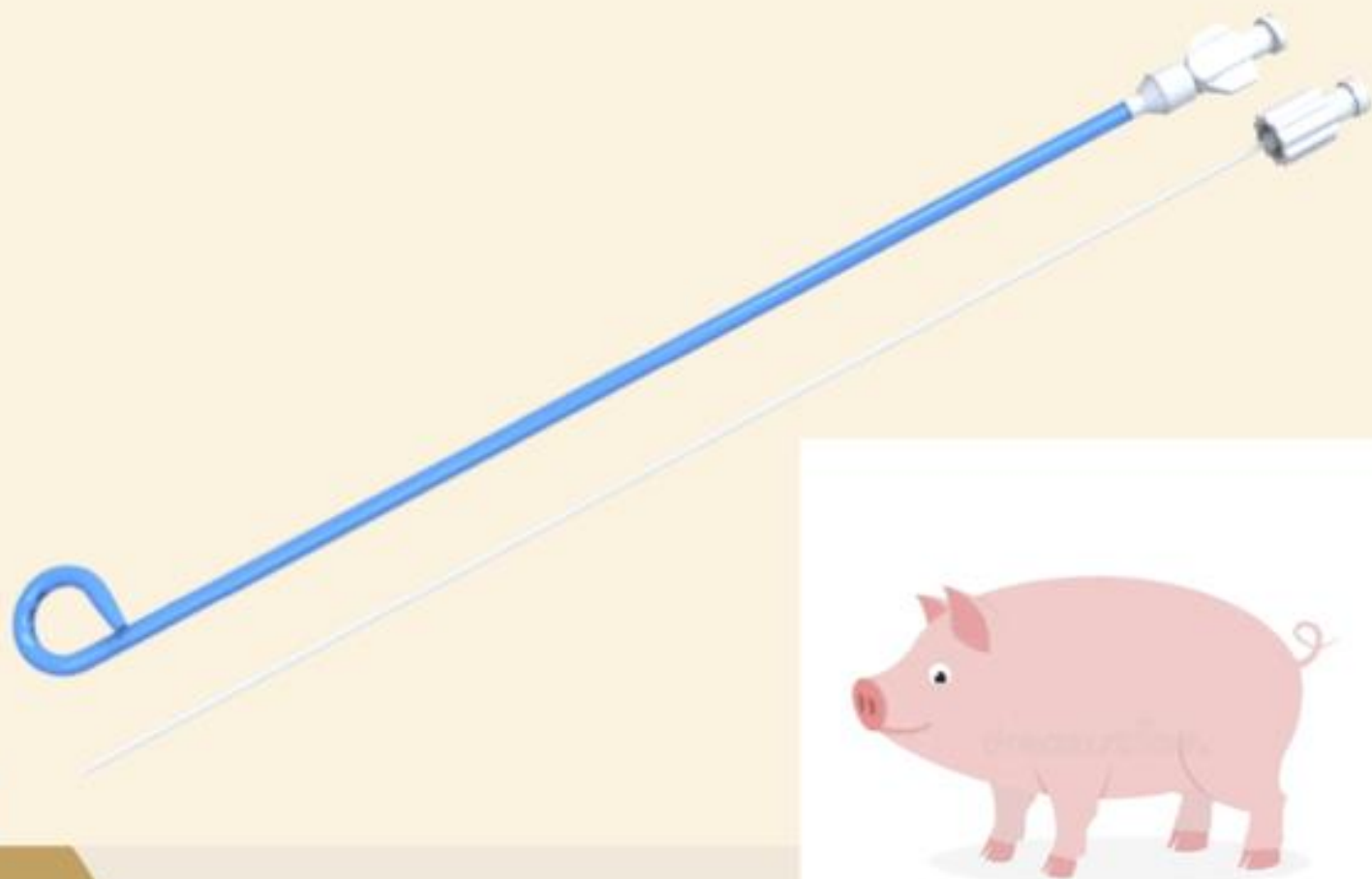
- สายระบายท่อไต
- การใส่สายระบายน้ำปัสสาวะ
 - ผ่านทางผิวหนัง
- สายที่ใส่ = Pigtail catheter
- ปลายสายด้านในจะอยู่ในกรวยไต (Renal pelvis)
- ใส่ PCN ในคนไข้ที่มีภาวะท่อไตอุดตันจากสาเหตุต่างๆ ที่ไม่สามารถใส่ DJ stent ได้

PCN

Percutaneous nephrostomy

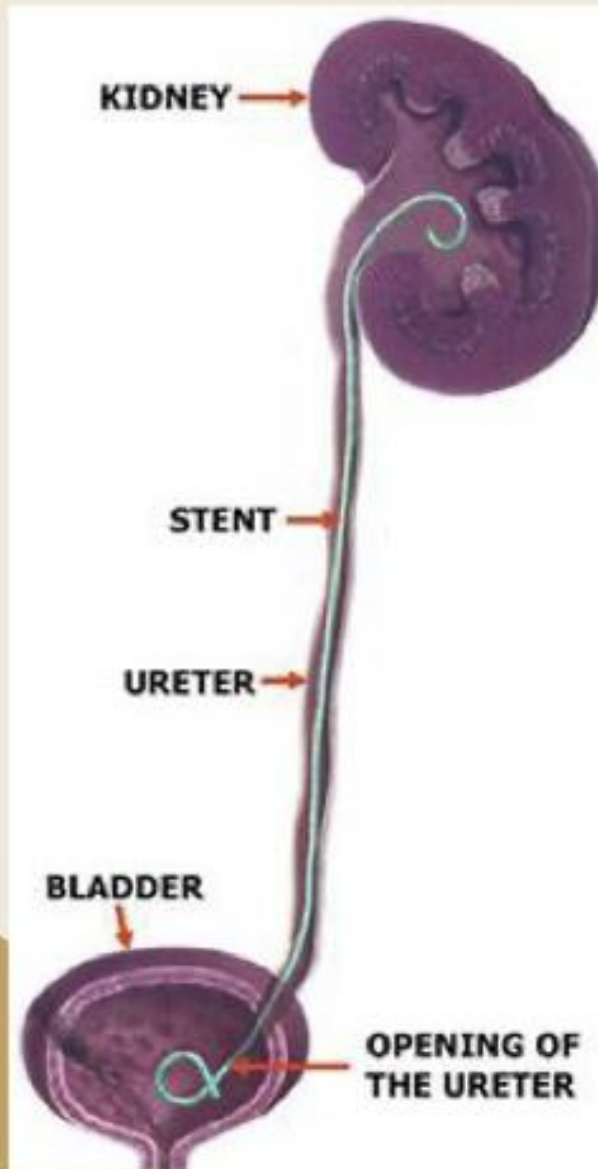


Pigtail catheter



- มีหลายขนาด 8, 10, 12, 14, ... Fr
- ใช้เพื่อ drainage ของเหลวต่างๆออกจากร่างกาย
- PCD = Percutaneous Drainage
 - drain อะไรก็ได้
- PCN = Percutaneous **Nephrostomy**
 - drain **ฉี่** เท่านั้น !!

DJ stent = Double J stent

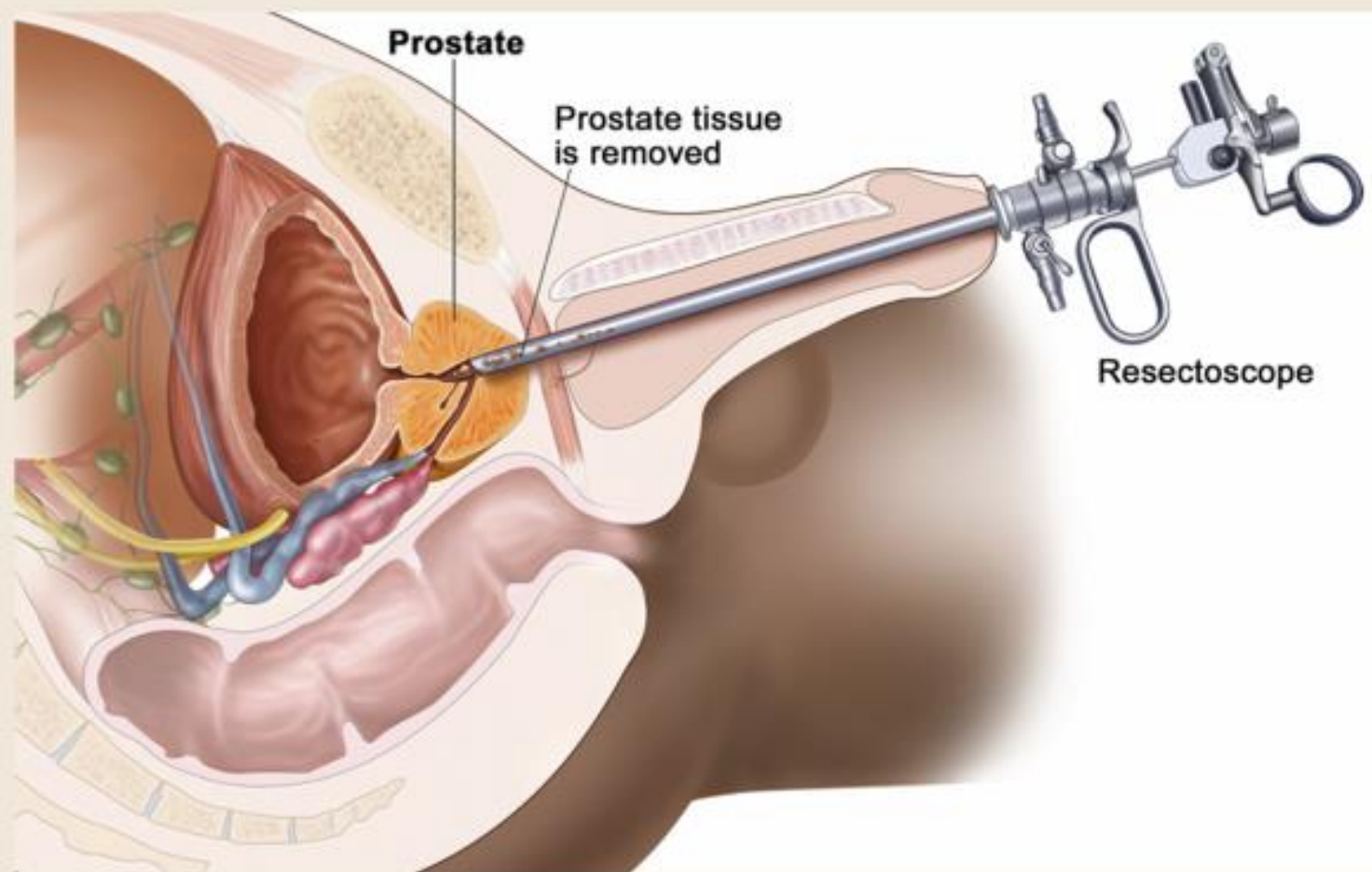


- สายระบายน้ำปัสสาวะมีสองปลาย
 - ปลายบน ในกรวยไต (renal pelvis)
 - ปลายล่าง ในกระเพาะปัสสาวะ (bladder)
- ต้องส่องกล้องกระเพาะปัสสาวะ (cystoscopy) เพื่อใส่ DJ stent
- ใส่ DJ stent ในคนไข้ที่มีภาวะท่อไตอุดตันจากสาเหตุต่างๆ

TUR-P

Transurethral Resection of Prostate

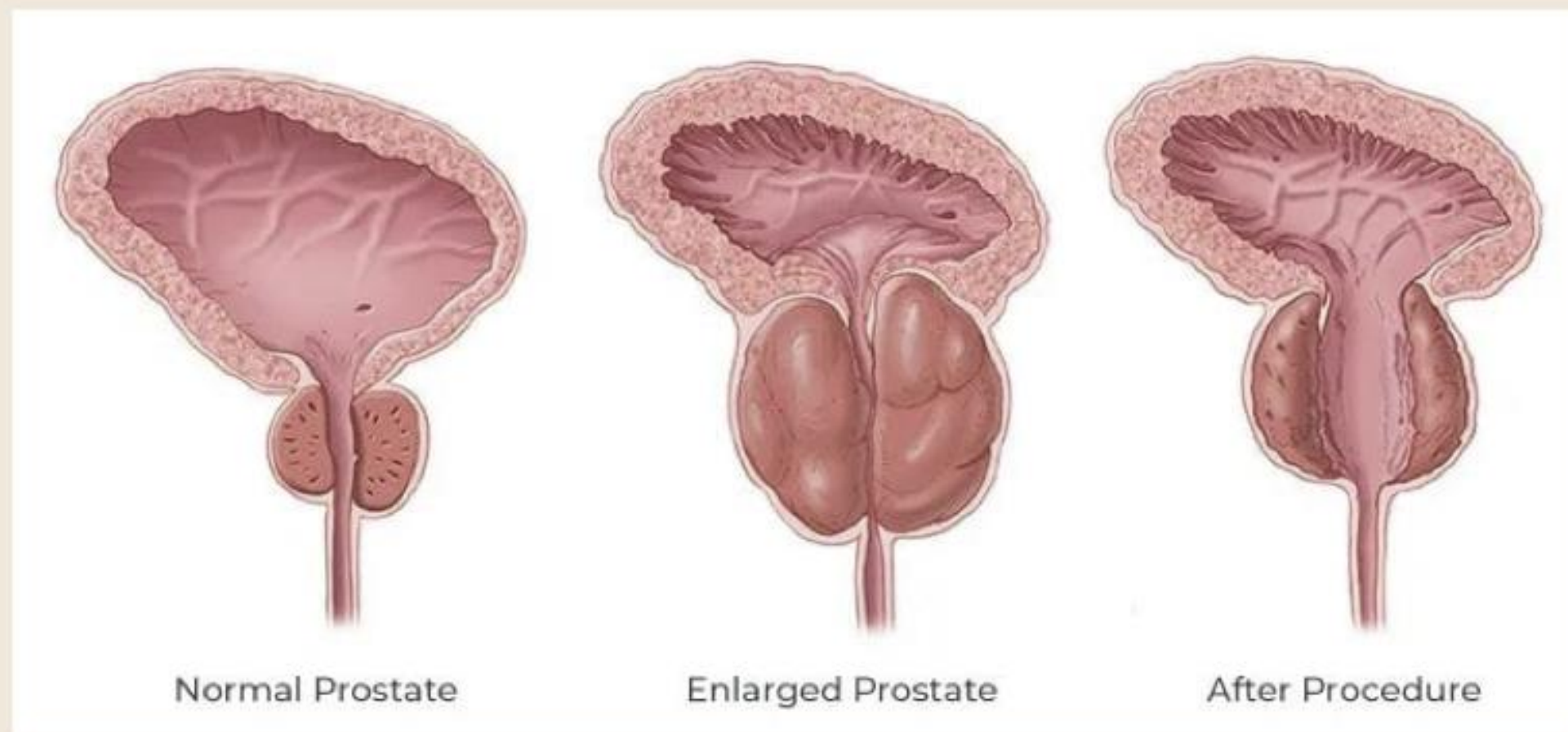
การผ่าตัดส่องกล้องทางท่อปัสสาวะเพื่อขูดเนื้อต่อมลูกหมาก
ใช้รักษาภาวะต่อมลูกหมากโต (BPH)



TUR-P

Transurethral Resection of Prostate

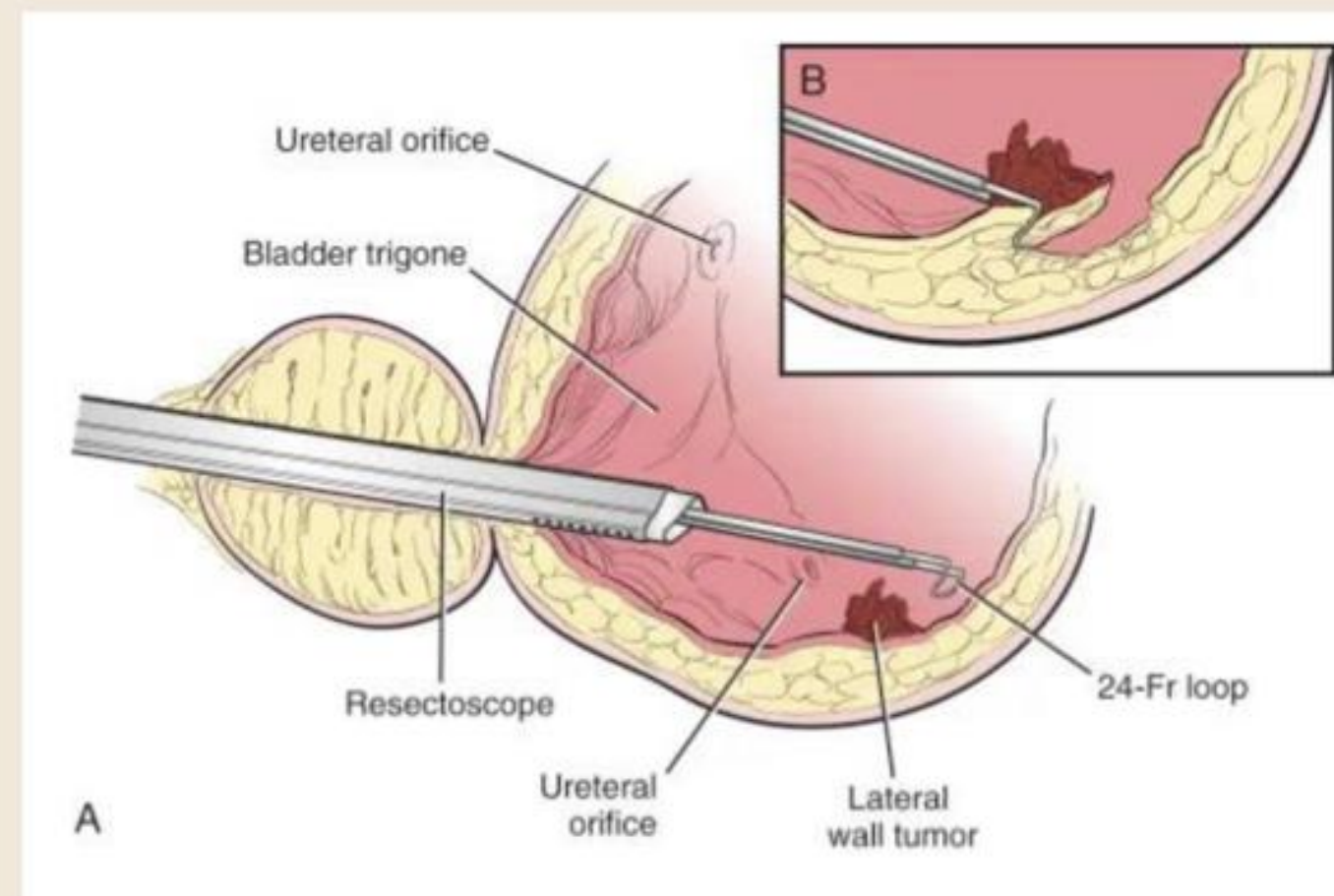
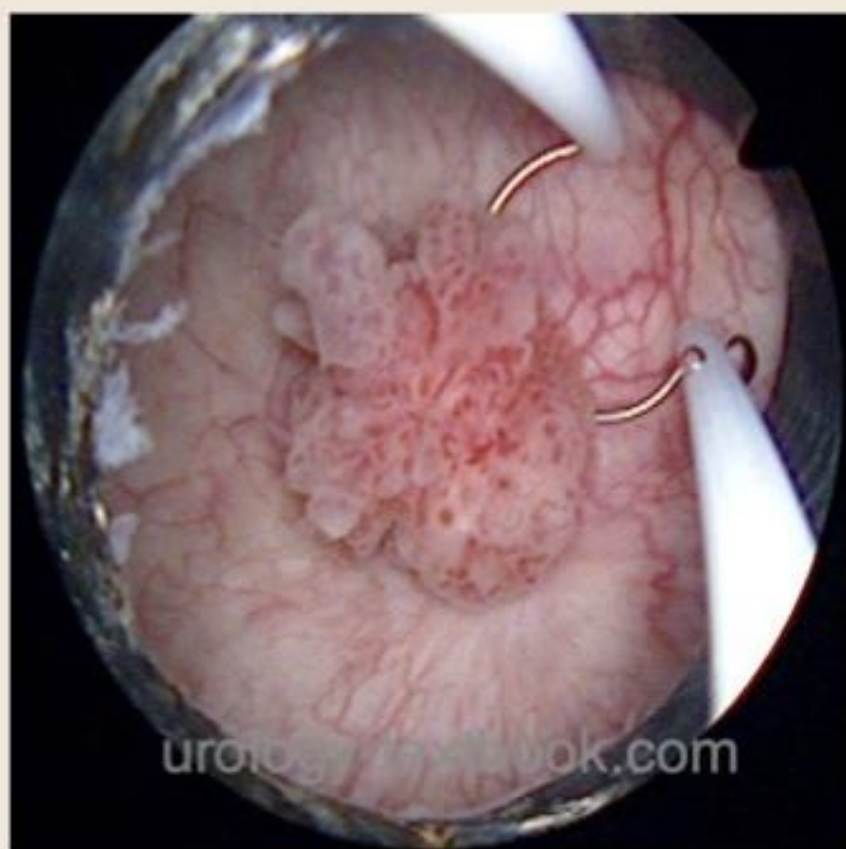
การผ่าตัดส่องกล้องทางท่อปัสสาวะเพื่อขูดเนื้อต่อมลูกหมาก
ใช้รักษาภาวะต่อมลูกหมากโต (BPH)



TUR-BT

Transurethral Resection of *Bladder tumor*

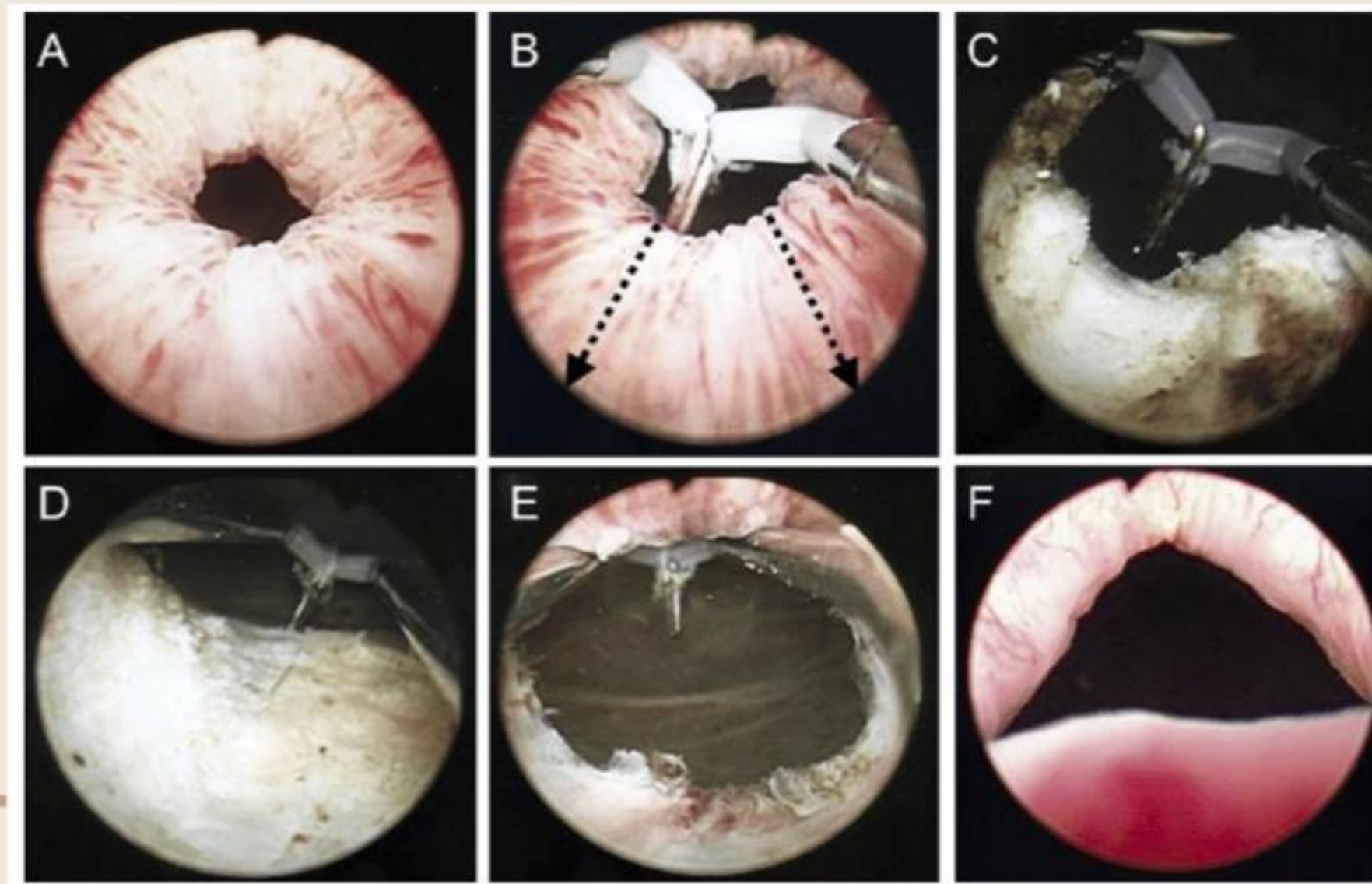
การผ่าตัดส่องกล้องทางท่อปัสสาวะเพื่อขูดเนื้องอกในกระเพาะปัสสาวะ
ใช้ในการรักษาโรคมะเร็งกระเพาะปัสสาวะ



TUI-BN

Transurethral *Incision* of *Bladder Neck*

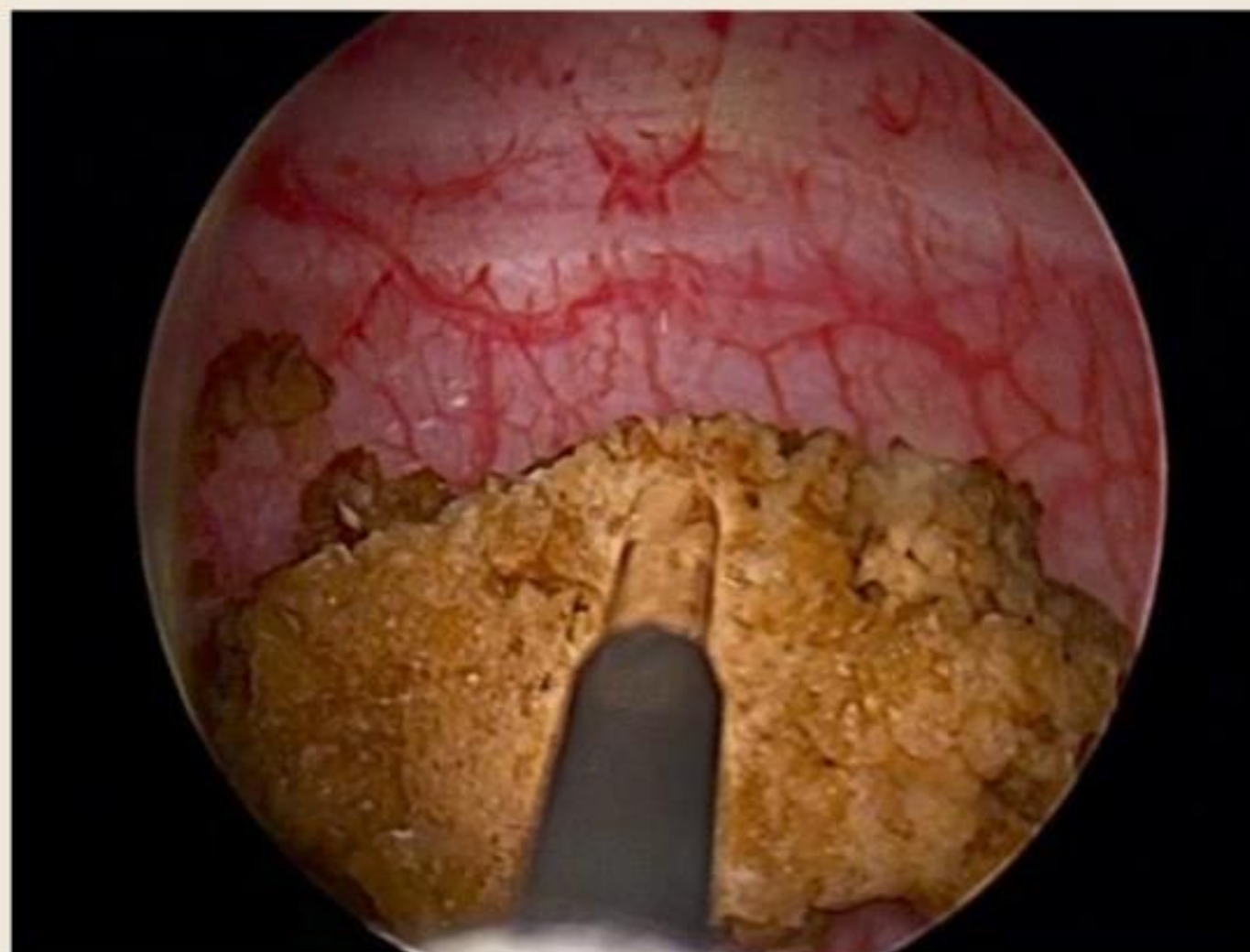
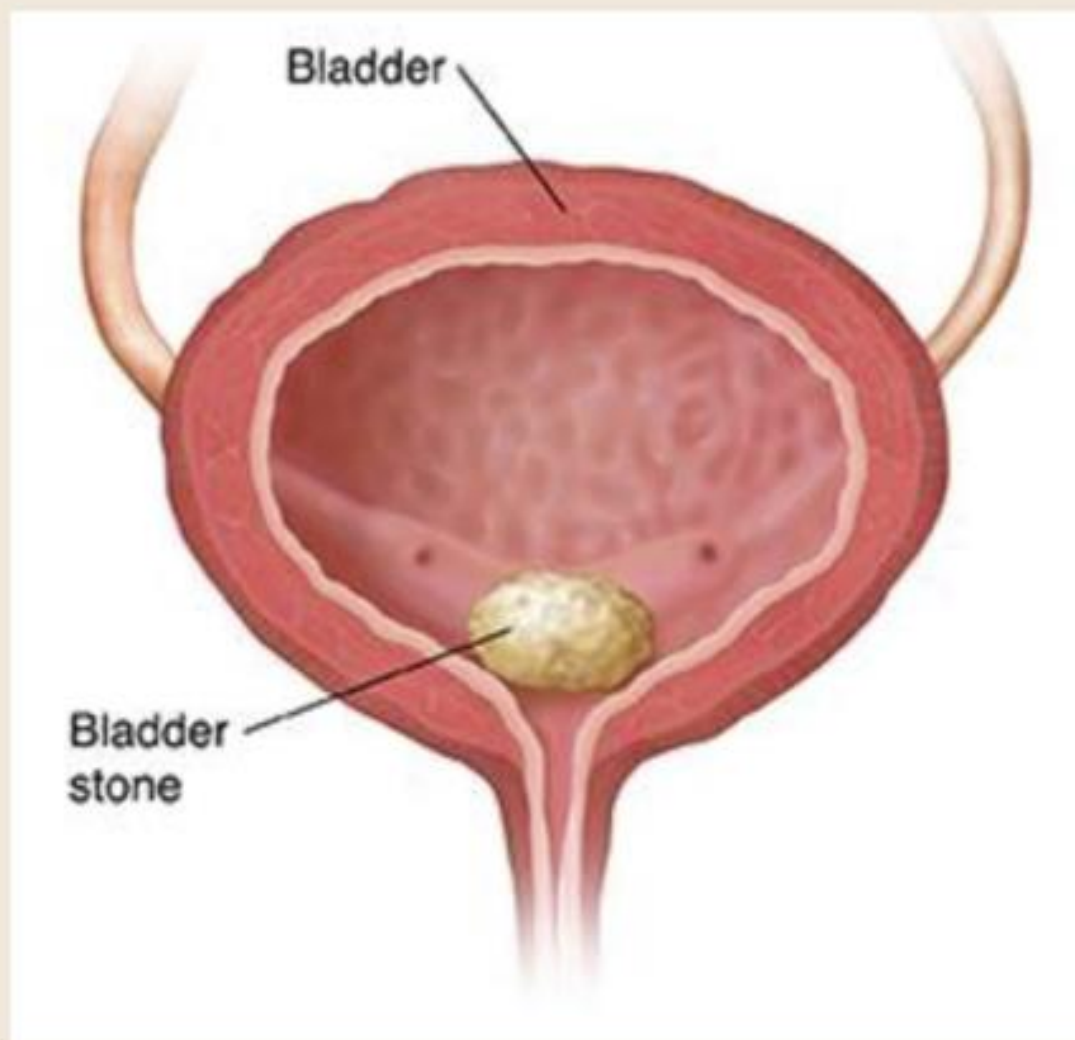
การผ่าตัดส่องกล้องทางท่อปัสสาวะกรีดคอคอของกระเพาะปัสสาวะ
ใช้ในการรักษาภาวะคอกระเพาะปัสสาวะตีบ (Contracture bladder neck: **CBN**)



TUL

Transurethral *Cystolitholapaxy*

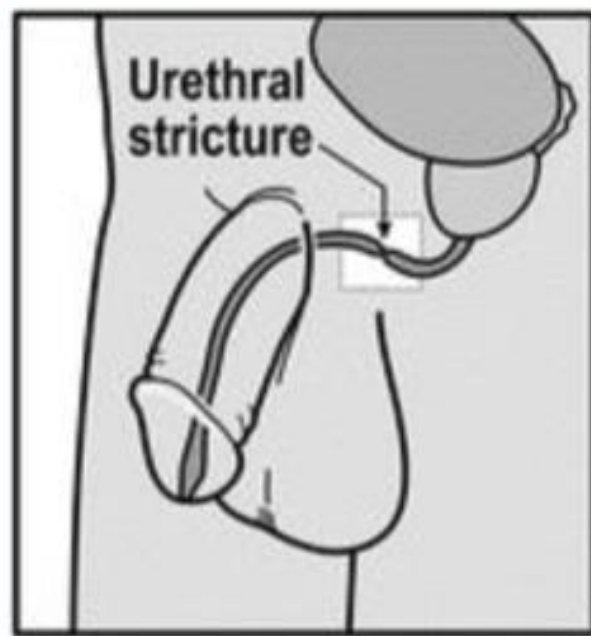
การผ่าตัดส่องกล้องทางท่อปัสสาวะเพื่อยิงสลายนิ่วในกระเพาะปัสสาวะ



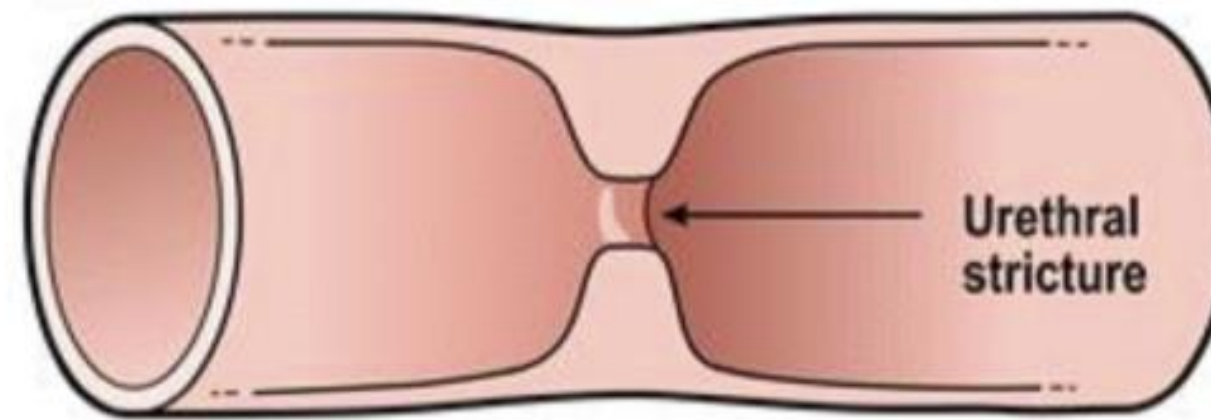
DVIU

Direct Vision Internal Urethrotomy

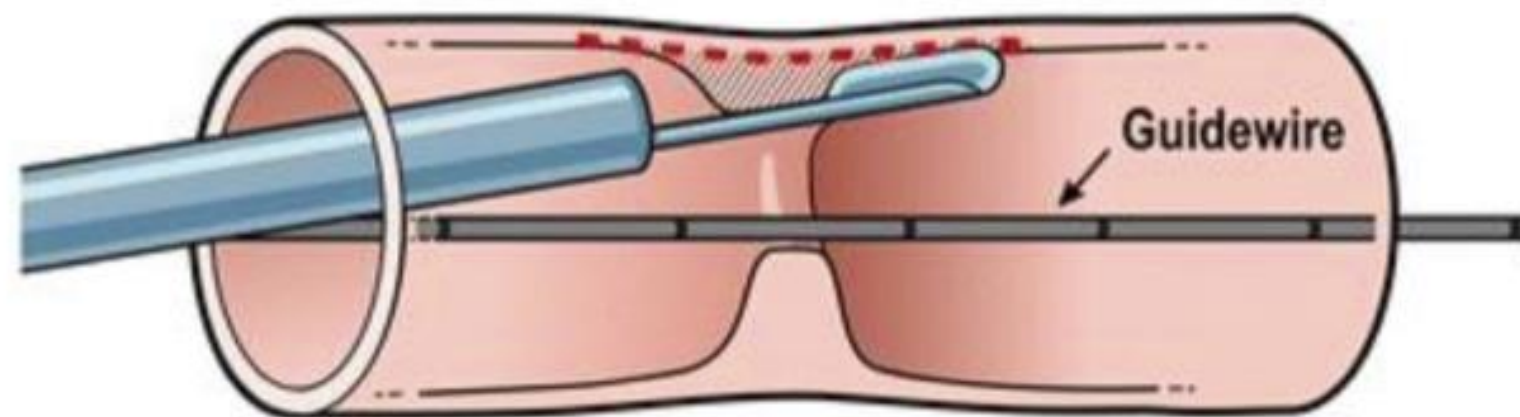
การผ่าตัดส่องกล้องทางท่อปัสสาวะเพื่อใช้มีดที่ติดอยู่กับกล้องกรีดขยายท่อปัสสาวะ
ใช้ในภาวะท่อปัสสาวะตีบ

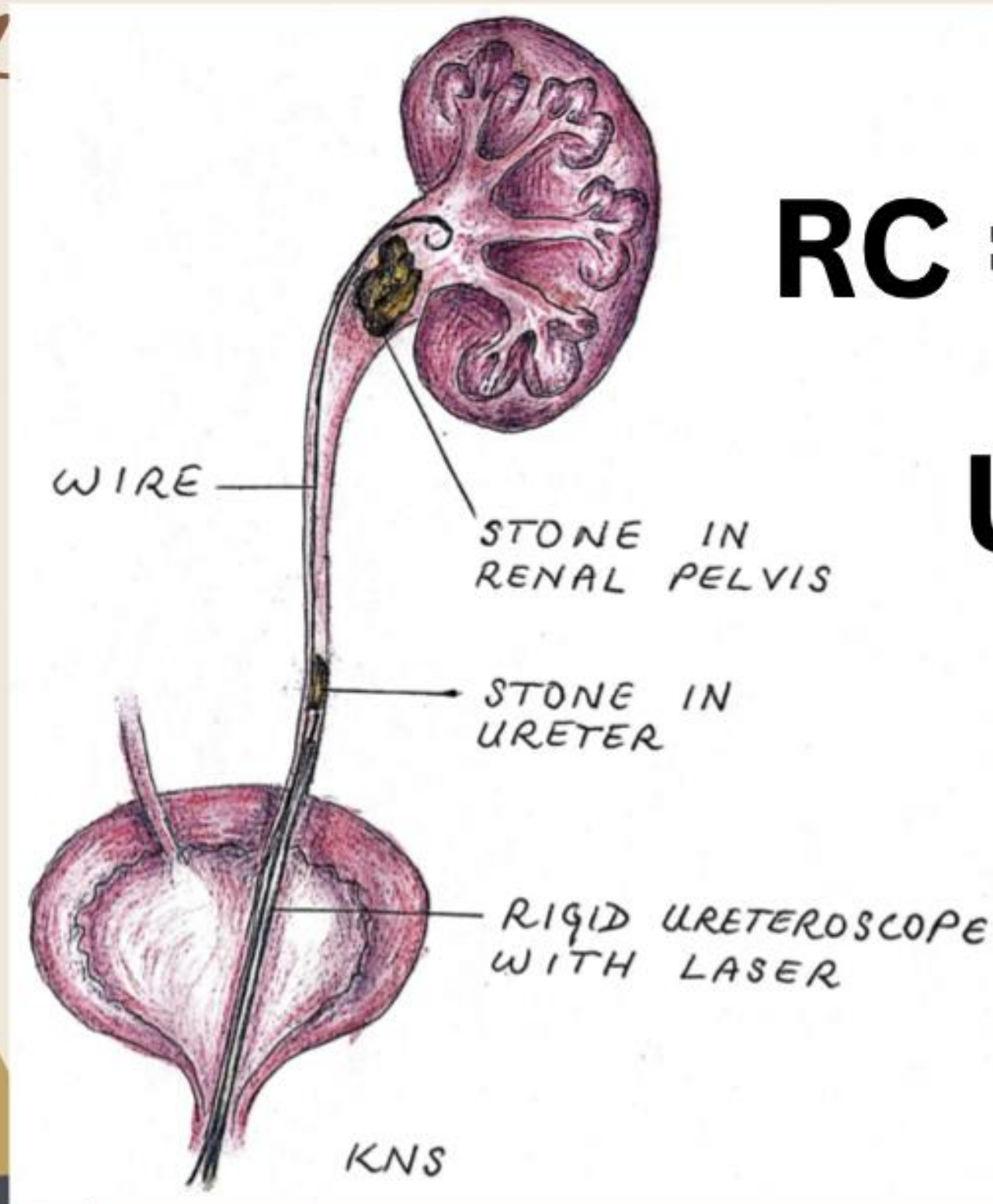


a



b





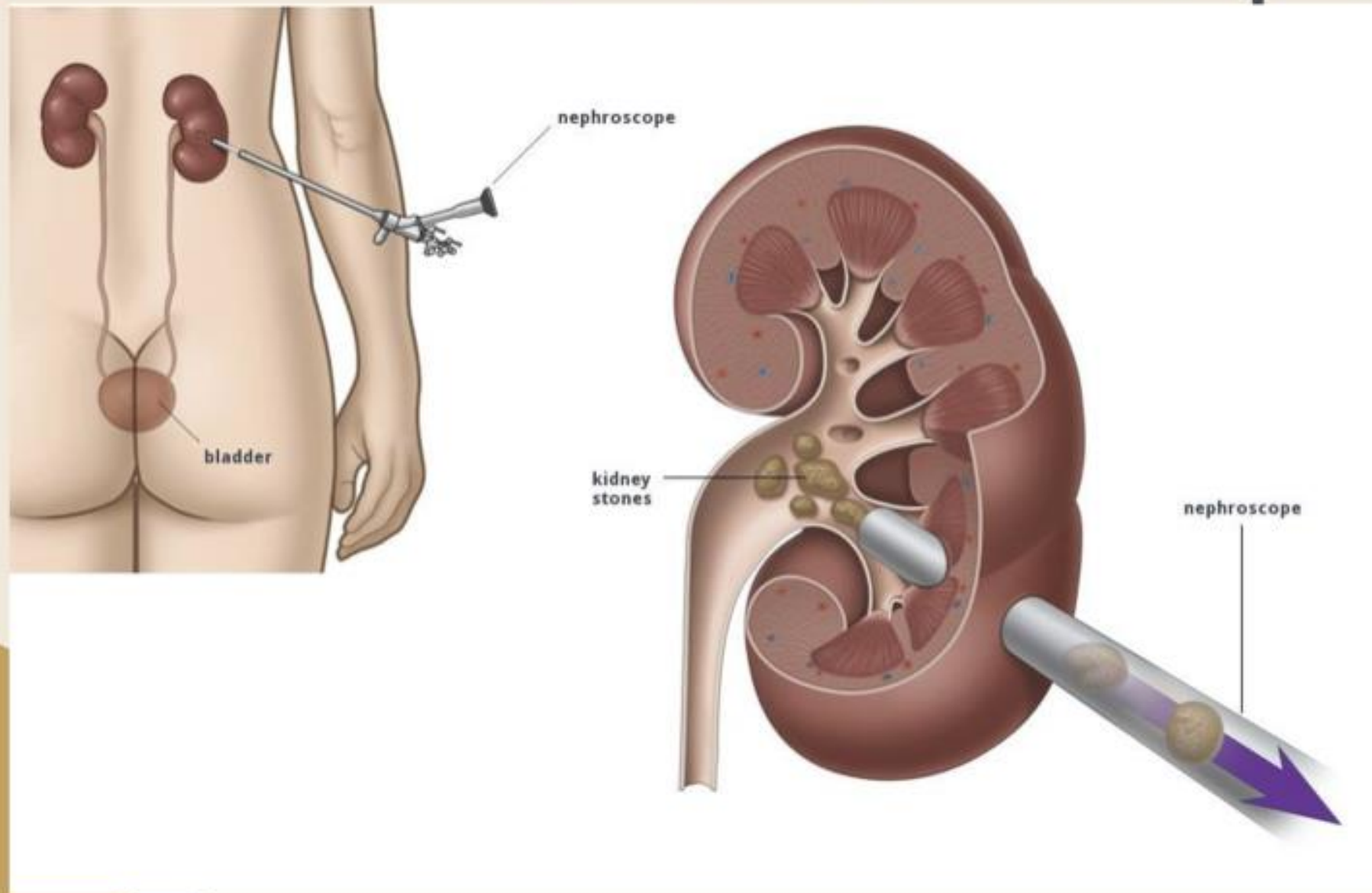
RC = Renal calculi

**UC = Ureteric
Calculi**

**VC = Vesical
calculi**

PCNL

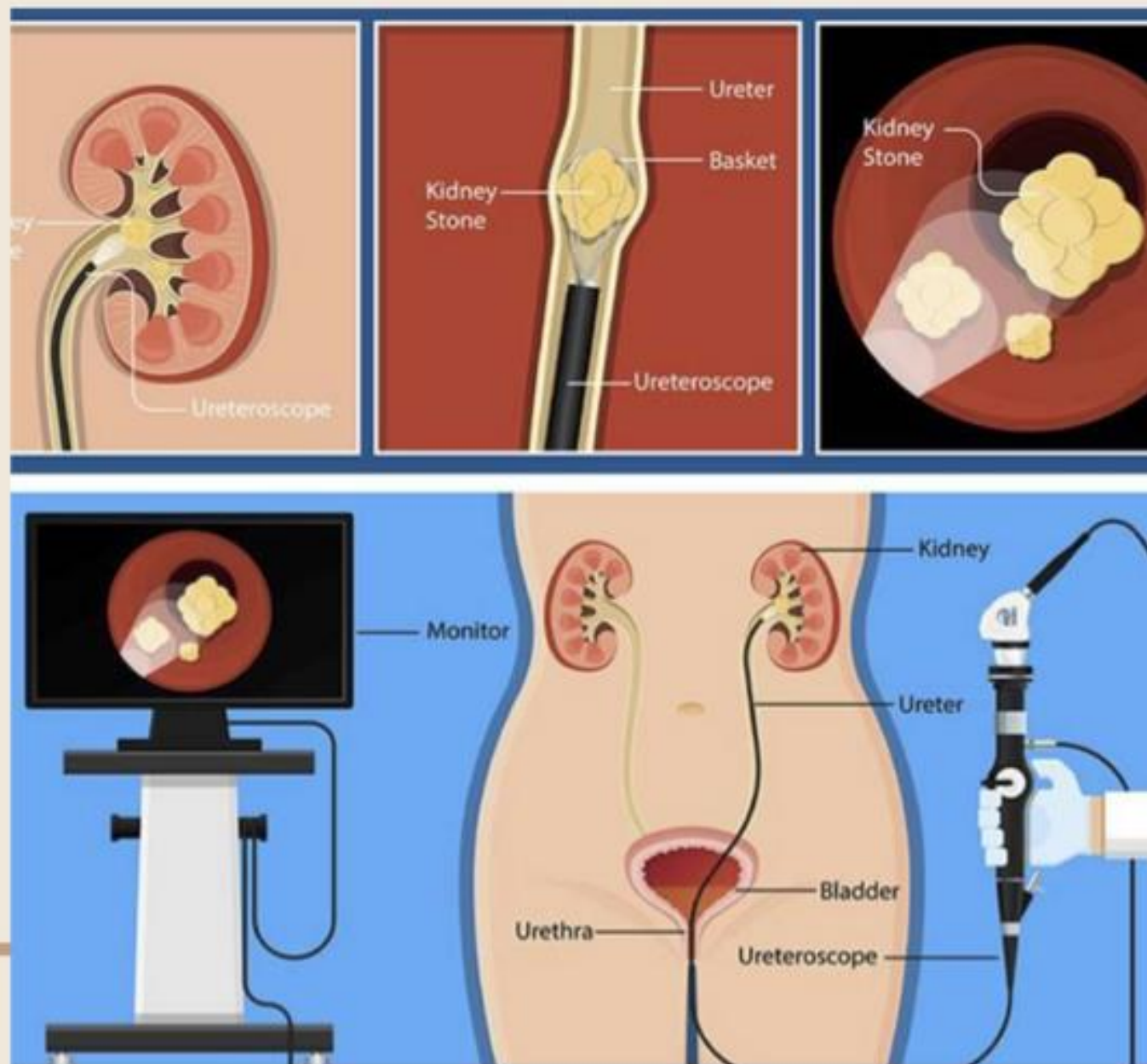
Percutaneous Nephrolithotomy



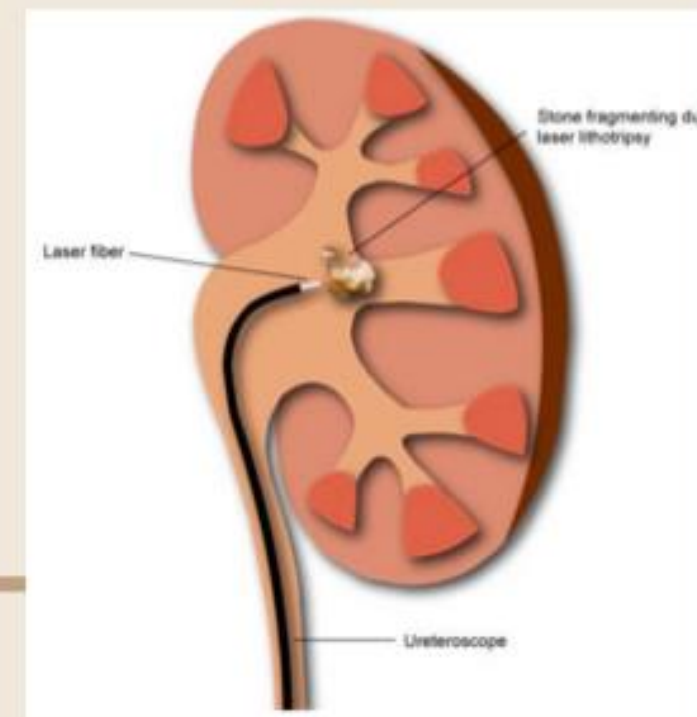
- การผ่าตัดส่องกล้องเพื่อสลาย
นิ่วในกรวยไต โดยการเจาะ
ผ่านผิวหนัง และเนื้อไต

RIRS

Retrograde IntraRenal Surgery



- การผ่าตัดส่องกล้องเข้าไปในกรวยไตเพื่อทำหัตถการในกรวยไต
 - เช่น ยิงสลายนิ่ว, ตัดเนื้องอก

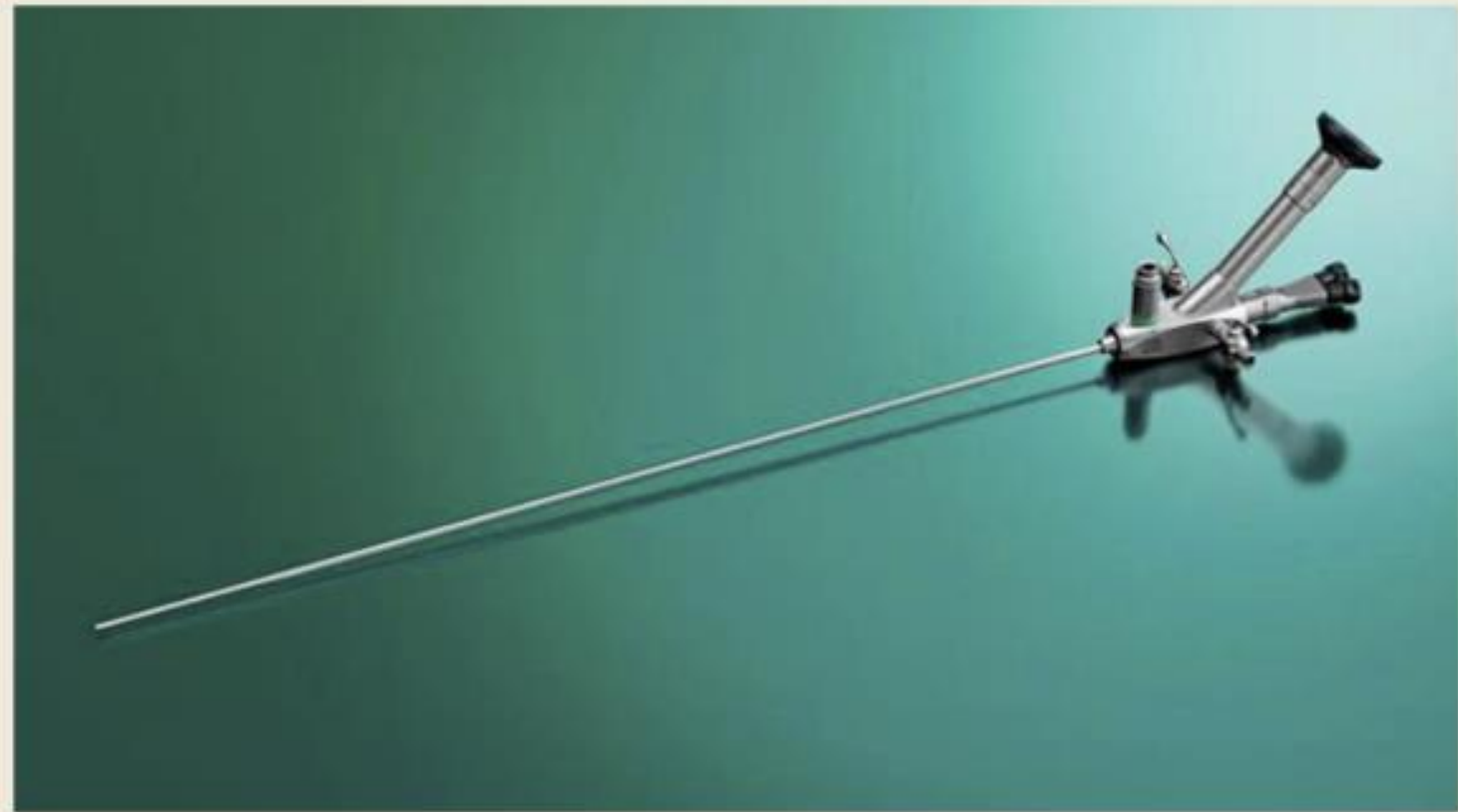




URSL

UreteroRenoScopic Lithotripsy

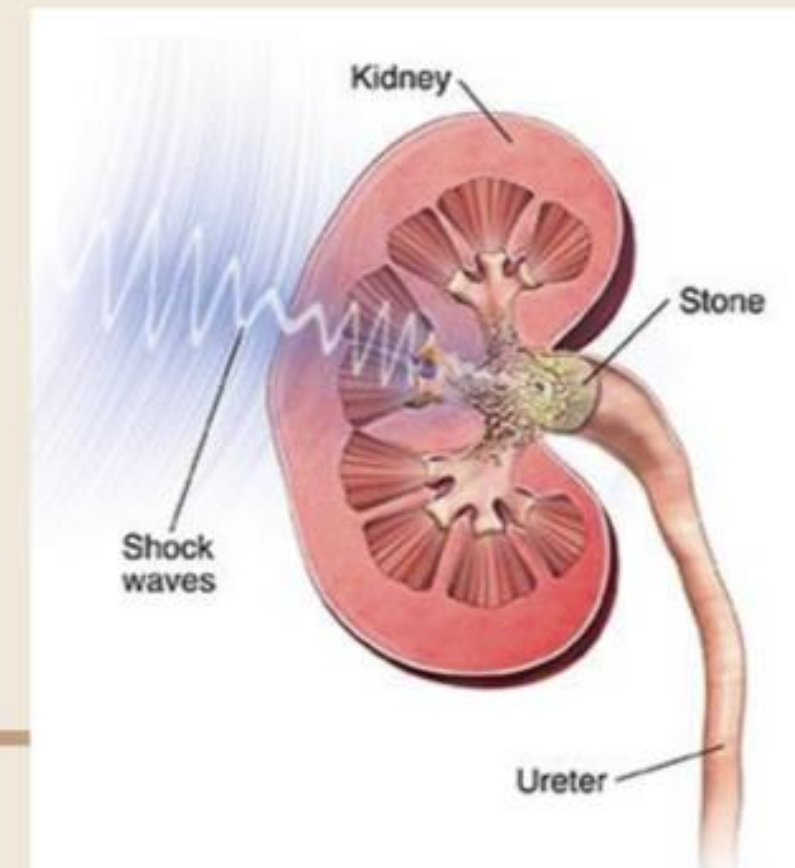
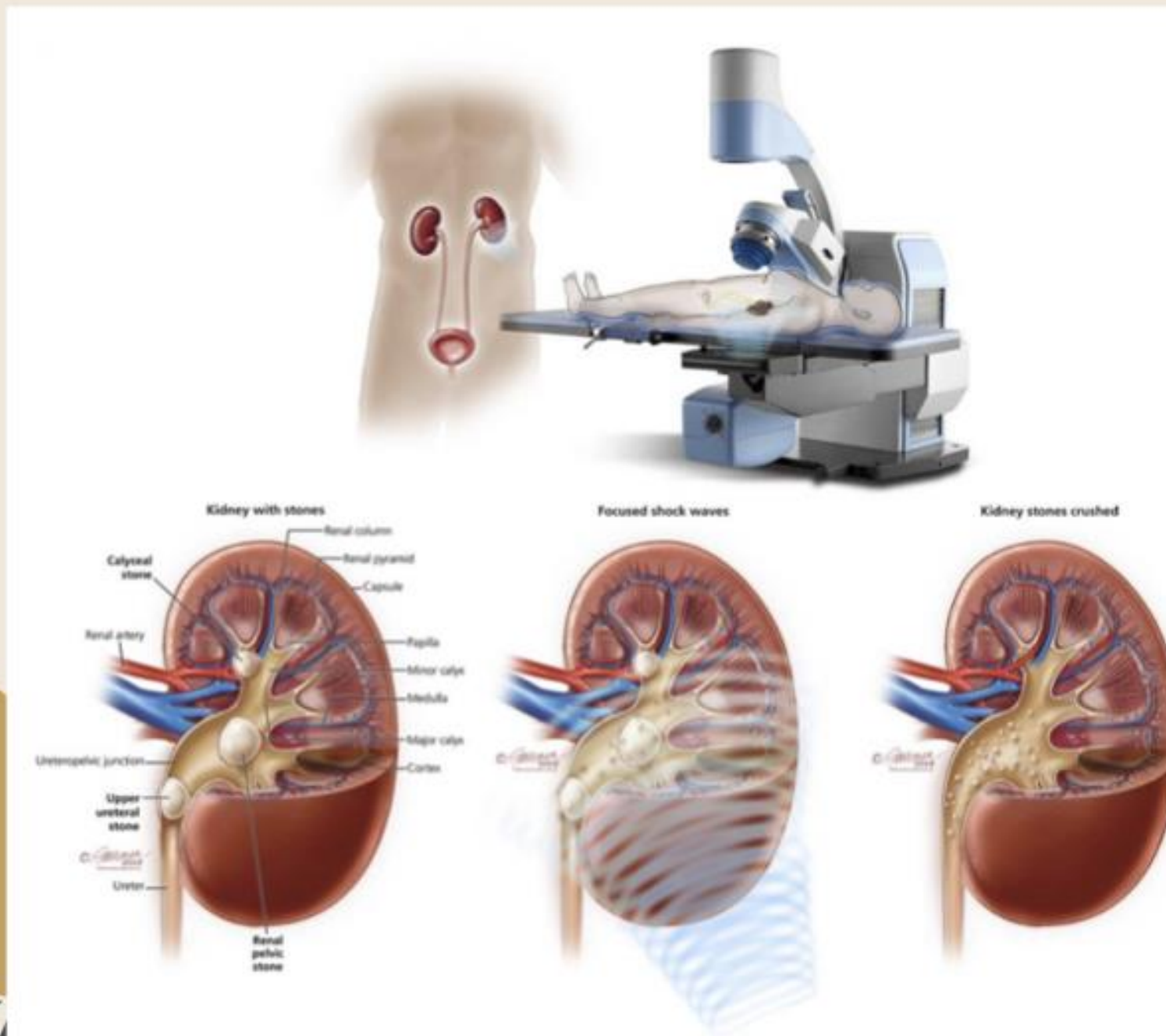
- การผ่าตัดส่องกล้องเพื่อยิงสลายนิ่วในท่อไต



ESWL

Extracorporeal Shock Wave Lithotripsy

- การสลายนิ่วด้วยคลื่นกระแทก
- ไม่ต้องดมยาสลบ ไม่ต้อง admit





CATHETERS

Foley catheter



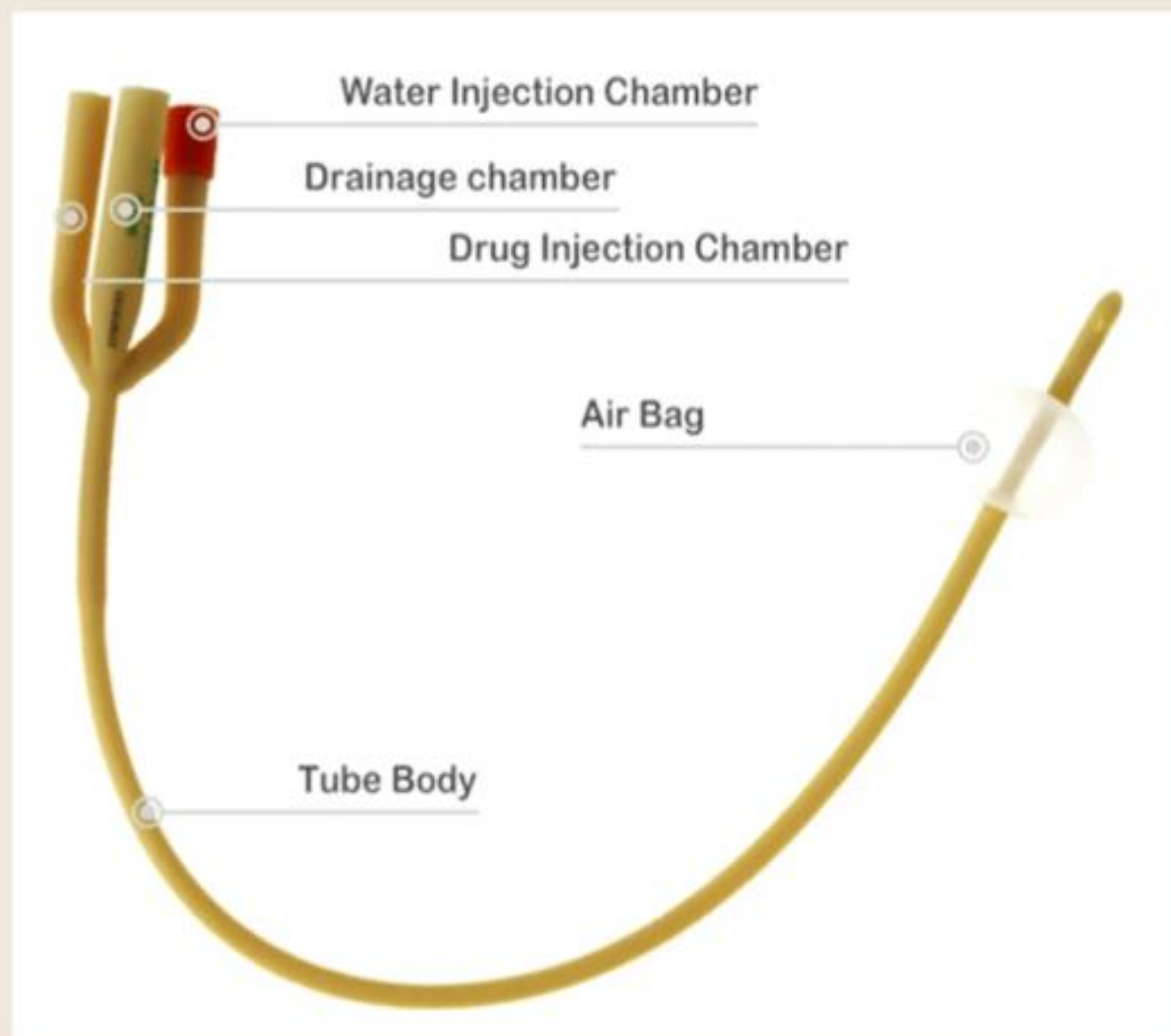
- ทำจาก latex
- สี่ = ขนาด = NG tube

Silicone Foley catheter



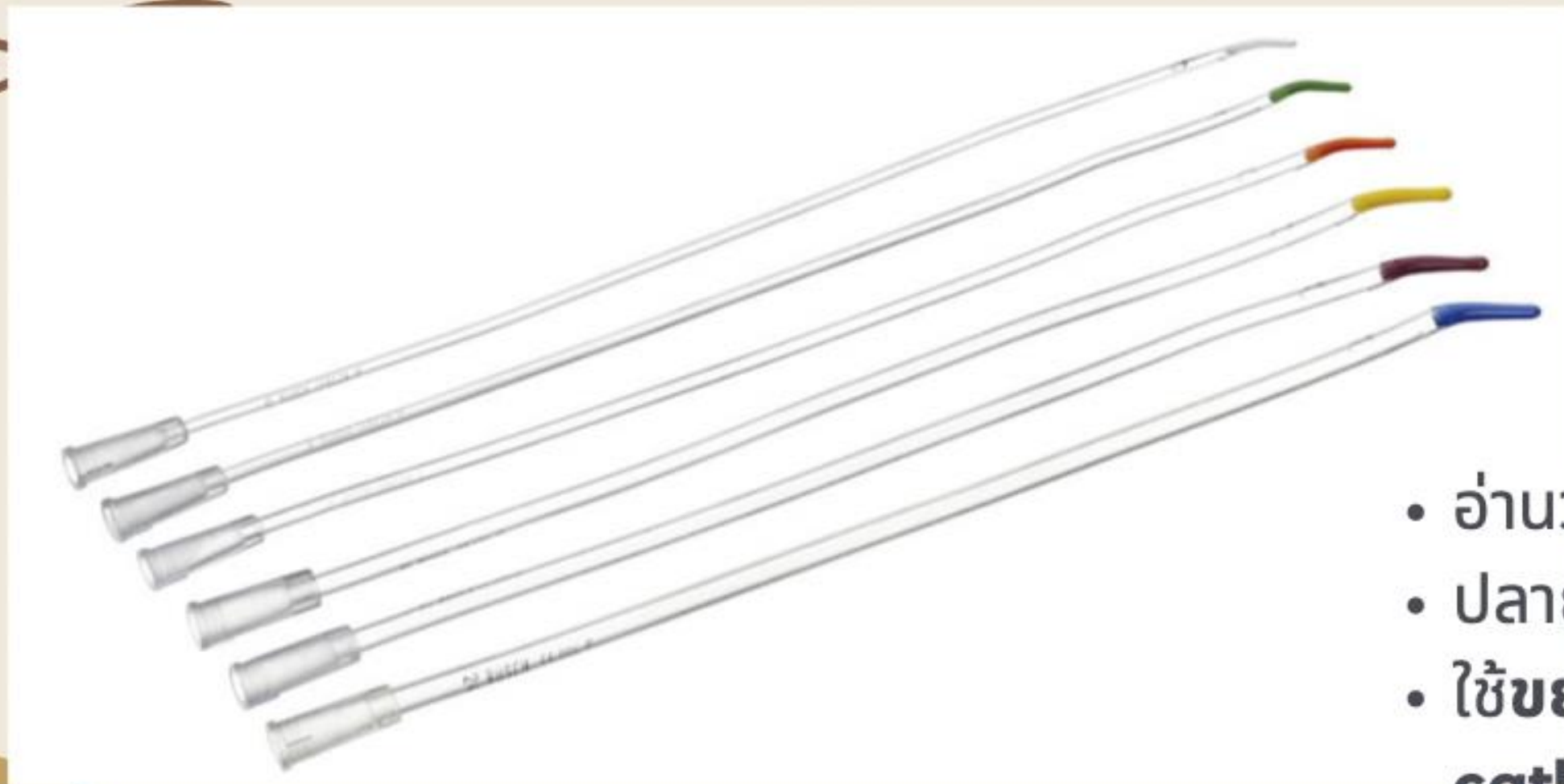
- ทำจาก silicone
- แพงกว่า
- อยู่ได้ 3 เดือน
- ซี = ขนาด = NG tube

3-way Foley catheter



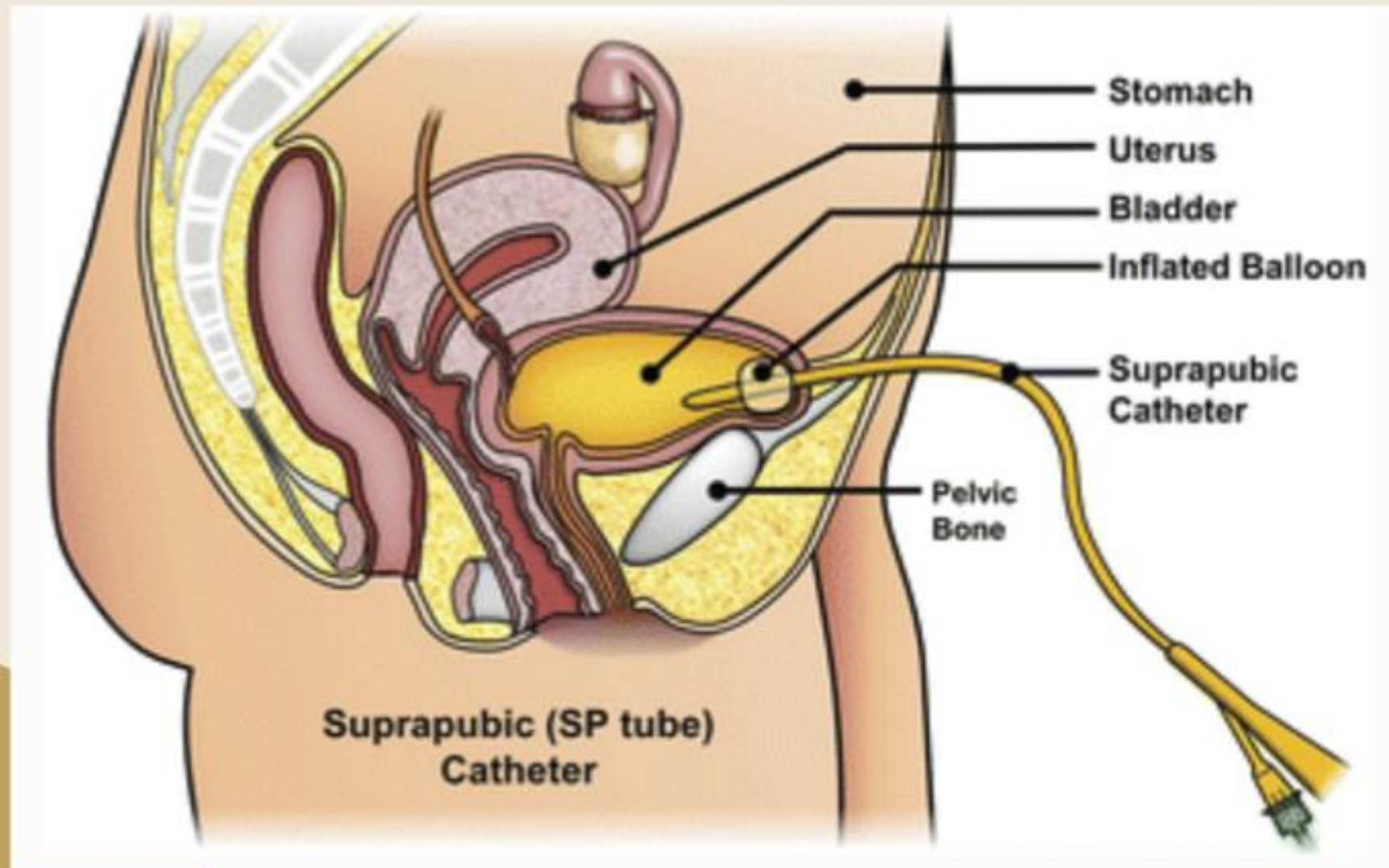
- Foley 3 ทาง
- ทางที่ 1 = balloon
- ทางที่ 2 รูใหญ่กว่า = ต่อ Urine bag
- ทางที่ 3 รูเล็กกว่า = ต่อ NSS (IV set)
- ใช้เมื่อ on **CBI = Continuous Bladder Irrigation**
- มีอยู่ที่ OR, ward URO

Tiemann catheter

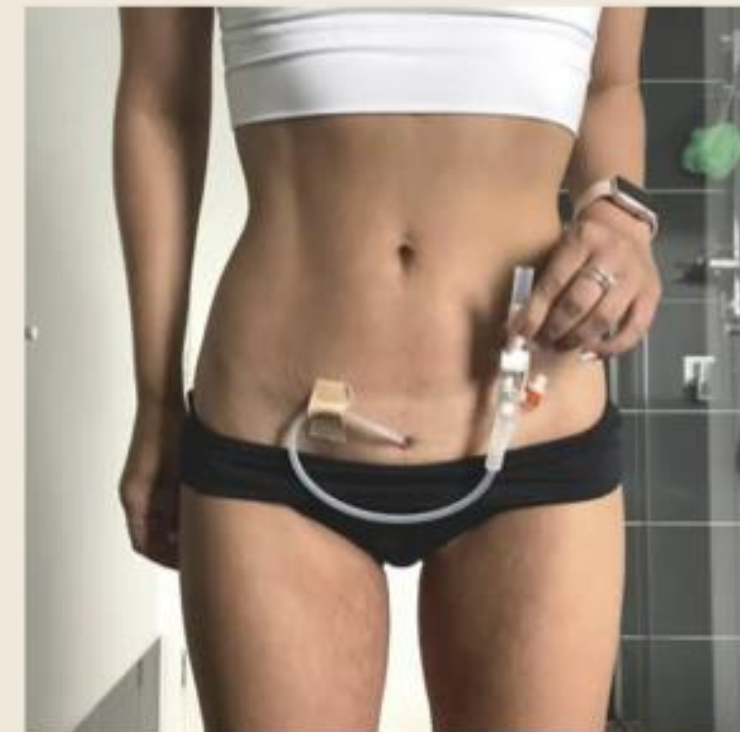


- อ่านว่า 'ทีแมน แคทเทเตอร์'
- ปลายแข็ง
- ใช้ขยายท่อปัสสาวะในเคส **difficult catheterization** ที่สงสัยสาเหตุจากท่อปัสสาวะตีบ
- มีอยู่ที่ OR, ward URO

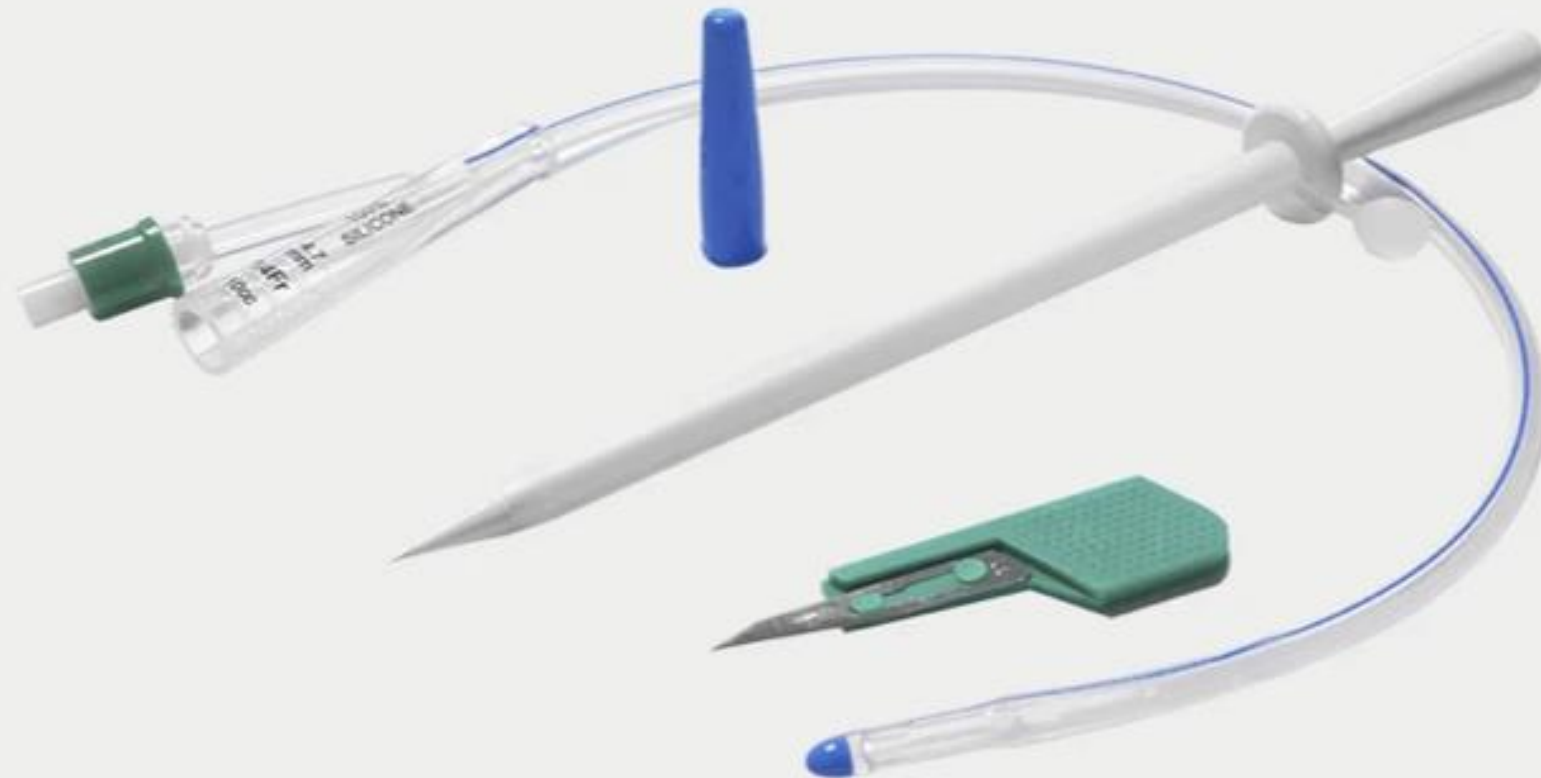
SPC : Suprapubic cystostomy



- การใส่สายสวนปัสสาวะในกระเพาะปัสสาวะ โดยเจาะผ่านทางผิวหนังบริเวณหน้าท้อง
- สายที่ใส่ = Foley catheter

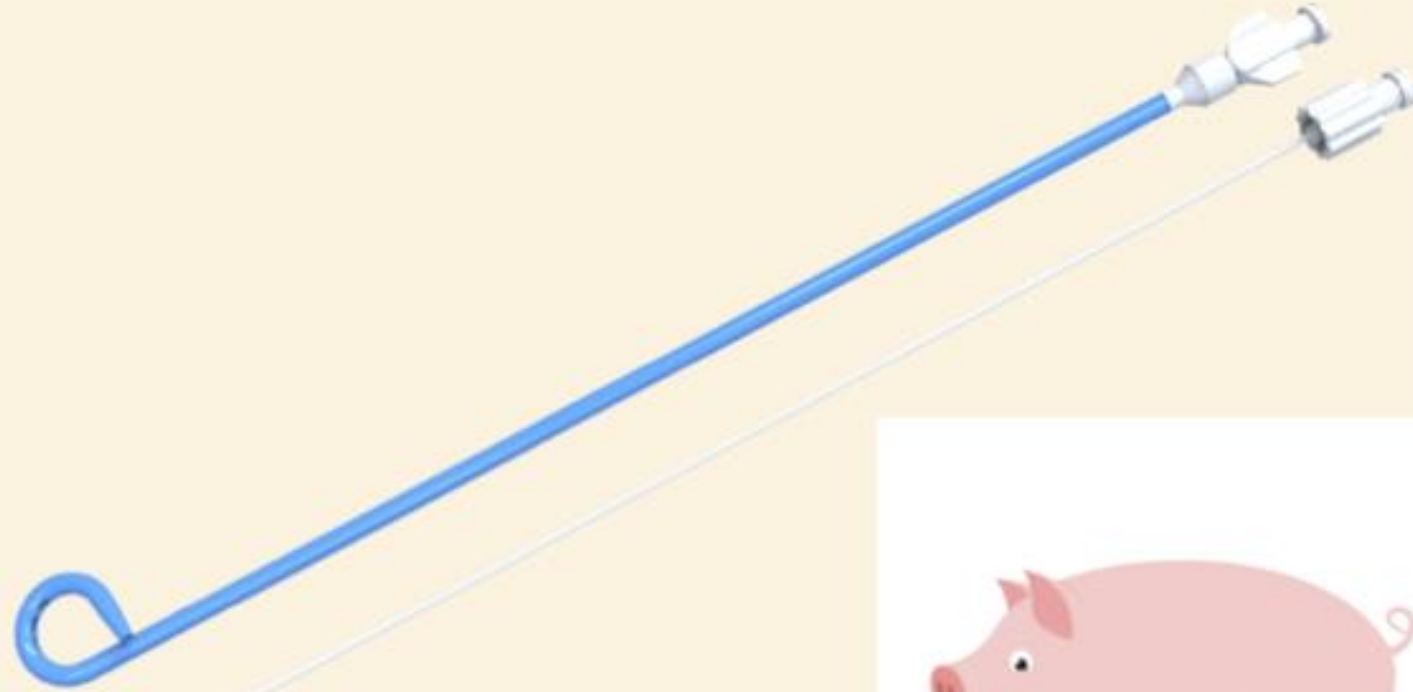


SPC : Suprapubic cystostomy



- **SPC set** ไว้สำหรับแทงสาย SPC
 - **Cystofix**
- ต้องใช้ **Ultrasound guided**
- สามารถทำได้ทั้ง **bedside/ER** ในผู้ป่วยที่ full bladder และไม่สามารถใส่สายสวนปัสสาวะทางท่อปัสสาวะได้
- มีอยู่ที่ **OR, ward URO**
- ปกติใช้ size 16 Fr หรือ 18 Fr

Pigtail catheter



- ลักษณะเด่นคือเป็นสายสีฟ้า (แล้วแต่ยี่ห้อ)
โผล่ออกมาจากผิวหนัง
- ถ้า drain urine = PCN
- ถ้า drain หลอดจากบริเวณต่างๆ = PCD
- ถ้า drain จากปอด = thoracostomy
- ถ้า drain จากถุงน้ำดี = cholecystostomy



International Urology Institute of the Land for Everyone

Question?



urosiriraj



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Thank you